

Rules of Differentiation: From Intuition to Mastery

A Step-by-Step Guide for HKDSE M1/M2 Students

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1 Limit

1.1 Limit as a substitution in function

Let us start with basic arithmetic. If you go to a market and buy x apples for \$3 each, the total cost is given by a simple rule: $f(x) = 3x$. If we want to know the cost of buying exactly 4 apples, we replace x with the number 4. We get $f(4) = 3(4) = 12$.

In mathematics, a **function** is simply a machine that takes an input number, follows a rule, and spits out an output number. When we ask, "What happens to the output when our input gets closer and closer to 4?", we are talking about a **limit**.

Why do we need a fancy new word like "limit" if we can just plug the number in? For well-behaved functions—like the polynomials you learned in Junior Secondary algebra—there are no traps, no division by zero, and no broken rules. The height of the graph at a point is exactly where the curve seems to be heading. Therefore, the most basic and intuitive way to find a limit is simply to substitute the number into the function!

Definition 1.1: Limit via Direct Substitution

Let $f(x)$ be a polynomial or a well-behaved algebraic function where no mathematical rules (like dividing by zero or taking the square root of a negative number) are violated at the point $x = c$.

The **limit** of $f(x)$ as x approaches c , written as:

$$\lim_{x \rightarrow c} f(x)$$

is evaluated simply by replacing every instance of x with the value c :

$$\lim_{x \rightarrow c} f(x) = f(c)$$

Practical Example 1.1

Evaluate the limit of the expression $4x^2 - 5x + 1$ as the input x approaches the value 3. In symbols:

$$\lim_{x \rightarrow 3} (4x^2 - 5x + 1)$$

Step-by-Step Solution**Guided Thinking & Hints****Step 1: Check the function type.**

The expression $f(x) = 4x^2 - 5x + 1$ is a polynomial.
There are no fractions or square roots.

Key Hint: Always scan for trouble first! Ask yourself: "Will plugging in this number cause a division by 0?" If no, you are safe to proceed.

Step 2: Apply direct substitution.

Replace every variable x with the target value 3:

$$\lim_{x \rightarrow 3} (4x^2 - 5x + 1) = 4(3)^2 - 5(3) + 1$$

Key Hint: Notice that once you substitute the number 3 into the expression, the symbol $\lim_{x \rightarrow 3}$ MUST disappear. Never write the limit symbol after substituting!

Step 3: Perform basic arithmetic.

Calculate exponents first, then multiplication, then addition/subtraction:

$$\begin{aligned} &= 4(9) - 15 + 1 \\ &= 36 - 15 + 1 \\ &= 22 \end{aligned}$$

Key Hint: Follow BODMAS/PEMDAS carefully. Weak algebra often stems from doing multiplication before squaring: $4(3)^2 \neq 12^2$. Calculate $3^2 = 9$ first!

Exercise 1.1.1 (Evaluate a linear limit)**Problem Statement:**

Evaluate the following limit by direct substitution:

$$\lim_{x \rightarrow 5} (3x - 7)$$

Exercise 1.1.2 (Evaluate a quadratic limit)**Problem Statement:**

Evaluate the following limit:

$$\lim_{x \rightarrow -2} (x^2 + 4x - 3)$$

Exercise 1.1.3 (Geometry and Limits)**Problem Statement:**

The area of a circle with radius r is given by the formula $A(r) = \pi r^2$. Evaluate $\lim_{r \rightarrow 4} A(r)$ and explain in one sentence what this value represents physically.

Exercise 1.1.4 (Expanding before substituting)**Problem Statement:**

Consider the limit $\lim_{x \rightarrow 2} (x + 3)(2x - 1)$.

1. Evaluate the limit by directly substituting $x = 2$ into the factored form.
 2. Expand the algebraic expression first to get a quadratic polynomial, then evaluate the limit. Confirm both methods yield the same result.
-

Exercise 1.1.5 (Unknown constant in a limit)**Problem Statement:**

It is given that $\lim_{x \rightarrow 3}(2x^2 - kx + 5) = 20$, where k is a constant. Find the value of k .

Exercise 1.1.6 (General polynomial evaluation)**Problem Statement:**

Let $P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$ be a general polynomial of degree n , where $a_0, a_1, \dots, a_n$ are real constants. Explain why $\lim_{x \rightarrow c} P(x) = P(c)$ is always true for any real number c .

Exercise 1.1.7 (When direct substitution fails)**Problem Statement:**

Consider the function $f(x) = \frac{x^2-4}{x-2}$. Attempt to evaluate $\lim_{x \rightarrow 2} f(x)$ using direct substitution. Explain what algebraic anomaly occurs and why Definition 1.1 cannot be used directly here.

1.2 Limit as a sequence of points

In Section 1.1, we saw that for stable, continuous functions, finding where a function is heading is as simple as plugging the target number directly into the variable. But what happens if the function behaves like a bridge with a missing plank?

Consider the function $f(x) = \frac{x^2-4}{x-2}$. If we try to find the limit as $x \rightarrow 2$ by substitution, we get $\frac{2^2-4}{2-2} = \frac{0}{0}$. In basic arithmetic, division by zero is strictly forbidden. The machine crashes, and we cannot find a value.

However, remember that a limit is not interested in what happens *exactly at* the target point, but rather where the curve is heading *as we approach* it. To build an intuitive grasp, we can feed the function a sequence of numbers that crawl closer and closer to 2 from both sides—values slightly less than 2 (like 1.9, 1.99, 1.999) and values slightly greater than 2 (like 2.1, 2.01, 2.001). By observing the resulting outputs, we can confidently predict where the graph is heading.

Definition 1.2: Sequential Approximation of a Limit

Let $f(x)$ be a function defined on an open interval around $x = c$, except possibly at c itself.

The limit of $f(x)$ as x approaches c is L , written as $\lim_{x \rightarrow c} f(x) = L$, if for every sequence of input values $\{x_n\}$ that converges to c (with $x_n \neq c$), the corresponding sequence of output values $\{f(x_n)\}$ converges to L .

In practice, we check this convergence from two sides:

- **Left-Hand Sequence:** x values rising toward c from below (e.g., $c-0.1, c-0.01, c-0.001, \dots$)
- **Right-Hand Sequence:** x values falling toward c from above (e.g., $c+0.1, c+0.01, c+0.001, \dots$)

If both sequences of outputs head toward the exact same real number L , then the limit is L .

Practical Example 1.2

Estimate the value of the following limit using sequential approximation:

$$\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1}$$

Step-by-Step Solution

Guided Thinking & Hints

Step 1: Check substitution first.

Substituting $x = 1$ yields:

$$\frac{1^2 - 1}{1 - 1} = \frac{0}{0}$$

Because this is an indeterminate form, we must investigate numerically.

Step 2: Construct the sequences.

We choose values approaching 1 from the left (less than 1) and right (greater than 1):

- Left-hand inputs: 0.9, 0.99, 0.999
- Right-hand inputs: 1.1, 1.01, 1.001

Step 3: Calculate the output values.

For Left-hand inputs:

$$f(0.9) = \frac{0.9^2 - 1}{0.9 - 1} = \frac{-0.19}{-0.1} = 1.9$$

$$f(0.99) = \frac{0.99^2 - 1}{0.99 - 1} = \frac{-0.0199}{-0.01} = 1.99$$

$$f(0.999) = \frac{0.999^2 - 1}{0.999 - 1} = 1.999$$

For Right-hand inputs:

$$f(1.1) = \frac{1.1^2 - 1}{1.1 - 1} = \frac{0.21}{0.1} = 2.1$$

$$f(1.01) = \frac{1.01^2 - 1}{1.01 - 1} = 2.01$$

$$f(1.001) = \frac{1.001^2 - 1}{1.001 - 1} = 2.001$$

Step 4: State the conclusion.

Since both sequences of outputs head towards 2, we conclude:

$$\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} = 2$$

Key Hint: Always try substitution first! If it gives a real number, you are finished. If it gives $\frac{0}{0}$, it is a signal to use sequential sequences.

Key Hint: By keeping your steps consistent (decreasing the offset by a power of 10 each time), the outputs will form a highly visible trend.

Key Hint: Notice that as the inputs get closer to 1, the outputs from both sides get closer and closer to 2. They are squeezing the target value from both sides!

Key Hint: Even though $f(1)$ does not exist, the limit as x approaches 1 is exactly 2.

Exercise 1.2.1 (Linear sequence confirmation)**Problem Statement:**

For the function $f(x) = 4x - 3$, evaluate $\lim_{x \rightarrow 2} f(x)$ by calculating the function values at the left-hand inputs $x = 1.9, 1.99, 1.999$ and right-hand inputs $x = 2.1, 2.01, 2.001$.

Exercise 1.2.2 (Analyzing rational sequences)**Problem Statement:**

Let $g(x) = \frac{2x^2-8}{x-2}$.

1. Evaluate $g(2.001)$ and $g(1.999)$ using a calculator.
 2. What integer value do these calculations suggest for $\lim_{x \rightarrow 2} g(x)$?
-

Exercise 1.2.3 (The reciprocal sequence)**Problem Statement:**

Consider the function $h(x) = \frac{1}{x}$. We want to evaluate the limit of $h(x)$ as x gets extremely large. Evaluate $h(10)$, $h(100)$, $h(1000)$, and $h(1000000)$. What real number does this sequence of outputs converge to as the inputs grow without bound?

Exercise 1.2.4 (Interpreting calculator round-off errors)**Problem Statement:**

A student attempts to estimate $\lim_{x \rightarrow 0} \frac{\sqrt{x+9}-3}{x}$ by plugging in extremely small values of x . Their calculator displays the following:

x	10^{-4}	10^{-8}	10^{-16}
$f(x)$	0.166662	0.166667	0.000000

Explain why the output for 10^{-16} drops suddenly to 0, and state the actual estimated value of the limit based on the first two terms.

Exercise 1.2.5 (Harmonic progression limit)**Problem Statement:**

Consider the sequence of points $x_n = \frac{1}{n}$ for $n = 1, 2, 3, \dots$. Explain why this sequence converges to 0 as $n \rightarrow \infty$. Use this to explain intuitively why any term of the form $\frac{k}{n^p}$ (where k is a constant and $p > 0$) must head to 0 as n grows infinitely large.

Exercise 1.2.6 (Broken sequences)**Problem Statement:**

Let $f(x) = \frac{|x-3|}{x-3}$. Evaluate $f(2.999)$ and $f(3.001)$. Explain why a single limit value $\lim_{x \rightarrow 3} f(x)$ does not exist based on these two results.

Exercise 1.2.7 (The oscillation limit failure)**Problem Statement:**

Let $f(x) = \cos\left(\frac{\pi}{x}\right)$.

1. Let $x_n = \frac{1}{2n}$ for $n = 1, 2, 3$. Evaluate $f(x_1)$, $f(x_2)$, and $f(x_3)$.
 2. Let $y_n = \frac{1}{2n+1}$ for $n = 1, 2, 3$. Evaluate $f(y_1)$, $f(y_2)$, and $f(y_3)$.
 3. Why does this prove that the sequential limit as $x \rightarrow 0$ cannot be determined?
-

1.3 Allowing discontinuity at the limit point

In Section 1.2, we discovered a fascinating mathematical phenomenon: a function can have a completely predictable limit at a point even when plugging that point into the function results in a math error like $\frac{0}{0}$. This brings us to one of the most important realizations in calculus: **the journey and the destination are two entirely separate concepts**.

To understand this intuitively, imagine you are driving along a newly built highway toward Exit 5. Every road sign, lane marking, and GPS signal confirms that the road leads directly to Exit 5. However, when you arrive exactly at Exit 5, you find that the off-ramp is blocked by construction (an undefined point), or perhaps a detour redirects you instantly to an overpass 10 meters above (a displaced value).

Does the blocked ramp change the fact that the highway *was leading* toward Exit 5? Of course not! In mathematics, a **discontinuity** is simply a "hole", a "jump", or a "break" in the graph at a specific input $x = c$. The **limit** only measures where the path *leads* as x gets infinitesimally close to c . It has no power to control, and completely ignores, what actually happens at the exact instant $x = c$.

Definition 1.3: Independence of Limits and Function Values

Let $f(x)$ be a function defined on an open interval around $x = c$, except possibly at c itself.

The existence and numerical value of the limit:

$$\lim_{x \rightarrow c} f(x) = L$$

is completely independent of the actual function value at $x = c$, denoted as $f(c)$.

When analyzing a function at a point $x = c$, exactly one of three structural behaviors will occur:

1. **Continuous:** $f(c)$ is defined, and $\lim_{x \rightarrow c} f(x) = f(c)$. (The path lands exactly where the dot is plotted).
2. **Removable Discontinuity (Hole):** $f(c)$ is undefined, yet $\lim_{x \rightarrow c} f(x) = L$ exists. (The path leads to a clear point, but the dot is missing).
3. **Displaced Discontinuity:** $f(c)$ is defined, but $\lim_{x \rightarrow c} f(x) \neq f(c)$. (The path leads to height L , but an isolated dot is plotted at a different height).

Practical Example 1.3

Consider the piecewise-defined function:

$$f(x) = \begin{cases} \frac{x^2-9}{x-3} & \text{if } x \neq 3 \\ 10 & \text{if } x = 3 \end{cases}$$

Evaluate the function value $f(3)$ and the limit $\lim_{x \rightarrow 3} f(x)$. Determine whether they are equal and state the geometric meaning of your result.

Step-by-Step Solution

Step 1: Determine the actual function value $f(3)$.

Look at the condition for $x = 3$ in the piecewise definition. The rule explicitly states that when x is exactly 3, the output is 10:

$$f(3) = 10$$

Step 2: Set up the limit calculation.

The limit symbol $\lim_{x \rightarrow 3}$ means we only consider x values that are getting closer and closer to 3, but ****never**** equal to 3. Therefore, we must use the branch where $x \neq 3$:

$$\lim_{x \rightarrow 3} f(x) = \lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$$

Step 3: Simplify and evaluate the limit.

Since $x \rightarrow 3$ implies $x \neq 3$, the quantity $(x - 3)$ is non-zero. This allows us to safely cancel the common factor:

$$\begin{aligned} \lim_{x \rightarrow 3} \frac{(x-3)(x+3)}{x-3} \\ = \lim_{x \rightarrow 3} (x + 3) \end{aligned}$$

$$= 3 + 3 = 6$$

Step 4: Compare and interpret.

We observe our two distinct values:

$$\lim_{x \rightarrow 3} f(x) = 6 \quad \text{while} \quad f(3) = 10$$

Since $6 \neq 10$, the limit does not equal the function value. Geometrically, the graph of $f(x)$ is a straight line $y = x + 3$ with an empty hole at $(3, 6)$, and a single solid dot plotted above the hole at $(3, 10)$.

Guided Thinking & Hints

Key Hint: To find $f(c)$, never perform any limit or algebraic simplification! Just read the exact rule assigned to that specific number.

Key Hint: This is the secret to piecewise limits: always choose the formula that applies to the neighborhood around the point, ignoring the isolated value at the point itself.

Key Hint: Why can we cancel $(x - 3)$ here but not when plugging in $x = 3$ directly? Because in a limit, $x - 3$ is a very small number like 0.0001, never zero! Dividing a number by itself is always 1.

Key Hint: This is a classic "displaced point discontinuity". The curve behaves perfectly everywhere else, but has a single artificial jump at that exact coordinate.

Exercise 1.3.1 (Evaluating a displaced piecewise limit)**Problem Statement:**

Let the function $g(x)$ be defined as:

$$g(x) = \begin{cases} 5x - 2 & \text{if } x \neq 2 \\ 20 & \text{if } x = 2 \end{cases}$$

Find the exact value of $g(2)$ and evaluate $\lim_{x \rightarrow 2} g(x)$.

Exercise 1.3.2 (Identifying holes in rational expressions)**Problem Statement:**

Consider the rational function $h(x) = \frac{x^2 - 25}{x - 5}$.

1. Explain why the function value $h(5)$ is completely undefined in arithmetic.
 2. Use sequential approximation (from Section 1.2) or algebraic factorization to find $\lim_{x \rightarrow 5} h(x)$.
 3. What coordinates mark the exact location of the "hole" on the graph of $y = h(x)$?
-

Exercise 1.3.3 (Repairing a missing point)**Problem Statement:**

An optical sensor records signal intensity using the formula $S(t) = \frac{t^2+3t-10}{t-2}$ for time $t \geq 0$, but the sensor experiences a system glitch that fails to record any data at exactly $t = 2$ seconds.

1. Evaluate the limit of the signal intensity as t approaches 2 seconds.
 2. Construct a new piecewise function, $S_{\text{repaired}}(t)$, that fills the missing data gap so that the signal becomes completely continuous at $t = 2$.
-

Exercise 1.3.4 (Tuning an unknown constant for continuity)**Problem Statement:**

In HKDSE M1/M2 exam questions, you are often asked to make a broken curve connect smoothly. Consider the function:

$$f(x) = \begin{cases} 2x^2 - 3x + k & \text{if } x \neq 4 \\ 15 & \text{if } x = 4 \end{cases}$$

Find the exact value of the constant k such that $\lim_{x \rightarrow 4} f(x) = f(4)$, thereby removing the discontinuity.

Exercise 1.3.5 (The universal removable hole theorem)**Problem Statement:**

Let $P(x)$ and $Q(x)$ be polynomials such that both $P(c) = 0$ and $Q(c) = 0$. Suppose $Q(x)$ can be written as $Q(x) = (x - c) \cdot R(x)$ where $R(c) \neq 0$. If $P(x)$ can also be factored as $P(x) = (x - c) \cdot K(x)$, prove algebraically that the rational function $F(x) = \frac{P(x)}{Q(x)}$ has a valid limit as $x \rightarrow c$, and express this limit value in terms of the polynomial factors K and R .

Exercise 1.3.6 (Why step functions cannot be repaired by substitution)**Problem Statement:**

Consider the step function:

$$f(x) = \begin{cases} 3 & \text{if } x < 1 \\ 8 & \text{if } x \geq 1 \end{cases}$$

A student claims: *"We can make this function continuous at $x = 1$ just by changing the definition at $x = 1$ to be the average of the two sides, so we set $f(1) = 5.5$."* Explain why the student's method fails to make the limit exist, using the concepts of left-hand and right-hand approaches.

Exercise 1.3.7 (The Dirichlet Function and nowhere-continuity)**Problem Statement:**

Consider the classic Dirichlet function $D(x)$, defined over all real numbers as:

$$D(x) = \begin{cases} 1 & \text{if } x \text{ is a rational number (can be written as a fraction)} \\ 0 & \text{if } x \text{ is an irrational number (like } \sqrt{2} \text{ or } \pi) \end{cases}$$

1. Choose any target point $x = c$. Explain why any tiny neighborhood around c will always contain an infinite number of rational points AND an infinite number of irrational points.
 2. Using the definition of sequential limits from Section 1.2, prove that $\lim_{x \rightarrow c} D(x)$ does not exist for ANY real number c .
 3. Conclude what this implies about the graph of $D(x)$: can it ever be drawn, even locally, without lifting your pencil?
-

1.4 Limit of rational function at finite points: Factorization application

In our previous discussions, we established that when direct substitution into a function yields the indeterminate form $\frac{0}{0}$, the limit may still exist as a perfectly well-behaved number. Up until now, our primary weapon for finding that number was numeric sequential approximation—plugging in values like 1.999 and 2.001 on a calculator. While intuitive, relying on calculators is tedious, prone to decimal rounding errors, and lacks mathematical rigor. How can we solve $\frac{0}{0}$ cleanly using algebra?

The key lies in a fundamental truth from Junior Secondary algebra: the ****Factor Theorem****. If substituting $x = c$ into a polynomial $P(x)$ results in 0, it is not an accident—it guarantees that $(x - c)$ is a hidden algebraic factor of $P(x)$! Therefore, when a rational function $\frac{P(x)}{Q(x)}$ produces $\frac{0}{0}$ at $x = c$, both the numerator and the denominator **MUST** contain the identical factor $(x - c)$. This common factor is the exact mathematical culprit causing the division by zero. By factoring both polynomials and canceling this common term, we effectively "remove the hole" from the algebraic expression, allowing us to evaluate the limit via simple direct substitution!

Theorem 1.4: Evaluation of Rational Limits via Factorization

Let $P(x)$ and $Q(x)$ be polynomials, and let c be a real number such that direct substitution yields an indeterminate form:

$$P(c) = 0 \quad \text{and} \quad Q(c) = 0$$

By the Factor Theorem, $(x - c)$ is a common factor of both polynomials, meaning there exist polynomials $P_1(x)$ and $Q_1(x)$ such that:

$$P(x) = (x - c)P_1(x) \quad \text{and} \quad Q(x) = (x - c)Q_1(x)$$

Provided that $x \rightarrow c$ implies $x \neq c$ (so $x - c \neq 0$), the common factor can be cancelled, yielding:

$$\lim_{x \rightarrow c} \frac{P(x)}{Q(x)} = \lim_{x \rightarrow c} \frac{(x - c)P_1(x)}{(x - c)Q_1(x)} = \lim_{x \rightarrow c} \frac{P_1(x)}{Q_1(x)} = \frac{P_1(c)}{Q_1(c)}$$

assuming $Q_1(c) \neq 0$. If $Q_1(c)$ is still 0, the factorization and cancellation process is repeated.

Student Proof Demonstration

Use this space to demonstrate the proof of the theorem above. Start from basic algebraic identities.

Practical Example 1.4

Evaluate the following limit using algebraic factorization:

$$\lim_{x \rightarrow 2} \frac{2x^2 - 3x - 2}{x^2 - 4}$$

Step-by-Step Solution**Step 1: Test direct substitution first.**

Substituting $x = 2$ into the expression gives:

$$\frac{2(2)^2 - 3(2) - 2}{2^2 - 4} = \frac{8 - 6 - 2}{4 - 4} = \frac{0}{0}$$

Since we get $\frac{0}{0}$, $(x - 2)$ must be a factor of both the top and the bottom.

Step 2: Factorize numerator and denominator.

For the numerator (using cross-method):

$$2x^2 - 3x - 2 = (2x + 1)(x - 2)$$

For the denominator (difference of two squares):

$$x^2 - 4 = (x - 2)(x + 2)$$

We rewrite the limit:

$$\lim_{x \rightarrow 2} \frac{(2x+1)(x-2)}{(x-2)(x+2)}$$

Step 3: Cancel the common factor.

Because $x \rightarrow 2$, we know $x \neq 2$, which guarantees that $(x - 2) \neq 0$. We can safely cancel $(x - 2)$:

$$= \lim_{x \rightarrow 2} \frac{2x+1}{x+2}$$

Step 4: Apply direct substitution.

Now that the division-by-zero trap is removed, substitute $x = 2$ directly:

$$= \frac{2(2)+1}{2+2}$$

$$= \frac{5}{4}$$

Guided Thinking & Hints

Key Hint: Never skip Step 1! If substitution does not give $\frac{0}{0}$, factoring is completely unnecessary. Here, $\frac{0}{0}$ confirms that $(x - 2)$ is hiding in both terms.

Key Hint: Weak at factoring quadratics? Use your calculator's equation solver! If the roots of $ax^2 + bx + c = 0$ are $x = 2$ and $x = -0.5$, the factors are $(x - 2)$ and $(2x + 1)$.

Key Hint: Notice that we MUST keep writing the $\lim_{x \rightarrow 2}$ symbol at every step until we actually substitute the number 2 for x .

Key Hint: Once you plug in the number, drop the limit symbol immediately! The final answer is a clean fraction: $\frac{5}{4}$ (or 1.25).

Exercise 1.4.1 (Factoring simple quadratics)**Problem Statement:**

Evaluate the following limit by factorizing both the numerator and the denominator:

$$\lim_{x \rightarrow 3} \frac{x^2 - x - 6}{x^2 - 9}$$

Exercise 1.4.2 (Piecewise continuity via difference of cubes)**Problem Statement:**

Consider the piecewise-defined function:

$$f(x) = \begin{cases} \frac{x^3 - 8}{x - 2} & \text{if } x \neq 2 \\ k & \text{if } x = 2 \end{cases}$$

1. Using the identity $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$, evaluate $\lim_{x \rightarrow 2} f(x)$.
 2. Find the value of the constant k such that $f(x)$ is continuous at $x = 2$.
-

Exercise 1.4.3 (Factoring by grouping for cubic polynomials)**Problem Statement:**

Evaluate the following limit without using a calculator:

$$\lim_{x \rightarrow 1} \frac{x^3 - 4x^2 + 5x - 2}{x^2 - 1}$$

Hint: Since $x = 1$ makes the numerator zero, use polynomial long division or synthetic division to divide the numerator by $(x - 1)$.

Exercise 1.4.4 (Reverse-engineering unknown coefficients)**Problem Statement:**

It is given that $\lim_{x \rightarrow -1} \frac{2x^2 + ax + b}{x^2 - 1} = 3$, where a and b are real constants.

1. Explain why the numerator $2x^2 + ax + b$ must equal 0 when $x = -1$.
 2. Hence, or otherwise, find the exact values of a and b .
-

Exercise 1.4.5 (General power difference factorization)**Problem Statement:**

For any positive integer $n \geq 2$, the algebraic identity for the difference of n -th powers is given by:

$$x^n - 1 = (x - 1)(x^{n-1} + x^{n-2} + \cdots + x^2 + x + 1)$$

Using this identity, prove that for any positive integer n :

$$\lim_{x \rightarrow 1} \frac{x^n - 1}{x - 1} = n$$

Exercise 1.4.6 (When factorization fails to remove the zero)**Problem Statement:**

Consider the limit:

$$\lim_{x \rightarrow 3} \frac{x^2 - 3x}{x^2 - 6x + 9}$$

1. Factorize both the numerator and the denominator, and cancel any common factors.
 2. Attempt to apply direct substitution to the simplified fraction. What mathematical anomaly occurs now?
 3. Explain geometrically why this limit fails to exist as a finite number, distinguishing it from a removable hole.
-

Exercise 1.4.7 (Multivariable indeterminacy and directional paths)**Problem Statement:**

In advanced calculus, rational limits can involve two variables, such as:

$$L = \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - y^2}{x^2 + y^2}$$

Here, the point (x, y) can approach the origin $(0, 0)$ along an infinite number of straight pathways.

1. Suppose we approach along the x -axis, meaning we set $y = 0$ first and then take $\lim_{x \rightarrow 0}$. What is the resulting value of L ?
 2. Suppose we approach along the y -axis, meaning we set $x = 0$ first and then take $\lim_{y \rightarrow 0}$. What is the resulting value of L ?
 3. Now suppose we approach along the diagonal line $y = mx$, where m is any constant slope. Substitute $y = mx$, factor out x^2 , and evaluate the limit.
 4. Synthesize your findings: Why does factoring fail to produce a single, unique limit for multivariable rational functions at the origin, and how does this mirror our 1D sequential limit rules from Section 1.2?
-

1.5 Limit of rational function at finite points: Rationalization application

In Section 1.4, we learned how to resolve the indeterminate form $\frac{0}{0}$ by factorizing polynomials to reveal and cancel the hidden common factor $(x - c)$. But what happens if our function is not a simple polynomial, but instead contains surds (square roots)?

Consider the limit $\lim_{x \rightarrow 3} \frac{\sqrt{x+1}-2}{x-3}$. If we attempt direct substitution, we get $\frac{\sqrt{4}-2}{3-3} = \frac{0}{0}$. We know that a common factor of $(x - 3)$ must be causing this division by zero, but because the variable x is trapped inside a square root symbol, standard factorization methods like the cross-method or difference of cubes fail completely!

To rescue the trapped variable and reveal the hidden factor, we must borrow a clever technique from Junior Secondary algebra: ****rationalization via conjugate multiplication****. By utilizing the famous difference of squares identity, $(a - b)(a + b) = a^2 - b^2$, we can multiply both the numerator and the denominator by the conjugate of the surd expression. This squashing action squares the square root, effectively destroying the radical sign without altering the actual value of the fraction. Once the surd is removed, the hidden factor $(x - c)$ magically appears, ready to be cancelled!

Theorem 1.5: Evaluation of Surd Limits via Rationalization

Let $f(x) = \frac{\sqrt{u(x)} - v(x)}{w(x)}$ be an algebraic function containing square roots, where $u(x)$, $v(x)$, and $w(x)$ are continuous functions such that direct substitution at $x = c$ yields an indeterminate form:

$$\sqrt{u(c)} - v(c) = 0 \quad \text{and} \quad w(c) = 0$$

By multiplying both the numerator and the denominator by the ****conjugate expression**** $(\sqrt{u(x)} + v(x))$, we apply the difference of squares identity to eliminate the radical sign in the numerator:

$$\lim_{x \rightarrow c} \frac{\sqrt{u(x)} - v(x)}{w(x)} = \lim_{x \rightarrow c} \frac{(\sqrt{u(x)} - v(x))(\sqrt{u(x)} + v(x))}{w(x)(\sqrt{u(x)} + v(x))} = \lim_{x \rightarrow c} \frac{u(x) - [v(x)]^2}{w(x)(\sqrt{u(x)} + v(x))}$$

This transformation guarantees that $(x - c)$ can now be factored out from the new polynomial numerator $u(x) - [v(x)]^2$ and cancelled with the corresponding $(x - c)$ factor inside $w(x)$, resolving the $\frac{0}{0}$ anomaly.

Student Proof Demonstration

Use this space to demonstrate the proof of the theorem above. Start from basic algebraic identities.

Practical Example 1.5

Evaluate the following limit using conjugate rationalization:

$$\lim_{x \rightarrow 3} \frac{\sqrt{x+1} - 2}{x - 3}$$

Step-by-Step Solution

Step 1: Test direct substitution first.

Substituting $x = 3$ into the expression gives:

$$\frac{\sqrt{3+1}-2}{3-3} = \frac{2-2}{0} = \frac{0}{0}$$

Since we get $\frac{0}{0}$ and there is a surd, we must rationalize!

Step 2: Multiply by the conjugate.

The conjugate of $(\sqrt{x+1}-2)$ is $(\sqrt{x+1}+2)$. We multiply both top and bottom by this exact term:

$$\lim_{x \rightarrow 3} \frac{\sqrt{x+1}-2}{x-3} \cdot \frac{\sqrt{x+1}+2}{\sqrt{x+1}+2}$$

Step 3: Expand the numerator ONLY.

Using $(a-b)(a+b) = a^2 - b^2$ on top:

$$\begin{aligned} &= \lim_{x \rightarrow 3} \frac{(\sqrt{x+1})^2 - 2^2}{(x-3)(\sqrt{x+1}+2)} \\ &= \lim_{x \rightarrow 3} \frac{(x+1)-4}{(x-3)(\sqrt{x+1}+2)} \end{aligned}$$

$$= \lim_{x \rightarrow 3} \frac{x-3}{(x-3)(\sqrt{x+1}+2)}$$

Step 4: Cancel the common factor.

Since $x \rightarrow 3$ implies $x \neq 3$, we can safely cancel the non-zero term $(x-3)$:

$$= \lim_{x \rightarrow 3} \frac{1}{\sqrt{x+1}+2}$$

Step 5: Apply direct substitution.

Now simply substitute $x = 3$:

$$= \frac{1}{\sqrt{3+1}+2}$$

$$= \frac{1}{2+2} = \frac{1}{4}$$

Guided Thinking & Hints

Key Hint: Never blindly multiply by the conjugate without checking substitution first! If the denominator isn't zero, direct substitution is all you need.

Key Hint: Why top AND bottom? Because dividing a number by itself equals 1. We are changing the *form* of the expression, never its mathematical value!

Key Hint: GOLDEN RULE: DO NOT expand the denominator! Leave it factored as $(x-3)(\dots)$. If you multiply it out, you will hide the exact $(x-3)$ factor you need to cancel!

Key Hint: Notice how the troublesome $(x-3)$ vanished from the bottom! The division-by-zero anomaly is completely eliminated.

Key Hint: Remember to drop the $\lim_{x \rightarrow 3}$ symbol the exact instant you replace x with the number 3. Our final limit is a clean fraction: $\frac{1}{4}$.

Exercise 1.5.1 (Rationalizing a simple square root)**Problem Statement:**

Evaluate the following limit by rationalizing the numerator:

$$\lim_{x \rightarrow 9} \frac{\sqrt{x} - 3}{x - 9}$$

Exercise 1.5.2 (Rationalization combined with quadratic factorization)**Problem Statement:**

Evaluate the following limit by using both conjugate multiplication on the numerator and difference of squares factorization on the denominator:

$$\lim_{x \rightarrow 4} \frac{\sqrt{x} - 2}{x^2 - 16}$$

Exercise 1.5.3 (Rationalizing the denominator)**Problem Statement:**

Sometimes the radical trap is situated in the denominator instead of the numerator. Evaluate the following limit without using a calculator:

$$\lim_{x \rightarrow 2} \frac{x - 2}{\sqrt{2x + 5} - 3}$$

Exercise 1.5.4 (Reverse-engineering surd constants for continuity)**Problem Statement:**

Consider the piecewise-defined function:

$$f(x) = \begin{cases} \frac{\sqrt{x+a}-2}{x-1} & \text{if } x \neq 1 \\ b & \text{if } x = 1 \end{cases}$$

where a and b are real constants. Find the exact values of a and b such that the function $f(x)$ is continuous at $x = 1$.

Exercise 1.5.5 (Rationalizing cube roots)**Problem Statement:**

While square roots use the difference of squares identity $(a - b)(a + b) = a^2 - b^2$, cube roots require the difference of cubes identity:

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

Letting $a = \sqrt[3]{x}$ and $b = 1$, use this identity to rationalize the numerator and evaluate:

$$\lim_{x \rightarrow 1} \frac{\sqrt[3]{x} - 1}{x - 1}$$

Exercise 1.5.6 (When surd rationalization fails due to sign traps)**Problem Statement:**

Consider the limit:

$$\lim_{x \rightarrow 0} \frac{\sqrt{x^2}}{x}$$

1. Explain why the identity $\sqrt{x^2} = x$ is strictly FALSE for negative real numbers, and state the correct identity involving absolute values.
 2. Attempt to evaluate the left-hand limit ($\lim_{x \rightarrow 0^-}$) and the right-hand limit ($\lim_{x \rightarrow 0^+}$).
 3. Conclude why conjugate rationalization cannot force this limit to exist.
-

Exercise 1.5.7 (Bridge to the Power Rule: General surd differentiation)**Problem Statement:**

In Section 2, we will define the derivative of a function $g(x)$ at $x = c$ as the limit:

$$g'(c) = \lim_{x \rightarrow c} \frac{g(x) - g(c)}{x - c}$$

Let $g(x) = \sqrt[n]{x}$ for any positive integer $n \geq 2$. We wish to find the slope of the n -th root function at $x = c > 0$.

1. Write down the limit expression for $g'(c)$ by substituting $g(x) = \sqrt[n]{x}$.
2. Recall the general factorization identity: $A^n - B^n = (A - B)(A^{n-1} + A^{n-2}B + \cdots + AB^{n-2} + B^{n-1})$. Let $A = \sqrt[n]{x}$ and $B = \sqrt[n]{c}$. Rationalize the numerator to cancel the $(x - c)$ term.
3. Apply direct substitution as $x \rightarrow c$ to prove that:

$$\lim_{x \rightarrow c} \frac{\sqrt[n]{x} - \sqrt[n]{c}}{x - c} = \frac{1}{n \cdot c^{\frac{n-1}{n}}} = \frac{1}{n} c^{\frac{1}{n}-1}$$

4. Notice how the exponent $\frac{1}{n}$ was brought down to the front, and the new exponent became $\frac{1}{n} - 1$. What famous calculus rule does this rigorous rationalization exercise prove in advance?

1.6 One-sided limit through piecewise-defined function

In Sections 1.4 and 1.5, we mastered the algebraic tools—factorization and rationalization—needed to repair functions that suffer from a single, isolated trouble point like $\frac{0}{0}$. However, in the real world, systems often change their governing rules abruptly at a specific threshold. Consider Hong Kong tax brackets, parking garage fee structures, or a physical electrical switch turning on at time $t = 0$. These situations cannot be described by a single formula; they require **piecewise-defined functions**.

When analyzing what happens as we approach a threshold or boundary point $x = c$ in a piecewise function, we encounter a new challenge: approaching from values smaller than c (moving rightward from the left) might follow a completely different mathematical rule than approaching from values larger than c (moving leftward from the right). To handle these split-personality functions without confusion, we must separate our investigation into **one-sided limits**: the left-hand limit and the right-hand limit.

Theorem 1.6: Two-Sided Limit Existence via One-Sided Limits

Let $f(x)$ be a function defined on an open interval containing $x = c$, except possibly at c itself.

We define the two **one-sided limits** as follows:

- **Left-Hand Limit:** The value L_1 that $f(x)$ approaches as x gets arbitrarily close to c using only values strictly less than c ($x < c$), denoted as:

$$\lim_{x \rightarrow c^-} f(x) = L_1$$

- **Right-Hand Limit:** The value L_2 that $f(x)$ approaches as x gets arbitrarily close to c using only values strictly greater than c ($x > c$), denoted as:

$$\lim_{x \rightarrow c^+} f(x) = L_2$$

The general (two-sided) limit $\lim_{x \rightarrow c} f(x)$ exists and equals a real number L if and only if both the left-hand limit and right-hand limit exist and are exactly equal to L :

$$\lim_{x \rightarrow c^-} f(x) = \lim_{x \rightarrow c^+} f(x) = L \iff \lim_{x \rightarrow c} f(x) = L$$

If $\lim_{x \rightarrow c^-} f(x) \neq \lim_{x \rightarrow c^+} f(x)$, the function experiences a **jump discontinuity** at $x = c$, and the two-sided limit $\lim_{x \rightarrow c} f(x)$ does not exist.

Student Proof Demonstration

Use this space to demonstrate the proof of the theorem above. Start from basic algebraic identities.

Practical Example 1.6

Consider the piecewise-defined function:

$$f(x) = \begin{cases} x^2 + 1 & \text{if } x < 2 \\ 10 & \text{if } x = 2 \\ 3x - 1 & \text{if } x > 2 \end{cases}$$

Evaluate the left-hand limit $\lim_{x \rightarrow 2^-} f(x)$ and right-hand limit $\lim_{x \rightarrow 2^+} f(x)$. Hence, determine whether the general limit $\lim_{x \rightarrow 2} f(x)$ exists.

Step-by-Step Solution**Guided Thinking & Hints****Step 1: Evaluate the left-hand limit ($\lim_{x \rightarrow 2^-}$).**

The notation $x \rightarrow 2^-$ means x is approaching 2 from values smaller than 2 (i.e., $x < 2$). Looking at our piecewise definition, we must select the first branch:

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} (x^2 + 1)$$

Since $x^2 + 1$ is a simple polynomial, substitute $x = 2$:
 $= 2^2 + 1 = 5$

Key Hint: The minus sign in 2^- is an indicator of DIRECTION (from below/left), NOT a negative sign! Never substitute $x = -2$. You are still substituting positive 2 into the selected formula.

Step 2: Evaluate the right-hand limit ($\lim_{x \rightarrow 2^+}$).

The notation $x \rightarrow 2^+$ means x is approaching 2 from values greater than 2 (i.e., $x > 2$). We must select the third branch:

$$\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} (3x - 1)$$

Substitute $x = 2$ directly into this branch:

$$= 3(2) - 1 = 5$$

Step 3: Compare both one-sided limits.

We observe that:

$$\lim_{x \rightarrow 2^-} f(x) = 5 \quad \text{and} \quad \lim_{x \rightarrow 2^+} f(x) = 5$$

Key Hint: Notice how we completely ignored the middle branch $f(2) = 10$! Remember from Section 1.3: limits never care about what happens AT the exact point, only what happens nearby.

Since the left-hand limit equals the right-hand limit ($5 = 5$), Theorem 1.6 guarantees that the two-sided limit exists.

Step 4: State the final conclusion.

Therefore, the general limit exists and is equal to 5:

$$\lim_{x \rightarrow 2} f(x) = 5$$

Key Hint: If two road builders start from opposite sides of a mountain and aim for height 5, their tunnels will meet perfectly, forming a seamless path!

Key Hint: Even though there is a displaced point high above at $(2, 10)$, the limit of the path as $x \rightarrow 2$ is securely established at 5.

Exercise 1.6.1 (Evaluating one-sided limits of a jump function)**Problem Statement:**

Let $g(x)$ be defined as:

$$g(x) = \begin{cases} 2x + 3 & \text{if } x < 1 \\ 7 - x & \text{if } x \geq 1 \end{cases}$$

1. Evaluate $\lim_{x \rightarrow 1^-} g(x)$.
 2. Evaluate $\lim_{x \rightarrow 1^+} g(x)$.
 3. Does the general limit $\lim_{x \rightarrow 1} g(x)$ exist? Explain briefly.
-

Exercise 1.6.2 (One-sided limits involving factorization)**Problem Statement:**

Consider the piecewise function:

$$h(x) = \begin{cases} \frac{x^2 - 16}{x - 4} & \text{if } x < 4 \\ 3x - 4 & \text{if } x > 4 \end{cases}$$

Evaluate $\lim_{x \rightarrow 4^-} h(x)$ using factorization, and evaluate $\lim_{x \rightarrow 4^+} h(x)$. Determine whether $\lim_{x \rightarrow 4} h(x)$ exists.

Exercise 1.6.3 (Unmasking absolute value functions)**Problem Statement:**

In calculus, absolute value expressions must be rewritten as piecewise functions before taking limits. Consider:

$$f(x) = \frac{|2x - 6|}{x - 3} \quad \text{for } x \neq 3$$

1. Rewrite $f(x)$ as a two-branch piecewise function by analyzing where $2x - 6 > 0$ and where $2x - 6 < 0$.
 2. Evaluate $\lim_{x \rightarrow 3^-} f(x)$ and $\lim_{x \rightarrow 3^+} f(x)$.
 3. Conclude whether $\lim_{x \rightarrow 3} f(x)$ exists.
-

Exercise 1.6.4 (Multi-boundary continuity system)**Problem Statement:**

An HKDSE M1/M2 exam question defines a smooth track using the function:

$$f(x) = \begin{cases} ax + b & \text{if } x \leq -1 \\ 2x^2 + 3 & \text{if } -1 < x < 2 \\ 3ax - 2b & \text{if } x \geq 2 \end{cases}$$

If $\lim_{x \rightarrow -1} f(x)$ and $\lim_{x \rightarrow 2} f(x)$ both exist as valid real numbers, set up a system of linear equations to find the exact values of the constants a and b .

Exercise 1.6.5 (Squeeze theorem on one side)**Problem Statement:**

Suppose that three functions satisfy the inequality $p(x) \leq q(x) \leq r(x)$ ONLY on the interval $a < x < c$ (to the left of c). If $\lim_{x \rightarrow c^-} p(x) = L$ and $\lim_{x \rightarrow c^-} r(x) = L$, explain intuitively why we can confidently state that $\lim_{x \rightarrow c^-} q(x) = L$, without needing to know anything about how these functions behave for $x \geq c$.

Exercise 1.6.6 (Wild oscillation on a single branch)**Problem Statement:**

Consider the hybrid function:

$$f(x) = \begin{cases} x + 2 & \text{if } x \leq 0 \\ \sin\left(\frac{1}{x}\right) & \text{if } x > 0 \end{cases}$$

1. Evaluate the left-hand limit $\lim_{x \rightarrow 0^-} f(x)$.
 2. Explain why the right-hand limit $\lim_{x \rightarrow 0^+} f(x)$ fails to exist.
 3. Can we say $\lim_{x \rightarrow 0} f(x) = 2$ because at least one side behaved nicely? State the rule from Theorem 1.6 that settles this.
-

Exercise 1.6.7 (Bridge to Section 2: Left and right derivatives at a corner)**Problem Statement:**

In Section 2, we will learn that for a function to be ****differentiable**** (to have a well-defined slope/tangent line) at $x = c$, not only must the function be continuous, but the limit of its secant slopes from the left and right must also be identical!

Let $f(x) = |x - 2|$. We wish to investigate its slope behavior at the corner point $x = 2$.

1. First, verify that $\lim_{x \rightarrow 2} f(x) = f(2) = 0$, confirming the function is continuous.
 2. The left-hand slope limit is defined as $\lim_{x \rightarrow 2^-} \frac{f(x) - f(2)}{x - 2}$. Substitute $f(x) = -(x - 2)$ for $x < 2$ and evaluate this limit. What is the slope approaching from the left?
 3. The right-hand slope limit is defined as $\lim_{x \rightarrow 2^+} \frac{f(x) - f(2)}{x - 2}$. Substitute $f(x) = x - 2$ for $x > 2$ and evaluate this limit. What is the slope approaching from the right?
 4. Graphically, $y = |x - 2|$ is a V-shaped curve with a sharp corner at $(2, 0)$. Using your results from parts (2) and (3), explain why a sharp corner prevents a function from having a single, unique tangent line, even when the curve is unbroken!
-

1.7 Continuity through one-sided limit

In Section 1.6, we learned how to analyze split-personality (piecewise) functions using one-sided limits. We discovered that a function can jump from one height to another at a boundary point, destroying the general limit. But what happens if the left-hand limit and the right-hand limit *do* meet at the exact same height, and moreover, that height is exactly where the physical dot $f(c)$ is plotted?

When the path approaching from the left, the path approaching from the right, and the actual destination point all align perfectly, there are no holes, no jumps, and no breaks. Imagine drawing a curve on a piece of paper: if your pencil never has to lift off the paper as it passes through $x = c$, the graph is unbroken. In mathematics, we call this seamless, uninterrupted structure *continuity*. Mastering continuity is essential because almost all advanced rules of differentiation require functions to be continuous before we can measure their rates of change!

Theorem 1.7: The Three-Part Continuity Test

Let $f(x)$ be a function defined on an open interval containing the point $x = c$.

The function $f(x)$ is said to be *continuous at $x = c$* if and only if all three of the following conditions are satisfied simultaneously:

1. **The Destination Exists:** The function value $f(c)$ is a defined real number (there is no missing hole at $x = c$).
2. **The Path is Unbroken:** The two-sided limit exists, meaning the left-hand limit equals the right-hand limit:

$$\lim_{x \rightarrow c^-} f(x) = \lim_{x \rightarrow c^+} f(x) = L$$

3. **The Path Meets the Destination:** The limit value exactly equals the function value:

$$\lim_{x \rightarrow c} f(x) = f(c)$$

If a function fails even one of these three tests at $x = c$, it is *discontinuous* at that point.

Student Proof Demonstration

Use this space to demonstrate the proof of the theorem above. Start from basic algebraic identities.

Practical Example 1.7

Determine whether the following piecewise function is continuous at the boundary point $x = 2$:

$$f(x) = \begin{cases} 2x^2 - 1 & \text{if } x < 2 \\ 7 & \text{if } x = 2 \\ 3x + 1 & \text{if } x > 2 \end{cases}$$

Step-by-Step Solution

Guided Thinking & Hints

Step 1: Test Condition 1 (Does $f(2)$ exist?).

Look at the middle branch of the piecewise definition where $x = 2$:

$$f(2) = 7$$

Since 7 is a defined real number, Condition 1 is satisfied.

Step 2: Test Condition 2 (Does the limit exist?).

We must evaluate both one-sided limits as x approaches 2:

Left-hand limit ($x < 2$, use top branch):

$$\begin{aligned}\lim_{x \rightarrow 2^-} f(x) &= \lim_{x \rightarrow 2^-} (2x^2 - 1) \\ &= 2(2)^2 - 1 = 8 - 1 = 7\end{aligned}$$

Right-hand limit ($x > 2$, use bottom branch):

$$\begin{aligned}\lim_{x \rightarrow 2^+} f(x) &= \lim_{x \rightarrow 2^+} (3x + 1) \\ &= 3(2) + 1 = 7\end{aligned}$$

Since $\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^+} f(x) = 7$, the two-sided limit exists:

$$\lim_{x \rightarrow 2} f(x) = 7$$

Step 3: Test Condition 3 (Limit = Function Value?).

We compare our results from Step 1 and Step 2:

$$\lim_{x \rightarrow 2} f(x) = 7 \quad \text{and} \quad f(2) = 7$$

Since $\lim_{x \rightarrow 2} f(x) = f(2)$, Condition 3 is satisfied.

Step 4: State the final conclusion.

Because all three conditions of Theorem 1.7 are satisfied simultaneously, the function $f(x)$ is continuous at $x = 2$.

Key Hint: Always check the simple function value first! If there is no equality sign ($\leq$, $\geq$, or $=$) for $x = 2$ anywhere in the definition, $f(2)$ is undefined and you can immediately stop: the function is discontinuous!

Key Hint: Remember from Section 1.6: if the left-hand limit does not equal the right-hand limit, the function has a jump discontinuity. Here, both sides equal 7, so the road tunnels meet!

Key Hint: This step checks if the dot is plotted directly on top of the meeting point of the two paths. Since $7 = 7$, there is no displaced point!

Key Hint: In an HKDSE exam, explicitly listing out all three checks is the most foolproof way to secure full method marks for continuity questions!

Exercise 1.7.1 (Three-part continuity check on a two-branch function)**Problem Statement:**

Let $g(x)$ be defined as:

$$g(x) = \begin{cases} 5 - x & \text{if } x \leq 3 \\ x^2 - 7 & \text{if } x > 3 \end{cases}$$

Use the Three-Part Continuity Test (Theorem 1.7) to determine whether $g(x)$ is continuous at $x = 3$. Show all three steps explicitly.

Exercise 1.7.2 (Continuity involving rationalization)**Problem Statement:**

Consider the piecewise function:

$$h(x) = \begin{cases} \frac{\sqrt{x}-2}{x-4} & \text{if } 0 \leq x < 4 \\ \frac{1}{4} & \text{if } x = 4 \\ \frac{x^2-16}{4x-16} & \text{if } x > 4 \end{cases}$$

Evaluate $\lim_{x \rightarrow 4^-} h(x)$ by rationalizing, and evaluate $\lim_{x \rightarrow 4^+} h(x)$ by factoring. Hence, determine if $h(x)$ is continuous at $x = 4$.

Exercise 1.7.3 (Tuning a single constant for seamless continuity)**Problem Statement:**

An engineer designs a mechanical transition modeled by the function:

$$f(x) = \begin{cases} x^2 - 3x + k & \text{if } x \leq 1 \\ 2x + 5 & \text{if } x > 1 \end{cases}$$

Find the exact value of the constant k such that the mechanical transition is completely continuous at the boundary $x = 1$.

Exercise 1.7.4 (Solving a two-variable continuity system)**Problem Statement:**

In HKDSE M1/M2 exams, you frequently encounter functions with two boundary points. Consider:

$$f(x) = \begin{cases} 3 & \text{if } x \leq -2 \\ ax^2 + bx - 1 & \text{if } -2 < x < 2 \\ 2x + 7 & \text{if } x \geq 2 \end{cases}$$

If $f(x)$ is continuous for all real numbers (and specifically at the boundaries $x = -2$ and $x = 2$), set up a simultaneous system of linear equations to find the exact values of a and b .

Exercise 1.7.5 (Algebra of continuous functions)**Problem Statement:**

Suppose two functions, $p(x)$ and $q(x)$, are both known to be continuous at $x = c$. Using the three conditions of Theorem 1.7 and standard limit properties, prove algebraically that their sum, $S(x) = p(x) + q(x)$, must also be continuous at $x = c$.

Exercise 1.7.6 (Classifying types of discontinuity)**Problem Statement:**

Consider the following two broken functions at $x = 1$:

$$f(x) = \begin{cases} x + 2 & \text{if } x \neq 1 \\ 9 & \text{if } x = 1 \end{cases} \quad \text{and} \quad g(x) = \begin{cases} 2 & \text{if } x < 1 \\ 5 & \text{if } x \geq 1 \end{cases}$$

1. Explain which specific condition of Theorem 1.7 is violated by $f(x)$ at $x = 1$. Why is this called a **removable discontinuity (hole)**?
 2. Explain which specific condition of Theorem 1.7 is violated by $g(x)$ at $x = 1$. Why is this called a **jump discontinuity**?
 3. Can either of these discontinuities be fixed just by redefining the single output value at $x = 1$? Explain.
-

Exercise 1.7.7 (The Intermediate Value Theorem (IVT) intuition)**Problem Statement:**

One of the most powerful consequences of continuity is the **Intermediate Value Theorem**. It states that if a function $f(x)$ is continuous on a closed interval $[a, b]$, and $f(a)$ and $f(b)$ have opposite signs (one is positive and one is negative), then the graph **MUST** cross the horizontal axis at least once, meaning there is at least one root c between a and b where $f(c) = 0$.

Consider the cubic equation $f(x) = x^3 + 2x - 5$.

1. Explain briefly why $f(x)$ is guaranteed to be continuous everywhere.
 2. Calculate $f(1)$ and $f(2)$.
 3. Using the Intermediate Value Theorem, explain why the equation $x^3 + 2x - 5 = 0$ must have at least one real solution between $x = 1$ and $x = 2$.
 4. Why would this theorem fail completely if the function had a jump discontinuity between $x = 1$ and $x = 2$? (Draw a quick mental sketch or explain in words how a jumping graph could skip over the zero line without ever touching it).
-

1.8 Limit to positive infinity of rational function

In Sections 1.1 through 1.7, we focused exclusively on limits where the input variable x approaches a finite, specific target number c . But what happens when we want to predict the long-term, eventual behavior of a system? For example, what is the concentration of a drug in a patient's bloodstream after an indefinitely long period of time, or what is the terminal velocity of a falling object as time continues without bound? In mathematics, we explore these ultimate steady states by evaluating limits as x approaches **positive infinity** ($x \rightarrow \infty$).

When we encounter a rational function like $f(x) = \frac{6x^2-4x+5}{3x^2+8x-2}$ and attempt to substitute an infinitely large number into the variable, both the numerator and the denominator grow indefinitely large, yielding the indeterminate form $\frac{\infty}{\infty}$. Unlike standard arithmetic quantities, infinity is a concept of boundless growth, not a static number—meaning $\frac{\infty}{\infty}$ is definitely NOT equal to 1! To tame these competing infinities, we use a classic algebraic leveling strategy: **dividing every term by the highest power of x present in the denominator**. Because any fixed number divided by an infinitely large quantity shrinks rapidly toward zero ($\lim_{x \rightarrow \infty} \frac{c}{x^p} = 0$), this division flattens all lower-degree terms into zeroes, leaving behind only the dominant leading terms that actually dictate the long-term destination of the curve!

Theorem 1.8: Evaluation of Rational Limits at Infinity via Highest Power Division

Let $f(x) = \frac{P(x)}{Q(x)}$ be a rational function where $P(x)$ and $Q(x)$ are polynomials. To evaluate the limit at positive infinity:

$$\lim_{x \rightarrow \infty} \frac{P(x)}{Q(x)}$$

we identify k , the **highest power of x present in the denominator** $Q(x)$, and divide both the numerator and the denominator by x^k :

$$\lim_{x \rightarrow \infty} \frac{P(x)}{Q(x)} = \lim_{x \rightarrow \infty} \frac{\frac{P(x)}{x^k}}{\frac{Q(x)}{x^k}}$$

By applying the **Fundamental Limit at Infinity**:

$$\lim_{x \rightarrow \infty} \frac{c}{x^p} = 0 \quad \text{for any real constant } c \text{ and rational number } p > 0$$

all fractional terms of the form $\frac{c}{x^p}$ vanish to 0, allowing us to safely evaluate the remaining simplified expression without encountering $\frac{\infty}{\infty}$.

Student Proof Demonstration

Use this space to demonstrate the proof of the theorem above. Start from basic algebraic identities.

Practical Example 1.8

Evaluate the long-term limit of the following rational function as x grows without bound:

$$\lim_{x \rightarrow \infty} \frac{6x^2 - 4x + 5}{3x^2 + 8x - 2}$$

Step-by-Step Solution

Step 1: Inspect the growth behavior.

As $x \rightarrow \infty$, both the numerator ($6x^2 \dots$) and the denominator ($3x^2 \dots$) grow indefinitely large. This produces the indeterminate form $\frac{\infty}{\infty}$.

Step 2: Identify the highest power in the denominator.

The denominator is $3x^2 + 8x - 2$. The highest exponent of the variable x present in this denominator is 2 (from the term $3x^2$). Therefore, we must divide by x^2 .

Step 3: Divide every term by x^2 .

We divide each individual term in top and bottom by x^2 :

$$\lim_{x \rightarrow \infty} \frac{\frac{6x^2}{x^2} - \frac{4x}{x^2} + \frac{5}{x^2}}{\frac{3x^2}{x^2} + \frac{8x}{x^2} - \frac{2}{x^2}}$$

Simplify the exponents using algebra:

$$= \lim_{x \rightarrow \infty} \frac{6 - \frac{4}{x} + \frac{5}{x^2}}{3 + \frac{8}{x} - \frac{2}{x^2}}$$

Step 4: Apply the Fundamental Limit at Infinity.

As $x \rightarrow \infty$, any constant divided by x or x^2 shrinks to 0:

$$= \frac{6-0+0}{3+0-0}$$

$$= \frac{6}{3} = 2$$

Guided Thinking & Hints

Key Hint: Never write $\frac{\infty}{\infty} = 1$! Infinity is an unbounded concept, not a standard number you can cancel out in division.

Key Hint: Always scan the **bottom polynomial** to choose your divisor! Dividing by the denominator's highest power guarantees that the bottom will never collapse into 0.

Key Hint: Take your time simplifying powers: $\frac{x}{x^2} = \frac{1}{x}$. We are preparing the expression so every leftover variable is trapped safely inside a denominator!

Key Hint: Think of $\frac{4}{x}$ as slicing 4 pizzas among 100 billion people: each person gets virtually zero pizza! Notice how only the leading coefficients (6 and 3) survived!

Exercise 1.8.1 (Equal degree polynomials at infinity)**Problem Statement:**

Evaluate the following limit by dividing every term by the highest power of x in the denominator:

$$\lim_{x \rightarrow \infty} \frac{10x^2 + 3x - 1}{5x^2 - 2x + 4}$$

Exercise 1.8.2 (Horizontal asymptotes in geometry)**Problem Statement:**

In coordinate geometry, the horizontal line $y = L$ is called a **horizontal asymptote** of a curve $y = f(x)$ if the curve gets arbitrarily close to height L as $x \rightarrow \infty$ or $x \rightarrow -\infty$. Consider the function:

$$g(x) = \frac{4x - 7}{2x^2 + 1}$$

1. Evaluate $\lim_{x \rightarrow \infty} g(x)$ by dividing numerator and denominator by x^2 .
 2. Write down the exact equation of the horizontal asymptote for the graph of $y = g(x)$.
-

Exercise 1.8.3 (Expanding before evaluating limits at infinity)**Problem Statement:**

Evaluate the following limit without using a calculator:

$$\lim_{x \rightarrow \infty} \frac{(3x - 2)(2x + 5)}{(x - 1)^2}$$

Hint: Expand both the numerator and the denominator into general polynomial form before identifying the highest power of x .

Exercise 1.8.4 (Reverse-engineering unknown constants at infinity)**Problem Statement:**

An HKDSE M1/M2 exam problem states that:

$$\lim_{x \rightarrow \infty} \frac{ax^3 + bx^2 + 5}{2x^2 - 3x + 1} = 4$$

where a and b are real constants.

1. Explain why the coefficient a MUST be equal to 0 for this limit to converge to a finite number like 4.
 2. Hence, or otherwise, find the exact value of the constant b .
-

Exercise 1.8.5 (The General Degree Rule for rational functions)**Problem Statement:**

Let $F(x) = \frac{a_n x^n + \dots + a_1 x + a_0}{b_m x^m + \dots + b_1 x + b_0}$ be a general rational function where $a_n \neq 0$ and $b_m \neq 0$. By dividing both top and bottom by x^m , prove algebraically that:

1. If the degree of the numerator is strictly less than the degree of the denominator ($n < m$), then $\lim_{x \rightarrow \infty} F(x) = 0$.
 2. If the degrees are exactly equal ($n = m$), then $\lim_{x \rightarrow \infty} F(x) = \frac{a_n}{b_m}$ (the ratio of leading coefficients).
-

Exercise 1.8.6 (Unbounded divergence when numerator degree is larger)**Problem Statement:**

Consider the limit where the numerator has a higher degree than the denominator:

$$\lim_{x \rightarrow \infty} \frac{x^3 + 1}{5x - 2}$$

1. Divide both numerator and denominator by x^1 (the highest power present in the denominator).
 2. Attempt to evaluate the limit as $x \rightarrow \infty$. Why does the expression fail to produce a finite real number?
 3. Explain why we write $\lim_{x \rightarrow \infty} \frac{x^3 + 1}{5x - 2} = \infty$, and clarify why this means the limit strictly **does not exist** according to our formal definition of a limit as a real number.
-

Exercise 1.8.7 (Biochemical saturation: Michaelis-Menten enzyme kinetics)**Problem Statement:**

In biochemistry, enzymes act as biological catalysts that speed up chemical reactions. The reaction velocity V of an enzyme depends on the concentration of the target substrate, denoted as $[S]$. The relationship is modeled by the famous **Michaelis-Menten equation**:

$$V([S]) = \frac{V_{\max} \cdot [S]}{K_m + [S]}$$

where $V_{\max}$ (maximum velocity) and K_m (Michaelis constant) are positive real constants, and $[S] \geq 0$ represents the substrate concentration.

1. Using our algebraic rule of dividing by the highest power of $[S]$ in the denominator, evaluate the theoretical steady-state limit:

$$\lim_{[S] \rightarrow \infty} V([S])$$

2. Explain what your result from part (1) represents physically: why can't the reaction velocity increase indefinitely as we add more and more substrate into the solution?
 3. Calculate the exact substrate concentration $[S]$ (expressed in terms of K_m) required for the reaction velocity to reach exactly half of its ultimate limiting speed ($0.5V_{\max}$). What profound physical meaning does the constant K_m hold?
-

1.9 Limit to negative infinity of rational function

In Section 1.8, we explored the eventual steady state of a function as the variable x travels indefinitely far to the right along the number line ($x \rightarrow \infty$). But what if we reverse our direction and investigate what happened in the indefinite past, or travel indefinitely far to the left? This brings us to limits at **negative infinity** ($x \rightarrow -\infty$), where we plug in numbers like -1000 , -1000000 , and beyond.

For simple rational functions (a polynomial divided by a polynomial), dividing by the highest power of x in the denominator still works because any fixed number divided by an infinitely large negative quantity still shrinks rapidly to zero ($\lim_{x \rightarrow -\infty} \frac{c}{x^p} = 0$). However, when dealing with negative numbers, weak students frequently make catastrophic sign errors: forgetting that odd powers preserve negative signs ($(-x)^3 = -x^3$) while even powers absorb them ($(-x)^2 = x^2$). To completely eliminate sign confusion and build a foolproof bridge to our upcoming topics on surds and exponentials, we introduce a highly reliable professional technique: **the variable substitution method**. By setting $u = -x$, a journey to the far left ($x \rightarrow -\infty$) is mathematically mirrored into a journey to the far right ($u \rightarrow +\infty$), allowing us to solve every negative infinity problem using the exact same rules we mastered in Section 1.8!

Theorem 1.9: Evaluation of Limits at Negative Infinity via Mirror Substitution

Let $f(x)$ be a function defined for extremely large negative values of x . To evaluate the limit as x approaches negative infinity:

$$\lim_{x \rightarrow -\infty} f(x)$$

we apply the **Mirror Substitution Principle** by introducing a new positive variable $u = -x$ (which implies $x = -u$). As x decreases without bound ($x \rightarrow -\infty$), our new variable u increases without bound ($u \rightarrow +\infty$). Therefore:

$$\lim_{x \rightarrow -\infty} f(x) = \lim_{u \rightarrow \infty} f(-u)$$

For a rational function $f(x) = \frac{P(x)}{Q(x)}$, this transformation converts the problem into a standard positive infinity limit:

$$\lim_{x \rightarrow -\infty} \frac{P(x)}{Q(x)} = \lim_{u \rightarrow \infty} \frac{P(-u)}{Q(-u)}$$

We then evaluate this expression by dividing every term by the highest power of u present in the denominator, applying $\lim_{u \rightarrow \infty} \frac{c}{u^p} = 0$.

Student Proof Demonstration

Use this space to demonstrate the proof of the theorem above. Start from basic algebraic identities.

Practical Example 1.9

Evaluate the following limit as x approaches negative infinity using the mirror substitution method:

$$\lim_{x \rightarrow -\infty} \frac{4x^3 - 5x + 2}{2x^3 + 3x^2 - 1}$$

Step-by-Step Solution

Step 1: Apply mirror substitution $u = -x$.

Replace the limit symbol $\lim_{x \rightarrow -\infty}$ with $\lim_{u \rightarrow \infty}$, and replace every variable x inside the fraction with $(-u)$:

$$= \lim_{u \rightarrow \infty} \frac{4(-u)^3 - 5(-u) + 2}{2(-u)^3 + 3(-u)^2 - 1}$$

Step 2: Simplify the signs using power rules.

Remember that odd powers stay negative and even powers become positive:

$$(-u)^3 = -u^3 \quad \text{and} \quad (-u)^2 = u^2$$

Substitute these simplified powers back in:

$$= \lim_{u \rightarrow \infty} \frac{4(-u^3) + 5u + 2}{2(-u^3) + 3(u^2) - 1}$$

$$= \lim_{u \rightarrow \infty} \frac{-4u^3 + 5u + 2}{-2u^3 + 3u^2 - 1}$$

Step 3: Identify highest power and divide.

The highest power of u in the denominator is u^3 .

Divide every single term by u^3 :

$$= \lim_{u \rightarrow \infty} \frac{\frac{-4u^3}{u^3} + \frac{5u}{u^3} + \frac{2}{u^3}}{\frac{-2u^3}{u^3} + \frac{3u^2}{u^3} - \frac{1}{u^3}}$$

$$= \lim_{u \rightarrow \infty} \frac{-4 + \frac{5}{u^2} + \frac{2}{u^3}}{-2 + \frac{3}{u} - \frac{1}{u^3}}$$

Step 4: Apply Fundamental Limit at Infinity.

As $u \rightarrow \infty$, all fractional terms with u in the denominator vanish to 0:

$$= \frac{-4 + 0 + 0}{-2 + 0 - 0}$$

$$= \frac{-4}{-2} = 2$$

Guided Thinking & Hints

Key Hint: Why substitute? Because your brain is much better at handling positive infinity! By swapping x for $(-u)$, you never have to worry about negative infinity traps again.

Key Hint: CRITICAL SIGN CHECK: Watch out for $-5(-u) = +5u$! Be extremely careful when multiplying negative coefficients with negative substitutions.

Key Hint: Notice that after our substitution in Step 1, we are just following the exact same procedure from Section 1.8: divide by the highest power of the variable on the bottom!

Key Hint: The negative signs on -4 and -2 cancel out cleanly, giving a final steady-state limit of positive 2.

Exercise 1.9.1 (Mirror substitution on a quadratic rational function)**Problem Statement:**

Evaluate the following limit by applying the mirror substitution $u = -x$:

$$\lim_{x \rightarrow -\infty} \frac{8x^2 - 3x + 1}{2x^2 + 5x - 4}$$

Exercise 1.9.2 (Comparing left and right horizontal asymptotes)**Problem Statement:**

Consider the rational function $g(x) = \frac{6x-1}{3x+2}$.

1. Evaluate the right-hand end behavior: $\lim_{x \rightarrow \infty} g(x)$.
 2. Evaluate the left-hand end behavior: $\lim_{x \rightarrow -\infty} g(x)$ using $u = -x$.
 3. What do your results prove about the horizontal asymptote of standard rational functions as we look infinitely far to the left versus to the right?
-

Exercise 1.9.3 (Factoring cubic polynomials before substitution)**Problem Statement:**

Evaluate the following limit without using a calculator:

$$\lim_{x \rightarrow -\infty} \frac{(1-2x)(3x^2+4)}{5x^3-x+2}$$

Hint: Expand the numerator first, then apply the substitution $u = -x$.

Exercise 1.9.4 (Reverse-engineering coefficients at negative infinity)**Problem Statement:**

It is given that:

$$\lim_{x \rightarrow -\infty} \frac{kx^3 - 4x^2 + 5}{3x^3 + 2x - 1} = -6$$

where k is a real constant. By using the mirror substitution $u = -x$, set up an algebraic equation to find the exact value of k .

Exercise 1.9.5 (Left-Right Asymptote Symmetry Theorem for Rational Functions)**Problem Statement:**

Let $F(x) = \frac{P(x)}{Q(x)}$ be any rational function where $P(x)$ and $Q(x)$ are pure polynomials (containing no square roots or absolute values). Prove algebraically that if $\lim_{x \rightarrow \infty} F(x) = L$ (where L is a finite real number), then it MUST always be true that $\lim_{x \rightarrow -\infty} F(x) = L$. Explain why this guarantees that pure rational functions can never have two different horizontal asymptotes.

Exercise 1.9.6 (Unbounded divergence to negative infinity vs positive infinity)**Problem Statement:**

Consider the two divergent limits:

$$L_1 = \lim_{x \rightarrow -\infty} \frac{x^3 + 1}{x^2 + 2} \quad \text{and} \quad L_2 = \lim_{x \rightarrow -\infty} \frac{x^4 + 1}{x^2 + 2}$$

1. Apply the substitution $u = -x$ to both L_1 and L_2 .
 2. Explain why L_1 shoots downward toward $-\infty$, whereas L_2 shoots upward toward $+\infty$ as we move infinitely far to the left.
 3. State the general rule determining the sign of infinity when the numerator has a higher degree than the denominator as $x \rightarrow -\infty$.
-

Exercise 1.9.7 (Bridge to Section 1.11: Why surds break left-right symmetry)**Problem Statement:**

In Level 5, we proved that pure polynomial rational functions always approach the exact same limit at $+\infty$ and $-\infty$. However, when we introduce ****surds (square roots)****, this symmetry can shatter completely!

Consider the surd rational function:

$$f(x) = \frac{\sqrt{x^2 + 4}}{x}$$

1. Evaluate the right-hand limit $\lim_{x \rightarrow \infty} f(x)$ by dividing numerator and denominator by x (for $x > 0$, notice that $x = \sqrt{x^2}$).
 2. Now evaluate the left-hand limit $\lim_{x \rightarrow -\infty} f(x)$ using the mirror substitution $u = -x$.
 3. Why did $\lim_{x \rightarrow -\infty} f(x)$ produce a completely different number with an opposite sign compared to $\lim_{x \rightarrow \infty} f(x)$?
 4. Explain why the identity $\sqrt{x^2} = x$ is strictly illegal when $x \rightarrow -\infty$, and show how using the true identity $\sqrt{x^2} = |x| = -x$ (for negative x) explains this two-asymptote phenomenon without needing substitution!
-

1.10 Limit to positive infinity with surds

In Sections 1.8 and 1.9, we mastered the technique of dividing by the highest power of x in the denominator to evaluate limits at infinity for pure polynomials. However, when we introduce **surds** (square roots) into the mix, two new fascinating mathematical challenges emerge.

First, when dividing by x inside a square root for positive infinity ($x \rightarrow \infty$), we must rely on the algebraic bridge $x = \sqrt{x^2}$ (for $x > 0$). This allows us to tuck the variable inside the radical sign: $\frac{1}{x}\sqrt{A} = \sqrt{\frac{A}{x^2}}$.

Second, what happens when we encounter an expression like $\lim_{x \rightarrow \infty} (\sqrt{4x^2 + 8x} - 2x)$? As x grows infinitely large, both $\sqrt{4x^2 + 8x}$ and $2x$ race toward infinity, creating the indeterminate difference form $\infty - \infty$. Which infinity wins the race? To decide, we cannot just guess! We must combine our **conjugate rationalization** skill from Section 1.5 with our **highest power division** skill from Section 1.8. By rationalizing the numerator, we transform the ambiguous subtraction $\infty - \infty$ into a manageable fraction of the form $\frac{\infty}{\infty}$, which can then be cleanly solved by dividing top and bottom by x !

Theorem 1.10: Evaluation of Surd Limits at Positive Infinity

Let $f(x)$ be an algebraic expression containing square roots evaluated as $x \rightarrow \infty$. Two primary transformation rules govern these evaluations:

Rule 1: Radical Ingestion (For fractional forms $\frac{\infty}{\infty}$)

For any positive number $x > 0$, the identity $x = \sqrt{x^2}$ holds. When dividing a surd expression by the highest power x^k , we absorb the divisor inside the radical by squaring its exponent:

$$\frac{\sqrt{P(x)}}{x^k} = \sqrt{\frac{P(x)}{x^{2k}}} \quad (\text{for } x > 0)$$

Rule 2: Infinite Difference Rationalization (For forms $\infty - \infty$)

If direct evaluation yields an indeterminate subtraction of two infinite quantities, we convert the expression into a quotient using conjugate multiplication:

$$\lim_{x \rightarrow \infty} (\sqrt{u(x)} - v(x)) = \lim_{x \rightarrow \infty} \frac{(\sqrt{u(x)} - v(x))(\sqrt{u(x)} + v(x))}{\sqrt{u(x)} + v(x)} = \lim_{x \rightarrow \infty} \frac{u(x) - [v(x)]^2}{\sqrt{u(x)} + v(x)}$$

Once rationalized into $\frac{\infty}{\infty}$, apply Rule 1 by dividing every term by the highest power of x present in the new denominator.

Student Proof Demonstration

Use this space to demonstrate the proof of the theorem above. Start from basic algebraic identities.

Practical Example 1.10

Evaluate the following limit at positive infinity:

$$\lim_{x \rightarrow \infty} \left(\sqrt{4x^2 + 12x} - 2x \right)$$

Step-by-Step Solution

Step 1: Check the form of the limit.

As $x \rightarrow \infty$, the term $\sqrt{4x^2 + 12x} \rightarrow \infty$ and $2x \rightarrow \infty$.
This gives the indeterminate form $\infty - \infty$.

Step 2: Rationalize by conjugate multiplication.

Treat the expression as a fraction over 1. Multiply top and bottom by the conjugate $(\sqrt{4x^2 + 12x} + 2x)$:

$$= \lim_{x \rightarrow \infty} \frac{(\sqrt{4x^2 + 12x} - 2x)(\sqrt{4x^2 + 12x} + 2x)}{\sqrt{4x^2 + 12x} + 2x}$$

Apply $(a - b)(a + b) = a^2 - b^2$ on top:

$$= \lim_{x \rightarrow \infty} \frac{(4x^2 + 12x) - 4x^2}{\sqrt{4x^2 + 12x} + 2x}$$

$$= \lim_{x \rightarrow \infty} \frac{12x}{\sqrt{4x^2 + 12x} + 2x}$$

Step 3: Identify highest power and divide.

The highest power of x in the denominator is x^1 (since $\sqrt{x^2} = x$). Divide top and bottom by x :

$$= \lim_{x \rightarrow \infty} \frac{\frac{12x}{x}}{\frac{\sqrt{4x^2 + 12x}}{x} + \frac{2x}{x}}$$

Step 4: Absorb x into the square root.

Since $x \rightarrow \infty$, x is positive, so we use Rule 1 to write $x = \sqrt{x^2}$ inside the radical:

$$= \lim_{x \rightarrow \infty} \frac{12}{\sqrt{\frac{4x^2 + 12x}{x^2}} + 2}$$

$$= \lim_{x \rightarrow \infty} \frac{12}{\sqrt{4 + \frac{12}{x}} + 2}$$

Step 5: Apply Fundamental Limit at Infinity.

As $x \rightarrow \infty$, the term $\frac{12}{x} \rightarrow 0$:

$$= \frac{12}{\sqrt{4+0}+2}$$

$$= \frac{12}{2+2} = \frac{12}{4} = 3$$

Guided Thinking & Hints

Key Hint: Never assume $\infty - \infty = 0$! Infinity is not a number; it is a rate of growth. One infinity might be much larger than the other!

Key Hint: Notice how rationalizing transformed our impossible $\infty - \infty$ subtraction into a standard $\frac{\infty}{\infty}$ division problem!

Key Hint: Why is $\sqrt{4x^2}$ considered degree 1? Because the square root cuts the exponent 2 in half! Always look at the effective degree.

Key Hint: RADICAL INGESTION: To bring x inside a square root sign, you must square it first! Thus, dividing by x outside becomes dividing by x^2 inside.

Key Hint: The competing infinities balanced out perfectly to arrive at a steady-state destination of 3!

Exercise 1.10.1 (Dividing by x with surds in rational fractions)

Problem Statement:

Evaluate the following limit by dividing numerator and denominator by x :

$$\lim_{x \rightarrow \infty} \frac{\sqrt{9x^2 + 5}}{2x - 3}$$

Exercise 1.10.2 (Rationalizing an infinite difference)

Problem Statement:

Evaluate the following limit by multiplying by the conjugate expression:

$$\lim_{x \rightarrow \infty} \left(\sqrt{x^2 + 6x} - x \right)$$

Exercise 1.10.3 (When the difference of squares leaves a quadratic numerator)**Problem Statement:**

Evaluate the following limit without using a calculator:

$$\lim_{x \rightarrow \infty} (\sqrt{9x^2 + x} - 3x)$$

Exercise 1.10.4 (Reverse-engineering a surd parameter at infinity)**Problem Statement:**

An HKDSE M1/M2 exam question defines a limit containing an unknown constant k :

$$\lim_{x \rightarrow \infty} (\sqrt{16x^2 + kx} - 4x) = 5$$

By rationalizing the expression and dividing by the highest power of x , set up an equation to find the exact value of k .

Exercise 1.10.5 (The Universal Surd Difference Formula)**Problem Statement:**

Let $a > 0$ and b be real constants. Prove algebraically that for any linear coefficient b :

$$\lim_{x \rightarrow \infty} \left(\sqrt{a^2 x^2 + bx} - ax \right) = \frac{b}{2a}$$

Hint: Apply conjugate rationalization, then divide top and bottom by x .

Exercise 1.10.6 (Why conjugate multiplication is unnecessary when leading terms don't cancel)**Problem Statement:**

Consider the limit:

$$\lim_{x \rightarrow \infty} \left(\sqrt{4x^2 + 3x} - x \right)$$

1. Explain why this expression is NOT an indeterminate form that requires rationalization. What is the effective coefficient of x inside the square root versus outside?
 2. Factor out x directly from both terms: $\lim_{x \rightarrow \infty} x \left(\sqrt{4 + \frac{3}{x}} - 1 \right)$.
 3. Evaluate the limit as $x \rightarrow \infty$ and explain why it diverges to $+\infty$. Why would blindly rationalizing this expression be a waste of time?
-

Exercise 1.10.7 (Higher-order radical differences: The Cube Root Challenge)**Problem Statement:**

While square root differences $\infty - \infty$ are resolved using $(A - B)(A + B) = A^2 - B^2$, cube root differences require the difference of cubes identity:

$$A^3 - B^3 = (A - B)(A^2 + AB + B^2) \implies A - B = \frac{A^3 - B^3}{A^2 + AB + B^2}$$

Consider the infinite cube root difference:

$$L = \lim_{x \rightarrow \infty} \left(\sqrt[3]{x^3 + 6x^2} - x \right)$$

1. Let $A = \sqrt[3]{x^3 + 6x^2}$ and $B = x$. Use the cube root rationalization formula above to rewrite the expression as a fraction.
 2. Simplify the numerator $A^3 - B^3$. What polynomial remains on top?
 3. What is the highest effective degree of x present in the denominator $(A^2 + AB + B^2)$?
Divide both numerator and denominator by this power of x .
 4. Evaluate the limit as $x \rightarrow \infty$ to show that $L = 2$.
-

1.11 Limit to negative infinity with surds

In Section 1.10, we learned to ingest variables into square roots using the identity $x = \sqrt{x^2}$. However, that identity is only true for positive x . When x is negative (which is the case when we analyze $x \rightarrow -\infty$), the algebra undergoes a dramatic change.

Because a principal square root must always yield a non-negative output, evaluating $\sqrt{(-5)^2}$ gives positive 5, not -5 . Therefore, the absolute correct algebraic identity is $\sqrt{x^2} = |x|$. For negative numbers, this means $\sqrt{x^2} = -x$, which implies $x = -\sqrt{x^2}$. Failing to account for this single negative sign is the absolute #1 cause of algebraic errors in HKDSE limit questions.

To completely eliminate sign confusion and make our calculations foolproof, we will consistently apply the ****Mirror Substitution Principle**** ($u = -x$). This transforms our negative infinity limit into a standard positive infinity limit where we can safely use $u = \sqrt{u^2}$ without any sign anxiety!

Theorem 1.11: Evaluation of Surd Limits at Negative Infinity

To evaluate the limit of a surd-containing function as $x \rightarrow -\infty$:

$$\lim_{x \rightarrow -\infty} f(x)$$

we apply the following three-step systematic transformation:

1. **Mirror Substitution:** Substitute $x = -u$. As $x \rightarrow -\infty$, the new variable $u \rightarrow \infty$.
2. **Radical Handling:** Since u is strictly positive as $u \rightarrow \infty$, we can utilize the standard positive identity:

$$u = \sqrt{u^2}$$

3. **Evaluation:** Divide the numerator and denominator by the highest power of u present in the denominator, evaluating the remaining terms using $\lim_{u \rightarrow \infty} \frac{c}{u^p} = 0$.

Student Proof Demonstration

Use this space to demonstrate the proof of the theorem above. Start from basic algebraic identities.

Practical Example 1.11

Evaluate the following limit at negative infinity using mirror substitution:

$$\lim_{x \rightarrow -\infty} \frac{\sqrt{9x^2 + 2}}{2x - 5}$$

Step-by-Step Solution

Step 1: Apply the mirror substitution.

Let $u = -x$, which means $x = -u$. As $x \rightarrow -\infty$, the variable $u \rightarrow \infty$:

$$\begin{aligned} &= \lim_{u \rightarrow \infty} \frac{\sqrt{9(-u)^2 + 2}}{2(-u) - 5} \\ &= \lim_{u \rightarrow \infty} \frac{\sqrt{9u^2 + 2}}{-2u - 5} \end{aligned}$$

Step 2: Identify highest power of u and divide.

The highest power of u in the denominator is u^1 . We divide top and bottom by u :

$$= \lim_{u \rightarrow \infty} \frac{\frac{\sqrt{9u^2 + 2}}{u}}{\frac{-2u - 5}{u}}$$

$$= \lim_{u \rightarrow \infty} \frac{\frac{\sqrt{9u^2 + 2}}{u}}{-2 - \frac{5}{u}}$$

Step 3: Ingest u into the square root.

Since $u \rightarrow \infty$, we know $u > 0$, so we can write $u = \sqrt{u^2}$ to bring the divisor inside the radical:

$$= \lim_{u \rightarrow \infty} \frac{\sqrt{\frac{9u^2 + 2}{u^2}}}{-2 - \frac{5}{u}}$$

$$= \lim_{u \rightarrow \infty} \frac{\sqrt{9 + \frac{2}{u^2}}}{-2 - \frac{5}{u}}$$

Step 4: Evaluate the limit.

As $u \rightarrow \infty$, both $\frac{2}{u^2} \rightarrow 0$ and $\frac{5}{u} \rightarrow 0$:

$$= \frac{\sqrt{9+0}}{-2-0}$$

$$= \frac{3}{-2} = -1.5$$

Guided Thinking & Hints

Key Hint: Notice that squaring negative u inside the radical gives positive u^2 , while the linear term in the denominator becomes negative: $2(-u) = -2u$.

Key Hint: Never split the denominator into multiple separate limits yet! Keep the entire fraction intact under a single limit symbol.

Key Hint: RADICAL INGESTION: Because u is positive, we do not need to worry about any external negative signs when putting it under the square root.

Key Hint: The final limit is negative! If we had incorrectly assumed $\sqrt{x^2} = x$ for negative numbers, we would have got positive 1.5—a fatal error.

Exercise 1.11.1 (Standard negative infinity surd fraction)**Problem Statement:**

Evaluate the following limit by applying the mirror substitution $u = -x$:

$$\lim_{x \rightarrow -\infty} \frac{\sqrt{x^2 + 4}}{3x - 1}$$

Exercise 1.11.2 (Rationalizing an infinite difference at negative infinity)**Problem Statement:**

Evaluate the following limit using both mirror substitution and conjugate rationalization:

$$\lim_{x \rightarrow -\infty} \left(\sqrt{x^2 + 4x} + x \right)$$

Exercise 1.11.3 (Polynomial-surd combination at negative infinity)**Problem Statement:**

Evaluate the following limit without using a calculator:

$$\lim_{x \rightarrow -\infty} \frac{x+2}{\sqrt{4x^2-x}}$$

Exercise 1.11.4 (Solving for unknown constants with negative infinity surds)**Problem Statement:**

An HKDSE M1/M2 exam problem states that:

$$\lim_{x \rightarrow -\infty} \frac{\sqrt{ax^2+5x}}{2x+1} = -2$$

where a is a positive constant. Find the exact value of a using the mirror substitution method.

Exercise 1.11.5 (General left-hand limit theorem for surd fractions)**Problem Statement:**

Let $a > 0$ and c be real constants. Prove algebraically that for any constant c :

$$\lim_{x \rightarrow -\infty} \frac{\sqrt{a^2x^2 + cx}}{x} = -a$$

Explain why the output is always negative, regardless of the sign of c .

Exercise 1.11.6 (Why direct division fails without substitution)**Problem Statement:**

A student attempts to evaluate $\lim_{x \rightarrow -\infty} \frac{\sqrt{16x^2}}{x}$ by directly dividing the top and bottom by x . The student writes:

$$\frac{\sqrt{16x^2}}{x} = \sqrt{\frac{16x^2}{x^2}} = \sqrt{16} = 4$$

and concludes the limit is 4. Explain algebraically why the student's step $\frac{\sqrt{16x^2}}{x} = \sqrt{\frac{16x^2}{x^2}}$ is completely illegal for negative values of x . Show how using the absolute value identity $\sqrt{x^2} = |x|$ fixes their method.

Exercise 1.11.7 (Shattering Left-Right Symmetry: Two Horizontal Asymptotes)**Problem Statement:**

Consider the algebraic curve:

$$y = \frac{2x}{\sqrt{x^2 + 1}}$$

1. Evaluate the right-hand end behavior of the curve: $\lim_{x \rightarrow \infty} y$.
 2. Evaluate the left-hand end behavior of the curve: $\lim_{x \rightarrow -\infty} y$ using $u = -x$.
 3. State the equations of the two distinct horizontal asymptotes of this curve.
 4. Sketch a quick mental coordinate grid or explain in words how a single unbroken curve can approach two completely different constant heights as we travel infinitely far to the left versus to the right.
-

1.12 Limit to positive infinity with exponential function

Up to this point, we have primarily dealt with variable terms of the form x^p , where the base is the variable and the exponent is a fixed number. We know these algebraic terms grow quickly, but there is a family of functions whose speed of growth dwarfs polynomials completely: **exponential functions**, where the base is a fixed constant greater than 1 (like $e \approx 2.71828$ or 2) and the variable sits high up in the exponent (like e^x or 2^x).

To put this in perspective, as x grows from 1 to 10, the polynomial term x^3 increases from 1 to 1,000. Meanwhile, the exponential term 2^x increases from 2 to 1,024. But if we let x grow to 100, x^3 becomes 1,000,000, while 2^{100} becomes approximately 1.27×10^{30} —a number so colossally large it exceeds the number of atoms in the human body by a factor of billions!

Because exponential functions grow infinitely faster than polynomials, they dominate limits at positive infinity. To evaluate indeterminate forms like $\frac{\infty}{\infty}$ involving exponentials, we use a similar logic to our polynomial rules: **we divide every term by the fastest-growing exponential term present in the denominator**.

Theorem 1.12: Behavior of Exponential Functions at Positive Infinity

Let b be a real number representing the base of an exponential function.

1. **Unbounded Growth (Base > 1):** If $b > 1$, then as $x \rightarrow \infty$, the exponential term grows without bound:

$$\lim_{x \rightarrow \infty} b^x = \infty \quad \text{and specifically} \quad \lim_{x \rightarrow \infty} e^x = \infty$$

2. **Decay to Zero (Base < 1):** If $0 < b < 1$, then as $x \rightarrow \infty$, the exponential term shrinks to zero:

$$\lim_{x \rightarrow \infty} b^x = 0 \quad \text{and specifically} \quad \lim_{x \rightarrow \infty} e^{-x} = \lim_{x \rightarrow \infty} \left(\frac{1}{e}\right)^x = 0$$

When evaluating rational fractions at positive infinity containing multiple exponential terms, we divide both the numerator and the denominator by the dominant base term b^x (where $b > 1$ is the largest base in the denominator) to force all other exponential terms to decay to 0.

Student Proof Demonstration

Use this space to demonstrate the proof of the theorem above. Start from basic algebraic identities.

Practical Example 1.12

Evaluate the following limit at positive infinity:

$$\lim_{x \rightarrow \infty} \frac{5 \cdot 3^x - 2^x}{4 \cdot 3^x + 7 \cdot 2^x}$$

Step-by-Step Solution**Step 1: Inspect the behavior.**

As $x \rightarrow \infty$, both $3^x \rightarrow \infty$ and $2^x \rightarrow \infty$. This results in the indeterminate form $\frac{\infty}{\infty}$.

Step 2: Identify the dominant denominator base.

The denominator is $4 \cdot 3^x + 7 \cdot 2^x$. The exponential term with the largest base is 3^x . We will divide every term in the fraction by 3^x .

Step 3: Divide every term by 3^x .

Divide each term in both the numerator and the denominator:

$$= \lim_{x \rightarrow \infty} \frac{5 \cdot \frac{3^x}{3^x} - \frac{2^x}{3^x}}{4 \cdot \frac{3^x}{3^x} + 7 \cdot \frac{2^x}{3^x}}$$

Step 4: Rewrite using fractional bases.

$$= \lim_{x \rightarrow \infty} \frac{5 - \left(\frac{2}{3}\right)^x}{4 + 7 \cdot \left(\frac{2}{3}\right)^x}$$

Step 5: Apply the decay limit.

Since $0 < \frac{2}{3} < 1$, we know $\lim_{x \rightarrow \infty} \left(\frac{2}{3}\right)^x = 0$:

$$= \frac{5-0}{4+7(0)}$$

$$= \frac{5}{4}$$

Guided Thinking & Hints

Key Hint: Although 3^x and 2^x both head to infinity, 3^x grows much faster than 2^x because its base is larger. We must identify the dominant terms.

Key Hint: Always divide by the largest base present in the **denominator**! This guarantees that the denominator will simplify to a stable non-zero constant instead of collapsing to 0.

Key Hint: Apply the algebraic power law: $\frac{2^x}{3^x} = \left(\frac{2}{3}\right)^x$. This groups the division into a single base term.

Key Hint: Notice that the base $\frac{2}{3}$ is strictly less than 1. This is the perfect setup for applying Theorem 1.12, Rule 2!

Key Hint: The slower-growing 2^x terms completely vanished, leaving only the coefficients of the dominant 3^x terms behind!

Exercise 1.12.1 (Standard natural exponential limit)**Problem Statement:**

Evaluate the following limit by dividing numerator and denominator by e^x :

$$\lim_{x \rightarrow \infty} \frac{3e^x + 4}{2e^x - 1}$$

Exercise 1.12.2 (Negative exponents transition)**Problem Statement:**

Evaluate the following limit by first converting all negative exponents into positive exponential fractions, and then finding the limit:

$$\lim_{x \rightarrow \infty} \frac{8 + 2e^{-x}}{4 - e^{-x}}$$

Exercise 1.12.3 (Competing bases of different magnitudes)**Problem Statement:**

Evaluate the following limit without using a calculator:

$$\lim_{x \rightarrow \infty} \frac{4^{x+1} - 5^x}{5^{x-1} + 3^x}$$

Hint: Use exponential index laws to separate the addition/subtraction in the powers, i.e., $4^{x+1} = 4 \cdot 4^x$ and $5^{x-1} = \frac{1}{5} \cdot 5^x$.

Exercise 1.12.4 (Solving for unknown constants with exponential limits)**Problem Statement:**

An HKDSE M1/M2 exam question states:

$$\lim_{x \rightarrow \infty} \frac{a \cdot 5^x + 2^{x+1}}{3 \cdot 5^{x-1} + b \cdot 2^x} = 10$$

where a and b are constants. Express a explicitly in terms of b , or find its numerical value if possible.

Exercise 1.12.5 (The Dominant Base Theorem)**Problem Statement:**

Let $a, b > 1$ be two real numbers such that $a < b$. Prove algebraically that for any non-zero real constants A, B, C, D :

$$\lim_{x \rightarrow \infty} \frac{A \cdot a^x + B \cdot b^x}{C \cdot a^x + D \cdot b^x} = \frac{B}{D} \quad (\text{where } D \neq 0)$$

Exercise 1.12.6 (When dividing by the wrong base collapses the system)**Problem Statement:**

Consider the limit evaluated in Practical Example 1.12: $\lim_{x \rightarrow \infty} \frac{5 \cdot 3^x - 2^x}{4 \cdot 3^x + 7 \cdot 2^x}$.

1. Attempt to evaluate this limit by dividing both the numerator and the denominator by 2^x (the smaller base) instead of 3^x .
 2. Show the resulting algebraic expression. Explain what mathematical difficulties arise when you attempt to take $x \rightarrow \infty$ on this new expression.
 3. Why does this confirm the rule that we must always divide by the *largest* base in the denominator?
-

Exercise 1.12.7 (Bridge to logistic growth in biology)**Problem Statement:**

In ecology and population biology, populations cannot grow exponentially forever due to limited resources. Instead, they are modeled by the **Logistic Growth Function**:

$$P(t) = \frac{K}{1 + Ae^{-rt}}$$

where $t \geq 0$ represents time, $K > 0$ is the carrying capacity, $r > 0$ is the growth rate, and $A > 0$ is a constant related to the initial population.

1. Evaluate the theoretical population limit as time approaches infinity: $\lim_{t \rightarrow \infty} P(t)$.
2. Explain what this limit value represents physically. Why is it independent of the growth rate r and the constant A ?
3. Suppose a population of rabbits on an isolated island is modeled by $P(t) = \frac{5000}{1 + 9e^{-0.2t}}$, where t is measured in years. Find the initial population $P(0)$, and find how many years it takes for the population to reach 80% of its ultimate carrying capacity.

1.13 Limit to negative infinity with exponential function

In Section 1.12, we witnessed the terrifying speed at which exponential functions grow when the variable x heads to positive infinity. But what happens when we reverse our direction and march toward negative infinity ($x \rightarrow -\infty$)?

Let us build our intuition using a simple physical process: half-life radioactive decay. If you start with a mass of a substance, its remaining mass over time is modeled by $M(t) = M_0 \cdot e^{-rt}$. As time goes forward into the future ($t \rightarrow \infty$), the mass decays rapidly toward 0 ($e^{-\infty} \rightarrow 0$). Conversely, if we look back into the indefinite past ($t \rightarrow -\infty$), the mass must have been theoretically infinite!

This reveals a profound asymmetry in exponential behavior: while e^x explodes to ∞ on the right side of the graph, it decays to a flat 0 on the left side of the graph. To avoid making disastrous sign mistakes when dealing with combinations of e^x and e^{-x} at negative infinity, we will once again deploy our absolute favorite mathematical shield: the **Mirror Substitution Principle** ($u = -x$). By converting our limit into $u \rightarrow \infty$, we can evaluate the expressions cleanly without getting tangled up in double-negative exponent rules!

Theorem 1.13: Behavior of Exponential Functions at Negative Infinity

Let b be a real number representing the base of an exponential function such that $b > 1$.

1. **Decay to Zero:** As $x \rightarrow -\infty$, the positive power of b shrinks to zero:

$$\lim_{x \rightarrow -\infty} b^x = 0 \quad \text{and specifically} \quad \lim_{x \rightarrow -\infty} e^x = 0$$

2. **Unbounded Growth:** As $x \rightarrow -\infty$, the negative power of b explodes to infinity:

$$\lim_{x \rightarrow -\infty} b^{-x} = \infty \quad \text{and specifically} \quad \lim_{x \rightarrow -\infty} e^{-x} = \infty$$

To evaluate limits of rational expressions containing exponential terms as $x \rightarrow -\infty$, we apply the mirror substitution $u = -x$ (such that $u \rightarrow \infty$), simplify the exponents, and divide by the largest base present in the new denominator.

Student Proof Demonstration

Use this space to demonstrate the proof of the theorem above. Start from basic algebraic identities.

Practical Example 1.13

Evaluate the following limit at negative infinity using mirror substitution:

$$\lim_{x \rightarrow -\infty} \frac{3 \cdot 5^x + 4 \cdot 2^{-x}}{2 \cdot 5^x - 7 \cdot 2^{-x}}$$

Step-by-Step Solution

Step 1: Apply the mirror substitution.

Let $u = -x$, which means $x = -u$. As $x \rightarrow -\infty$, the variable $u \rightarrow \infty$:

$$\begin{aligned} &= \lim_{u \rightarrow \infty} \frac{3 \cdot 5^{-u} + 4 \cdot 2^{-(-u)}}{2 \cdot 5^{-u} - 7 \cdot 2^{-(-u)}} \\ &= \lim_{u \rightarrow \infty} \frac{3 \cdot 5^{-u} + 4 \cdot 2^u}{2 \cdot 5^{-u} - 7 \cdot 2^u} \end{aligned}$$

Step 2: Rewrite negative exponents as fractions.

Convert 5^{-u} to $\frac{1}{5^u}$ to make the division visible:

$$= \lim_{u \rightarrow \infty} \frac{3 \cdot \left(\frac{1}{5}\right)^u + 4 \cdot 2^u}{2 \cdot \left(\frac{1}{5}\right)^u - 7 \cdot 2^u}$$

Step 3: Identify the dominant base on the bottom.

The denominator contains $(1/5)^u$ and 2^u . Since $2 > 1/5$, the term 2^u is the dominant base. We divide every term in top and bottom by 2^u :

$$= \lim_{u \rightarrow \infty} \frac{3 \cdot \frac{(1/5)^u}{2^u} + 4 \cdot \frac{2^u}{2^u}}{2 \cdot \frac{(1/5)^u}{2^u} - 7 \cdot \frac{2^u}{2^u}}$$

Step 4: Combine fractional bases and simplify.

Apply the power law $\frac{A^u}{B^u} = \left(\frac{A}{B}\right)^u$:

$$= \lim_{u \rightarrow \infty} \frac{3 \cdot \left(\frac{1}{10}\right)^u + 4}{2 \cdot \left(\frac{1}{10}\right)^u - 7}$$

Step 5: Apply the decay limit.

Since $0 < 1/10 < 1$, we know $\lim_{u \rightarrow \infty} \left(\frac{1}{10}\right)^u = 0$:

$$= \frac{3(0)+4}{2(0)-7}$$

$$= -\frac{4}{7}$$

Guided Thinking & Hints

Key Hint: Watch the double negatives closely! 2^{-x} becomes $2^{-(-u)} = 2^u$. This is where most students make a mistake if they don't substitute.

Key Hint: This step is excellent for building visual clarity. We now see a competition between a decaying fraction $(1/5)^u \rightarrow 0$ and a growing exponential $2^u \rightarrow \infty$.

Key Hint: Always scan the denominator to find the fastest-growing term. Dividing by 2^u forces the dominant growth terms to simplify to stable constants.

Key Hint: To combine, note that: $\frac{(1/5)^u}{2^u} = \left(\frac{1/5}{2}\right)^u = \left(\frac{1}{10}\right)^u$. This is basic fraction multiplication.

Key Hint: The decaying terms successfully vanish to 0, leaving behind the ratio of the coefficients of the dominant terms: $-\frac{4}{7}$.

Exercise 1.13.1 (Standard natural exponential at negative infinity)**Problem Statement:**

Evaluate the following limit by using the mirror substitution $u = -x$:

$$\lim_{x \rightarrow -\infty} \frac{5e^x - 2}{3e^x + 7}$$

Exercise 1.13.2 (Rationalizing exponential fractions)**Problem Statement:**

Evaluate the following limit by rewriting negative powers and dividing by the dominant base:

$$\lim_{x \rightarrow -\infty} \frac{3 \cdot 2^x + 4 \cdot 3^x}{2 \cdot 2^x - 5 \cdot 3^x}$$

Exercise 1.13.3 (Variable bases at negative infinity)**Problem Statement:**

Evaluate the following limit without using a calculator:

$$\lim_{x \rightarrow -\infty} \frac{4^{-x} + 3^x}{2^{-x} - 5^x}$$

Exercise 1.13.4 (Solving for unknown boundaries)**Problem Statement:**

An HKDSE M1/M2 exam question defines a modeling curve for heat loss:

$$\lim_{x \rightarrow -\infty} \frac{a \cdot e^{-x} + 12}{3 \cdot e^{-x} + b \cdot e^x} = 8$$

Find the exact value of the constant a . Explain why the value of b does not affect the outcome of this limit.

Exercise 1.13.5 (General Base Asymmetry Theorem)**Problem Statement:**

Let $a, b > 1$. Prove algebraically that:

$$\lim_{x \rightarrow \infty} \frac{a^x}{a^x + b^x} + \lim_{x \rightarrow -\infty} \frac{a^x}{a^x + b^x} = 1$$

Exercise 1.13.6 (The dangers of direct intuition without substitution)**Problem Statement:**

A student attempts to quickly solve $\lim_{x \rightarrow -\infty} \frac{e^{-x}+2}{e^x-3}$ and argues: "Since e^x goes to 0 as $x \rightarrow -\infty$, we can just substitute 0 for all exponential terms, yielding $\frac{0+2}{0-3} = -\frac{2}{3}$."

1. Use mirror substitution to find the correct value of the limit.
 2. Identify the exact logical error in the student's assumption about the term e^{-x} .
 3. Why does this prove that we cannot blindly replace all exponential terms with 0?
-

Exercise 1.13.7 (Bridge to hyperbolic functions in physics)**Problem Statement:**

In physics, the shapes of hanging power lines and cables are modeled by the hyperbolic cosine function, defined as $\cosh(x) = \frac{e^x + e^{-x}}{2}$. A related function is the **hyperbolic tangent**, defined as:

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

This function is widely used to model terminal velocity with air resistance.

1. Evaluate the right-hand limit: $\lim_{x \rightarrow \infty} \tanh(x)$.
 2. Evaluate the left-hand limit: $\lim_{x \rightarrow -\infty} \tanh(x)$ using $u = -x$.
 3. What do these results tell us about the boundary limits of $\tanh(x)$?
 4. Sketch a quick mental graph or explain in words why this function is ideal for modeling physical transitions that saturate between two distinct directional states (such as maximum forward speed vs maximum reverse speed).
-

1.14 Limit to positive infinity with logarithmic function

We have now analyzed polynomial functions (which grow at a steady, algebraic speed) and exponential functions (which explode with extreme speed). To complete our toolbelt of basic functions, we must investigate the turtle of the mathematical world: **logarithmic functions**, such as $f(x) = \ln(x)$ (logarithm to base e).

Logarithmic functions are the inverse of exponential functions. While e^x reaches astronomical values almost instantly, $\ln(x)$ takes an incredibly long time to grow. For example, to make $\ln(x)$ reach a modest height of 10, you must feed it an input of $x = e^{10} \approx 22,026$. To make it reach 20, x must be $e^{20} \approx 485,165,195$!

Even though logarithmic growth is incredibly slow, it does grow without bound—meaning $\lim_{x \rightarrow \infty} \ln(x) = \infty$. However, because its rate of growth is so lethargic, any positive power of x (even a very small one like $x^{0.01}$ or $\sqrt{x}$) will eventually grow infinitely faster than $\ln(x)$. We can exploit this hierarchy to evaluate limits at positive infinity: in any direct race between a logarithm and a polynomial, the polynomial term in the denominator will always drag the ratio down to 0.

Theorem 1.14: Growth Hierarchy of Logarithms at Positive Infinity

Let $\ln(x)$ denote the natural logarithmic function.

1. **Unbounded but Slow Growth:** As $x \rightarrow \infty$, the logarithm approaches infinity:

$$\lim_{x \rightarrow \infty} \ln(x) = \infty$$

2. **The Log-Polynomial Power Rule:** For any positive rational power $p > 0$:

$$\lim_{x \rightarrow \infty} \frac{\ln(x)}{x^p} = 0$$

This theorem holds true because any polynomial growth rate x^p eventually dwarfs logarithmic growth.

When evaluating limits at positive infinity containing logarithmic terms, we divide by the highest power of x to isolate the $\frac{\ln(x)}{x^p}$ components, which cleanly vanish to 0.

Student Proof Demonstration

Use this space to demonstrate the proof of the theorem above. Start from basic algebraic identities.

Practical Example 1.14

Evaluate the following limit at positive infinity:

$$\lim_{x \rightarrow \infty} \frac{3 \ln(x^4) + 5x}{2x - 7}$$

Step-by-Step Solution

Step 1: Simplify the logarithm using index laws.

Apply the logarithmic power rule, $\ln(a^b) = b \ln(a)$, to bring the exponent of x to the front:

$$3 \ln(x^4) = 3(4 \ln(x)) = 12 \ln(x)$$

Rewrite the limit expression:

$$= \lim_{x \rightarrow \infty} \frac{12 \ln(x) + 5x}{2x - 7}$$

Step 2: Identify the highest power of x in the denominator.

The denominator is $2x - 7$. The highest power of x is x^1 . We divide every term in both the numerator and denominator by x :

$$= \lim_{x \rightarrow \infty} \frac{12 \cdot \frac{\ln(x)}{x} + \frac{5x}{x}}{\frac{2x}{x} - \frac{7}{x}}$$

Step 3: Simplify the fractions.

$$= \lim_{x \rightarrow \infty} \frac{12 \left(\frac{\ln(x)}{x} \right) + 5}{2 - \frac{7}{x}}$$

Step 4: Apply the growth hierarchy limits.

As $x \rightarrow \infty$, we apply $\lim_{x \rightarrow \infty} \frac{\ln(x)}{x} = 0$ and $\lim_{x \rightarrow \infty} \frac{7}{x} = 0$:

$$= \frac{12(0) + 5}{2 - 0}$$

$$= \frac{5}{2}$$

Guided Thinking & Hints

Key Hint: Always use logarithmic properties to simplify your terms before doing calculus! Turning $\ln(x^4)$ into $4 \ln(x)$ makes identifying the variables much easier.

Key Hint: Just like in Section 1.8, we divide by the highest algebraic power of x on the bottom to stabilize the denominator.

Key Hint: Notice how we isolated the term $\frac{\ln(x)}{x}$. This matches the exact template of Theorem 1.14, Rule 2!

Key Hint: Even though the logarithm grows to infinity, its speed is so slow that dividing it by x forces it to vanish to 0, leaving only the algebraic coefficients $\frac{5}{2}$ (or 2.5).

Exercise 1.14.1 (Standard log-linear limit)**Problem Statement:**

Evaluate the following limit by dividing the numerator and denominator by x :

$$\lim_{x \rightarrow \infty} \frac{\ln(x) - 4x}{3x + 2}$$

Exercise 1.14.2 (Using log properties before evaluating)**Problem Statement:**

Evaluate the following limit by first combining the logarithmic subtraction into a single fraction using the property $\ln(a) - \ln(b) = \ln\left(\frac{a}{b}\right)$:

$$\lim_{x \rightarrow \infty} \ln(3x^2 + 5) - \ln(x^2 + 1)$$

Exercise 1.14.3 (Logarithms paired with surds)**Problem Statement:**

Evaluate the following limit without using a calculator:

$$\lim_{x \rightarrow \infty} \frac{\ln(x) + \sqrt{x}}{2\sqrt{x} - 5}$$

Hint: Identify the highest power of x in the denominator as $x^{1/2} = \sqrt{x}$, and apply the hierarchy rules.

Exercise 1.14.4 (Tuning logarithmic boundaries)**Problem Statement:**

An HKDSE M1/M2 exam question defines a function for information processing capacity:

$$f(x) = \frac{a \ln(x^3) + 8x}{b \ln(x) + 2x}$$

Given that $\lim_{x \rightarrow \infty} f(x) = 4$, determine the relationship between the constants a and b , or find their numerical values if possible.

Exercise 1.14.5 (General Log-Polynomial Theorem)**Problem Statement:**

Let k, c be non-zero real constants and let $p > 0$. Prove algebraically that for any base $b > 1$:

$$\lim_{x \rightarrow \infty} \frac{\log_b(x^k)}{cx^p} = 0$$

Exercise 1.14.6 (When the logarithm is in the denominator)**Problem Statement:**

Consider the limit:

$$\lim_{x \rightarrow \infty} \frac{3x + 2}{\ln(x)}$$

1. Explain intuitively why this limit diverges to infinity rather than collapsing to 0. Which term is growing faster?
 2. Divide the top and bottom by x and show the resulting expression.
 3. Explain why applying the limit as $x \rightarrow \infty$ to your expression in part (2) yields $\frac{\infty}{0}$, confirming the divergence.
-

Exercise 1.14.7 (Bridge to Section 2: Limit of the difference quotient of $\ln(x)$)**Problem Statement:**

In Section 2, we will prove that the derivative of $g(x) = \ln(x)$ at $x = c > 0$ is given by the limit of its secant slopes:

$$g'(c) = \lim_{h \rightarrow 0} \frac{\ln(c+h) - \ln(c)}{h}$$

We wish to evaluate this limit analytically.

1. Use the logarithmic subtraction property to rewrite the numerator as a single natural logarithm.
2. Bring the coefficient $\frac{1}{h}$ inside the logarithm as an exponent. Show that the expression becomes:

$$\lim_{h \rightarrow 0} \ln \left[\left(1 + \frac{h}{c} \right)^{\frac{1}{h}} \right]$$

3. Let $w = \frac{h}{c}$, which means $h = cw$. Note that as $h \rightarrow 0$, $w \rightarrow 0$. Substitute w into your expression to show it becomes:

$$\lim_{w \rightarrow 0} \ln \left[(1+w)^{\frac{1}{cw}} \right] = \ln \left[\left(\lim_{w \rightarrow 0} (1+w)^{\frac{1}{w}} \right)^{\frac{1}{c}} \right]$$

4. Recall the famous definition of the mathematical constant e : $\lim_{w \rightarrow 0} (1+w)^{\frac{1}{w}} = e$. Substitute this limit into your expression to prove that:

$$g'(c) = \frac{1}{c}$$

What fundamental derivative rule of logarithmic functions does this rigorous limit calculation prove?

1.15 The inequality $\ln(x) < P(x) < E(x)$ and mixed rational limit

We have reached the grand finale of Section 1! In our journey through limits, we have analyzed three distinct mathematical kingdoms: the incredibly slow logarithmic functions ($\ln x$), the steady algebraic polynomial functions (x^k), and the explosively fast exponential functions (e^x).

When these three kingdoms collide in a single fraction—such as $\lim_{x \rightarrow \infty} \frac{e^x + x^5}{\ln(x) + 3^x}$ —how do we determine which function dictates the final limit? Rather than memorizing isolated algebraic tricks, we can visualize this as a grand race at positive infinity. On the starting line, a logarithmic turtle, a polynomial horse, and an exponential jet engine begin to move. As $x \rightarrow \infty$, the gaps between their positions grow so astronomically wide that the slower participants become completely insignificant. In this section, we formalize this race using a master growth hierarchy inequality:

$$\ln(x) \ll x^p \ll b^x \quad (\text{for } p > 0, b > 1)$$

By mastering this hierarchy, we can instantly analyze any complex, mixed-function limit at infinity simply by identifying and isolating the single dominant champion of the numerator and the denominator.

Theorem 1.15: The Grand Growth Hierarchy at Positive Infinity

Let $p, q > 0$ be any positive real powers, and let $a, b > 1$ be exponential bases.

As $x \rightarrow \infty$, the master growth hierarchy states:

$$\ln(x) \ll x^p \ll x^{p+q} \ll a^x \ll b^x \quad (\text{assuming } a < b)$$

where the symbol $\ll$ means "grows infinitely slower than". This hierarchy implies the following limit behavior:

1. **Logarithm vs. Polynomial:** $\lim_{x \rightarrow \infty} \frac{\ln(x)}{x^p} = 0$
2. **Polynomial vs. Exponential:** $\lim_{x \rightarrow \infty} \frac{x^p}{a^x} = 0$
3. **Logarithm vs. Exponential:** $\lim_{x \rightarrow \infty} \frac{\ln(x)}{a^x} = 0$

When evaluating a mixed rational limit at positive infinity, we divide every term in the fraction by the single fastest-growing term present in the entire denominator to resolve the indeterminacy.

Student Proof Demonstration

Use this space to demonstrate the proof of the theorem above. Start from basic algebraic identities.

Practical Example 1.15

Evaluate the following mixed rational limit at positive infinity:

$$\lim_{x \rightarrow \infty} \frac{e^x + x^4 - \ln(x)}{2e^x + 5x^2}$$

Step-by-Step Solution

Step 1: Identify the growth rates of all terms.

The numerator contains e^x (exponential), x^4 (polynomial), and $\ln(x)$ (logarithmic).

The denominator contains e^x (exponential) and x^2 (polynomial).

Step 2: Identify the dominant term in the denominator.

The denominator is $2e^x + 5x^2$. The term with the fastest growth rate in this expression is e^x . We will divide every term in the fraction by e^x :

$$= \lim_{x \rightarrow \infty} \frac{\frac{e^x}{e^x} + \frac{x^4}{e^x} - \frac{\ln(x)}{e^x}}{2\frac{e^x}{e^x} + 5\frac{x^2}{e^x}}$$

Step 3: Simplify the expression.

$$= \lim_{x \rightarrow \infty} \frac{1 + \left(\frac{x^4}{e^x}\right) - \left(\frac{\ln(x)}{e^x}\right)}{2 + 5\left(\frac{x^2}{e^x}\right)}$$

Step 4: Apply the growth hierarchy limits.

Using Theorem 1.15, as $x \rightarrow \infty$, we know:

$$\lim_{x \rightarrow \infty} \frac{x^4}{e^x} = 0, \lim_{x \rightarrow \infty} \frac{\ln(x)}{e^x} = 0, \text{ and } \lim_{x \rightarrow \infty} \frac{x^2}{e^x} = 0.$$

Substitute these limits:

$$= \frac{1+0-0}{2+5(0)}$$

$$= \frac{1}{2}$$

Guided Thinking & Hints

Key Hint: Before doing any algebra, perform a mental scan. According to the hierarchy, the exponential e^x will eventually dwarf the polynomial and logarithmic terms. It is the dominant term on both top and bottom!

Key Hint: Always divide by the dominant term of the **denominator**! This stabilizes the bottom of the fraction into a non-zero real constant, avoiding division by zero.

Key Hint: Look at how beautifully the fraction is structured now. Every non-dominant term has been transformed into a ratio comparing a slower term with a faster term!

Key Hint: Since e^x grows infinitely faster than x^4 and $\ln(x)$, the slower terms contribute absolutely nothing to the limit at infinity, leaving only the coefficients of e^x : $\frac{1}{2}$!

Exercise 1.15.1 (Standard mixed rational limit)**Problem Statement:**

Evaluate the following limit by dividing every term by the dominant term in the denominator:

$$\lim_{x \rightarrow \infty} \frac{x^5 + e^x}{3e^x - \ln(x)}$$

Exercise 1.15.2 (Hierarchy with negative exponents)**Problem Statement:**

Evaluate the following limit by first converting negative exponents to positive fractions and identifying the dominant terms:

$$\lim_{x \rightarrow \infty} \frac{x^2 + e^{-x} + 3 \ln(x)}{4x^2 - 5 \ln(x)}$$

Exercise 1.15.3 (Competing bases and powers)**Problem Statement:**

Evaluate the following limit without using a calculator:

$$\lim_{x \rightarrow \infty} \frac{2 \cdot 5^x + x^{100}}{5^{x-1} + e^x + \ln(x^8)}$$

Exercise 1.15.4 (Mixed coefficient system)**Problem Statement:**

An HKDSE M1/M2 exam question defines a model of internet bandwidth demand:

$$D(t) = \frac{a \cdot e^t + bt^3 + \ln(t)}{4 \cdot e^t + 2t^2}$$

Given that $\lim_{t \rightarrow \infty} D(t) = 3$, find the exact value of the constant a . Explain why the value of b does not affect this long-term demand limit.

Exercise 1.15.5 (The Dominant Ratio Theorem)**Problem Statement:**

Let $A(x)$, $B(x)$, $C(x)$, and $D(x)$ be functions such that $A(x) \ll B(x)$ and $C(x) \ll D(x)$ as $x \rightarrow \infty$. If $B(x)$ and $D(x)$ are of the same growth order (meaning $\lim_{x \rightarrow \infty} \frac{B(x)}{D(x)} = k$ for some non-zero constant k), prove algebraically that:

$$\lim_{x \rightarrow \infty} \frac{A(x) + B(x)}{C(x) + D(x)} = k$$

Exercise 1.15.6 (Evaluating infinite ratios containing non-growing terms)**Problem Statement:**

Consider the limit:

$$\lim_{x \rightarrow \infty} \frac{x^2 \sin(x) + e^x}{e^x + \ln(x)}$$

1. Explain why the term $\sin(x)$ does not grow as $x \rightarrow \infty$, and state its upper and lower bounds.
 2. Divide the top and bottom by the dominant term e^x .
 3. Use the Squeeze Theorem to prove that $\lim_{x \rightarrow \infty} \frac{x^2 \sin(x)}{e^x} = 0$.
 4. Hence, evaluate the overall limit.
-

Exercise 1.15.7 (The ultimate growth comparison: Factorials vs. Exponentials)**Problem Statement:**

In computer science and discrete mathematics, we often study growth rates that exceed even standard exponential functions. One such function is the **factorial function**, defined for positive integers as:

$$n! = n \cdot (n-1) \cdot (n-2) \cdots 3 \cdot 2 \cdot 1$$

Consider the limit of the ratio of a base- b exponential to a factorial as $n \rightarrow \infty$:

$$L = \lim_{n \rightarrow \infty} \frac{b^n}{n!} \quad (\text{where } b > 1)$$

1. Let k be a fixed integer such that $k > b$. For any $n > k$, write out the product expansion of the fraction:

$$\frac{b^n}{n!} = \left(\frac{b}{1} \cdot \frac{b}{2} \cdots \frac{b}{k} \right) \cdot \left(\frac{b}{k+1} \cdot \frac{b}{k+2} \cdots \frac{b}{n} \right)$$

2. Let $C = \left(\frac{b}{1} \cdot \frac{b}{2} \cdots \frac{b}{k} \right)$ be a fixed constant. Explain why each term in the second set of brackets satisfies the inequality:

$$\frac{b}{k+m} < \frac{b}{k}$$

3. Use these relations to prove the inequality:

$$0 < \frac{b^n}{n!} < C \cdot \left(\frac{b}{k} \right)^{n-k}$$

4. Since $k > b$, we know $0 < \frac{b}{k} < 1$. Apply the Squeeze Theorem as $n \rightarrow \infty$ to prove that $L = 0$.
5. What does this imply about the growth rate of factorials compared to exponentials? Where would $n!$ fit in our master growth hierarchy?

2 Derivative

2.1 Slope of straight line

In Section 1, we spent a great deal of time mastering the concept of a limit—learning how to inspect where a path is heading as we get infinitely close to a point. Why did we do this? The answer lies in our desire to understand **change**.

In our everyday lives, everything is constantly changing: the speed of a car, the temperature of a room, the stock market, or the population of a city. To study these changes mathematically, we must start with the simplest type of change: a steady, constant change. Imagine walking up a straight, uniform ramp. No matter where you stand on that ramp, every step forward lifts you by the exact same height. This constant steepness is what we call the **slope** or **gradient** of a straight line.

In junior secondary mathematics, you learned that slope is simply "rise over run". It tells us how much our vertical output (y) changes for every single unit we move to the right (x). Because straight lines are perfectly uniform, this rate of change is identical everywhere along the line. Before we can measure the changing steepness of curves, we must firmly anchor our understanding in the algebraic and geometric mechanics of this constant slope.

Proposition 2.1: Constancy of the Slope of a Straight Line

Let L be a non-vertical straight line in the coordinate plane. Let $P_1(x_1, y_1)$, $P_2(x_2, y_2)$, and $P_3(x_3, y_3)$ be any three distinct points lying on L .

The **slope** m of the line segment connecting any two points on the line is constant. That is:

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{y_3 - y_2}{x_3 - x_2}$$

where $x_2 \neq x_1$ and $x_3 \neq x_2$. This constant ratio represents the rate of change of y with respect to x .

Student Proof Demonstration

Use this space to demonstrate the proof of the theorem above. Start from basic algebraic identities.

Practical Example 2.1

Find the slope m of the straight line passing through the points $A(1, 3)$ and $B(4, 9)$, and find the coordinates of a third point C on this line that has an x -coordinate of 6.

Step-by-Step Solution**Step 1: Assign coordinates.**

Let the first point be $P_1(x_1, y_1) = (1, 3)$ and the second point be $P_2(x_2, y_2) = (4, 9)$.

Step 2: Calculate the slope m .

Apply the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$:

$$m = \frac{9-3}{4-1}$$

$$m = \frac{6}{3} = 2$$

Step 3: Set up the equation for the third point C .

Let the coordinates of point C be $(6, y_3)$. Since C lies on the same line, the slope between $B(4, 9)$ and $C(6, y_3)$ must also equal 2:

$$\frac{y_3 - 9}{6 - 4} = 2$$

Step 4: Solve for the missing coordinate.

Simplify the denominator:

$$\frac{y_3 - 9}{2} = 2$$

Multiply both sides by 2:

$$y_3 - 9 = 4$$

Add 9 to both sides:

$$y_3 = 13$$

Therefore, the coordinates of point C are $(6, 13)$.

Guided Thinking & Hints

Key Hint: Label your points clearly on your paper! Getting the coordinates swapped (like mixing up x_1 and y_1) is the most common arithmetic slip-up for beginners.

Key Hint: Think of the result geometrically: a slope of 2 means that for every 1 step we take to the right, the line rises up by 2 units.

Key Hint: Since the slope is constant everywhere along a straight line, we can pair our new point with any of the previous points (A or B) to set up this equation.

Key Hint: Check your answer using basic arithmetic! Starting at $B(4, 9)$, moving 2 steps right to $x = 6$ means our height must rise by $2 \times 2 = 4$ units. $9 + 4 = 13$. It matches perfectly!

Exercise 2.1.1 (Finding basic positive and negative slopes)**Problem Statement:**

Find the slope of the straight line passing through each of the following pairs of points:

1. $A(2, 5)$ and $B(5, 14)$
 2. $C(-1, 4)$ and $D(3, -4)$
-

Exercise 2.1.2 (Slope of parallel and perpendicular lines)**Problem Statement:**

Recall from Junior Secondary Coordinate Geometry that:

- Parallel lines have identical slopes ($m_1 = m_2$).
- Perpendicular lines have slopes that multiply to -1 ($m_1 \cdot m_2 = -1$).

A straight line L_1 passes through $P(1, 2)$ and $Q(3, 8)$.

1. Find the slope of a line L_2 that is parallel to L_1 .
 2. Find the slope of a line L_3 that is perpendicular to L_1 .
-

Exercise 2.1.3 (Finding a missing coordinate from a given slope)**Problem Statement:**

A straight line has a slope of $-\frac{3}{4}$. It passes through the points $A(-2, 5)$ and $B(6, k)$. Find the exact value of the constant k .

Exercise 2.1.4 (Collinear points and linear modeling)**Problem Statement:**

Three points, $P(2, -3)$, $Q(5, k)$, and $R(11, 15)$, are **collinear** (meaning they all lie on the exact same straight line).

1. By comparing the slope of segment PR with segment PQ , find the value of k .
 2. Find the equation of the straight line passing through these points, expressing your answer in the slope-intercept form $y = mx + c$.
-

Exercise 2.1.5 (The general slope-intercept form derivation)**Problem Statement:**

Let a straight line have a constant slope m , and let it cross the vertical y -axis at the point $(0, c)$ (where c is the y -intercept). By choosing any arbitrary point (x, y) on the line, use the slope definition to prove algebraically that the relationship between x and y must satisfy the classic equation:

$$y = mx + c$$

Exercise 2.1.6 (Zero slope versus undefined slope)**Problem Statement:**

Consider the following two straight lines:

- Line L_1 passing through $A(2, 5)$ and $B(7, 5)$.
 - Line L_2 passing through $C(3, 1)$ and $D(3, 6)$.
1. Calculate the slope of L_1 . Describe the geometric orientation of this line.
 2. Attempt to calculate the slope of L_2 using our formula. What mathematical barrier arises?
 3. Explain why we say the slope of a vertical line is "undefined", and distinguish this from a slope of "zero".
-

Exercise 2.1.7 (The bridge to curves: Average rate of change)**Problem Statement:**

Suppose a moving particle's position $s(t)$ (in meters) at time t (in seconds) is modeled by the non-linear quadratic function:

$$s(t) = t^2 + 1$$

Because this is a curve, its actual steepness changes at every instant. However, we can approximate its behavior by drawing a straight secant line between two time coordinates.

1. Calculate the positions of the particle at $t_1 = 1$ and $t_2 = 3$.
 2. Find the slope of the straight line connecting the points $(1, s(1))$ and $(3, s(3))$. This slope represents the **average velocity** of the particle over the interval $[1, 3]$.
 3. Now, calculate the average velocity over the tighter intervals $[1, 2]$ and $[1, 1.1]$.
 4. Synthesize your findings: As the second time coordinate gets closer and closer to 1, what value does the average velocity seem to approach? How does this link the slope of a straight line with our limit concepts from Section 1?
-

2.2 Slope of Secant line

In Section 2.1, we reviewed the constant slope of a straight line, defined by the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. This formula is highly reliable, but it has a massive limitation: it only works when the path is perfectly straight. What happens when we are dealing with a curve?

If you drive a car along a winding, curved road, your speed and direction are changing at every fraction of a second. If we pick two distant times, say $t_1 = 1$ and $t_2 = 3$, we can draw a straight line that cuts directly through the curve at those two points. In geometry, a straight line that intersects a curve at two or more distinct points is called a **secant line**.

The slope of this secant line does not tell us our exact speed at any single instant, but it gives us the **average rate of change** over that interval. By calculating the slope of the secant line, we create an algebraic bridge: we use our simple, junior-level "rise over run" formula on a straight secant line to approximate the complex, senior-level behavior of a curve!

Definition 2.2: Slope of a Secant Line

Let $y = f(x)$ be a continuous curve in the coordinate plane. Let $P(c, f(c))$ be a fixed point on the curve, and let $Q(c + h, f(c + h))$ be a neighboring point on the curve, where $h \neq 0$ represents the horizontal distance (or run) between the two points.

The straight line passing through P and Q is called the **secant line**. The slope of this secant line, denoted as m_{sec} , is given by the formula:

$$m_{\text{sec}} = \frac{f(c + h) - f(c)}{h}$$

This quotient represents the **average rate of change** of the function $f(x)$ over the interval $[c, c + h]$.

Practical Example 2.2

Consider the quadratic curve $f(x) = x^2$. Find the slope of the secant line passing through the curve at the fixed point where $x = 1$ and a neighboring point where the horizontal distance is $h = 0.5$.

Step-by-Step Solution

Guided Thinking & Hints

Step 1: Identify the components from the problem.

The function is $f(x) = x^2$.

The starting point is at $c = 1$.

The step size is $h = 0.5$.

Key Hint: Write down your variables clearly first. Here, the neighboring x -coordinate is $c + h = 1 + 0.5 = 1.5$.

Step 2: Find the coordinates of points P and Q .

For the fixed point P :

$x_1 = 1 \implies y_1 = f(1) = 1^2 = 1$. Thus, $P(1, 1)$.

For the neighboring point Q :

Key Hint: Remember that to find the y -coordinate, you must plug the x -value back into the original curve's equation! Weak algebra students often confuse the step size h with the actual y -value.

$x_2 = 1.5 \implies y_2 = f(1.5) = 1.5^2 = 2.25$. Thus, $Q(1.5, 2.25)$.

Step 3: Apply the secant slope formula.

Substitute the values into $m_{\text{sec}} = \frac{f(c+h)-f(c)}{h}$:

$$m_{\text{sec}} = \frac{2.25-1}{0.5}$$

Step 4: Calculate the final slope.

$$m_{\text{sec}} = \frac{1.25}{0.5} = 2.5$$

Key Hint: This formula is exactly the same as $\frac{y_2-y_1}{x_2-x_1}$. We are just using functional notation to represent the vertical rise and horizontal run.

Key Hint: Dividing by 0.5 is the same as multiplying by 2. Our secant line rises at a rate of 2.5 units vertically for every 1 unit horizontally.

Therefore, the slope of the secant line over the interval $[1, 1.5]$ is 2.5.

Exercise 2.2.1 (Secant slope on a quadratic curve)

Problem Statement:

Consider the curve $f(x) = 3x^2 + 2$.

1. Find the coordinates of the points on the curve corresponding to $x_1 = 2$ and $x_2 = 4$.
2. Calculate the slope of the secant line connecting these two points.

Exercise 2.2.2 (Average speed of a falling object)**Problem Statement:**

A heavy stone is dropped from a cliff. Its distance fallen $s(t)$ (in meters) after t seconds is modeled by the equation $s(t) = 5t^2$.

1. Find the average speed of the stone between $t = 2$ seconds and $t = 5$ seconds by calculating the slope of the secant line.
 2. What are the physical units of this secant slope?
-

Exercise 2.2.3 (Secant slope with a general step size h)**Problem Statement:**

Let $f(x) = x^2$.

1. Find the coordinate of the neighboring point at $x = 3 + h$.
2. Show that the slope of the secant line passing through $x = 3$ and $x = 3 + h$ can be simplified algebraically to:

$$m_{\text{sec}} = 6 + h$$

Exercise 2.2.4 (Secant slope on a rational curve)**Problem Statement:**

Consider the rational curve $f(x) = \frac{4}{x}$ for $x > 0$.

1. Find the slope of the secant line connecting the points on the curve where $x = 2$ and $x = 2 + h$.
 2. Simplify your algebraic expression into a single fraction.
 3. Evaluate the value of this slope when the step size is $h = 0.1$.
-

Exercise 2.2.5 (General quadratic secant slope formula)**Problem Statement:**

Let $f(x) = ax^2 + bx + c$ be a general quadratic curve. Prove algebraically that the slope of the secant line connecting $x = w$ and $x = w + h$ is given by:

$$m_{\text{sec}} = 2aw + b + ah$$

Exercise 2.2.6 (The vanishing interval trap)**Problem Statement:**

Consider the curve $f(x) = x^3$. Suppose we attempt to calculate the slope of the secant line at $x = 2$ by setting the step size to exactly $h = 0$.

1. Write down the resulting expression for the slope of the secant line when $h = 0$.
 2. Explain why this expression is mathematically illegal and fails to yield a value.
 3. Why does this confirm that a secant line strictly requires *two distinct points* to exist?
-

Exercise 2.2.7 (Secant slopes on oscillating trigonometric curves)**Problem Statement:**

Consider the trigonometric curve $f(x) = \sin(x)$. We wish to study the behavior of its secant line starting at $x = 0$.

1. Write down the formula for the slope of the secant line passing through $x = 0$ and $x = h$.
2. Using a calculator in radian mode, evaluate the value of this secant slope for $h = 1.0, 0.5, 0.1, 0.01$, and 0.001 .
3. What specific real number does the slope of the secant line seem to approach as h gets closer and closer to 0?
4. Now, use the trigonometric sum-to-product identity $\sin(A) - \sin(B) = 2 \sin\left(\frac{A-B}{2}\right) \cos\left(\frac{A+B}{2}\right)$ to show that the secant slope starting at any general point $x = c$ can be written as:

$$m_{\text{sec}} = \cos\left(c + \frac{h}{2}\right) \cdot \left[\frac{\sin(h/2)}{h/2}\right]$$

Assuming that $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$, find what limit this general secant slope approaches as $h \rightarrow 0$.

2.3 First principle: Taking limit of secant line to a point

In Section 2.2, we calculated the slope of a straight secant line connecting two distinct points on a curve, $P(c, f(c))$ and $Q(c+h, f(c+h))$. We noticed a major structural barrier: if we tried to find the slope at the single point P by setting the step size h to exactly 0, our "rise over run" formula collapsed into the mathematically illegal indeterminate form $\frac{0}{0}$.

But wait! In Section 1, we spent fifteen chapters mastering a spectacular mathematical super-power designed specifically to conquer the $\frac{0}{0}$ trap: **the limit**.

Instead of setting h to exactly 0 immediately, we keep h as a tiny, non-zero variable. This allows us to perform clean algebraic simplifications to cancel out the division by h . Once the zero-denominator threat is removed, we take the limit as h approaches 0. As the second point Q crawls infinitely close to P , the secant line rotates and stabilizes, transforming into a line that grazes the curve at exactly one point. This line is the **tangent line**, and its slope is the **instantaneous rate of change**—also known as the **derivative** of the function! This ultimate process is called finding the derivative from **First Principles**.

Definition 2.3: The Derivative from First Principles

Let $y = f(x)$ be a continuous function. The **derivative** of $f(x)$ at $x = c$, denoted by $f'(c)$ or $\left.\frac{df}{dx}\right|_{x=c}$, is the instantaneous rate of change of the function at that point. It is defined as the limit of the secant slopes as the horizontal run h approaches 0:

$$f'(c) = \lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h}$$

provided this limit exists. If the limit exists, we say the function is **differentiable** at $x = c$.

The resulting value $f'(c)$ represents the exact geometric slope of the tangent line to the curve at the coordinate $(c, f(c))$.

Student Proof Demonstration

Use this space to demonstrate the proof of the theorem above. Start from basic algebraic identities.

Practical Example 2.3

Using the first principle of differentiation, find the exact slope of the tangent line to the quadratic curve $f(x) = x^2$ at the point where $x = 3$.

Step-by-Step Solution

Step 1: Set up the limit of the difference quotient.

We apply Definition 2.3 with $f(x) = x^2$ and $c = 3$:

$$\begin{aligned} f'(3) &= \lim_{h \rightarrow 0} \frac{f(3+h) - f(3)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(3+h)^2 - 3^2}{h} \end{aligned}$$

Step 2: Expand the numerator algebraically.

Expand the quadratic term $(3+h)^2$:

$$= \lim_{h \rightarrow 0} \frac{(9+6h+h^2)-9}{h}$$

Step 3: Simplify and cancel constant terms.

Notice that $9 - 9 = 0$. The constants vanish:

$$= \lim_{h \rightarrow 0} \frac{6h+h^2}{h}$$

Step 4: Factor out and cancel h .

Factor out h from the numerator terms:

$$= \lim_{h \rightarrow 0} \frac{h(6+h)}{h}$$

Since $h \rightarrow 0$ implies $h \neq 0$, cancel the common factor h :

$$= \lim_{h \rightarrow 0} (6+h)$$

Step 5: Evaluate the remaining limit.

Substitute $h = 0$ into the simplified expression:

$$= 6 + 0 = 6$$

Thus, the slope of the tangent line to $y = x^2$ at $x = 3$ is exactly 6.

Guided Thinking & Hints

Key Hint: Always start by writing down the general definition before substituting numbers. This keeps your structure organized and ensures you do not forget the h in the denominator.

Key Hint: Be extremely careful with expanding terms like $(c+h)^2$. A very common junior error is writing $(3+h)^2 = 9+h^2$. Remember the middle term: $2ab = 6h$!

Key Hint: If your constant terms do not cancel out completely at this step, you have made an algebraic expansion error! Re-check your expansion.

Key Hint: This is the magic moment! By canceling h , we have successfully removed the division-by-zero anomaly that was causing the $\frac{0}{0}$ trap.

Key Hint: The limit symbol $\lim_{h \rightarrow 0}$ disappears the exact instant you substitute $h = 0$. The curve's instantaneous steepness at $x = 3$ is exactly 6!

Exercise 2.3.1 (First principles on a basic quadratic)**Problem Statement:**

Let $f(x) = 2x^2$. Use the first principle of differentiation to find the exact value of $f'(1)$.

Exercise 2.3.2 (Tangents on multi-term quadratics)**Problem Statement:**

Consider the curve $g(x) = x^2 + 3x$.

1. Evaluate the derivative $g'(2)$ using the first principle of differentiation.
 2. Find the equation of the straight tangent line to the curve at the point where $x = 2$, expressing your answer in the form $y = mx + c$.
-

Exercise 2.3.3 (First principles on a rational function)**Problem Statement:**

Let $h(x) = \frac{1}{x}$. We wish to find the slope of the curve at $x = 2$.

1. Write down the difference quotient expression $\frac{h(2+h)-h(2)}{h}$.
 2. Simplify the numerator by finding a common denominator.
 3. Evaluate the limit as $h \rightarrow 0$ to find $h'(2)$.
-

Exercise 2.3.4 (First principles involving surds)**Problem Statement:**

Consider the square-root curve $f(x) = \sqrt{x}$.

1. Write down the limit of the difference quotient for $f'(4)$.
 2. Evaluate the limit by multiplying the numerator and the denominator by the conjugate expression $(\sqrt{4+h} + 2)$.
 3. State the exact slope of the tangent line to the curve at $x = 4$.
-

Exercise 2.3.5 (The general derivative of a linear function)**Problem Statement:**

Let $f(x) = mx + k$ be a general straight line, where m and k are real constants. Using the first principle of differentiation, prove algebraically that for any point $x = c$:

$$f'(c) = m$$

Explain why this result makes perfect intuitive sense geometrically.

Exercise 2.3.6 (Non-differentiability at a sharp corner)**Problem Statement:**

Consider the absolute value function $f(x) = |x|$. We want to investigate if the derivative exists at the origin $x = 0$.

1. Write down the difference quotient for $f'(0)$: $\frac{|0+h|-|0|}{h} = \frac{|h|}{h}$.
 2. Evaluate the left-hand limit of this quotient: $\lim_{h \rightarrow 0^-} \frac{|h|}{h}$.
 3. Evaluate the right-hand limit of this quotient: $\lim_{h \rightarrow 0^+} \frac{|h|}{h}$.
 4. Using your results and Theorem 1.6, explain why $f(x) = |x|$ is not differentiable at $x = 0$.
-

Exercise 2.3.7 (The bizarre case of the continuous but nowhere-differentiable curve)**Problem Statement:**

Consider the function $f(x) = x \sin\left(\frac{1}{x}\right)$ for $x \neq 0$, and let us define $f(0) = 0$ at the origin.

1. Use the Squeeze Theorem (from Section 1) to prove that $\lim_{x \rightarrow 0} f(x) = 0$, confirming that the curve is continuous at the origin.
2. Now, set up the first principles limit for the derivative at the origin:

$$f'(0) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h}$$

Simplify this expression algebraically.

3. Explain why the resulting limit fails to exist as $h \rightarrow 0$.
 4. What does this result prove? Can a function be completely continuous (unbroken) at a point, yet completely fail to have a defined slope at that same point?
-

2.4 The power rule of differentiation

In Section 2.3, we used the first principle of differentiation to find the slope of curves. While mathematically beautiful, calculating limits of difference quotients is incredibly exhausting. If we wanted to find the slope of $f(x) = x^{10}$ at $x = 2$, we would have to expand $(2 + h)^{10}$ algebraically—a nightmare of expanding ten sets of brackets!

Fortunately, mathematicians are professional pattern-seekers. Let us look closely at the derivatives we have already found using first principles:

- The derivative of x^1 is 1.
- The derivative of x^2 is $2x$.
- The derivative of x^3 is $3x^2$.

Do you see the pattern? The original power of x seems to "tumble down" to the front to become a multiplying coefficient, and the new exponent of x is decreased by exactly 1. This spectacular shortcut is known as the **Power Rule**. It is the single most frequently used shortcut in all of calculus, allowing us to differentiate complex variable powers in less than two seconds!

Theorem 2.4: The Power Rule of Differentiation

Let $f(x) = x^n$ be a power function, where the exponent n is any real number (including positive integers, negative integers, and fractions).

The derivative of $f(x)$ with respect to x is given by:

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

Furthermore, by combining this with the Constant Multiple Rule, we have:

$$\frac{d}{dx}(c \cdot x^n) = c \cdot nx^{n-1}$$

where c is any real constant.

Student Proof Demonstration

Use this space to demonstrate the proof of the theorem above. Start from basic algebraic identities.

Practical Example 2.4

Find the derivative of the following function with respect to x :

$$f(x) = \frac{3}{x^2} + 4\sqrt{x}$$

Step-by-Step Solution

Guided Thinking & Hints

Step 1: Convert all terms into standard power form.

We rewrite the fractions and radical signs as exponents of base x :

$$\frac{3}{x^2} = 3x^{-2}$$

$$4\sqrt{x} = 4x^{\frac{1}{2}}$$

So, the rewritten function is:

$$f(x) = 3x^{-2} + 4x^{\frac{1}{2}}$$

Key Hint: CRITICAL PREPARATION

STEP: The Power Rule only works when your terms look exactly like x^n . Before doing any calculus, always rewrite denominators using negative indices, and surds using fractional indices!

Step 2: Apply the Power Rule to the first term.

Differentiate $3x^{-2}$: bring the power -2 to the front and subtract 1 from the power:

$$\frac{d}{dx}(3x^{-2}) = 3 \cdot (-2)x^{-2-1} = -6x^{-3}$$

Step 3: Apply the Power Rule to the second term.

Differentiate $4x^{\frac{1}{2}}$: bring the power $\frac{1}{2}$ to the front and subtract 1:

$$\frac{d}{dx}(4x^{\frac{1}{2}}) = 4 \cdot \left(\frac{1}{2}\right)x^{\frac{1}{2}-1} = 2x^{-\frac{1}{2}}$$

Step 4: Combine the terms and format the final answer.

We write down the derivative:

$$f'(x) = -6x^{-3} + 2x^{-\frac{1}{2}}$$

Convert back to standard positive radical form:

$$f'(x) = -\frac{6}{x^3} + \frac{2}{\sqrt{x}}$$

Key Hint: Be extremely careful when subtracting 1 from negative numbers! A very common algebra error is writing $-2-1 = -1$. Remember: going further negative yields -3 !

Key Hint: To subtract 1 from a fraction, find a common denominator:
 $\frac{1}{2} - 1 = \frac{1}{2} - \frac{2}{2} = -\frac{1}{2}$. Keep your fraction arithmetic sharp!

Key Hint: While the expression with negative exponents is mathematically correct, HKDSE markers strongly prefer final answers written with positive indices and radical signs.

Exercise 2.4.1 (Direct polynomial differentiation)**Problem Statement:**

Differentiate the following functions with respect to x :

1. $f(x) = x^8$
 2. $g(x) = 5x^3 - 4x + 7$
-

Exercise 2.4.2 (Tangent line slope via Power Rule)

Problem Statement:

Consider the curve $y = 3x^2 - 5x + 2$.

1. Find the derivative $\frac{dy}{dx}$ using the Power Rule.
 2. Find the equation of the straight tangent line to the curve at the point where $x = 2$.
-

Exercise 2.4.3 (Differentiating negative and fractional powers)**Problem Statement:**

Find the derivative of the following function, expressing your final answer using positive indices and radical signs:

$$h(x) = \frac{2}{x^4} - \frac{6}{\sqrt[3]{x}}$$

Exercise 2.4.4 (Finding coordinates with a target gradient)**Problem Statement:**

A curve is given by $y = \frac{1}{3}x^3 - 2x^2 - 3x + 8$. Find the coordinates of the two points on the curve where the tangent line is perfectly horizontal.

Exercise 2.4.5 (The derivative of a reciprocal power)**Problem Statement:**

Let n be a positive integer, and let $f(x) = \frac{1}{x^n}$. By rewriting the function in power form and applying the Power Rule, prove that:

$$f'(x) = -\frac{n}{x^{n+1}}$$

Exercise 2.4.6 (Recognizing base-exponent traps)**Problem Statement:**

Consider the following three functions:

$$f_1(x) = x^5, \quad f_2(x) = 5^x, \quad f_3(x) = x^x$$

1. For which of these functions is the Power Rule (nx^{n-1}) valid? Differentiate it.
 2. For the other two functions, explain why the Power Rule is completely illegal. Identify where the variable is located in each case.
-

Exercise 2.4.7 (Rigorously proving the Power Rule for positive integers)**Problem Statement:**

To prove the Power Rule $\frac{d}{dx}(x^n) = nx^{n-1}$ for any positive integer $n \geq 1$ from first principles, we must expand $(x+h)^n$ using the ****Binomial Theorem****:

$$(x+h)^n = x^n + nx^{n-1}h + \frac{n(n-1)}{2}x^{n-2}h^2 + \cdots + nxh^{n-1} + h^n$$

1. Set up the first principles derivative formula: $f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^n - x^n}{h}$.
 2. Substitute the Binomial Theorem expansion into the numerator and cancel the x^n terms.
 3. Factor out h from the remaining terms and cancel it with the denominator.
 4. Evaluate the limit as $h \rightarrow 0$ to complete the proof.
-

2.5 The exponential rule of differentiation

In Section 2.4, we mastered the Power Rule, which allows us to differentiate functions where a variable base is raised to a constant power (like x^5). But what happens if we swap their positions? Consider $f(x) = 5^x$, where a constant base is raised to a variable exponent. As we discussed in the Section 2.4 failure analysis, applying the Power Rule here is completely illegal!

To find the rate of change of an exponential function, we must investigate its behavior from first principles. If we look at the natural exponential function $f(x) = e^x$, where $e \approx 2.71828$ is Euler's number, we discover one of the most beautiful and unique secrets in all of mathematics. The rate at which the curve grows is exactly equal to the height of the curve itself! In other words, e^x is its own derivative. If you have a population of bacteria growing exponentially, the speed at which the population increases at any instant is directly proportional to the number of bacteria present at that exact moment.

Theorem 2.5: The Exponential Rule of Differentiation

Let $f(x) = e^x$ be the natural exponential function.

1. **The Natural Exponential Derivative:** The derivative of e^x with respect to x is itself:

$$\frac{d}{dx}(e^x) = e^x$$

2. **General Exponential Derivative (Base $b > 0$):** For any positive constant base $b \neq 1$, the derivative of b^x is scaled by the natural logarithm of the base:

$$\frac{d}{dx}(b^x) = b^x \ln(b)$$

Student Proof Demonstration

Use this space to demonstrate the proof of the theorem above. Start from basic algebraic identities.

Practical Example 2.5

Find the derivative of the following function with respect to x :

$$f(x) = 3e^x - 5 \cdot 2^x$$

Step-by-Step Solution

Step 1: Check the term types.

The term $3e^x$ is a natural exponential function with a constant multiplier.

The term $5 \cdot 2^x$ is a general exponential function with base 2 and a constant multiplier.

Step 2: Differentiate the first term.

Apply the Natural Exponential Rule to $3e^x$. Since e^x is its own derivative, the term remains unchanged:

$$\frac{d}{dx}(3e^x) = 3e^x$$

Step 3: Differentiate the second term.

Apply the General Exponential Rule to $5 \cdot 2^x$, where $b = 2$:

$$\frac{d}{dx}(5 \cdot 2^x) = 5 \cdot 2^x \ln(2)$$

Step 4: Write down the combined derivative.

Combine the differentiated terms:

$$f'(x) = 3e^x - 5 \cdot 2^x \ln(2)$$

Guided Thinking & Hints

Key Hint: Always categorize your terms first! Notice that the variables are in the exponent, which immediately tells you to use the Exponential Rule instead of the Power Rule.

Key Hint: This is the easiest derivative in calculus! Any constant multiplier c simply remains attached to the front of e^x .

Key Hint: Don't forget the scaling factor! For any base other than e , you must multiply the term by the natural logarithm of the base, $\ln(b)$.

Key Hint: Keep the final expression exact. Do not turn $\ln(2)$ into a decimal approximation on your calculator; HKDSE markers expect exact values!

Exercise 2.5.1 (Differentiating standard exponentials)**Problem Statement:**

Differentiate the following functions with respect to x :

1. $f(x) = 4e^x$

2. $g(x) = 7^x$

Exercise 2.5.2 (Power Rule and Exponential Rule combination)**Problem Statement:**

Find the derivative of the following mixed function:

$$h(x) = 3x^4 - 2e^x + 5 \cdot 10^x$$

Exercise 2.5.3 (Manipulating exponents before differentiating)**Problem Statement:**

Find the derivative of the following function:

$$f(x) = e^{x+2} + 3^{x-1}$$

Hint: Use exponential index laws ($e^{x+2} = e^2 \cdot e^x$ and $3^{x-1} = \frac{1}{3} \cdot 3^x$) to isolate the basic terms before differentiating.

Exercise 2.5.4 (Finding coordinates of a specific tangent slope)**Problem Statement:**

Consider the curve $y = 3e^x - 4x$.

1. Find the derivative $\frac{dy}{dx}$.
 2. Find the exact coordinates of the point on the curve where the tangent line is parallel to the straight line $y = 8x + 5$.
-

Exercise 2.5.5 (Linking general exponential base derivatives to natural base)**Problem Statement:**

Let $b > 0$ be a constant base. By using the natural logarithm identity $b = e^{\ln(b)}$, we can rewrite any exponential function as:

$$b^x = \left(e^{\ln(b)}\right)^x = e^{x \ln(b)}$$

Assuming that $\frac{d}{dx}(e^{kx}) = ke^{kx}$ for any constant k , prove algebraically that the derivative of b^x is indeed $b^x \ln(b)$.

Exercise 2.5.6 (The constant base trap)**Problem Statement:**

A student attempts to evaluate the derivative of $f(x) = e^2$ and writes:

$$f'(x) = e^2$$

Explain why the student's answer is completely wrong. What is the derivative of any constant, and why is e^2 considered a constant rather than an exponential function?

Exercise 2.5.7 (Proving the derivative of e^x from First Principles)**Problem Statement:**

To prove that the derivative of $f(x) = e^x$ is indeed e^x from first principles, we set up the limit:

$$f'(x) = \lim_{h \rightarrow 0} \frac{e^{x+h} - e^x}{h}$$

1. Use index laws to factor out e^x from the numerator. Show that the expression becomes:

$$f'(x) = e^x \cdot \left(\lim_{h \rightarrow 0} \frac{e^h - 1}{h} \right)$$

2. Let us evaluate the limit $L = \lim_{h \rightarrow 0} \frac{e^h - 1}{h}$ numerically. Choose $h = 0.1, 0.01, 0.001$, and 0.0001 . Use your calculator to complete the table of values below (to 5 decimal places):

h	0.1	0.01	0.001	0.0001
$\frac{e^h - 1}{h}$				

3. What limit does this sequence suggest for L as $h \rightarrow 0$?
4. Now, use the compound interest definition of e , $e = \lim_{n \rightarrow \infty} (1 + \frac{1}{n})^n$, by letting $h = \frac{1}{n}$ to show analytically why $L = 1$, completing the proof.

2.6 The logarithmic rule of differentiation

We have now mastered the derivatives of polynomials (x^n) and exponential curves (e^x). To round out our mastery of fundamental transcendental functions, we must find the derivative of the natural logarithm, $f(x) = \ln(x)$ (where $x > 0$).

Recall from our growth hierarchy discussion in Section 1.14 that the logarithm is the ultimate "turtle" of mathematics. It grows incredibly slowly, creeping upward without bound. Geometrically, this suggests that as x gets larger and larger, the slope of its tangent line must become flatter and flatter, approaching zero. Conversely, as x approaches zero from the right, the graph plunges vertically downward, implying the slope must become infinitely steep!

What simple algebraic formula behaves exactly like this—starting infinitely steep near $x = 0$ and flattening to 0 as $x \rightarrow \infty$? As we demonstrated in the research discussion of Section 1.14, the answer is the simple reciprocal function, $\frac{1}{x}$. This elegant connection bridges transcendental logarithmic curves back into the clean world of basic rational functions!

Theorem 2.6: The Logarithmic Rule of Differentiation

Let x be a strictly positive real number ($x > 0$).

1. **The Natural Logarithm Derivative:** The derivative of the natural logarithm of x with respect to x is:

$$\frac{d}{dx}(\ln(x)) = \frac{1}{x}$$

2. **General Logarithm Derivative (Base b):** For any positive constant base $b \neq 1$, the derivative of $\log_b(x)$ is scaled in the denominator by the natural logarithm of the base:

$$\frac{d}{dx}(\log_b(x)) = \frac{1}{x \ln(b)}$$

Student Proof Demonstration

Use this space to demonstrate the proof of the theorem above. Start from basic algebraic identities.

Practical Example 2.6

Find the derivative of the following function with respect to x :

$$f(x) = 4 \ln(x) + \log_3(x)$$

Step-by-Step Solution**Step 1: Inspect the term types.**

The term $4 \ln(x)$ is a natural logarithm with a constant multiplier.

The term $\log_3(x)$ is a logarithm to base 3.

Step 2: Differentiate the first term.

Apply the Natural Logarithm Rule to $4 \ln(x)$. The constant multiplier 4 remains attached:

$$\frac{d}{dx}(4 \ln(x)) = 4 \cdot \frac{1}{x} = \frac{4}{x}$$

Step 3: Differentiate the second term.

Apply the General Logarithm Rule to $\log_3(x)$, where $b = 3$:

$$\frac{d}{dx}(\log_3(x)) = \frac{1}{x \ln(3)}$$

Step 4: Write down the combined derivative.

Combine the differentiated terms:

$$f'(x) = \frac{4}{x} + \frac{1}{x \ln(3)}$$

Guided Thinking & Hints

Key Hint: Always make sure the variable x is strictly positive before differentiating! Logarithms are completely undefined for negative numbers or zero.

Key Hint: Notice that the logarithmic term completely vanishes and turns into a simple algebraic term. The derivative of $\ln(x)$ is just its reciprocal!

Key Hint: For any base other than e , we must multiply the denominator of our reciprocal by $\ln(b)$ (change of base formula). Do not put this $\ln(b)$ on top!

Key Hint: We keep the expression exact. Leave $\ln(3)$ as a constant in the denominator rather than decimalizing it.

Exercise 2.6.1 (Differentiating standard logarithms)**Problem Statement:**

Differentiate the following functions with respect to x :

1. $f(x) = 5 \ln(x)$

2. $g(x) = \log_5(x)$

Exercise 2.6.2 (Combining rules)**Problem Statement:**

Find the derivative of the following mixed function:

$$h(x) = \frac{1}{x^3} + 3e^x - 2 \ln(x)$$

Exercise 2.6.3 (Using log laws before differentiating)**Problem Statement:**

Find the derivative of the following function:

$$f(x) = \ln(5x^3)$$

Hint: Use logarithmic properties ($\ln(ab) = \ln(a) + \ln(b)$ and $\ln(x^c) = c\ln(x)$) to expand and simplify the function completely before applying the differentiation rules.

Exercise 2.6.4 (Finding points of intersection with perpendicular tangents)**Problem Statement:**

Consider the curve $y = \ln(x)$ for $x > 0$.

1. Find the equation of the tangent line to the curve at the point where $x = 1$.
 2. Find the coordinates of the point on the curve where the tangent line is perpendicular to the straight line $y = -2x + 7$.
-

Exercise 2.6.5 (General base derivation using Change of Base rule)**Problem Statement:**

Recall the algebraic Change of Base formula:

$$\log_b(x) = \frac{\ln(x)}{\ln(b)} = \frac{1}{\ln(b)} \cdot \ln(x)$$

Using this identity and the Constant Multiple Rule, prove that the derivative of $\log_b(x)$ is indeed $\frac{1}{x \ln(b)}$.

Exercise 2.6.6 (The domain restriction trap)**Problem Statement:**

Consider the function $f(x) = \ln(-x)$.

1. What is the domain of this function? For what values of x is this function defined?
 2. A student claims that since we cannot take the logarithm of a negative number, the derivative of $f(x)$ must not exist anywhere. Explain why this claim is incorrect.
 3. Assuming the derivative of $\ln(-x)$ is $\frac{1}{x}$ (which we will prove rigorously in Section 2.13 using the Chain Rule), evaluate $f'(-5)$.
-

Exercise 2.6.7 (Logarithmic derivative of the absolute value function)**Problem Statement:**

In higher-level integration, we often write the antiderivative of $\frac{1}{x}$ as $\ln|x| + C$. Let us define the absolute value logarithm function:

$$f(x) = \ln|x| \quad \text{for } x \neq 0$$

1. Express $f(x)$ as a two-branch piecewise function for $x > 0$ and $x < 0$.
 2. Differentiate the first branch ($x > 0$) directly.
 3. Given that the derivative of $\ln(-x)$ for $x < 0$ is also $\frac{1}{x}$, state the derivative for the second branch.
 4. Combine your results: What is the derivative of $\ln|x|$ for any non-zero real number x ? Explain how this extends the domain of our differentiation rule to the entire real number line (except $x = 0$).
-

2.7 The Linear rule of differentiation

Up to this point, we have learned how to differentiate individual functions in isolation: polynomials (x^n), exponentials (b^x), and logarithms ($\log_b x$). But real-world mathematical models are rarely this simple. More often, they are constructed by mixing these functions together—adding them up, subtracting them, or scaling them with constants (for example, $f(x) = 3x^2 + 5e^x - 2\ln(x)$).

Can we still use our shortcut rules when functions are joined together by addition and subtraction?

The answer is a resounding yes! This convenient property is called the **Linearity of the Derivative** (or the Sum and Difference Rule). Because limits themselves behave linearly (as we saw in Section 1), the derivative—which is just a limit of secant slopes—inherits this property perfectly. Linearity allows us to differentiate any long algebraic expression term-by-term. We can simply ignore the additions and subtractions, focus on differentiating one individual piece at a time, and then glue them back together!

Theorem 2.7: The Linearity of Differentiation

Let $f(x)$ and $g(x)$ be differentiable functions, and let a and b be any real constants.

The derivative operator is **linear**, satisfying two fundamental properties:

1. **The Constant Multiple Rule:** The derivative of a constant times a function is the constant times the derivative of the function:

$$\frac{d}{dx} (c \cdot f(x)) = c \cdot f'(x)$$

2. **The Sum and Difference Rule:** The derivative of a sum or difference of two functions is the sum or difference of their individual derivatives:

$$\frac{d}{dx} (f(x) \pm g(x)) = f'(x) \pm g'(x)$$

Combining these two rules yields the **Linear Combination Rule**:

$$\frac{d}{dx} (a \cdot f(x) \pm b \cdot g(x)) = a \cdot f'(x) \pm b \cdot g'(x)$$

Student Proof Demonstration

Use this space to demonstrate the proof of the theorem above. Start from basic algebraic identities.

Practical Example 2.7

Differentiate the following multi-term function with respect to x :

$$f(x) = 5x^3 - 2e^x + \frac{4}{x} - 7$$

Step-by-Step Solution**Step 1: Rewrite all terms in standard differentiable form.**

Convert fractions and roots into exponents of x :

$$\frac{4}{x} = 4x^{-1}$$

So, we rewrite the function as:

$$f(x) = 5x^3 - 2e^x + 4x^{-1} - 7$$

Step 2: Differentiate term-by-term.

By the Linear Rule, we can write:

$$f'(x) = \frac{d}{dx}(5x^3) - \frac{d}{dx}(2e^x) + \frac{d}{dx}(4x^{-1}) - \frac{d}{dx}(7)$$

Step 3: Apply individual derivative rules.

Differentiate each piece:

$$\frac{d}{dx}(5x^3) = 5(3x^2) = 15x^2$$

$$\frac{d}{dx}(2e^x) = 2e^x$$

$$\frac{d}{dx}(4x^{-1}) = 4(-1x^{-2}) = -4x^{-2}$$

$$\frac{d}{dx}(7) = 0 \text{ (The derivative of any constant is 0).}$$

Step 4: Combine and format the final answer.

Combine all the pieces:

$$f'(x) = 15x^2 - 2e^x - 4x^{-2}$$

Rewrite using positive exponents:

$$f'(x) = 15x^2 - 2e^x - \frac{4}{x^2}$$

Guided Thinking & Hints

Key Hint: Always prepare your algebraic terms first before applying any differentiation rules. Write division by x as x^{-1} .

Key Hint: Linearity means we can treat each term as a completely separate mini-problem! You don't have to differentiate the whole expression in one giant leap.

Key Hint: Ensure you match each term with its correct rule:
 Term 1: Power Rule.
 Term 2: Exponential Rule.
 Term 3: Power Rule.
 Term 4: Constant Rule.

Key Hint: Don't forget that the constant term -7 completely vanishes! The final simplified rate of change is $15x^2 - 2e^x - \frac{4}{x^2}$.

Exercise 2.7.1 (Standard linear combination differentiation)**Problem Statement:**

Differentiate the following functions with respect to x :

1. $f(x) = 3x^4 - 5x^2 + 8$
 2. $g(x) = 2e^x + 6\ln(x) - \frac{1}{x}$
-

Exercise 2.7.2 (Finding the equation of a tangent with multiple terms)**Problem Statement:**

Consider the curve $y = x^2 - 3e^x + 4$.

1. Find the derivative $\frac{dy}{dx}$ using the Linear Rule.
 2. Find the exact equation of the tangent line to the curve at the point where $x = 0$.
-

Exercise 2.7.3 (Differentiating polynomials disguised as rational fractions)**Problem Statement:**

Differentiate the following function with respect to x :

$$h(x) = \frac{2x^3 - 5x^2 + 3}{x}$$

Hint: Split the fraction into three separate terms by dividing each term in the numerator by the denominator x before differentiating.

Exercise 2.7.4 (Solving for unknown constants using simultaneous linear systems)**Problem Statement:**

A curve is given by the equation $y = ax^3 + bx^2 - 6x + 5$, where a and b are constants.

1. Find the derivative $\frac{dy}{dx}$.
 2. It is given that the curve has stationary points (where the tangent line is perfectly horizontal, i.e., $\frac{dy}{dx} = 0$) at both $x = 1$ and $x = -2$. Set up a system of simultaneous equations to find the exact values of a and b .
-

Exercise 2.7.5 (Rigorous proof of the Sum Rule using First Principles)**Problem Statement:**

Let $S(x) = f(x) + g(x)$, where $f(x)$ and $g(x)$ are differentiable functions. Using the first principle of differentiation, prove algebraically that:

$$S'(x) = f'(x) + g'(x)$$

Exercise 2.7.6 (Recognizing product and quotient traps)**Problem Statement:**

A student is asked to differentiate $f(x) = x^2 \cdot \ln(x)$ and writes:

$$f'(x) = \frac{d}{dx}(x^2) \cdot \frac{d}{dx}(\ln(x)) = 2x \cdot \frac{1}{x} = 2$$

1. Explain why the student's step is completely wrong. Does linearity apply to products or divisions?
 2. Show that if $f'(x) = 2$ were true, the slope of the curve at $x = e$ would be 2. In Section 2.8, we will learn the Product Rule: $\frac{d}{dx}(u \cdot v) = u \cdot v' + v \cdot u'$. Use this true rule to find the correct value of $f'(e)$ and show it does not equal 2.
-

Exercise 2.7.7 (The Linear Algebra connection: Differentiation as a Linear Transformation)**Problem Statement:**

In university linear algebra, a **vector space** is a set of elements (such as functions) that can be added together and scaled by constants. A **linear transformation** T is an operator that satisfies:

$$T(a\mathbf{u} + b\mathbf{v}) = aT(\mathbf{u}) + bT(\mathbf{v})$$

1. Let V be the vector space of all infinitely differentiable real-valued functions, and let $D = \frac{d}{dx}$ be the differentiation operator. Explain how Theorem 2.7 proves that D is a linear transformation.
 2. A vector $\mathbf{v}$ is called an **eigenvector** (or eigenfunction) of a transformation T with **eigenvalue** λ if $T(\mathbf{v}) = \lambda\mathbf{v}$. Show that the exponential function $f(x) = e^{kx}$ is an eigenfunction of the derivative operator D , and state its eigenvalue.
 3. The **kernel** (or null space) of a linear transformation contains all vectors $\mathbf{v}$ such that $T(\mathbf{v}) = \mathbf{0}$. Find the kernel of the derivative operator D . What set of functions does this represent physically?
-

2.8 The Product rule of differentiation

In Section 2.7, we analyzed a major warning sign: while addition and subtraction allow us to differentiate term-by-term, multiplication absolutely does not! If $f(x) = u(x) \cdot v(x)$, the derivative $f'(x)$ is **not** $u'(x) \cdot v'(x)$.

To build an intuitive grasp of why this is true, let us think about geometry. Imagine a rectangular farm with a width of u meters and a length of v meters. The total area of the farm is $A = u \cdot v$ square meters. Now, suppose both the width and length grow over time. The width increases by a small amount Δu , and the length increases by a small amount Δv .

The new area is:

$$(u + \Delta u)(v + \Delta v) = u \cdot v + u \cdot \Delta v + v \cdot \Delta u + \Delta u \cdot \Delta v$$

Subtracting the original area ($u \cdot v$), we see the change in area consists of three new regions:

1. A thin strip along the length with area $u \cdot \Delta v$.
2. A thin strip along the width with area $v \cdot \Delta u$.
3. A tiny corner square with area $\Delta u \cdot \Delta v$.

As we take the limit as the change becomes infinitely small, the tiny corner term vanishes. We are left with exactly two dominant growing strips: $u \cdot dv + v \cdot du$. This beautiful geometric reality is the foundation of the **Product Rule**!

Theorem 2.8: The Product Rule of Differentiation

Let $u(x)$ and $v(x)$ be two differentiable functions.

The derivative of their product with respect to x is given by:

$$\frac{d}{dx} (u(x) \cdot v(x)) = u(x) \cdot v'(x) + v(x) \cdot u'(x)$$

Using shorthand notation where u and v represent the functions, this is commonly written as:

$$(uv)' = uv' + vu'$$

In words: "The first times the derivative of the second, plus the second times the derivative of the first."

Student Proof Demonstration

Use this space to demonstrate the proof of the theorem above. Start from basic algebraic identities.

Practical Example 2.8

Differentiate the following function with respect to x using the Product Rule:

$$f(x) = x^2 \cdot e^x$$

Step-by-Step Solution

Step 1: Identify and assign $u(x)$ and $v(x)$.

We split the product into two distinct functions:

$$\text{Let } u(x) = x^2$$

$$\text{Let } v(x) = e^x$$

Step 2: Differentiate u and v separately.

Find their individual derivatives:

$$u'(x) = \frac{d}{dx}(x^2) = 2x$$

$$v'(x) = \frac{d}{dx}(e^x) = e^x$$

Step 3: Assemble using the Product Rule formula.

Substitute the parts into $(uv)' = uv' + vu'$:

$$f'(x) = (x^2) \cdot (e^x) + (e^x) \cdot (2x)$$

Step 4: Factor and simplify.

Factor out the common terms (x and e^x) to get a clean final answer:

$$f'(x) = xe^x(x + 2)$$

Guided Thinking & Hints

Key Hint: Always write down your assignments clearly on the side. This prevents you from mixing up the terms as you apply the formula.

Key Hint: Apply the correct rule to each term! Here, u uses the Power Rule, and v uses the Exponential Rule.

Key Hint: Watch the order carefully:
First $\times \frac{d}{dx}$ (Second) + Second $\times \frac{d}{dx}$ (First).

Key Hint: EXAM TIP: DSE markers highly value fully factored answers! Factoring out common exponential terms like e^x makes finding stationary points in Section 3 much easier.

Exercise 2.8.1 (Standard algebraic-exponential product)**Problem Statement:**

Differentiate the following functions with respect to x using the Product Rule:

1. $f(x) = x^5 \cdot e^x$
 2. $g(x) = (3x^2 - 2x) \cdot \ln(x)$
-

Exercise 2.8.2 (Finding equations of tangents using products)**Problem Statement:**

Consider the curve $y = x \cdot 2^x$.

1. Find the derivative $\frac{dy}{dx}$ using the Product Rule.
 2. Find the exact equation of the straight tangent line to the curve at the point where $x = 1$.
-

Exercise 2.8.3 (Differentiating radical-logarithmic products)**Problem Statement:**

Differentiate the following function, simplifying your final answer into a single factored fraction:

$$h(x) = \sqrt{x} \cdot \ln(x)$$

Exercise 2.8.4 (Locating stationary points on a product curve)**Problem Statement:**

Consider the curve $y = (x^2 - 5x + 4) \cdot e^x$.

1. Find $\frac{dy}{dx}$ and simplify your answer by factoring.
 2. Find the coordinates of the two stationary points (where $\frac{dy}{dx} = 0$) on this curve.
-

Exercise 2.8.5 (The Triple Product Rule derivation)**Problem Statement:**

Let $u(x)$, $v(x)$, and $w(x)$ be three differentiable functions. By treating $u \cdot v$ as a single grouped function and applying the standard Product Rule, prove that the derivative of their triple product is:

$$\frac{d}{dx}(u \cdot v \cdot w) = u' \cdot v \cdot w + u \cdot v' \cdot w + u \cdot v \cdot w'$$

Exercise 2.8.6 (Product expansion vs. Product Rule)**Problem Statement:**

Consider the function $f(x) = x^2(x^3 + 5x)$.

1. Differentiate $f(x)$ by expanding the expression first into a single polynomial and then applying the Power Rule.
 2. Differentiate $f(x)$ directly using the Product Rule.
 3. Show that both methods yield identical answers. Explain why expanding first is usually less prone to algebraic mistakes for simple polynomials.
-

Exercise 2.8.7 (Rigorously proving the Product Rule from First Principles)**Problem Statement:**

To prove Theorem 2.8 from first principles without relying on geometry, we set up the limit:

$$\frac{d}{dx}(u(x)v(x)) = \lim_{h \rightarrow 0} \frac{u(x+h)v(x+h) - u(x)v(x)}{h}$$

1. To separate the variables, we perform a clever algebraic trick: adding and subtracting the identical term $u(x+h)v(x)$ in the middle of the numerator. Rewrite the limit with this modification.
2. Group the terms in the numerator and show that the expression splits into:

$$\lim_{h \rightarrow 0} \left[u(x+h) \cdot \frac{v(x+h) - v(x)}{h} + v(x) \cdot \frac{u(x+h) - u(x)}{h} \right]$$

3. Apply limit properties and the definitions of $u'(x)$ and $v'(x)$ to complete the proof.
 4. Explain why the assumption that $u(x)$ is continuous (meaning $\lim_{h \rightarrow 0} u(x+h) = u(x)$) is required to complete the final step.
-

2.9 The Quotient rule of differentiation

In Section 2.8, we mastered the Product Rule to handle multiplication. But what happens when we need to differentiate division? Consider the function $f(x) = \frac{u(x)}{v(x)}$, such as $f(x) = \frac{x^2-1}{x+2}$. Just like products, we cannot simply differentiate the top and the bottom separately. The derivative of a quotient is absolutely **not** the quotient of the derivatives!

To understand why, we can think of division as the mathematical sibling of multiplication. If $y = \frac{u}{v}$, then we can rewrite this as $y \cdot v = u$. By applying the Product Rule to this multiplication and using some algebraic rearrangements, we can isolate the rate of change of y . This elegant derivation reveals a subtraction in the numerator: because division represents sharing, growing the denominator v actually **pulls down** the overall value of the fraction. This is why the derivative of the bottom term subtracted from the top term is what we call the **Quotient Rule**.

Theorem 2.9: The Quotient Rule of Differentiation

Let $u(x)$ and $v(x)$ be two differentiable functions such that $v(x) \neq 0$.

The derivative of their quotient with respect to x is given by:

$$\frac{d}{dx} \left(\frac{u(x)}{v(x)} \right) = \frac{v(x) \cdot u'(x) - u(x) \cdot v'(x)}{[v(x)]^2}$$

Using shorthand notation where u and v represent the functions, this is commonly written as:

$$\left(\frac{u}{v} \right)' = \frac{vu' - uv'}{v^2}$$

In words: **"The bottom times the derivative of the top, minus the top times the derivative of the bottom, all over the bottom squared."**

Student Proof Demonstration

Use this space to demonstrate the proof of the theorem above. Start from basic algebraic identities.

Practical Example 2.9

Differentiate the following rational function with respect to x :

$$f(x) = \frac{x^2 - 1}{x + 2}$$

Step-by-Step Solution

Step 1: Identify and assign $u(x)$ and $v(x)$.

Let the numerator be:

$$u(x) = x^2 - 1$$

Let the denominator be:

$$v(x) = x + 2$$

Step 2: Calculate their individual derivatives.

Differentiate both parts:

$$u'(x) = 2x$$

$$v'(x) = 1$$

Step 3: Assemble using the Quotient Rule.

Substitute into the formula:

$$f'(x) = \frac{(x+2) \cdot (2x) - (x^2-1) \cdot (1)}{(x+2)^2}$$

Step 4: Expand and simplify the numerator.

Expand the top terms:

$$f'(x) = \frac{(2x^2+4x)-(x^2-1)}{(x+2)^2}$$

Distribute the negative sign carefully:

$$f'(x) = \frac{2x^2+4x-x^2+1}{(x+2)^2}$$

Combine like terms:

$$f'(x) = \frac{x^2+4x+1}{(x+2)^2}$$

Guided Thinking & Hints

Key Hint: Always treat the top function as u and the bottom function as v . Order is extremely critical in the Quotient Rule because of the subtraction sign in the numerator!

Key Hint: Both of these derivatives are simple applications of the Power Rule and Constant Rule from Section 2.4.

Key Hint: GOLDEN RULE: Always write out the expanded formula with protective parentheses before simplifying! Parentheses prevent catastrophic distribution errors.

Key Hint: Watch out for the double-negative trap:
 $-(x^2 - 1) = -x^2 + 1$.
 Leave the denominator factored as $(x + 2)^2$.
 DSE markers never require you to expand the denominator!

Exercise 2.9.1 (Standard algebraic quotient)**Problem Statement:**

Differentiate the following functions with respect to x using the Quotient Rule:

1. $f(x) = \frac{5x}{2x+3}$
 2. $g(x) = \frac{x^2+3}{x-1}$
-

Exercise 2.9.2 (Finding equations of tangents using quotients)**Problem Statement:**

Consider the curve $y = \frac{e^x}{x}$.

1. Find the derivative $\frac{dy}{dx}$ using the Quotient Rule.
 2. Find the exact equation of the straight tangent line to the curve at the point where $x = 1$.
-

Exercise 2.9.3 (Differentiating logarithmic quotients)**Problem Statement:**

Differentiate the following function, simplifying your final answer into a single clean fraction:

$$h(x) = \frac{\ln(x)}{x^2}$$

Exercise 2.9.4 (Analyzing turning points on a rational curve)**Problem Statement:**

Consider the curve $y = \frac{x^2 - 4x}{x - 2}$.

1. Evaluate the derivative $\frac{dy}{dx}$ using the Quotient Rule and simplify the numerator completely.
 2. Explain why the derivative simplifies to a constant, and what this implies about the shape of this curve.
-

Exercise 2.9.5 (The Reciprocal Derivative Rule derivation)**Problem Statement:**

Let $g(x)$ be a differentiable function such that $g(x) \neq 0$. By setting the numerator $u(x) = 1$ and applying the Quotient Rule, prove that:

$$\frac{d}{dx} \left(\frac{1}{g(x)} \right) = -\frac{g'(x)}{[g(x)]^2}$$

Exercise 2.9.6 (Choosing the best path: Quotient Rule vs. index manipulation)**Problem Statement:**

Consider the function $f(x) = \frac{3x^4 - 2x + 1}{x^2}$.

1. Differentiate $f(x)$ using the Quotient Rule, grouping and simplifying the final fraction.
 2. Differentiate $f(x)$ by first splitting the fraction into three separate terms using index laws (i.e., $f(x) = 3x^2 - 2x^{-1} + x^{-2}$) and applying the Power Rule.
 3. Confirm both methods yield equivalent answers. Why is the second method generally faster and less error-prone?
-

Exercise 2.9.7 (Rigorously proving the Quotient Rule from the Product Rule)**Problem Statement:**

Rather than using first-principles limits, we can prove Theorem 2.9 algebraically using the Product Rule and the Chain Rule (or reciprocal power rule). Let $f(x) = \frac{u(x)}{v(x)}$.

1. Rewrite the quotient as a product: $f(x) = u(x) \cdot [v(x)]^{-1}$.
 2. Apply the Product Rule to this expression.
 3. Use the Power Rule on the term $[v(x)]^{-1}$ to show that its derivative with respect to x is $-[v(x)]^{-2} \cdot v'(x)$.
 4. Factor out the common denominator $[v(x)]^{-2}$ (which is $\frac{1}{v^2}$) and combine the terms to complete the proof of the Quotient Rule.
-

2.10 Derivatives of trigonometric functions

In our previous chapters, we analyzed the rates of change of algebraic, exponential, and logarithmic functions. To unlock the full power of calculus, we must venture into the world of periodic motion. From the swinging of a clock's pendulum to the alternating current powering your home, systems that oscillate are mathematically modeled using trigonometric functions: $\sin(x)$ and $\cos(x)$.

What are the derivatives of these wave-like functions?

Let us look at the graph of $y = \sin(x)$ starting at $x = 0$. At the origin, the sine wave climbs upward at its steepest rate, which turns out to have a slope of exactly 1. As it reaches its peak at $x = \frac{\pi}{2}$, the curve flattens out, giving a slope of 0. It then slides downward, reaching its steepest descent at $x = \pi$ with a slope of -1 . If we plot these slopes $(1, 0, -1, 0, 1)$ on a new coordinate grid, they trace out a perfect, familiar wave: the cosine curve! This beautiful geometric dance shows that differentiating $\sin(x)$ yields $\cos(x)$, and differentiating $\cos(x)$ yields $-\sin(x)$.

Theorem 2.10: Trigonometric Rules of Differentiation

Let x be an angle measured strictly in **radians**.

The derivatives of the primary trigonometric functions are given by:

1. **Derivative of Sine:**

$$\frac{d}{dx}(\sin(x)) = \cos(x)$$

2. **Derivative of Cosine:**

$$\frac{d}{dx}(\cos(x)) = -\sin(x)$$

3. **Derivative of Tangent:**

$$\frac{d}{dx}(\tan(x)) = \sec^2(x)$$

Student Proof Demonstration

Use this space to demonstrate the proof of the theorem above. Start from basic algebraic identities.

Practical Example 2.10

Differentiate the following function with respect to x using the Product Rule and trigonometric rules:

$$f(x) = x^3 \cdot \sin(x)$$

Step-by-Step Solution**Step 1: Identify and assign the functions.**

This is a product of an algebraic term and a trigonometric term.

$$\text{Let } u(x) = x^3$$

$$\text{Let } v(x) = \sin(x)$$

Step 2: Differentiate each part.

Apply the appropriate rules:

$$u'(x) = 3x^2 \text{ (Power Rule)}$$

$$v'(x) = \cos(x) \text{ (Trigonometric Rule)}$$

Step 3: Assemble using the Product Rule.

Substitute the parts into the formula:

$$f'(x) = (x^3) \cdot (\cos(x)) + (\sin(x)) \cdot (3x^2)$$

Step 4: Factor out common terms.

Identify the common algebraic factor x^2 and group the terms:

$$f'(x) = x^2(x \cos(x) + 3 \sin(x))$$

Guided Thinking & Hints

Key Hint: Always scan the overall structure first! Since the terms are multiplied, we must deploy the Product Rule $(uv)' = uv' + vu'$.

Key Hint: Double-check your signs! The derivative of positive sine is positive cosine. Only cosine's derivative introduces a negative sign.

Key Hint: Keep your terms organized: First $\times \frac{d}{dx}$ (Second) + Second $\times \frac{d}{dx}$ (First).

Key Hint: Factoring algebraic terms out of trigonometric combinations creates a clean, professional, and examination-ready final presentation.

Exercise 2.10.1 (Standard trigonometric combinations)**Problem Statement:**

Differentiate the following functions with respect to x :

1. $f(x) = 4 \sin(x) - 5 \cos(x)$
 2. $g(x) = e^x \cdot \cos(x)$
-

Exercise 2.10.2 (Finding equations of tangents with trigonometric waves)**Problem Statement:**

Consider the curve $y = \sin(x) + \cos(x)$ over the interval $[0, \pi]$.

1. Find the derivative $\frac{dy}{dx}$.
 2. Find the exact equation of the tangent line to the curve at the point where $x = \frac{\pi}{4}$.
-

Exercise 2.10.3 (Differentiating the tangent function from first identities)**Problem Statement:**

By rewriting the tangent function as a quotient, $\tan(x) = \frac{\sin(x)}{\cos(x)}$, and applying the Quotient Rule, prove that its derivative is indeed $\sec^2(x)$.

Exercise 2.10.4 (Locating stationary points on a wave curve)**Problem Statement:**

Consider the curve $y = \frac{\sin(x)}{e^x}$ over the interval $[0, \pi]$.

1. Find $\frac{dy}{dx}$ using the Quotient Rule and simplify.
 2. Find the exact coordinate of the single stationary point (where $\frac{dy}{dx} = 0$) in this interval.
-

Exercise 2.10.5 (Secant, Cosecant, and Cotangent derivatives)**Problem Statement:**

Using the Reciprocal Rule derived in Section 2.9 (Level 5) or the Quotient Rule, prove the following identities:

1. $\frac{d}{dx}(\sec(x)) = \sec(x)\tan(x)$
 2. $\frac{d}{dx}(\csc(x)) = -\csc(x)\cot(x)$
 3. $\frac{d}{dx}(\cot(x)) = -\csc^2(x)$
-

Exercise 2.10.6 (The absolute necessity of Radian Mode in calculus)**Problem Statement:**

Suppose an angle is measured in degrees (D°) instead of radians (x). The conversion formula is $D = \frac{180}{\pi}x$. This means $\sin(D^\circ) = \sin\left(\frac{\pi}{180}D\right)$.

1. Assuming the Chain Rule derivative of $\sin(kD)$ is $k\cos(kD)$, write down the derivative of $\sin(D^\circ)$ with respect to D .
 2. Why does this prove that the simple derivative $\frac{d}{dx}(\sin(x)) = \cos(x)$ is ****completely false**** if x is measured in degrees?
 3. Explain why all calculus operations must be performed strictly in radian mode.
-

Exercise 2.10.7 (Rigorously proving the derivative of $\sin(x)$ from First Principles)**Problem Statement:**

To prove that the derivative of $f(x) = \sin(x)$ is indeed $\cos(x)$ from first principles, we must use the trigonometric sum identity:

$$\sin(x + h) = \sin(x) \cos(h) + \cos(x) \sin(h)$$

1. Set up the first principles limit expression: $f'(x) = \lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin(x)}{h}$.
2. Substitute the sum identity and group the terms containing $\sin(x)$ together to show the limit splits into:

$$f'(x) = \sin(x) \cdot \left(\lim_{h \rightarrow 0} \frac{\cos(h) - 1}{h} \right) + \cos(x) \cdot \left(\lim_{h \rightarrow 0} \frac{\sin(h)}{h} \right)$$

3. Given the two fundamental trigonometric limits:

$$\lim_{h \rightarrow 0} \frac{\sin(h)}{h} = 1 \quad \text{and} \quad \lim_{h \rightarrow 0} \frac{\cos(h) - 1}{h} = 0$$

evaluate the expression to complete the proof.

2.11 The Chain Rule of differentiation

Up to this point, we have learned how to differentiate combinations of functions that are added, subtracted, multiplied, or divided. But there is one more vital way to combine functions: nesting them inside one another. This is called function composition, written as $f(g(x))$. For example, consider the function $y = (3x^2 + 1)^5$. This is not a simple power of x , but rather a quadratic function $g(x) = 3x^2 + 1$ nested inside an outer power function $f(u) = u^5$.

Expanding $(3x^2 + 1)^5$ algebraically to use the Power Rule would be tedious and painful. Instead, we can think of this nested structure as a chain of gears. If you turn a bicycle pedal (the input x), it turns the front sprocket (the inner function $u = g(x)$), which in turn drives the rear wheel (the outer function $y = f(u)$).

How fast does the rear wheel spin relative to your foot? It is simply the product of the gear ratios! If the front sprocket spins 3 times faster than your foot, and the wheel spins 4 times faster than the sprocket, then the wheel spins $3 \times 4 = 12$ times faster than your foot. In calculus, this multiplying gear ratio behavior is called the **Chain Rule**. It tells us that to differentiate a composite function, we multiply the rate of change of the outer function by the rate of change of the inner function!

Theorem 2.11: The Chain Rule of Differentiation

Let $y = f(u)$ be a differentiable function of u , and let $u = g(x)$ be a differentiable function of x .

The derivative of the composite function $y = f(g(x))$ with respect to x is given by:

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

In prime notation, this is written as:

$$\frac{d}{dx}(f(g(x))) = f'(g(x)) \cdot g'(x)$$

In words: "Differentiate the outer function (leaving the inner function untouched), then multiply by the derivative of the inner function."

Student Proof Demonstration

Use this space to demonstrate the proof of the theorem above. Start from basic algebraic identities.

Practical Example 2.11

Differentiate the following composite function with respect to x using the Chain Rule:

$$y = \ln(x^2 + 3)$$

Step-by-Step Solution**Step 1: Identify the outer and inner functions.**

The inner function is:

$$u = g(x) = x^2 + 3$$

The outer function is:

$$y = f(u) = \ln(u)$$

Step 2: Differentiate both layers separately.

Find their derivatives:

$$\frac{dy}{du} = \frac{d}{du}(\ln(u)) = \frac{1}{u}$$

$$\frac{du}{dx} = \frac{d}{dx}(x^2 + 3) = 2x$$

Step 3: Multiply the derivatives using the Chain Rule.

Substitute into the formula $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$:

$$\frac{dy}{dx} = \frac{1}{u} \cdot 2x$$

Step 4: Substitute back and format.

Replace u with the original inner function $x^2 + 3$ to write the answer in terms of x :

$$\frac{dy}{dx} = \frac{1}{x^2+3} \cdot 2x = \frac{2x}{x^2+3}$$

Guided Thinking & Hints

Key Hint: Think of the function as layers of an onion. The outermost layer is the natural logarithm, and the innermost core is the quadratic expression.

Key Hint: Apply the correct standard rules to each layer! $\ln(u)$ becomes $\frac{1}{u}$ (Logarithmic Rule), and $x^2 + 3$ becomes $2x$ (Power Rule).

Key Hint: This step represents the gear multiplication. We multiply the rates of change of both layers.

Key Hint: GOLDEN SHORTCUT: For any logarithmic composite of the form $y = \ln(g(x))$, the Chain Rule guarantees that its derivative is always $\frac{g'(x)}{g(x)}$!

Exercise 2.11.1 (Standard power composite)**Problem Statement:**

Differentiate the following functions with respect to x using the Chain Rule:

1. $f(x) = (2x^3 - 5)^4$
 2. $g(x) = e^{3x^2}$
-

Exercise 2.11.2 (Finding coordinates with horizontal tangents)**Problem Statement:**

Consider the curve $y = \sin(2x)$ over the interval $[0, \pi]$.

1. Find the derivative $\frac{dy}{dx}$ using the Chain Rule.
 2. Find the coordinates of all points on this curve where the tangent line is perfectly horizontal.
-

Exercise 2.11.3 (Differentiating radical composites)**Problem Statement:**

Differentiate the following function with respect to x , and express your final answer using positive indices and radical signs:

$$h(x) = \sqrt{x^2 + 4}$$

Exercise 2.11.4 (Locating stationary points on an exponential composite)**Problem Statement:**

Consider the curve $y = e^{-x^2+4x}$.

1. Find $\frac{dy}{dx}$ using the Chain Rule.
 2. Find the coordinate of the single stationary point (where the gradient is zero).
 3. Determine whether this stationary point is a maximum or a minimum by evaluating the y -values of nearby points.
-

Exercise 2.11.5 (The General Power Rule derivation)**Problem Statement:**

Let $g(x)$ be a differentiable function, and let n be any real number. Let $f(x) = [g(x)]^n$. Using the Chain Rule, prove that:

$$f'(x) = n[g(x)]^{n-1} \cdot g'(x)$$

This extremely useful identity is called the ****General Power Rule**** (or the Extended Power Rule).

Exercise 2.11.6 (Parentheses positioning traps)**Problem Statement:**

Consider the following two highly similar trigonometric functions:

$$f(x) = \sin^2(x) \quad \text{and} \quad g(x) = \sin(x^2)$$

1. Rewrite $f(x)$ using explicit nested parentheses to identify the outer and inner functions. Find $f'(x)$ using the Chain Rule.
 2. Identify the outer and inner functions for $g(x)$, and find $g'(x)$.
 3. Explain how a mistake in reading parentheses can lead to a complete structural failure when differentiating composition functions.
-

Exercise 2.11.7 (The General Logarithmic and Exponential Composite Identities)**Problem Statement:**

In university calculus, we establish general shortcuts for compound transcendental functions to skip manual substitution steps. Let $u(x)$ be a differentiable function.

1. Prove that $\frac{d}{dx} (e^{u(x)}) = u'(x)e^{u(x)}$.
 2. Prove that $\frac{d}{dx} (\ln |u(x)|) = \frac{u'(x)}{u(x)}$.
 3. Differentiate the complex compound function $y = e^{\sin(x^2)}$ using your shortcut identities. Outline each step of the "chain" clearly.
-

2.12 Second derivative

In our journey through Section 2, we have treated the derivative $f'(x)$ as a magic tool that measures the slope of a curve at any given point. But remember: the derivative $f'(x)$ is itself a function! Since it is a function, it has its own graph, its own values, and crucially, its own rate of change.

What happens if we take the derivative of the derivative?

This process gives us the **second derivative**, denoted as $f''(x)$ or $\frac{d^2y}{dx^2}$. To understand this physically, think of driving a car. Your position along the road is $s(t)$. The first derivative of your position with respect to time is your velocity, $v(t) = s'(t)$, which measures how fast your position is changing. The second derivative is your acceleration, $a(t) = s''(t)$, which measures how fast your velocity is changing! When you step on the gas pedal, you are not directly changing your position; you are changing your velocity, creating acceleration. Geometrically, the second derivative tells us how the steepness of a curve is bending—a vital concept called **concavity** which we will explore to classify turning points.

Definition 2.12: The Second Derivative

Let $y = f(x)$ be a twice-differentiable function of x .

The **second derivative** of $f(x)$ with respect to x is the derivative of the first derivative:

$$f''(x) = \frac{d}{dx} (f'(x))$$

Alternative notations for the second derivative include:

$$y'', \quad \frac{d^2y}{dx^2}, \quad \frac{d^2}{dx^2}[f(x)]$$

Physically, if $f(x)$ represents position at time x , then $f'(x)$ represents velocity and $f''(x)$ represents instantaneous acceleration.

Student Proof Demonstration

Use this space to demonstrate the proof of the theorem above. Start from basic algebraic identities.

Practical Example 2.12

Find the second derivative of the following function with respect to x :

$$f(x) = x^4 - 3e^{2x} + \ln(x) \quad (x > 0)$$

Step-by-Step Solution

Step 1: Calculate the first derivative $f'(x)$.

Differentiate each term once using our standard rules:

$$\frac{d}{dx}(x^4) = 4x^3$$

$$\frac{d}{dx}(3e^{2x}) = 3(2e^{2x}) = 6e^{2x} \text{ (using Chain Rule)}$$

$$\frac{d}{dx}(\ln(x)) = x^{-1}$$

So, the first derivative is:

$$f'(x) = 4x^3 - 6e^{2x} + x^{-1}$$

Step 2: Differentiate a second time to find $f''(x)$.

Apply the derivative operator to $f'(x)$:

$$f''(x) = \frac{d}{dx}(4x^3 - 6e^{2x} + x^{-1})$$

Step 3: Calculate the individual term derivatives.

Differentiate each term of $f'(x)$:

$$\frac{d}{dx}(4x^3) = 12x^2$$

$$\frac{d}{dx}(6e^{2x}) = 6(2e^{2x}) = 12e^{2x}$$

$$\frac{d}{dx}(x^{-1}) = -1x^{-2}$$

Step 4: Combine and format the final answer.

Combine the results:

$$f''(x) = 12x^2 - 12e^{2x} - x^{-2}$$

Convert negative exponents back to standard fraction form:

$$f''(x) = 12x^2 - 12e^{2x} - \frac{1}{x^2}$$

Guided Thinking & Hints

Key Hint: To make differentiating a second time as easy as possible, write any rational terms using negative indices (like writing $\frac{1}{x}$ as x^{-1}) rather than leaving them as fractions!

Key Hint: The second derivative is literally just doing the first derivative process all over again on your previous answer.

Key Hint: Don't forget the Chain Rule factor of 2 again when differentiating the exponential term! $6 \times 2 = 12$.

Key Hint: The final simplified second derivative is $12x^2 - 12e^{2x} - \frac{1}{x^2}$. This represents the acceleration of the curve's height!

Exercise 2.12.1 (Standard polynomial second derivative)**Problem Statement:**

Find the first and second derivatives of the following functions:

1. $f(x) = 2x^5 - 4x^3 + 7x - 3$
 2. $g(x) = \frac{4}{x^2}$
-

Exercise 2.12.2 (Trigonometric acceleration)**Problem Statement:**

Consider the oscillating position of a spring modeled by $y = \sin(2x)$.

1. Find the velocity function $\frac{dy}{dx}$ and the acceleration function $\frac{d^2y}{dx^2}$.
2. Show that the acceleration function satisfies the differential equation:

$$\frac{d^2y}{dx^2} + 4y = 0$$

Exercise 2.12.3 (Second derivative of a product)**Problem Statement:**

Find the second derivative $f''(x)$ of the following function:

$$f(x) = x^2 \cdot e^x$$

Simplify your final answer by factoring out any common terms.

Exercise 2.12.4 (Locating points of inflection)**Problem Statement:**

In HKDSE exams, a point where the second derivative is zero ($\frac{d^2y}{dx^2} = 0$) and the concavity changes is called a ****point of inflection****. Consider the cubic curve $y = 2x^3 - 9x^2 + 12x - 5$.

1. Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$.
 2. Find the exact coordinates of the point of inflection on this curve.
-

Exercise 2.12.5 (The General Product Rule for Second Derivatives)**Problem Statement:**

Let $u(x)$ and $v(x)$ be twice-differentiable functions. By starting with the first derivative $(uv)' = uv' + vu'$ and differentiating a second time, prove that:

$$(uv)'' = u''v + 2u'v' + uv''$$

Observe the striking similarity between this result and the algebraic expansion of $(a+b)^2 = a^2 + 2ab + b^2$.

Exercise 2.12.6 (The notation mismatch trap)**Problem Statement:**

A student is asked to find the second derivative of $y = x^3$ and writes:

$$\left(\frac{dy}{dx}\right)^2 = (3x^2)^2 = 9x^4$$

1. Explain the fundamental difference in meaning between the notation $\frac{d^2y}{dx^2}$ and $\left(\frac{dy}{dx}\right)^2$.
 2. Calculate the actual value of $\frac{d^2y}{dx^2}$ for $y = x^3$, and show that it does not equal the student's result.
 3. Why does this mismatch highlight the importance of writing derivative exponents in the correct positions?
-

Exercise 2.12.7 (Bridge to Section 3: Concavity and the Second Derivative Test)**Problem Statement:**

The second derivative $f''(x)$ reveals the **concavity** of a curve:

- If $f''(c) > 0$, the slope is increasing. The curve bends upward like a cup ($\cup$), meaning it is **concave up**.
- If $f''(c) < 0$, the slope is decreasing. The curve bends downward like a dome ($\cap$), meaning it is **concave down**.

This forms the basis of the **Second Derivative Test** for classifying local extrema.

1. Consider the quadratic curve $f(x) = x^2 - 4x + 5$. Find the coordinate of its stationary point (where $f'(x) = 0$).
 2. Evaluate $f''(x)$ at this stationary point. Explain why its positive sign proves the stationary point is a local minimum.
 3. Now consider the curve $g(x) = 12x - x^3$. Locate both of its stationary points.
 4. Use the Second Derivative Test to classify each stationary point of $g(x)$ as either a local maximum or a local minimum.
-

2.13 Implicit differentiation

Up to this point, every function we have differentiated has been written in an **explicit** form, $y = f(x)$. The variable y sat comfortably isolated on the left-hand side, while all the algebraic operations involving x stayed on the right. This setup is highly convenient, but many geometric paths cannot be written this way.

Consider a perfect circle centered at the origin with a radius of 5. Its equation is $x^2 + y^2 = 25$. If we try to solve for y explicitly, we get $y = \pm\sqrt{25 - x^2}$. This split definition is clunky, painful to differentiate, and breaks down at the boundary points!

Is there a way to find the slope of a curve without isolating y first?

Yes! We can treat y as an "implicit" function of x (meaning y is secretly a function of x hiding inside the equation). Instead of rearranging, we differentiate both sides of the equation with respect to x term-by-term. Whenever we differentiate a term containing y , we apply the Chain Rule: we differentiate with respect to y first, and then multiply by $\frac{dy}{dx}$. This brilliant shortcut is called **implicit differentiation**, and it allows us to find the exact tangent slope of circles, ellipses, and complex algebraic curves with ease!

Theorem 2.13: The Method of Implicit Differentiation

Let $F(x, y) = C$ be an equation defining a curve implicitly, where y is a differentiable function of x .

To find the derivative $\frac{dy}{dx}$ without isolating y :

1. Differentiate both sides of the equation with respect to x term-by-term.
2. Apply the **Chain Rule for y -terms**:

$$\frac{d}{dx} (g(y)) = \frac{d}{dy} [g(y)] \cdot \frac{dy}{dx} = g'(y) \cdot \frac{dy}{dx}$$

3. Apply the Product Rule or Quotient Rule to terms containing both x and y (e.g., $\frac{d}{dx} (xy) = x\frac{dy}{dx} + y$).
4. Collect all terms containing $\frac{dy}{dx}$ on one side of the equation, factor out $\frac{dy}{dx}$, and solve for it algebraically.

Student Proof Demonstration

Use this space to demonstrate the proof of the theorem above. Start from basic algebraic identities.

Practical Example 2.13

Find the derivative $\frac{dy}{dx}$ for the curve defined by the equation $x^2 + y^2 = 25$, and evaluate the slope of the tangent line at the point $(3, 4)$.

Step-by-Step Solution

Step 1: Differentiate both sides with respect to x .

Apply the derivative operator $\frac{d}{dx}()$ to every term:
 $\frac{d}{dx}(x^2) + \frac{d}{dx}(y^2) = \frac{d}{dx}(25)$

Step 2: Apply the rules (and Chain Rule for y).

$$\frac{d}{dx}(x^2) = 2x$$

$$\frac{d}{dx}(y^2) = 2y \cdot \frac{dy}{dx} \text{ (using Chain Rule)}$$

$$\frac{d}{dx}(25) = 0$$

So, we get the differentiated equation:

$$2x + 2y \frac{dy}{dx} = 0$$

Step 3: Solve for $\frac{dy}{dx}$ algebraically.

Move $2x$ to the right-hand side:

$$2y \frac{dy}{dx} = -2x$$

Divide both sides by $2y$ (assuming $y \neq 0$):

$$\frac{dy}{dx} = \frac{-2x}{2y} = -\frac{x}{y}$$

Step 4: Evaluate the slope at the point $(3, 4)$.

Substitute $x = 3$ and $y = 4$ into our derivative formula:

$$\left. \frac{dy}{dx} \right|_{(3,4)} = -\frac{3}{4} = -0.75$$

Guided Thinking & Hints

Key Hint: Never forget to differentiate the right-hand side! A very common student mistake is leaving the constant 25 unchanged instead of turning it into 0.

Key Hint: IMPLICIT KEY: Think of y as a box containing x . Differentiating $(\text{box})^2$ gives $2(\text{box}) \times \frac{d}{dx}(\text{box})$, which translates exactly to $2y \frac{dy}{dx}$!

Key Hint: Notice that our derivative formula contains both x and y ! This is perfectly normal for implicit curves since multiple points can share the same x -coordinate.

Key Hint: Geometrically, a radius drawn to $(3, 4)$ has a slope of $\frac{4}{3}$. Since the tangent line is perpendicular to the radius, its slope must be the negative reciprocal: $-\frac{3}{4}$. It matches perfectly!

Exercise 2.13.1 (Standard ellipse slope)**Problem Statement:**

A curve is defined implicitly by the ellipse equation:

$$2x^2 + 3y^2 = 14$$

1. Find $\frac{dy}{dx}$ in terms of x and y using implicit differentiation.
 2. Find the exact slope of the tangent line to the ellipse at the coordinate point $(1, 2)$.
-

Exercise 2.13.2 (Implicit curves with product terms)**Problem Statement:**

Consider the algebraic curve $x^2 + xy + y^2 = 7$.

1. Differentiate the equation with respect to x . *Hint: Apply the Product Rule to the term xy so that $\frac{d}{dx}(xy) = x\frac{dy}{dx} + y$.*
 2. Solve for $\frac{dy}{dx}$ explicitly.
 3. Find the equation of the tangent line to the curve at the point $(1, 2)$.
-

Exercise 2.13.3 (Implicit differentiation with natural logarithms)**Problem Statement:**

Find the expression for $\frac{dy}{dx}$ for the curve defined by:

$$\ln(y) + x^2y = 5$$

Exercise 2.13.4 (Locating points with horizontal and vertical tangents)**Problem Statement:**

Consider the curve defined by the equation $y^2 - 2x^2 = 8$.

1. Find $\frac{dy}{dx}$ using implicit differentiation.
 2. Explain why this curve has no horizontal tangents (where $\frac{dy}{dx} = 0$).
 3. Find the coordinates of the points on the curve where the tangent line is vertical (meaning the denominator of $\frac{dy}{dx}$ is 0).
-

Exercise 2.13.5 (The Derivative of Inverse Functions)**Problem Statement:**

Let $y = f^{-1}(x)$ be the inverse of a differentiable function $f(x)$. This implies $f(y) = x$. By using implicit differentiation on $f(y) = x$ with respect to x , prove that:

$$\frac{dy}{dx} = \frac{1}{f'(y)}$$

Use this general theorem to prove that the derivative of $y = \arcsin(x)$ is $\frac{1}{\sqrt{1-x^2}}$ by differentiating $\sin(y) = x$.

Exercise 2.13.6 (The isolated y illusion)**Problem Statement:**

A student attempts to find the derivative of the circular arc $y = \sqrt{25 - x^2}$ using the Chain Rule. Another student finds the derivative of the circle $x^2 + y^2 = 25$ using implicit differentiation and gets $\frac{dy}{dx} = -\frac{x}{y}$.

1. Differentiate $y = (25 - x^2)^{1/2}$ explicitly and simplify.
 2. Substitute $y = \sqrt{25 - x^2}$ into the second student's implicit formula $-\frac{x}{y}$.
 3. Show that both methods yield identical mathematical expressions. Why is the implicit path generally considered cleaner and less prone to radical-sign mistakes?
-

Exercise 2.13.7 (Second derivative of an implicit curve)**Problem Statement:**

We wish to find the second derivative $\frac{d^2y}{dx^2}$ for the circle $x^2 + y^2 = 25$.

1. Recall from the practical example that the first derivative is $\frac{dy}{dx} = -\frac{x}{y}$.
2. Differentiate this expression with respect to x using the Quotient Rule. Note that your answer will contain the term $\frac{dy}{dx}$.
3. Substitute the expression $\frac{dy}{dx} = -\frac{x}{y}$ into your second derivative equation to eliminate the first derivative term. Show that it simplifies to:

$$\frac{d^2y}{dx^2} = -\frac{y^2 + x^2}{y^3}$$

4. Use the original circle equation ($x^2 + y^2 = 25$) to simplify the numerator of your second derivative. State the clean, final expression for $\frac{d^2y}{dx^2}$. What does the negative sign tell us about the concavity of the top half of a circle ($y > 0$)?
-

2.14 Logarithmic differentiation

In Section 2.13, we discovered the power of implicit differentiation to solve curves where the variables are closely intertwined. However, there is another category of functions that seem completely impossible to differentiate using our current tools.

Consider the function $y = x^x$, where a variable base is raised to a variable exponent.

- We cannot use the Power Rule ($\frac{d}{dx}(x^n) = nx^{n-1}$) because the exponent is not a constant.
- We cannot use the Exponential Rule ($\frac{d}{dx}(b^x) = b^x \ln(b)$) because the base is not a constant.

When both the floor and the ceiling of a term are moving, how do we measure its slope?

To solve this, we can pull down the variable exponent using natural logarithms! By taking the natural logarithm ($\ln$) of both sides of the equation first, we can use the identity $\ln(a^b) = b \ln(a)$ to transform the impossible power into a simple product of functions. We then apply implicit differentiation to evaluate the derivative. This extremely elegant and advanced technique is called **logarithmic differentiation**, and it is the ultimate tool for handling variable-to-variable powers and massive, clustered rational expressions!

Theorem 2.14: The Method of Logarithmic Differentiation

Let $y = f(x)$ be a differentiable function where $f(x) > 0$.

To find $\frac{dy}{dx}$ using logarithmic differentiation:

1. Take the natural logarithm of both sides of the equation:

$$\ln(y) = \ln[f(x)]$$

2. Use logarithmic expansion properties to simplify the right-hand side:

$$\ln(A \cdot B) = \ln(A) + \ln(B)$$

$$\ln\left(\frac{A}{B}\right) = \ln(A) - \ln(B)$$

$$\ln(A^k) = k \ln(A)$$

3. Differentiate both sides with respect to x implicitly:

$$\frac{1}{y} \frac{dy}{dx} = \frac{d}{dx} (\ln[f(x)])$$

4. Multiply both sides by y to isolate the derivative, and substitute back $y = f(x)$:

$$\frac{dy}{dx} = f(x) \cdot \frac{d}{dx} (\ln[f(x)])$$

Student Proof Demonstration

Use this space to demonstrate the proof of the theorem above. Start from basic algebraic identities.

Practical Example 2.14

Find the derivative $\frac{dy}{dx}$ for the variable-to-variable power curve $y = x^x$ (for $x > 0$).

Step-by-Step Solution

Step 1: Take the natural logarithm of both sides.

Apply the $\ln$ operator:

$$\ln(y) = \ln(x^x)$$

Step 2: Pull the exponent down.

Apply the power rule of logarithms, $\ln(A^b) = b \ln(A)$:

$$\ln(y) = x \ln(x)$$

Step 3: Differentiate implicitly.

Differentiate both sides with respect to x :

$$\frac{d}{dx}(\ln(y)) = \frac{d}{dx}(x \ln(x))$$

Apply the Chain Rule on the left, and the Product Rule on the right:

$$\frac{1}{y} \frac{dy}{dx} = (x) \cdot \left(\frac{1}{x}\right) + (\ln(x)) \cdot (1)$$

$$\frac{1}{y} \frac{dy}{dx} = 1 + \ln(x)$$

Step 4: Solve for $\frac{dy}{dx}$ and substitute y .

Multiply both sides of the equation by y :

$$\frac{dy}{dx} = y(1 + \ln(x))$$

Substitute back the original $y = x^x$ to write the final answer in terms of x only:

$$\frac{dy}{dx} = x^x(1 + \ln(x))$$

Guided Thinking & Hints

Key Hint: Taking the logarithm on both sides is the key step that unlocks the variable exponent.

Key Hint: Notice how the exponent x has tumbled down to the front! The expression is now a simple multiplication of two basic functions.

Key Hint: The derivative of $\ln(y)$ is always $\frac{1}{y} \frac{dy}{dx}$ when using implicit differentiation. Be careful with the Product Rule on the right!

Key Hint: Always substitute back the original definition of y at the end so your derivative is expressed entirely in terms of the input variable x .

Exercise 2.14.1 (Variable base and exponent)**Problem Statement:**

Evaluate the derivative of the following function using logarithmic differentiation:

$$y = x^{\sin(x)} \quad (x > 0)$$

Exercise 2.14.2 (Clustered product differentiation)**Problem Statement:**

Consider the clustered product function:

$$y = (x^2 + 1)^3 \cdot e^{4x}$$

While this can be differentiated using the standard Product Rule combined with the Chain Rule, evaluate it instead using logarithmic differentiation to see how it simplifies the calculation.

Exercise 2.14.3 (Logarithmic differentiation of complex quotients)**Problem Statement:**

Find the derivative $\frac{dy}{dx}$ for the function:

$$y = \frac{(x+1)^4 \sqrt{x-2}}{(2x+3)^3} \quad (x > 2)$$

Hint: Expand the natural logarithm completely using sum, difference, and power laws before differentiating.

Exercise 2.14.4 (Locating stationary points of x^x)**Problem Statement:**

Consider the curve $y = x^x$ for $x > 0$.

1. Write down the derivative $\frac{dy}{dx}$.
 2. Find the exact coordinates of the single stationary point on this curve.
 3. Explain why the curve has a local minimum at this point by evaluating the y -value at nearby points.
-

Exercise 2.14.5 (The General Exponential derivative proof)**Problem Statement:**

Let $y = [f(x)]^{g(x)}$ where $f(x) > 0$. Prove algebraically using logarithmic differentiation that the general derivative is given by:

$$\frac{dy}{dx} = [f(x)]^{g(x)} \left[g'(x) \ln[f(x)] + \frac{g(x)f'(x)}{f(x)} \right]$$

Exercise 2.14.6 (The base-exponent trap redux)**Problem Statement:**

A student is asked to differentiate $y = 2^3$ and writes:

$$\ln(y) = 3 \ln(2) \implies \frac{1}{y} \frac{dy}{dx} = 0 \implies \frac{dy}{dx} = 0$$

Another student writes:

$$y' = 3 \cdot 2^2 = 12$$

1. Explain why the first student's result is correct, even if their method was unnecessarily complicated.
 2. Explain the exact mistake made by the second student.
 3. Why must we always check if a term is actually a variable function before choosing a differentiation technique?
-

Exercise 2.14.7 (Comparing growth speeds of variable-exponent functions)**Problem Statement:**

Consider the three functions:

$$f(x) = x^2, \quad g(x) = 2^x, \quad h(x) = x^x \quad (x > 1)$$

1. Write down the derivatives $f'(x)$, $g'(x)$, and $h'(x)$.
 2. Evaluate the rates of change of all three functions at $x = 10$.
 3. Compare the values: which function is growing the fastest, and which is growing the slowest?
 4. Discuss how the structure of x^x explains why its derivative grows significantly faster than both polynomial and exponential functions at large values of x .
-

2.15 Leibniz Rule of differentiation

In Section 2.8, we mastered the Product Rule to calculate the first derivative of two multiplied functions: $(uv)' = u'v + uv'$. Later, in Section 2.12, we explored the second derivative and proved that differentiating a product twice yields $(uv)'' = u''v + 2u'v' + uv''$. If we were to go a step further and find the third derivative, we would obtain $(uv)''' = u'''v + 3u''v' + 3u'v'' + uv'''$.

If you look closely at the coefficients of these higher-order derivatives—1, 2, 1 for the second derivative and 1, 3, 3, 1 for the third derivative—you will notice a breathtaking connection to algebra. These are the exact same numbers that appear in **Pascal's Triangle** and the **Binomial Expansion** of $(a + b)^n$!

Instead of laboriously executing the Product Rule over and over again to find higher-order derivatives, we can utilize this beautiful algebraic symmetry. Named after Gottfried Wilhelm Leibniz, the **Leibniz Rule of differentiation** allows us to write down the n -th derivative of any product function instantly using combinations and binomial coefficients. It serves as an elegant bridge connecting calculus with combinatorics!

Theorem 2.15: The Leibniz Rule of Differentiation

Let $u(x)$ and $v(x)$ be functions that are differentiable up to n times.

The n -th derivative of their product with respect to x , denoted as $(uv)^{(n)}$ or $\frac{d^n}{dx^n}[u(x)v(x)]$, is given by:

$$(uv)^{(n)} = \sum_{k=0}^n \binom{n}{k} u^{(n-k)} v^{(k)}$$

where:

- $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ represents the standard binomial coefficient.
- $u^{(j)}$ and $v^{(j)}$ denote the j -th derivatives of u and v respectively, with the convention that the 0-th derivative $u^{(0)}$ is simply the original function.

Student Proof Demonstration

Use this space to demonstrate the proof of the theorem above. Start from basic algebraic identities.

Practical Example 2.15

Find the 3rd derivative of the function $f(x) = x^2e^x$ with respect to x using the Leibniz Rule.

Step-by-Step Solution

Step 1: Set up the components and identify n .

We want the 3rd derivative, so $n = 3$.

Let $u(x) = x^2$ and $v(x) = e^x$.

Step 2: Find the derivatives of u and v up to order 3.

For $u = x^2$:

$$u^{(0)} = x^2, \quad u^{(1)} = 2x, \quad u^{(2)} = 2, \quad u^{(3)} = 0$$

For $v = e^x$:

$$v^{(0)} = e^x, \quad v^{(1)} = e^x, \quad v^{(2)} = e^x, \quad v^{(3)} = e^x$$

Step 3: Expand the Leibniz Rule formula for $n = 3$.

Write out the summation terms explicitly:

$$(uv)^{(3)} = \binom{3}{0}u^{(3)}v^{(0)} + \binom{3}{1}u^{(2)}v^{(1)} + \binom{3}{2}u^{(1)}v^{(2)} + \binom{3}{3}u^{(0)}v^{(3)}$$

Substitute the binomial coefficients $\binom{3}{0} = 1, \binom{3}{1} = 3, \binom{3}{2} = 3, \binom{3}{3} = 1$:

$$(uv)^{(3)} = 1 \cdot u^{(3)}v^{(0)} + 3 \cdot u^{(2)}v^{(1)} + 3 \cdot u^{(1)}v^{(2)} + 1 \cdot u^{(0)}v^{(3)}$$

Step 4: Substitute the values and simplify.

$$(uv)^{(3)} = 1(0)(e^x) + 3(2)(e^x) + 3(2x)(e^x) + 1(x^2)(e^x)$$

Combine the remaining non-zero terms:

$$(uv)^{(3)} = 6e^x + 6xe^x + x^2e^x$$

Factor out the common term e^x :

$$f^{(3)}(x) = e^x(x^2 + 6x + 6)$$

Guided Thinking & Hints

Key Hint: Always choose the polynomial term as $u(x)$ if its derivatives eventually collapse to zero. This will make almost all terms in the Leibniz expansion vanish, saving you immense calculation time!

Key Hint: Notice how $u^{(k)} = 0$ for all $k \geq 3$. This means any term in our expansion containing $u^{(3)}$ or higher will instantly become zero!

Key Hint: This behaves exactly like the binomial expansion $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$, where the powers are replaced by the order of the derivatives.

Key Hint: Using the Leibniz Rule bypassed having to expand and factor three successive Product Rules. It keeps the algebra incredibly clean and direct!

Exercise 2.15.1 (Linear-exponential higher order derivative)**Problem Statement:**

Let $f(x) = xe^x$. Use the Leibniz Rule to find the exact expression for the 4-th derivative, $f^{(4)}(x)$.

Exercise 2.15.2 (Higher order derivatives of trigonometric products)**Problem Statement:**

Evaluate the 3-rd derivative of the function $g(x) = x \cos(x)$ using the Leibniz Rule. Recall that x is measured in radians.

Exercise 2.15.3 (Locating polynomials of order n)**Problem Statement:**

Find the exact 50-th derivative, $\frac{d^{50}}{dx^{50}}[u(x)v(x)]$, for the product:

$$f(x) = x^2 \sin(x)$$

Hint: Find the derivatives of $u(x) = x^2$ first, and determine which terms in the expansion survive.

Exercise 2.15.4 (Mathematical induction verification of a pattern)**Problem Statement:**

Let $y = x^2 e^{2x}$.

1. Find the first derivative $\frac{dy}{dx}$ and the second derivative $\frac{d^2y}{dx^2}$.
2. Use the Leibniz Rule to prove that the n -th derivative of y is given by:

$$y^{(n)} = 2^{n-2} e^{2x} (4x^2 + 4nx + n(n-1))$$

Exercise 2.15.5 (Rigorous proof of the Leibniz Rule using Mathematical Induction)**Problem Statement:**

Let $P(n)$ be the statement that the n -th derivative of the product uv is:

$$(uv)^{(n)} = \sum_{k=0}^n \binom{n}{k} u^{(n-k)} v^{(k)}$$

Using Pascal's Identity, $\binom{n}{k-1} + \binom{n}{k} = \binom{n+1}{k}$, prove by mathematical induction that $P(n)$ is true for all positive integers $n \geq 1$.

Exercise 2.15.6 (The order selection trap)**Problem Statement:**

A student is asked to find the 4-th derivative of $f(x) = x^4 \ln(x)$ and sets $u(x) = \ln(x)$ and $v(x) = x^4$.

1. Find the derivatives of $u(x) = \ln(x)$ up to order 4. Do they ever collapse to 0?
 2. Find the derivatives of $v(x) = x^4$ up to order 4.
 3. Explain why the student's assignment of u and v makes the Leibniz calculation extremely tedious, whereas choosing $u(x) = x^4$ and $v(x) = \ln(x)$ makes the evaluation much simpler.
-

Exercise 2.15.7 (Patterns of the n -th derivative of a logarithmic product)**Problem Statement:**

Consider the product function $f(x) = x^n \ln(x)$ where $x > 0$ and n is a positive integer.

1. Use the Leibniz Rule to write a general expression for the n -th derivative $f^{(n)}(x)$.
 2. Simplify the expression by evaluating the derivatives and show that the term containing $\ln(x)$ simplifies to $n! \ln(x)$.
 3. Evaluate the exact value of $f^{(n)}(1)$ for $n = 5$.
-

3 Applications of the Derivative

3.1 Tangents and Normals

Throughout Section 2, we defined the derivative $f'(x)$ geometrically as the slope of the tangent line grazing a curve at a single point. But we did not explore the complete linear equations of these lines, nor did we look at their perpendicular partners: **normals**.

Imagine a race car speeding around a curved track. If the car suddenly loses traction on a patch of oil at a specific point, it will spin off the track, traveling along a perfect straight path. That path is the **tangent line**, representing the instantaneous direction of motion.

Now, imagine the protective concrete barrier built to absorb crashes on the outside of that curve. To maximize safety, the barrier must be positioned exactly perpendicular to the car's direction of travel at the impact point. This perpendicular line is called the **normal line**.

In coordinate geometry, finding the equations of these two straight lines is the ultimate way to study the local linear behavior of curves. By linking our new derivative techniques with the classic point-slope equation from junior secondary school, we can construct these lines at any point on any differentiable curve.

Proposition 3.1: Equations of Tangents and Normals

Let $y = f(x)$ be a differentiable curve, and let $P(x_0, y_0)$ be a fixed point lying on the curve, where $y_0 = f(x_0)$.

1. **The Tangent Line:** The slope of the tangent line at P is $m_{\text{tan}} = f'(x_0)$. The equation of the tangent line is:

$$y - y_0 = f'(x_0)(x - x_0)$$

2. **The Normal Line:** The normal line is perpendicular to the tangent line. Assuming $f'(x_0) \neq 0$, its slope is the negative reciprocal: $m_{\text{norm}} = -\frac{1}{f'(x_0)}$. The equation of the normal line is:

$$y - y_0 = -\frac{1}{f'(x_0)}(x - x_0)$$

Student Proof Demonstration

Use this space to demonstrate the proof of the theorem above. Start from basic algebraic identities.

Practical Example 3.1

Find the equations of the tangent line and the normal line to the cubic curve $f(x) = x^3 - 3x + 2$ at the point where $x = 2$.

Step-by-Step Solution

Guided Thinking & Hints

Step 1: Find the coordinate point (x_0, y_0) .

The given x -coordinate is $x_0 = 2$.

Find the corresponding y -coordinate by plugging $x = 2$ into the original function:

$$y_0 = f(2) = (2)^3 - 3(2) + 2$$

$$y_0 = 8 - 6 + 2 = 4$$

So, the point of contact is $P(2, 4)$.

Key Hint: Always find the complete (x, y) coordinate point first! A common rookie mistake is trying to write the line equation using the slope value as the y -coordinate.

Step 2: Find the tangent slope m_{tan} .

First, find the general derivative $f'(x)$ using the Power Rule:

$$f'(x) = 3x^2 - 3$$

Evaluate the derivative at $x_0 = 2$:

$$m_{\text{tan}} = f'(2) = 3(2)^2 - 3 = 12 - 3 = 9$$

Step 3: Construct the tangent line equation.

Using the point-slope form with $P(2, 4)$ and $m_{\text{tan}} = 9$:

$$y - 4 = 9(x - 2)$$

Expand and simplify:

$$y - 4 = 9x - 18 \implies y = 9x - 14$$

Step 4: Construct the normal line equation.

The normal slope is the negative reciprocal of the tangent slope:

$$m_{\text{norm}} = -\frac{1}{m_{\text{tan}}} = -\frac{1}{9}$$

Using the point-slope form with $P(2, 4)$ and $m_{\text{norm}} = -\frac{1}{9}$:

$$y - 4 = -\frac{1}{9}(x - 2)$$

Multiply both sides by 9 and expand:

$$9y - 36 = -x + 2 \implies x + 9y - 38 = 0$$

Key Hint: Remember, the derivative formula gives you the slope at *any* point. You must substitute the specific x -coordinate of your target point to get a numerical slope!

Key Hint: You can express your final lines in slope-intercept form ($y = mx + c$) or general form ($Ax + By + C = 0$). Check what the exam question asks for!

Key Hint: Since normal slopes are often fractions, multiplying the entire equation by the denominator is a clean way to write the final answer in general form without clunky decimals.

Exercise 3.1.1 (Standard polynomial tangents and normals)**Problem Statement:**

Consider the quadratic curve $y = 2x^2 - 5x + 1$.

1. Find the equation of the tangent line to the curve at the point where $x = 3$. Express your answer in the form $y = mx + c$.
 2. Find the equation of the normal line to the curve at the same point.
-

Exercise 3.1.2 (Tangents and normals to exponential curves)**Problem Statement:**

Find the equations of the tangent and normal lines to the natural exponential curve $y = e^{2x}$ at the point where it crosses the vertical y -axis.

Exercise 3.1.3 (Tangents to rational curves with coordinate limits)**Problem Statement:**

Consider the rational curve $y = \frac{x+2}{x-1}$ for $x > 1$.

1. Find the derivative $\frac{dy}{dx}$ using the Quotient Rule.
 2. Find the equation of the tangent line to the curve at the point where the curve intersects the horizontal line $y = 2$.
-

Exercise 3.1.4 (Implicit curves and normal line equations)**Problem Statement:**

An elliptical curve is defined implicitly by the equation:

$$3x^2 - 2xy + y^2 = 9$$

Find the exact equation of the normal line to the curve at the point $P(2, 3)$. Express your answer in the general form $Ax + By + C = 0$.

Exercise 3.1.5 (The point-to-tangent distance formula)**Problem Statement:**

Let $y = ax^2$ be a standard parabola. Prove that the tangent line to this parabola at any arbitrary point (k, ak^2) crosses the horizontal x -axis at exactly half the distance from the origin to the vertical projection of the point of contact. That is, show the x -intercept of the tangent line is always $\frac{k}{2}$.

Exercise 3.1.6 (Tangents on vertical walls)**Problem Statement:**

Consider the radical curve $y = \sqrt{x}$ for $x \geq 0$.

1. Attempt to find the equation of the tangent line to the curve at the origin $(0, 0)$. What mathematical breakdown occurs when evaluating the slope?
 2. Sketch or describe the geometric behavior of the curve as x approaches 0 from the right.
 3. State the physical equation of this tangent line. Why does our point-slope formula fail to generate it?
-

Exercise 3.1.7 (The family of orthogonal trajectories)**Problem Statement:**

In advanced physics and vector calculus, **orthogonal trajectories** are families of curves that intersect another family of curves at perfect right angles (90°) at every point of intersection.

1. Consider the family of ellipses defined by $x^2 + 2y^2 = C$, where $C > 0$ is a constant. Use implicit differentiation to show that the tangent slope at any point (x, y) on any of these ellipses is given by:

$$\frac{dy}{dx} = -\frac{x}{2y}$$

2. Since the orthogonal trajectories must intersect these curves perpendicularly, write down the required differential equation for the orthogonal family's slope, $\frac{dy_{\text{orth}}}{dx}$.
3. Solve this differential equation by separating variables:

$$\int \frac{1}{y} dy = \int \frac{2}{x} dx$$

Show that the family of orthogonal trajectories is a family of parabolas of the form $y = kx^2$, where k is a real constant.

3.2 Tangents parallel/perpendicular to other straight lines

In Section 3.1, we mastered the mechanics of finding the equation of a tangent or normal line when we were handed a specific starting point of contact (x_0, y_0) . Geometrically, that is a highly direct path. But in examinations and real-world engineering, you are rarely given the point of contact on a silver platter.

Instead, you might be told: "Find the tangent to this parabolic solar sail that is perfectly parallel to the incoming beam of light," or "Find the normal line that runs perpendicular to a reference grid line."

How do we work backward to find the point of contact when we only know the **steepness or direction** we want to achieve?

The secret lies in recognizing that the derivative $f'(x)$ is a "slope-generating factory." If we are given a target line, we can easily extract its slope m_{target} . To find a tangent parallel to this line, we set our slope-generator equal to the target slope ($f'(x) = m_{\text{target}}$) and solve for the unknown coordinate x . To find a tangent perpendicular to the line, we use the negative reciprocal relation ($f'(x) = -\frac{1}{m_{\text{target}}}$). This shift in perspective transforms our calculus engine from a forward calculator into a backward pathfinder!

Proposition 3.2: Parallel and Perpendicular Tangents

Let $y = f(x)$ be a differentiable curve, and let L_1 be a given straight line with a known slope m_{target} .

To find the point(s) of contact $P(x_c, y_c)$ where the tangent line to the curve is:

1. **Parallel to L_1 :** Solve the algebraic equation for x_c :

$$f'(x_c) = m_{\text{target}}$$

2. **Perpendicular to L_1 :** Solve the algebraic equation for x_c (assuming $m_{\text{target}} \neq 0$):

$$f'(x_c) = -\frac{1}{m_{\text{target}}}$$

Once the coordinate x_c is found, evaluate $y_c = f(x_c)$ to establish the full point of contact, and then construct the line equation using the point-slope form.

Student Proof Demonstration

Use this space to demonstrate the proof of the theorem above. Start from basic algebraic identities.

Practical Example 3.2

Find the equation(s) of the tangent line(s) to the curve $f(x) = \frac{1}{3}x^3 - x^2 - 2x + 5$ that are perfectly parallel to the straight line $y = 10 - 2x$.

Step-by-Step Solution

Step 1: Extract the target slope.

The given target line is $y = -2x + 10$.

Comparing this to $y = mx + c$, the target slope is:

$$m_{\text{target}} = -2$$

Step 2: Set up and solve the slope equation.

Find the derivative $f'(x)$:

$$f'(x) = x^2 - 2x - 2$$

Since the tangent must be parallel, we set $f'(x) = m_{\text{target}}$:

$$x^2 - 2x - 2 = -2$$

Solve for x :

$$x^2 - 2x = 0 \implies x(x - 2) = 0$$

Thus, $x_1 = 0$ or $x_2 = 2$.

Step 3: Find the corresponding y -coordinates.

For $x_1 = 0$:

$$y_1 = f(0) = \frac{1}{3}(0)^3 - (0)^2 - 2(0) + 5 = 5.$$

Point of contact: $P_1(0, 5)$.

For $x_2 = 2$:

$$y_2 = f(2) = \frac{1}{3}(8) - (4) - 4 + 5 = \frac{8}{3} - 3 = -\frac{1}{3}.$$

Point of contact: $P_2(2, -\frac{1}{3})$.

Step 4: Construct the tangent line equations.

Both tangents share the slope $m = -2$.

Tangent 1 at $P_1(0, 5)$:

$$y - 5 = -2(x - 0) \implies y = -2x + 5$$

Tangent 2 at $P_2(2, -\frac{1}{3})$:

$$y - (-\frac{1}{3}) = -2(x - 2) \implies y + \frac{1}{3} = -2x + 4$$

Multiply by 3:

$$3y + 1 = -6x + 12 \implies 6x + 3y - 11 = 0$$

Guided Thinking & Hints

Key Hint: Always extract the target slope first. If the target line is written in general form like $Ax + By + C = 0$, rearrange it into $y = -\frac{A}{B}x - \frac{C}{B}$ to find the slope!

Key Hint: Because the curve is a cubic wave, it bends back and forth. It is perfectly natural to find *multiple* points along the curve that share the exact same steepness!

Key Hint: Double-check which function you use to find the y -value! You must substitute your x -values back into the **original curve** $f(x)$, NOT the derivative $f'(x)$ or the target line.

Key Hint: Write down both final equations clearly. We have successfully found two parallel protective straight paths grazing our cubic curve!

Exercise 3.2.1 (Standard parallel tangents)**Problem Statement:**

Consider the quadratic curve $y = 3x^2 - 8x + 2$. Find the exact equation of the tangent line to this curve that is perfectly parallel to the straight line $y = 4x - 7$.

Exercise 3.2.2 (Perpendicular tangents to rational curves)**Problem Statement:**

Let a curve be given by $f(x) = x + \frac{4}{x}$ for $x > 0$. Find the equation of the tangent line to the curve that is perpendicular to the straight line $2x + 6y - 5 = 0$.

Exercise 3.2.3 (Finding unknown coefficients from target slopes)**Problem Statement:**

A curve is given by $y = ax^2 + bx + 3$, where a and b are constant coefficients. It is known that the tangent line to the curve at the point where $x = 1$ is parallel to the line $y = 2x + 5$, and the tangent line at the point where $x = -2$ is perpendicular to the line $x - 4y + 1 = 0$. Find the exact values of the constants a and b .

Exercise 3.2.4 (Double parallel tangents and normal distance)**Problem Statement:**

Consider the cubic curve $y = x^3 - 3x^2 + 1$.

1. Find the equations of the two distinct tangent lines to the curve, L_1 and L_2 , that are both parallel to the horizontal line $y = 5$.
 2. Find the vertical distance between these two parallel tangent lines.
-

Exercise 3.2.5 (General perpendicular points on a parabola)**Problem Statement:**

Let $y = kx^2$ be a standard parabola, where $k > 0$. Let L be a given line with a non-zero slope m . Prove that the coordinates of the unique point of contact on the parabola where the tangent line is perpendicular to L is always:

$$P\left(-\frac{1}{2km}, \frac{1}{4k^2m^2}\right)$$

Exercise 3.2.6 (The impossible target slope trap)**Problem Statement:**

Consider the natural logarithmic curve $y = 2\ln(x)$ for $x > 0$. A student is asked to find a tangent line to this curve that is parallel to the straight line $y = -3x + 4$.

1. Set up the slope equation for this problem.
 2. Solve the equation for the contact coordinate x . What mathematical conflict arises with the domain of our function?
 3. Explain why it is geometrically impossible to find any tangent with a negative slope on this logarithmic curve.
-

Exercise 3.2.7 (Infinite families of parallel tangents on wave curves)**Problem Statement:**

Consider the sinusoidal wave curve $y = 2\sin(x)$ over the entire real domain. We wish to study all tangent lines to this wave that have a slope of 1.

1. Find the general expression for the x -coordinates of all points of contact where the tangent slope is exactly 1. Express your answer using an integer parameter $n \in \mathbb{Z}$.
2. Find the corresponding y -coordinates for these points. Note that your answer will depend on whether n is even or odd.
3. Show that these points of contact fall into two parallel families of points:

$$P_k = \left(\frac{\pi}{3} + 2k\pi, \sqrt{3}\right) \quad \text{and} \quad Q_k = \left(-\frac{\pi}{3} + 2k\pi, -\sqrt{3}\right) \quad \text{for } k \in \mathbb{Z}$$

4. Write down the equations for the two parallel families of tangent lines.
-

3.3 Tangents passing through external points

In Section 3.2, we solved problems where we worked backward from a given target slope. Now, we tackle the most mathematically sophisticated geometric puzzle in linear calculus: finding a tangent line that must pass through a specific **external point** (x_1, y_1) that does not lie on the curve.

To visualize this, imagine standing on a hilltop at point $A(x_1, y_1)$ and shining a straight laser beam toward a curved reflective sculpture $y = f(x)$ below you. You want the laser beam to graze the sculpture perfectly.

Since the starting point A is not on the curve, we cannot simply evaluate $f'(x_1)$ to find the slope. Doing so would find the slope of the curve **at** x_1 , which is completely irrelevant to our laser line!

Instead, we must introduce an unknown variable, say a , to represent the x -coordinate of the secret **point of contact** $P(a, f(a))$ where the line will graze the curve. We can then express the slope of this tangent line in two completely different ways: algebraically using the slope formula between two coordinate points, and analytically using the derivative. By setting these two representations equal to each other, we can solve for a and crack open the entire system!

Proposition 3.3: Tangents from an External Point

Let $y = f(x)$ be a differentiable curve, and let $A(x_1, y_1)$ be a given external point that does **not** lie on the curve.

To find the equation(s) of the tangent line(s) passing through A :

1. Let the unknown point of contact on the curve be $P(a, f(a))$.
2. Express the slope of the tangent line, m , in two ways:

$$m = f'(a) \quad (\text{Analytic Slope})$$

$$m = \frac{f(a) - y_1}{a - x_1} \quad (\text{Geometric Slope})$$

3. Set the two expressions equal to solve for the unknown contact coordinate a :

$$f'(a) = \frac{f(a) - y_1}{a - x_1}$$

4. Once the values of a are resolved, calculate the slope $m = f'(a)$ and construct the tangent line equations using the point-slope form.

Student Proof Demonstration

Use this space to demonstrate the proof of the theorem above. Start from basic algebraic identities.

Practical Example 3.3

Find the equation(s) of the tangent line(s) to the parabola $f(x) = x^2 + 3$ that pass through the external point $A(1, 0)$.

Exercise 3.3.1 (Parabolic tangents from origin)**Problem Statement:**

Find the equations of the two tangent lines to the parabola $y = x^2 + 4$ that pass through the origin $(0, 0)$.

Exercise 3.3.2 (Tangents to exponential curves from external source)**Problem Statement:**

Find the equation of the tangent line to the natural exponential curve $y = e^x$ that passes through the external point $A(2, 0)$.

Exercise 3.3.3 (Tangents to rational curves from coordinate points)**Problem Statement:**

Consider the hyperbola $y = \frac{4}{x}$ for $x > 0$. Find the exact equation of the tangent line to the curve that passes through the external point $A(-2, 8)$.

Exercise 3.3.4 (HKDSE-style double tangent intersection)**Problem Statement:**

Consider the cubic curve $y = x^3 - 3x^2 + 4$. Two distinct tangent lines are drawn to the curve from the external point $P(1, 0)$ on the x -axis.

1. Show that the x -coordinates of the points of contact satisfy the equation $2x^3 - 6x^2 + 6x - 4 = 0$.
 2. By factoring out $(x - 2)$, solve the equation to find the exact coordinates of the two points of contact.
 3. Find the equations of both tangent lines.
-

Exercise 3.3.5 (The general boundary of external tangency)**Problem Statement:**

Let $y = x^2$ be a standard parabola. Let $A(h, k)$ be any point in the coordinate plane. By setting up the external slope balance equation, prove that:

1. If $k < h^2$, there are exactly two distinct real tangent lines that can be drawn from A to the parabola.
2. If $k = h^2$, there is exactly one tangent line (since A lies directly on the curve).
3. If $k > h^2$, no real tangent lines can be drawn from A to the parabola.

Discuss how this partitions the coordinate plane into regions of varying tangency potential.

Exercise 3.3.6 (The zero-denominator division anomaly)**Problem Statement:**

Consider the curve $y = \sqrt{x}$ for $x \geq 0$. We wish to find a tangent line passing through the external point $A(0, 2)$.

1. Set up the slope balance equation with contact coordinate $a > 0$.
 2. Solve the equation to find a .
 3. Now, suppose we attempted to find a tangent line passing through the point $B(4, 2)$ which lies directly on the curve, but we accidentally used the external point formula with point B . Show how a division-by-zero anomaly occurs in your equation, and discuss how to resolve it.
-

Exercise 3.3.7 (The orthoptic property of parabolas)**Problem Statement:**

In coordinate geometry, the locus of points from which two tangent lines to a curve intersect at a perfect right angle (90°) is called the **orthoptic curve**.

1. Let $y = ax^2$ be a parabola (where $a \neq 0$). Let $P(h, k)$ be an external point from which two perpendicular tangents are drawn to the parabola.
2. Let the points of contact be (x_1, ax_1^2) and (x_2, ax_2^2) . Write down the quadratic equation in terms of the contact variable x derived from the slope balance equation passing through $P(h, k)$.
3. Since the two slopes $m_1 = 2ax_1$ and $m_2 = 2ax_2$ must multiply to -1 , use Vieta's formulas on your quadratic equation to prove that:

$$k = -\frac{1}{4a}$$

4. What geometric line does this constant y -value represent? Discuss how this proves that the orthoptic curve of any parabola is its **directrix**.

3.4 Newton's method on approximating real roots

In Sections 3.1 through 3.3, we developed a complete geometric toolkit for constructing tangent lines. We saw that a tangent line acts as a local "linear mirror" of a curve, matching its position and steepness at a single point. Now, we will use this local linear behavior to solve one of the oldest challenges in algebra: finding the roots of equations.

Suppose you want to find the exact x -intercept of a highly complex polynomial or transcendental curve, such as $x^5 + 2x - 3 = 0$ or $x - \cos(x) = 0$. In junior secondary school, you learned formulas to solve quadratic equations, but no simple formulas exist for higher-degree polynomials or mixed trigonometric-algebraic systems.

How can we approximate these roots with incredible speed and precision?

We turn to **Newton's Method** (also known as the Newton-Raphson method). Instead of guessing blindly, we pick an initial estimate x_0 near the true root. We construct the tangent line to the curve at x_0 and trace it down to where it crosses the horizontal x -axis. Because the tangent line closely mimics the curve, its x -intercept x_1 will be significantly closer to the true root than our initial guess! By repeating this process iteratively, we construct a sequence of values $(x_0, x_1, x_2, \dots)$ that rapidly converges to the exact root.

Theorem 3.4: Newton's Method Iterative Formula

Let $f(x)$ be a differentiable function on an interval containing a real root r such that $f(r) = 0$.

To approximate the root r starting from an initial estimate x_0 :

1. Find the first derivative $f'(x)$.
2. Generate successive approximations x_{n+1} using the iterative formula:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

for $n = 0, 1, 2, \dots$, provided $f'(x_n) \neq 0$.

3. Repeat this cycle until the difference between successive terms $|x_{n+1} - x_n|$ falls below your desired tolerance (precision level).

Student Proof Demonstration

Use this space to demonstrate the proof of the theorem above. Start from basic algebraic identities.

Practical Example 3.4

Use Newton's Method to approximate the real root of the cubic equation $x^3 - 2x - 5 = 0$. Take $x_0 = 2$ as your initial estimate, and perform two iterations to find the approximation x_2 (express your intermediate and final steps to 4 decimal places).

Exercise 3.4.1 (Standard polynomial root approximation)**Problem Statement:**

Let $f(x) = x^3 - x - 1$.

1. Using $x_0 = 1$ as your initial guess, calculate x_1 and x_2 using Newton's Method. Keep your answers to 4 decimal places.
 2. Evaluate $f(x_2)$ to check the accuracy of your second estimate.
-

Exercise 3.4.2 (Approximating transcendental roots)**Problem Statement:**

Consider the transcendental equation $\cos(x) - x = 0$, where x is measured in radians.

1. Write down the Newton's Method iteration formula for this function.
 2. Using $x_0 = 0.5$ as your starting estimate, perform two iterations to find x_2 to 4 decimal places.
-

Exercise 3.4.3 (Evaluating square roots without the radical sign)**Problem Statement:**

Newton's Method can be used to construct manual algorithms for finding roots. To calculate the value of $\sqrt{6}$ without using a radical sign, we can solve for the positive root of the polynomial $f(x) = x^2 - 6 = 0$.

1. Set up the iteration formula for $f(x) = x^2 - 6$, and simplify the right-hand side into a single elegant fraction.
 2. Using $x_0 = 2.5$ as an initial guess, perform two iterations to find x_2 . Express your steps using exact fractions.
 3. Show that the square of your fraction x_2 is incredibly close to 6.
-

Exercise 3.4.4 (HKDSE-style intersection of curves)**Problem Statement:**

Consider the exponential curve $y = e^{-x}$ and the linear line $y = x$.

1. Sketch both curves on a coordinate plane and explain why they must intersect at exactly one point.
 2. Show that finding their point of intersection is equivalent to finding the root of $f(x) = xe^x - 1 = 0$.
 3. Using the initial guess $x_0 = 0.5$, use Newton's Method to find the intersection x -coordinate to 4 decimal places after two iterations.
-

Exercise 3.4.5 (The algebraic derivation of Newton's formula)**Problem Statement:**

Let $y = f(x)$ be a differentiable curve, and let x_n be your current estimate. By writing down the equation of the tangent line to the curve at the point $(x_n, f(x_n))$ and solving for its x -intercept (the point where $y = 0$), prove algebraically that this intercept is indeed:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Exercise 3.4.6 (The turning point and cycle traps)**Problem Statement:**

Newton's Method is incredibly fast, but it is not infallible.

1. **The Slope Anomaly:** Consider $f(x) = x^3 - 3x + 1 = 0$. Suppose we choose our starting guess to be $x_0 = 1$. What happens when you attempt to calculate x_1 ? Explain the geometric breakdown.
 2. **The Cyclic Oscillation Trap:** Consider the function $f(x) = x^3 - 5x = 0$. Suppose we choose $x_0 = 1$. Show that $x_1 = -1$ and $x_2 = 1$. Describe the behavioral failure of the sequence and why it fails to converge to any root.
-

Exercise 3.4.7 (The Speed of Convergence: Quadratic Behavior)**Problem Statement:**

In numerical analysis, we study the **rate of convergence** of algorithms. Newton's Method is celebrated because under favorable conditions, it exhibits **quadratic convergence**. This means that if $e_n = |x_n - r|$ is the error at step n , then:

$$e_{n+1} \approx M \cdot e_n^2 \quad \text{where } M = \left| \frac{f''(r)}{2f'(r)} \right|$$

Roughly speaking, the number of correct decimal places doubles with every single step!

1. Let $f(x) = x^2 - 2 = 0$, which has the root $r = \sqrt{2} \approx 1.41421356$.
 2. Using the iteration formula $x_{n+1} = \frac{1}{2} \left(x_n + \frac{2}{x_n} \right)$, and starting with $x_0 = 1.5$:
 3. Evaluate x_1, x_2, x_3 , and x_4 , keeping your values to 10 decimal places.
 4. Calculate the error $e_n = |x_n - r|$ for $n = 0, 1, 2, 3, 4$.
 5. Tabulate your results and show how the number of matching decimal places behaves at each step. Discuss how this validates the quadratic convergence theorem.
-

3.5 Instantaneous rate of change

In Section 3.4, we explored how the local linear approximation of a tangent line can help us hunt down the roots of equations. We now transition from pure coordinate geometry to physical and physical-economic modeling. We begin with **Section 3.5: Instantaneous rate of change**.

When you look at the speedometer of a car, it tells you how fast you are moving at that exact, fleeting second. This is not your average speed over a journey, but your **instantaneous speed**—mathematically, the derivative of your position with respect to time.

Similarly, in business, if you are producing thousands of smartphones, you want to know the exact cost of producing just **one more** unit. This is the **marginal cost**—the instantaneous rate of change of total cost with respect to quantity.

In this section, we take the abstract derivative rules we mastered in Section 2 and apply them to direct physical and economic models. Whenever we are asked how fast one quantity is changing with respect to another **at a specific instant**, we are being asked to find and evaluate a first derivative.

Proposition 3.5: Physical and Economic Rates of Change

Let $y = f(x)$ be a differentiable function modeling a physical or economic quantity.

1. **Kinematics (Motion):** Let $s(t)$ represent the position of a particle at time t .
 - The **instantaneous velocity** is $v(t) = s'(t) = \frac{ds}{dt}$.
 - The **instantaneous acceleration** is $a(t) = v'(t) = s''(t) = \frac{d^2s}{dt^2}$.
2. **Marginal Analysis (Economics):** Let $C(q)$ be the total cost of producing q units of a commodity.
 - The **marginal cost** is $MC = C'(q) = \frac{dC}{dq}$. This approximates the cost of producing the $(q + 1)$ -th unit.
3. **General Science:** The instantaneous rate of change of any quantity y with respect to x at $x = x_0$ is simply $f'(x_0)$.

Student Proof Demonstration

Use this space to demonstrate the proof of the theorem above. Start from basic algebraic identities.

Practical Example 3.5

The position of a particle moving along a straight line is modeled by the function $s(t) = t^3 - 6t^2 + 9t + 2$ for $t \geq 0$, where s is in meters and t is in seconds.

1. Find the velocity $v(t)$ and acceleration $a(t)$ at any time t .
2. Determine the time intervals when the particle is momentarily at rest, and find its position at these moments.

Exercise 3.5.1 (Standard projectile motion)**Problem Statement:**

A model rocket is launched vertically upward. Its height $h(t)$ (in meters) after t seconds is given by:

$$h(t) = -5t^2 + 40t + 1.5 \quad (t \geq 0)$$

1. Find the instantaneous velocity of the rocket at $t = 2$ seconds.
 2. Find the acceleration of the rocket at any time t . Explain the physical meaning of your result.
-

Exercise 3.5.2 (Marginal cost analysis in economics)**Problem Statement:**

The total daily cost $C(q)$ (in HKD) of producing q high-end headphone units is modeled by:

$$C(q) = 0.02q^3 - 0.9q^2 + 50q + 1200$$

1. Find the marginal cost function $MC = C'(q)$.
 2. Calculate the marginal cost when $q = 20$.
 3. Calculate the actual cost of producing the 21-st unit, i.e., $C(21) - C(20)$, and compare it to your result in part (2).
-

Exercise 3.5.3 (Kinematics of a particle changing directions)**Problem Statement:**

A particle moves along a straight line such that its displacement $s(t)$ (in meters) at time t (in seconds) is given by:

$$s(t) = 2t^3 - 15t^2 + 24t \quad (t \geq 0)$$

Find the velocity of the particle at the exact instant its acceleration is zero.

Exercise 3.5.4 (HKDSE M1-style radioactive decay rates)**Problem Statement:**

The mass $M(t)$ (in grams) of a radioactive isotope remaining after t years is modeled by:

$$M(t) = 80e^{-0.05t} \quad (t \geq 0)$$

1. Find the instantaneous rate of decay of the mass at $t = 10$ years. Express your answer as a negative rate of change.
 2. Find the time t at which the mass is decaying at a rate of -2 grams per year. Express your answer in terms of natural logarithms.
-

Exercise 3.5.5 (Marginal Revenue and Profit Maximisation)**Problem Statement:**

Let $R(q)$ be the revenue function and $C(q)$ be the cost function for producing q units of a product. The profit function is given by $P(q) = R(q) - C(q)$. Prove that the profit is maximized at a production level where the **marginal revenue** equals the **marginal cost** ($R'(q) = C'(q)$), assuming $P''(q) < 0$.

Exercise 3.5.6 (The instantaneous velocity paradox)**Problem Statement:**

Suppose a particle's position is given by the continuous but non-differentiable piecewise function:

$$s(t) = \begin{cases} 4t & \text{if } 0 \leq t < 2 \\ 8 & \text{if } t \geq 2 \end{cases}$$

1. Calculate the velocity of the particle for $t < 2$ and $t > 2$.
 2. Attempt to calculate the instantaneous velocity at the boundary $t = 2$.
 3. Discuss the physical impossibility of this motion. Why does a sharp corner in a displacement-time graph represent an infinite, physically impossible acceleration?
-

Exercise 3.5.7 (The relationship between velocity, acceleration, and speed)**Problem Statement:**

In physics, **speed** is the magnitude of velocity, $|v(t)|$. A particle is **speeding up** if its speed is increasing, and **slowing down** if its speed is decreasing.

1. Prove that a particle is speeding up if and only if its velocity $v(t)$ and acceleration $a(t)$ have the **same sign** (i.e., $v(t) \cdot a(t) > 0$).
 2. Prove that a particle is slowing down if and only if $v(t)$ and $a(t)$ have **opposite signs** (i.e., $v(t) \cdot a(t) < 0$).
 3. Let $s(t) = t^3 - 9t^2 + 24t$. Determine the exact time intervals during which the particle is speeding up, and the intervals during which it is slowing down.
-

3.6 Implicit rate of change (Related Rates)

In Section 3.5, we calculated direct physical rates of change where a single dependent variable changed with respect to an independent variable (like position over time). Now, we step up to a more advanced, multi-variable arena: **implicit rates of change**, universally known in calculus as **Related Rates**.

Imagine pumping air into a spherical balloon. As you blow air inside at a constant rate, the volume of the balloon expands. But what happens to the radius? Because a sphere grows wider in all directions, its radius does not expand at a constant rate; it grows rapidly at first when the balloon is tiny, and then slows down as the balloon becomes large.

Even though volume (V) and radius (r) are different physical quantities, they are locked together by the geometric formula $V = \frac{4}{3}\pi r^3$. Because both variables are secretly functions of time (t), we can differentiate their geometric relationship implicitly with respect to t . By applying the Chain Rule, we can relate their rates of change ($\frac{dV}{dt}$ and $\frac{dr}{dt}$). This allows us to use the known rate of one variable to instantly calculate the hidden rate of the other!

Proposition 3.6: The Related Rates Method

Let two variables, x and y , be related by an equation $F(x, y) = 0$, where both $x(t)$ and $y(t)$ are implicit, differentiable functions of time t .

To calculate a hidden rate of change with respect to time:

1. Write down the geometric or physical formula that links the variables (e.g., Pythagorean theorem, volume/area formulas, or trigonometric ratios).
2. Differentiate both sides of the equation implicitly with respect to time t by applying the Chain Rule:

$$\frac{d}{dt}(g(x)) = g'(x) \cdot \frac{dx}{dt}$$

3. Substitute all known values and rates of change into the resulting rate equation ***only after*** completing the differentiation.
4. Solve algebraically for the single remaining unknown rate.

Student Proof Demonstration

Use this space to demonstrate the proof of the theorem above. Start from basic algebraic identities.

Practical Example 3.6

Air is being pumped into a spherical balloon at a constant rate of $100 \text{ cm}^3/\text{s}$. Find the rate at which the radius of the balloon is increasing at the exact instant the diameter is 50 cm.

Exercise 3.6.1 (Expanding circular oil slick)**Problem Statement:**

Oil spills from a ruptured tanker and spreads in a circular pattern. The area of the oil slick is expanding at a constant rate of $6\pi \text{ m}^2/\text{hr}$. Find the rate at which the radius of the circle is growing at the exact instant the radius is 3 meters.

Exercise 3.6.2 (The sliding ladder problem)**Problem Statement:**

A 5-meter long ladder leans against a vertical wall. The base of the ladder is pulled away from the wall at a constant rate of 1.5 m/s . Find the rate at which the top of the ladder is sliding down the wall at the exact instant the base is 3 meters away from the wall.

Exercise 3.6.3 (Filling a conical water tank)**Problem Statement:**

A water tank has the shape of an inverted circular cone (pointing downwards) with a top radius of 2 meters and a total height of 6 meters. Water is being poured into the tank at a constant rate of $0.2 \text{ m}^3/\text{min}$.

1. Use similar triangles to express the radius r of the water surface in terms of its height h .
 2. Express the volume of the water in the tank entirely in terms of h .
 3. Find the rate at which the water level is rising at the exact instant the water is 3 meters deep.
-

Exercise 3.6.4 (HKDSE M1-style angular rates of tracking radar)**Problem Statement:**

A spacecraft is launched vertically from a pad. A radar tracking station is positioned on the ground 3 km away from the launchpad. At a certain instant, the spacecraft is at an altitude of 4 km and is rising vertically at a speed of 0.5 km/s .

1. Write down the trigonometric relation linking the radar's angle of elevation θ with the spacecraft's altitude y .
 2. Find the rate at which the radar antenna must rotate (i.e., $\frac{d\theta}{dt}$ in radians per second) at this exact instant.
-

Exercise 3.6.5 (The general adiabatic gas expansion law)**Problem Statement:**

In thermodynamics, an ideal gas undergoing adiabatic expansion satisfies the law:

$$PV^\gamma = C$$

where P is pressure, V is volume, and γ and C are constant physical parameters. By differentiating this equation implicitly with respect to time t , prove that the rates of change of pressure and volume are related by the formula:

$$\frac{dP}{dt} = -\gamma \left(\frac{P}{V} \right) \frac{dV}{dt}$$

Exercise 3.6.6 (The infinite speed anomaly at the wall)**Problem Statement:**

Consider again the sliding ladder problem from Level 2, where $x^2 + y^2 = L^2$ and the base is pulled out at a constant rate $\frac{dx}{dt} = v$.

1. Solve for the downward rate of change of the top of the ladder, $-\frac{dy}{dt}$, in terms of x , y , and v .
 2. Evaluate the limit of this downward rate as the ladder gets incredibly close to hitting the floor (i.e., as $y \rightarrow 0$ and $x \rightarrow L$).
 3. Explain why the resulting infinite speed is physically impossible, and discuss the mechanical breakdown that occurs in real-life physics before the ladder hits the ground.
-

Exercise 3.6.7 (The multi-vessel drainage problem)**Problem Statement:**

A cylindrical container of radius R is draining water through a small circular hole of radius r_0 in its bottom. According to **Torricelli's Law**, the velocity of the exiting water is $v = \sqrt{2gh}$, where h is the current depth of the water and $g \approx 9.81 \text{ m/s}^2$ is the acceleration due to gravity.

1. Use the conservation of volume flow rate ($A_{\text{cylinder}} \frac{dh}{dt} = -A_{\text{hole}} v_{\text{exit}}$) to show that the water level changes according to the differential equation:

$$\frac{dh}{dt} = - \left(\frac{r_0}{R} \right)^2 \sqrt{2gh}$$

2. Now, suppose the cylindrical container is draining into a second, identical empty cylinder positioned directly below it. Let $h_1(t)$ be the water height in the top cylinder, and $h_2(t)$ be the height in the bottom cylinder.
 3. Set up a system of equations relating $\frac{dh_1}{dt}$ and $\frac{dh_2}{dt}$.
 4. Show that at any instant, the total sum of the heights is constant, and find the rate of change of $h_2(t)$ in terms of the current height $h_1(t)$.
-

3.7 Monotonicity (Increasing and Decreasing Functions)

In Sections 3.5 and 3.6, we calculated specific rates of change—analyzing how rapidly physical quantities shift over time. Now, we turn our analytical lens back to the pure geometry of functions. We want to understand the overall trends of a curve across its entire domain: where is it climbing upward, where is it sliding downward, and where does it temporarily flatline?

This property of climbing or falling is called **monotonicity**.

If you think of walking along a mountain path from left to right, some intervals force you to climb uphill (where your altitude increases), while other intervals lead you downhill (where your altitude decreases).

In calculus, we do not need to plot thousands of points to map these hills and valleys. The first derivative $f'(x)$ tells us the exact slope of the tangent line at any point.

- If $f'(x) > 0$, the tangent line points upward, meaning the function is **increasing**.
- If $f'(x) < 0$, the tangent line points downward, meaning the function is **decreasing**.

By systematically analyzing the sign of $f'(x)$, we can partition any continuous curve into clear intervals of growth and decay.

Theorem 3.7: Monotonicity and the First Derivative

Let $f(x)$ be a continuous function on a closed interval $[a, b]$ and differentiable on the open interval (a, b) .

1. **Strictly Increasing:** If $f'(x) > 0$ for all $x \in (a, b)$, then $f(x)$ is strictly increasing on $[a, b]$. Geometrically, as you move left to right, the curve strictly climbs upward:

$$x_1 < x_2 \implies f(x_1) < f(x_2)$$

2. **Strictly Decreasing:** If $f'(x) < 0$ for all $x \in (a, b)$, then $f(x)$ is strictly decreasing on $[a, b]$. Geometrically, as you move left to right, the curve strictly slides downward:

$$x_1 < x_2 \implies f(x_1) > f(x_2)$$

Student Proof Demonstration

Use this space to demonstrate the proof of the theorem above. Start from basic algebraic identities.

Practical Example 3.7

Find the exact intervals on which the cubic function $f(x) = 2x^3 - 3x^2 - 12x + 5$ is strictly increasing and the intervals on which it is strictly decreasing.

Exercise 3.7.1 (Standard polynomial monotonicity)**Problem Statement:**

Find the exact intervals on which the quadratic function $f(x) = 3x^2 - 12x + 7$ is:

1. Strictly increasing.
 2. Strictly decreasing.
-

Exercise 3.7.2 (Monotonicity of mixed exponential curves)**Problem Statement:**

Consider the function $g(x) = xe^{-x}$ for all real numbers x .

1. Find the derivative $g'(x)$ using the Product Rule.
 2. Determine the intervals of monotonicity for $g(x)$.
-

Exercise 3.7.3 (Monotonicity of rational curves with asymptotes)**Problem Statement:**

Consider the rational function $h(x) = x + \frac{9}{x}$ defined for $x \neq 0$. Find the intervals on which $h(x)$ is strictly increasing, and the intervals on which it is strictly decreasing. Be careful to exclude the vertical asymptote from your intervals.

Exercise 3.7.4 (HKDSE M2-style parameter bounds for constant growth)**Problem Statement:**

A cubic function is defined by $f(x) = \frac{1}{3}x^3 + kx^2 + 9x - 2$, where k is a real constant.

1. Find $f'(x)$.
 2. Find the range of values of k such that $f(x)$ is strictly increasing for all real numbers x . *Hint: For $f'(x)$ to be strictly positive everywhere, its corresponding quadratic discriminant Δ must satisfy a specific inequality.*
-

Exercise 3.7.5 (Proving Monotonicity via the Mean Value Theorem)**Problem Statement:**

The ****Mean Value Theorem (MVT)**** states that if $f(x)$ is continuous on $[a, b]$ and differentiable on (a, b) , then there exists some $c \in (a, b)$ such that:

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Use the MVT to prove Theorem 3.7 (Part 1): that if $f'(x) > 0$ for all $x \in (a, b)$, then for any $x_1, x_2 \in [a, b]$ with $x_1 < x_2$, it must hold that $f(x_1) < f(x_2)$.

Exercise 3.7.6 (The zero-derivative point exception)**Problem Statement:**

Consider the cubic function $f(x) = x^3$.

1. Calculate $f'(x)$ and evaluate its value at $x = 0$.
 2. Explain why $f(x) = x^3$ is still considered strictly increasing on the ***entire*** real line $\mathbb{R}$, despite having a slope of exactly 0 at the origin.
 3. Discuss the difference between $f'(x) > 0$ (a sufficient condition for growth) and the formal algebraic definition of strictly increasing.
-

Exercise 3.7.7 (Monotonicity of complex transcendental functions)**Problem Statement:**

We wish to study the monotonicity of the function $f(x) = \frac{\ln(x)}{x}$ defined on the domain $x > 0$.

1. Differentiate the function using the Quotient Rule.
2. Determine the exact intervals of monotonicity for this function.
3. Use your results to prove which of the following two transcendental numbers is larger without evaluating them on a calculator:

$$e^\pi \quad \text{versus} \quad \pi^e$$

Hint: Compare the values of $f(e)$ and $f(\pi)$ using the monotonicity of $f(x)$ on the interval $[e, \infty)$.

3.8 Stationary points and local extrema

In Section 3.7, we learned how to map out the rising and falling intervals of a curve using the sign of the first derivative. But what happens at the exact transition points where the curve stops climbing and begins to slide downward, or vice versa?

At these boundary peaks and valleys, the curve must momentarily flatten out entirely. This means the slope of the tangent line drops to exactly 0. We call these resting points **stationary points**.

Stationary points are of paramount importance because they are the primary candidates for **local extrema**—the relative high points (local maxima) and relative low points (local minima) of a function.

To determine whether a stationary point is a peak, a valley, or simply a temporary pause on a slope, we use the **First Derivative Test**. By observing how the sign of $f'(x)$ behaves immediately before and after the stationary point, we can decipher the local shape of the curve with absolute mathematical certainty.

Theorem 3.8: Stationary Points and the First Derivative Test

Let $f(x)$ be a continuous function. A point $(c, f(c))$ is a **stationary point** of $f(x)$ if:

$$f'(c) = 0$$

To classify a stationary point at $x = c$, analyze the sign of $f'(x)$ on either side of c :

1. **Local Maximum:** If $f'(x)$ changes from **positive** to **negative** as x increases through c , then $(c, f(c))$ is a local maximum. (The curve climbs up, flattens, then slides down: $\nearrow \rightarrow \searrow$).
2. **Local Minimum:** If $f'(x)$ changes from **negative** to **positive** as x increases through c , then $(c, f(c))$ is a local minimum. (The curve slides down, flattens, then climbs up: $\searrow \rightarrow \nearrow$).
3. **Stationary Point of Inflexion:** If $f'(x)$ does not change sign (remaining positive on both sides or negative on both sides), then $(c, f(c))$ is a stationary point of inflexion. ($\nearrow \rightarrow \nearrow$ or $\searrow \rightarrow \searrow$).

Student Proof Demonstration

Use this space to demonstrate the proof of the theorem above. Start from basic algebraic identities.

Practical Example 3.8

Locate all stationary points of the function $f(x) = x^3 - 3x^2 - 9x + 15$, and classify each of them using the First Derivative Test.

Exercise 3.8.1 (Standard polynomial turning points)**Problem Statement:**

Consider the quadratic curve $y = -2x^2 + 8x + 3$.

1. Find the coordinates of the unique stationary point on this curve.
 2. Use the First Derivative Test to prove whether this point is a local maximum or a local minimum.
-

Exercise 3.8.2 (Stationary points of rational functions)**Problem Statement:**

Let $f(x) = x + \frac{4}{x}$ be defined for $x \neq 0$. Find and classify all stationary points of $f(x)$ using the First Derivative Test.

Exercise 3.8.3 (Locating stationary points of exponential products)**Problem Statement:**

Consider the product function $g(x) = x^2e^x$.

1. Find $g'(x)$ and factor out the common terms.
 2. Find the exact coordinates of all stationary points.
 3. Classify each stationary point using the First Derivative Test.
-

Exercise 3.8.4 (HKDSE M1-style parametric turning points)**Problem Statement:**

Consider the curve $y = \frac{\ln(x)}{x^2}$ for $x > 0$.

1. Show that $\frac{dy}{dx} = \frac{1-2\ln(x)}{x^3}$.
 2. Find the exact coordinates of the stationary point on this curve.
 3. Construct a sign-change table to determine the nature of this stationary point.
-

Exercise 3.8.5 (Stationary points of cubic polynomials)**Problem Statement:**

A general cubic polynomial is given by $f(x) = ax^3 + bx^2 + cx + d$, where $a > 0$.

1. Write down the quadratic equation for its derivative $f'(x) = 0$.
 2. Prove that if $b^2 - 3ac < 0$, the cubic curve has ****no**** stationary points anywhere on the real line.
 3. Prove that if $b^2 - 3ac > 0$, the curve has exactly two distinct stationary points: one local maximum and one local minimum.
-

Exercise 3.8.6 (The constant slope anomaly)**Problem Statement:**

Consider the linear function $y = 3x - 5$.

1. Attempt to find any stationary points for this function.
 2. Explain why a linear function can never have a local extremum.
 3. If a student writes that the maximum of $y = 3x - 5$ on the interval $[1, 5]$ is at the stationary point, explain the logical failure and identify the actual location of the maximum.
-

Exercise 3.8.7 (Stationary points of variable-base functions)**Problem Statement:**

Let $f(x) = x^x$ be defined for $x > 0$.

1. Recall from Section 2.14 that the derivative is $f'(x) = x^x(1 + \ln(x))$.
 2. Show that the curve has exactly one stationary point, and find its exact coordinates.
 3. Use the First Derivative Test to prove that this stationary point is a local minimum.
 4. Discuss the global behavior of x^x as $x \rightarrow 0^+$ and $x \rightarrow \infty$ to explain why this local minimum is also the absolute lowest point of the entire curve.
-

3.9 Stationary points: Second derivative test

In Section 3.8, we mastered the First Derivative Test. That test is incredibly robust and intuitive, but it relies on a major assumption: we must be able to easily select and test numerical values in the open intervals surrounding our stationary points. This works perfectly when our function is completely known and numerically simple.

But what happens when we are dealing with complex parametric functions, symbolic proofs, or "pointwise unknown" cases where we only have coordinate data at a single, isolated point?

In these advanced scenarios, constructing a sign-change interval chart is algebraically impossible.

To bypass this bottleneck, we use the ****Second Derivative Test****. Instead of testing a neighborhood of points *around* the critical value $x = c$, we look directly *at* the single point $x = c$ and evaluate its second derivative, $f''(c)$. Since the second derivative measures the acceleration or bending of the curve (concavity), it can instantly determine whether a flat spot is a peak or a valley. If the curve is bending downward like a dome ($\cap$) at the stationary point, it must be a peak. If it is bending upward like a cup ($\cup$), it must be a valley.

Theorem 3.9: The Second Derivative Test

Let $f(x)$ be a function such that $f'(c) = 0$ (meaning $x = c$ is a stationary point), and let the second derivative $f''(x)$ be continuous on an open interval containing c .

1. **Local Maximum (Concave Down):** If $f''(c) < 0$, then $f(x)$ has a local maximum at $x = c$.
2. **Local Minimum (Concave Up):** If $f''(c) > 0$, then $f(x)$ has a local minimum at $x = c$.
3. **Test Inconclusive:** If $f''(c) = 0$, the test fails completely. The stationary point could be a local maximum, a local minimum, or a stationary point of inflexion. (*In this event, you must fall back on the First Derivative Test.*)

Student Proof Demonstration

Use this space to demonstrate the proof of the theorem above. Start from basic algebraic identities.

Practical Example 3.9

A parametric curve is defined by $f(x) = x^3 - 3kx^2 + 5$, where $k > 0$ is a positive constant parameter. Locate the stationary points of the curve, and use the Second Derivative Test to classify their nature symbolically.

Exercise 3.9.1 (Standard polynomial verification)**Problem Statement:**

Consider the function $f(x) = 2x^3 - 9x^2 + 12x - 3$.

1. Find the stationary points of $f(x)$.
 2. Classify their nature using the Second Derivative Test.
-

Exercise 3.9.2 (Classification of trigonometric turning points)**Problem Statement:**

Consider the curve $y = \sin(2x)$ over the interval $(0, \pi)$.

1. Use the first derivative to locate the stationary coordinates on this open interval.
 2. Apply the Second Derivative Test to classify each coordinate as a local maximum or a local minimum.
-

Exercise 3.9.3 (Symbolic classification of a general quadratic)**Problem Statement:**

Let $f(x) = ax^2 + bx + c$ be a general quadratic function, where $a \neq 0$.

1. Find the single critical point of this function.
 2. Evaluate $f''(x)$.
 3. Use the Second Derivative Test to prove why the sign of a alone determines whether the vertex of a parabola is a maximum or a minimum.
-

Exercise 3.9.4 (Parametric classification with exponential components)**Problem Statement:**

A family of curves is defined by $y = x^2e^{-kx}$, where $k > 0$ is a positive constant parameter.

1. Find $\frac{dy}{dx}$ and show that the stationary points occur at $x = 0$ and $x = \frac{2}{k}$.
 2. Find $\frac{d^2y}{dx^2}$.
 3. Apply the Second Derivative Test to classify both stationary points symbolically.
-

Exercise 3.9.5 (Pointwise unknown data analysis)**Problem Statement:**

An experimental aircraft's flight computer logs continuous positional data. At a key flight milestone, the sensor suite records the following isolated pointwise telemetry data:

$$f(c) = 1500, \quad f'(c) = 0, \quad f''(c) = -9.8$$

where $f(t)$ represents height in meters.

1. Explain why it is impossible to construct a first-derivative sign table for this flight milestone.
 2. State whether the aircraft reached a local peak, a local valley, or a point of transition at this milestone. Justify your answer mathematically.
-

Exercise 3.9.6 (The inconclusive second derivative breakdown)**Problem Statement:**

Consider the three basic power functions:

$$f(x) = x^4, \quad g(x) = -x^4, \quad h(x) = x^3$$

1. Show that all three functions have a stationary point at the origin $x = 0$, and show that the second derivative evaluates to 0 at the origin for all three curves.
 2. Describe the actual physical/geometric nature of the origin for each of the three curves.
 3. Discuss how this demonstrates that the condition $f''(c) = 0$ provides zero information about the classification of a stationary point, and explain why the First Derivative Test is mathematically superior in failure scenarios.
-

Exercise 3.9.7 (The Hessian Matrix and the Multivariable Extension)**Problem Statement:**

In university-level multivariable calculus, the Second Derivative Test is extended to functions of two variables, $f(x, y)$, using a matrix of second-order partial derivatives called the **Hessian Matrix**:

$$H = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix}$$

To classify a critical point (a, b) where $f_x(a, b) = 0$ and $f_y(a, b) = 0$:

- We calculate the determinant $D = \det(H) = f_{xx}f_{yy} - (f_{xy})^2$.
 - If $D > 0$ and $f_{xx} > 0$, the point is a local minimum.
 - If $D > 0$ and $f_{xx} < 0$, the point is a local maximum.
 - If $D < 0$, the point is a **saddle point** (concave up in one direction, concave down in another).
1. Let $f(x, y) = x^2 - y^2$ represent a hyperbolic paraboloid (a saddle shape). Show that $(0, 0)$ is a critical point.
 2. Construct the Hessian Matrix for $f(x, y) = x^2 - y^2$ at the origin.
 3. Calculate the determinant D , and use the multivariable second derivative test to classify the origin. Explain how the geometric structure of a saddle point links to your result.
-

3.10 Concavity and points of inflection

In Section 3.9, we saw how the sign of the second derivative at a single point ($f''(c)$) can act as an instant shortcut for classifying stationary points. Now, we expand our view to analyze the global bending of a curve across its entire domain.

If you drive a car along a winding road, you are constantly rotating the steering wheel. Sometimes you hold the wheel to the left, tracing out a leftward curve; other times you hold it to the right, tracing a rightward curve. The exact moment you transition from steering left to steering right—where your steering wheel passes momentarily through the perfectly straight, neutral center—is the physical equivalent of a **point of inflection**.

In calculus, this steering direction is called **concavity**.

- When a curve is "hollowed out" upward like a bowl, we say it is **concave up** ($f''(x) > 0$).
- When a curve is arched downward like an umbrella, we say it is **concave down** ($f''(x) < 0$).

A point of inflection is the precise coordinate where the curve transitions from one concavity to another. Finding these points allows us to pinpoint the exact structural pivots of a function's graph.

Definition 3.10: Concavity and Points of Inflection

Let $f(x)$ be a twice-differentiable function on an interval.

1. **Concave Up ($\cup$):** The graph of $f(x)$ is concave up on an interval if $f''(x) > 0$ for all x in that interval. Geometrically, the tangent lines lie *below* the curve.
2. **Concave Down ($\cap$):** The graph of $f(x)$ is concave down on an interval if $f''(x) < 0$ for all x in that interval. Geometrically, the tangent lines lie *above* the curve.
3. **Point of Inflection:** A point $(c, f(c))$ on the curve is a **point of inflection** if:
 - $f(x)$ is continuous at $x = c$.
 - The second derivative $f''(c) = 0$ (or is undefined).
 - The concavity of the curve **changes sign** (from positive to negative, or vice versa) as x passes through c .

Student Proof Demonstration

Use this space to demonstrate the proof of the theorem above. Start from basic algebraic identities.

Practical Example 3.10

Find the intervals of concavity and locate all points of inflection for the cubic curve $f(x) = x^3 - 6x^2 + 9x + 2$.

Exercise 3.10.1 (Standard polynomial concavity mapping)**Problem Statement:**

Consider the quartic curve $y = x^4 - 6x^2 + 8$.

1. Find the second derivative $\frac{d^2y}{dx^2}$.
 2. Determine the intervals of concavity.
 3. Find the coordinates of all points of inflection.
-

Exercise 3.10.2 (Concavity of exponential bells)**Problem Statement:**

Consider the normal distribution curve $f(x) = e^{-x^2}$.

1. Show that $f''(x) = 2(2x^2 - 1)e^{-x^2}$.
 2. Find the exact coordinates of the points of inflection on this curve.
-

Exercise 3.10.3 (Logarithmic concavity boundaries)**Problem Statement:**

Consider the function $g(x) = x^2 \ln(x)$ defined for $x > 0$. Find the interval on which the curve is concave up, and locate any points of inflection.

Exercise 3.10.4 (HKDSE-style fractional curve inflection analysis)**Problem Statement:**

Consider the curve $y = \frac{x}{x^2+3}$.

1. Show that the second derivative is given by $\frac{d^2y}{dx^2} = \frac{2x(x^2-9)}{(x^2+3)^3}$.
 2. Find all points of inflection on the curve.
 3. Determine the intervals of concavity.
-

Exercise 3.10.5 (The point-of-inflection symmetry of cubics)**Problem Statement:**

Let $f(x) = ax^3 + bx^2 + cx + d$ be a general cubic polynomial (where $a \neq 0$).

1. Show that $f(x)$ has exactly one point of inflection, and prove that its x -coordinate is always $x_I = -\frac{b}{3a}$.
 2. Recall that the stationary points x_1 and x_2 of a cubic satisfy $3ax^2 + 2bx + c = 0$. Use the midpoint formula to prove that if a cubic has two stationary points, the point of inflection lies exactly halfway between them.
-

Exercise 3.10.6 (The flat-spot inflection trap)**Problem Statement:**

A student is asked to find the points of inflection for $f(x) = x^4$ and writes:

$$f''(x) = 12x^2 \implies f''(0) = 0 \implies (0, 0) \text{ is a point of inflection}$$

1. Evaluate the sign of $f''(x)$ for $x < 0$ and $x > 0$.
 2. Explain the student's logical error. Why does the curve $y = x^4$ have no points of inflection despite having $f''(x) = 0$ at the origin?
 3. What does this highlight about the difference between a necessary condition ($f''(c) = 0$) and a sufficient condition (sign change) for inflection points?
-

Exercise 3.10.7 (The point of diminishing returns in logistics models)**Problem Statement:**

In economics, biology, and epidemiology, growth is often modeled using the **Logistic Function**:

$$P(t) = \frac{L}{1 + e^{-k(t-t_0)}}$$

where L is the carrying capacity (maximum population), $k > 0$ is the growth rate, and t_0 is the time of maximum growth.

1. Show that the second derivative of $P(t)$ can be factored into the form:

$$P''(t) = k^2 P(t) \left(1 - \frac{2P(t)}{L} \right)$$

2. Find the point of inflection by setting $P''(t) = 0$ (since $P(t) > 0$ always). Show that this occurs exactly when the population reaches half its carrying capacity ($P(t) = \frac{L}{2}$).
 3. Explain why this point of inflection is called the "point of diminishing returns" in economics. How does the concavity transition from $t < t_0$ to $t > t_0$ explain a shift from accelerating growth to decelerating growth?
-

3.11 Curve sketching

Up to this point in Section 3, we have developed a suite of highly precise diagnostic tools. We can find intercepts, locate where a curve climbs or falls (Section 3.7), identify peaks and valleys (Sections 3.8 and 3.9), and map out exactly how and where a curve bends (Section 3.10).

Now, we bring all of these isolated pieces together to achieve the ultimate goal of differential calculus: **Curve Sketching**.

Before computers and graphing calculators existed, mathematicians relied entirely on calculus to visualize complex mathematical relationships. By combining our derivative tests with the study of **asymptotes** (lines that a curve approaches infinitely closely but never quite touches), we can sketch an exceptionally accurate portrait of any continuous function. This process is not about plotting hundreds of individual coordinates, but rather about capturing the definitive structural skeleton of a curve.

Theorem 3.11: Systematic Strategy for Curve Sketching

To sketch the graph of a function $y = f(x)$ systematically, evaluate the following:

1. **Domain and Intercepts:** Identify the allowed inputs. Find the y -intercept (set $x = 0$) and any x -intercepts (solve $f(x) = 0$).
2. **Asymptotes:**
 - **Vertical:** Occur at $x = c$ where $\lim_{x \rightarrow c^\pm} f(x) = \pm\infty$ (usually where the denominator is 0).
 - **Horizontal:** Occur at $y = L$ where $\lim_{x \rightarrow \pm\infty} f(x) = L$.
 - **Oblique (Slanted):** Occur for rational functions $\frac{P(x)}{Q(x)}$ where the degree of $P(x)$ is exactly one higher than $Q(x)$. Find $y = mx + c$ using polynomial long division.
3. **Monotonicity and Local Extrema:** Find $f'(x)$. Set $f'(x) = 0$ to locate stationary points. Use a sign table to identify intervals of growth/decay and classify local maxima/minima.
4. **Concavity and Inflection Points:** Find $f''(x)$. Set $f''(x) = 0$ to identify candidate transition boundaries. Chart the signs to map concavity and locate points of inflection.

Student Proof Demonstration

Use this space to demonstrate the proof of the theorem above. Start from basic algebraic identities.

Practical Example 3.11

Analyze and sketch the rational function $f(x) = \frac{x^2-3}{x-2}$ for $x \neq 2$.

Exercise 3.11.1 (Sketching basic rational functions)**Problem Statement:**

Consider the rational function $f(x) = \frac{x+1}{x-1}$ for $x \neq 1$.

1. Find all horizontal and vertical asymptotes.
 2. Locate any coordinate intercepts.
 3. Show that the curve has no stationary points.
 4. Sketch the complete curve.
-

Exercise 3.11.2 (Curve sketching with transcendental elements)**Problem Statement:**

Consider the exponential-polynomial curve $g(x) = xe^{-x}$.

1. Evaluate the horizontal limit $\lim_{x \rightarrow \infty} xe^{-x}$ and identify the horizontal asymptote.
 2. Find the coordinates of the unique local maximum and the unique point of inflection.
 3. Sketch the curve over the domain $x \geq -1$.
-

Exercise 3.11.3 (Locating oblique asymptotes and graphing)**Problem Statement:**

Consider the rational function:

$$h(x) = \frac{x^2 - x - 2}{x} \quad (x \neq 0)$$

1. Find the vertical asymptote and the oblique asymptote.
 2. Find all turning points and analyze their nature.
 3. Sketch the curve, showing all key structural features.
-

Exercise 3.11.4 (HKDSE M2-style complete curve analysis)**Problem Statement:**

Let $f(x) = \frac{x^2+1}{x^2-4}$ for $x \neq \pm 2$.

1. State the equations of all vertical and horizontal asymptotes.
 2. Locate all stationary points and point of inflections, verifying their classifications.
 3. Build a comprehensive structural summary table.
 4. Sketch the curve.
-

Exercise 3.11.5 (Classifying the shapes of general cubics)**Problem Statement:**

Let $f(x) = x^3 + px + q$ be a simplified cubic curve.

1. Show that the point of inflection is always located at $(0, q)$.
 2. Show that if $p \geq 0$, the curve is strictly increasing everywhere and has no local extrema.
 3. Prove that if $p < 0$, the curve has exactly one local maximum and one local minimum, sitting symmetrically on opposite sides of the point of inflection.
-

Exercise 3.11.6 (The missing asymptote trap)**Problem Statement:**

A student is analyzing the rational function $f(x) = \frac{x^2-4}{x-2}$ and writes:

Vertical asymptote at $x = 2$ since the denominator is zero.

1. Simplify the expression for $f(x)$ algebraically for $x \neq 2$.
 2. Evaluate the limit $\lim_{x \rightarrow 2} f(x)$. Does the curve approach infinity at $x = 2$?
 3. Explain the student's conceptual error. Describe the actual geometric behavior of the graph at $x = 2$, and discuss how to represent it on a sketch.
-

Exercise 3.11.7 (Graphing complex implicit curves: The Folium of Descartes)**Problem Statement:**

Consider the famous implicit algebraic curve known as the **Folium of Descartes**:

$$x^3 + y^3 - 3axy = 0 \quad (a > 0)$$

1. Convert this equation to polar coordinates by setting $x = r \cos(\theta)$ and $y = r \sin(\theta)$, and show that:

$$r(\theta) = \frac{3a \sin(\theta) \cos(\theta)}{\sin^3(\theta) + \cos^3(\theta)}$$

2. Show that the curve is symmetric with respect to the line $y = x$.
3. Prove that the curve has an oblique asymptote given by the equation:

$$y = -x - a$$

4. Use these structural markers to sketch the curve, highlighting the singular loop in the first quadrant.
-

3.12 Global extrema on a closed interval

In Sections 3.8 and 3.9, we developed two powerful diagnostic tests to classify stationary points as local maxima or local minima. These "local" peaks and valleys represent the highest and lowest points *in their immediate neighborhoods*. However, they do not necessarily tell us the absolute highest or lowest points over an entire bounded territory.

When we transition to finding the **global (or absolute) extrema** over a defined range, the rules of the game change.

Imagine hiking along a mountain trail that begins at a trailhead checkpoint A and ends at a checkpoint B . The absolute highest altitude you reach on your hike might be the peak of a hill you climb along the way (a local maximum). But what if the trail is a continuous upward slope? In that case, the highest altitude of your entire journey is reached at the very end of the trail: checkpoint B (the boundary endpoint).

In calculus, this fundamental truth is guaranteed by the **Extreme Value Theorem**. On any closed, bounded interval $[a, b]$, a continuous function is guaranteed to reach an absolute maximum and an absolute minimum. To find them, we do not just look at where the derivative is zero; we must run a direct "computational tournament" comparing all local turning points and the boundary endpoints!

Theorem 3.12: The Extreme Value Theorem and Closed Interval Method

Let $f(x)$ be a continuous function on a closed interval $[a, b]$.

To find the absolute maximum and absolute minimum values of $f(x)$ on $[a, b]$:

1. Find all critical points of $f(x)$ on the open interval (a, b) where $f'(x) = 0$ (or is undefined). Ignore any critical points that lie outside $[a, b]$.
2. Calculate the function values $f(c)$ for each critical point c in the interval.
3. Calculate the boundary endpoint values: $f(a)$ and $f(b)$.
4. Compare all evaluated values:
 - The **largest** of these values is the **global maximum** value of f on $[a, b]$.
 - The **smallest** of these values is the **global minimum** value of f on $[a, b]$.

Student Proof Demonstration

Use this space to demonstrate the proof of the theorem above. Start from basic algebraic identities.

Practical Example 3.12

Find the absolute maximum and absolute minimum values of the cubic function $f(x) = x^3 - 3x^2 + 2$ on the closed interval $[-1, 4]$.

Exercise 3.12.1 (Standard polynomial tournament)**Problem Statement:**

Find the absolute maximum and absolute minimum values of the quadratic function:

$$f(x) = 2x^2 - 8x + 5 \quad \text{on the closed interval } [0, 3]$$

Exercise 3.12.2 (Trigonometric boundary tournament)**Problem Statement:**

Find the absolute maximum and absolute minimum values of the function:

$$g(x) = x + 2\cos(x) \quad \text{on the closed interval } [0, \pi]$$

Recall that x is measured in radians. Keep your final comparative values exact.

Exercise 3.12.3 (Radical and fractional tournaments)**Problem Statement:**

Find the global maximum and global minimum values of the function:

$$h(x) = x\sqrt{4-x^2} \quad \text{on the closed interval } [-2, 2]$$

Exercise 3.12.4 (HKDSE M1-style exponential model boundaries)**Problem Statement:**

The concentration of a drug in a patient's bloodstream t hours after injection is modeled by:

$$C(t) = 15te^{-0.5t} \quad \text{for } 0 \leq t \leq 8$$

1. Find the exact time t at which the drug concentration is at its absolute maximum.
 2. Find the absolute minimum concentration of the drug over this 8-hour window.
-

Exercise 3.12.5 (The algebraic boundary threshold of quadratic systems)**Problem Statement:**

Let $f(x) = x^2 - 2kx$ be a quadratic function evaluated on the closed interval $[0, 2]$, where k is a positive parameter. Prove that:

1. If $0 < k < 1$, the absolute minimum of $f(x)$ on $[0, 2]$ occurs at the critical point $x = k$.
2. If $k \geq 1$, the absolute minimum of $f(x)$ on $[0, 2]$ occurs at the boundary endpoint $x = 2$.

Discuss how changing the parameter k sweeps the local minimum point out of the closed interval's boundaries.

Exercise 3.12.6 (The continuity requirement breakdown)**Problem Statement:**

Consider the piecewise defined function:

$$f(x) = \begin{cases} x^2 & \text{if } -1 \leq x < 2 \\ 1 & \text{if } x = 2 \end{cases}$$

1. Show that $f(x)$ is defined on the closed interval $[-1, 2]$.
 2. Attempt to find the absolute maximum value of $f(x)$ on this interval. Explain why the value 4 is approached but never actually reached.
 3. Why does this demonstrate that the **continuity** requirement in the Extreme Value Theorem is not just a pedantic formality, but a strict logical prerequisite?
-

Exercise 3.12.7 (The Extreme Value Theorem and non-bounded domains)**Problem Statement:**

The Extreme Value Theorem states that continuity on a *closed and bounded* interval $[a, b]$ guarantees absolute extrema. If the interval is not bounded (e.g., $[a, \infty)$), the theorem does not apply. However, we can often prove the existence of global extrema by evaluating limits at infinity. Let $f(x) = \frac{x^2}{x^3+2}$ be defined on the semi-infinite closed interval $[0, \infty)$.

1. Find all critical points of $f(x)$ on $(0, \infty)$.
 2. Evaluate the left-hand boundary endpoint $f(0)$.
 3. Evaluate the limit at the infinite boundary: $\lim_{x \rightarrow \infty} f(x)$.
 4. Use your results to prove that $f(x)$ must possess an absolute maximum value on $[0, \infty)$, and find its exact coordinates.
 5. Does $f(x)$ possess an absolute minimum on $[0, \infty)$? State its value and justify your reasoning.
-

3.13 Global extrema on an open interval

In Section 3.12, we mastered the Closed Interval Method, which guarantees the existence of absolute extrema under the protective umbrella of the Extreme Value Theorem. But what happens when our domain is an **open interval** (a, b) ?

Without the boundary endpoints safely locked in, the Extreme Value Theorem no longer applies. The "boundary tournament" is broken because we can get infinitely close to the edges a and b without ever actually touching them.

Imagine walking along a path that descends toward a sheer canyon wall. As you get closer to the edge, your altitude drops continuously. If the path is open-ended, there is no "lowest point" on the path because for any step you take, you can always take another tiny half-step closer to the edge to reach an even lower height.

To evaluate global extrema on an open interval, we must look beyond static coordinates. We must combine our local turning points with **boundary limits**. By taking the limits of the function as x approaches the open boundaries ($\lim_{x \rightarrow a^+} f(x)$ and $\lim_{x \rightarrow b^-} f(x)$), we can determine whether the curve escapes to infinity or flattens out, allowing us to safely declare whether a global peak or valley exists.

Theorem 3.13: The Open Interval Method

Let $f(x)$ be a continuous function on an open interval (a, b) , where a and b can be finite numbers or $\pm\infty$.

To find the global extrema of $f(x)$ on (a, b) :

1. Find all critical points of $f(x)$ on the open interval (a, b) where $f'(x) = 0$ (or is undefined).
2. Calculate the limits of the function at the open boundaries:

$$L_a = \lim_{x \rightarrow a^+} f(x) \quad \text{and} \quad L_b = \lim_{x \rightarrow b^-} f(x)$$

3. Compare the critical value heights $f(c)$ to the boundary limits L_a and L_b :
 - If $f(c) \geq L_a$ and $f(c) \geq L_b$ for some critical point c , then $f(c)$ is the **global maximum** value.
 - If $f(c) \leq L_a$ and $f(c) \leq L_b$ for some critical point c , then $f(c)$ is the **global minimum** value.
 - If the function limits go to ∞ or $-\infty$ at either boundary, then those corresponding global extrema **do not exist**.

Student Proof Demonstration

Use this space to demonstrate the proof of the theorem above. Start from basic algebraic identities.

Practical Example 3.13

Find the absolute maximum and absolute minimum values (if they exist) of the function $f(x) = x + \frac{4}{x}$ on the open interval $(0, \infty)$.

Exercise 3.13.1 (Rational curve on an infinite open domain)**Problem Statement:**

Find the absolute maximum and absolute minimum values (if they exist) of the rational function:

$$f(x) = \frac{x}{x^2 + 1} \quad \text{on the open interval } (-\infty, \infty)$$

Exercise 3.13.2 (Logarithmic open interval tournament)**Problem Statement:**

Consider the function $g(x) = \frac{\ln(x)}{x}$ defined on the open interval $(0, \infty)$.

1. Find the coordinates of the unique critical point.
 2. Evaluate the open boundary limits $\lim_{x \rightarrow 0^+} g(x)$ and $\lim_{x \rightarrow \infty} g(x)$.
 3. Find the absolute maximum and absolute minimum values of $g(x)$ on this domain (if they exist).
-

Exercise 3.13.3 (Minimizing sum of reciprocals)**Problem Statement:**

Let $h(x) = 2x + \frac{8}{x^2}$ be defined on the open interval $(0, \infty)$. Find the exact coordinates of the absolute minimum point of $h(x)$ on this domain. Show that no absolute maximum exists.

Exercise 3.13.4 (HKDSE M2-style parametric open boundaries)**Problem Statement:**

A curve is given by $f(x) = x^2 e^{-kx}$ on the open interval $(0, \infty)$, where $k > 0$ is a positive constant parameter.

1. Locate all critical points on this open interval.
 2. Find the open boundary limits: $\lim_{x \rightarrow 0^+} f(x)$ and $\lim_{x \rightarrow \infty} f(x)$.
 3. Prove that the curve achieves an absolute maximum value on this interval, and state its coordinates in terms of k .
-

Exercise 3.13.5 (The Single-Peak Global Theorem)**Problem Statement:**

Let $f(x)$ be a continuous function on an open interval (a, b) that possesses **exactly one** critical point $x = c$ which is a local minimum. Prove that $f(c)$ must be the absolute minimum of $f(x)$ on (a, b) . *Hint: Use the Mean Value Theorem or the intermediate sign behavior of $f'(x)$ to show that the function must strictly decrease to the left of c and strictly increase to the right of c .*

Exercise 3.13.6 (The missing ceiling trap on open intervals)**Problem Statement:**

A student is asked to find the absolute maximum of $y = 3x - 2$ on the finite open interval $(1, 4)$ and writes:

$$\lim_{x \rightarrow 4^-} (3x - 2) = 10 \implies \text{The absolute maximum is } 10$$

1. Explain the conceptual error in the student's statement. Does the curve ever actually achieve a height of 10 on the interval $(1, 4)$?
 2. If the interval is changed to the half-open (clopen) interval $(1, 4]$, discuss how the existence of the maximum changes.
 3. Why does this exercise highlight the necessity of distinguishing between a limit and an attained maximum value?
-

Exercise 3.13.7 (The Extreme Value Theorem extension for semi-continuous functions)**Problem Statement:**

In advanced mathematical analysis, the concept of **coercivity** allows us to prove the existence of global minimizers on unbounded open sets. A continuous function $f : \mathbb{R} \rightarrow \mathbb{R}$ is called **coercive** if:

$$\lim_{|x| \rightarrow \infty} f(x) = \infty$$

This guarantees that as you travel infinitely far in any direction, the function blows up to positive infinity.

1. Let $f(x) = x^4 - 4x$. Prove that $f(x)$ is coercive.
 2. Show that $f(x)$ has exactly one critical point on $(-\infty, \infty)$, and calculate its coordinates.
 3. Use the definition of coercivity to prove that $f(x)$ must possess an absolute minimum on $(-\infty, \infty)$, and state its exact coordinates.
 4. Does $f(x)$ possess a global maximum on $(-\infty, \infty)$? Explain your reasoning using limits.
-

3.14 Global extrema on a clopen interval

In Section 3.12, we evaluated global extrema on closed intervals where both boundaries are strictly included. In Section 3.13, we navigated open intervals where both boundaries are strictly excluded. Now, we confront the hybrid territory of ****clopen** (half-open, half-closed) intervals******, such as $[a, b)$ or $(a, b]$, as well as semi-infinite intervals like $[a, \infty)$.

These mixed domains frequently appear in real-world modeling. For instance, a factory's production timeline might start at a definitive initial time $t = 0$ (a closed endpoint $[0, \dots)$, but continue indefinitely into the future as $t \rightarrow \infty$ (an open end $\dots, \infty)$).

How do we run a global tournament on a domain that is half-gated and half-infinite?

We use a hybrid approach! We identify all interior critical points and evaluate their heights. Then, for the closed side of the interval, we calculate the exact coordinate height at the endpoint. For the open side, we evaluate the boundary limit. By pitting the critical heights, the solid endpoint height, and the open-ended limits against each other, we can safely locate the absolute peaks and valleys of any semi-bounded system.

Theorem 3.14: The Clopen Interval Method

Let $f(x)$ be a continuous function on a clopen interval, such as $[a, b)$ or $[a, \infty)$.

To find the global extrema of $f(x)$ on $[a, b)$:

1. Find all critical points c of $f(x)$ on the interior open interval (a, b) where $f'(x) = 0$ (or is undefined).
2. Evaluate the function value at the closed endpoint: $f(a)$.
3. Evaluate the limit of the function approaching the open boundary:

$$L = \lim_{x \rightarrow b^-} f(x) \quad (\text{or } \lim_{x \rightarrow \infty} f(x))$$

4. Compare the critical heights $f(c)$, the endpoint height $f(a)$, and the boundary limit L :
 - If a critical height or the endpoint $f(a)$ is larger than or equal to all other values and limits, it is the ****global maximum****.
 - If a critical height or the endpoint $f(a)$ is smaller than or equal to all other values and limits, it is the ****global minimum****.
 - If the boundary limit L is the largest (or smallest) value and is not matched by an attained coordinate, that corresponding global extremum ****does not exist****.

Student Proof Demonstration

Use this space to demonstrate the proof of the theorem above. Start from basic algebraic identities.

Practical Example 3.14

Find the absolute maximum and absolute minimum values (if they exist) of the function $f(x) = x^2e^{-x}$ on the semi-infinite clopen interval $[0, \infty)$.

Exercise 3.14.1 (Finite clopen interval tournament)**Problem Statement:**

Find the absolute maximum and absolute minimum values (if they exist) of the function:

$$f(x) = x^3 - 3x \quad \text{on the clopen interval } [-1, 2)$$

Exercise 3.14.2 (Transcendental semi-infinite clopen tournament)**Problem Statement:**

Consider the function $g(x) = (x-3)e^{-x}$ defined on the semi-infinite clopen interval $[0, \infty)$.

1. Locate all interior critical points.
 2. Evaluate the closed boundary height $g(0)$ and the open boundary limit $\lim_{x \rightarrow \infty} g(x)$.
 3. Find the absolute maximum and absolute minimum values of $g(x)$ on this domain.
-

Exercise 3.14.3 (Logarithmic clopen boundaries)**Problem Statement:**

Find the global maximum and global minimum values (if they exist) of the function:

$$h(x) = \ln(x) - x \quad \text{on the clopen interval } (0, 2]$$

Exercise 3.14.4 (HKDSE M1-style asset depreciation model)**Problem Statement:**

The market value $V(t)$ (in thousands of HKD) of a piece of industrial machinery over time t (in years) is modeled by:

$$V(t) = \frac{10t + 40}{t^2 + 4} \quad \text{for } t \in [0, \infty)$$

1. Find the value of the machinery at the moment of purchase ($t = 0$).
 2. Use the Clopen Interval Method to find the maximum value of the machinery and the exact year t this peak is achieved.
 3. Find the long-term steady-state limit of the value as $t \rightarrow \infty$.
 4. State the absolute minimum value of the machinery over its entire operational life-time.
-

Exercise 3.14.5 (The semi-infinite peak-valley threshold)**Problem Statement:**

Let $f(x) = xe^{-kx}$ be defined on the semi-infinite clopen domain $[0, \infty)$, where $k > 0$ is a positive constant parameter.

1. Prove that $f(x)$ always possesses a global maximum on this domain, and find its exact coordinates in terms of k .
 2. Prove that $f(x)$ always possesses a global minimum on this domain, and state its value.
-

Exercise 3.14.6 (The unachieved boundary trap on clopen domains)**Problem Statement:**

Consider the simple linear function $f(x) = 4 - 2x$ evaluated on the clopen interval $[0, 2)$.

1. Find the closed endpoint value $f(0)$.
 2. Evaluate the open boundary limit $\lim_{x \rightarrow 2^-} f(x)$.
 3. A student claims that the absolute minimum of the function on this interval is 0. Explain the mathematical error in this claim. Does a global minimum exist on this interval?
-

Exercise 3.14.7 (The economics of optimal harvesting on semi-infinite horizons)**Problem Statement:**

In resource economics, the net present value $J(T)$ of harvesting a forest timber stand at time $T \in [0, \infty)$ is modeled by the clopen objective function:

$$J(T) = (e^{\sqrt{T}} - c)e^{-rT}$$

where $c > 0$ is the harvesting cost and $r > 0$ is the continuous discount interest rate. Let $c = 2$ and $r = 0.1$.

1. Evaluate the initial value $J(0)$.
2. Find the limit $\lim_{T \rightarrow \infty} J(T)$. Explain the economic meaning of this limit.
3. Set up the transcendental equation for the critical points of $J(T)$ by setting $J'(T) = 0$. Show that any interior critical point T must satisfy the equation:

$$\frac{e^{\sqrt{T}}}{2\sqrt{T}} - 0.1e^{\sqrt{T}} + 0.2 = 0$$

4. Using the initial guess $T_0 = 16$, perform one iteration of Newton's Method (Section 3.4) to approximate the optimal harvesting year T_1 to 2 decimal places.
-

3.15 Applied optimization: Geometric problems

In Sections 3.12 through 3.14, we mastered the core mathematical machinery of finding global extrema across various interval types. Now, we are ready to deploy these techniques to solve real-world design, architecture, and manufacturing puzzles. We begin with **Section 3.15: Geometric Optimization**.

Imagine you are a packaging engineer tasked with designing a new cardboard shipping box. Your company wants the box to hold the largest possible volume of goods to maximize shipping efficiency, but the manufacturing plant has a strict limit on the total surface area of cardboard available for each box.

How do you find the exact dimensions that maximize volume under a fixed surface area constraint?

To tackle this, we translate the physical setup into two mathematical equations:

1. **The Constraint Equation:** A fixed geometric relationship that limits our system (e.g., surface area, perimeter, or raw material dimensions).
2. **The Objective Function:** The quantity we desperately want to optimize (e.g., maximizing volume or minimizing perimeter).

Because geometric shapes initially involve multiple variables (like length, width, and height), we use our constraint equation to substitute variables, reducing the objective function to a single-variable equation. We then run our global extrema tests to locate the perfect geometric configuration!

Proposition 3.15: The Optimization Modeling Workflow

To solve any applied geometric optimization problem systematically:

1. **Draw and Label:** Sketch the geometric scenario. Label all dimensions with variables (e.g., x, y, r, h).
2. **Identify the Target (Objective):** Write a formula for the quantity S to be maximized or minimized.
3. **Identify the Limit (Constraint):** Write an equation linking the variables to a constant physical limit.
4. **Reduce to One Variable:** Solve the constraint equation for one variable and substitute it into the objective function: $S(x)$.
5. **Establish the Physical Domain:** Determine the allowed range of x based on physical reality (e.g., lengths must be strictly positive: $x > 0$).
6. **Find the Global Extremum:** Set $S'(x) = 0$, solve for the critical points, and verify the absolute maximum/minimum using global extrema interval tests.

Student Proof Demonstration

Use this space to demonstrate the proof of the theorem above. Start from basic algebraic identities.

Practical Example 3.15

A farmer has 400 meters of fencing material. They want to enclose a rectangular pasture adjacent to a long, straight stone wall, meaning no fencing is needed along the wall. Find the dimensions of the pasture that maximize the enclosed area, and calculate this maximum area.

Exercise 3.15.1 (Maximizing the area of a closed pen)**Problem Statement:**

A rectangular pen is to be fenced off in an open field. The total length of the fencing material available is 120 meters.

1. Write down the constraint equation and the objective function.
 2. Find the dimensions of the rectangle that maximize the enclosed area.
 3. Show that for any fixed perimeter, the rectangle with the maximum area is always a perfect square.
-

Exercise 3.15.2 (The classic open-top box problem)**Problem Statement:**

A sheet of cardboard measuring 12 cm by 12 cm is used to make an open-top box by cutting equal square corners of side-length x and folding up the flaps.

1. Show that the volume $V(x)$ of the resulting box is given by $V(x) = x(12 - 2x)^2$.
 2. State the physical domain of x .
 3. Find the value of x that maximizes the volume, and state this maximum volume.
-

Exercise 3.15.3 (Minimizing material of a cylindrical can)**Problem Statement:**

A cylindrical soda can must be designed to hold a fixed volume of $250\pi \text{ cm}^3$ of liquid.

1. Express the height h of the cylinder in terms of its radius r .
 2. Show that the total surface area $A(r)$ of the cylinder is given by $A(r) = 2\pi r^2 + \frac{500\pi}{r}$.
 3. Use the Open Interval Method on $(0, \infty)$ to find the dimensions (radius r and height h) that minimize the total surface area of the can.
-

Exercise 3.15.4 (HKDSE M2-style window light optimization)**Problem Statement:**

A Norman window consists of a rectangle surmounted by a semicircle, as shown in the diagram. The total perimeter of the window is fixed at 6 meters.

1. Let x be the width of the rectangular base, and y be the height of the rectangle. Show that the perimeter is modeled by $x + 2y + \frac{\pi x}{2} = 6$.
 2. Show that the total area $A(x)$ of the window is given by $A(x) = 3x - \frac{x^2}{2} - \frac{\pi x^2}{8}$.
 3. Find the width x that maximizes the amount of light entering the window.
-

Exercise 3.15.5 (The optimal aspect ratio of bounded cylinders)**Problem Statement:**

Prove that for any closed cylinder of fixed volume V , the total surface area is minimized if and only if the height of the cylinder is exactly equal to its diameter ($h = 2r$).

Exercise 3.15.6 (The unbounded domain breakdown)**Problem Statement:**

A student is asked to find the dimensions of a rectangle with a fixed area of 36 m^2 that minimizes the perimeter. They write:

Constraint: $xy = 36 \implies y = \frac{36}{x}$. Objective: $P(x) = 2x + \frac{72}{x}$ on the domain $(0, \infty)$.

1. Find the critical point of $P(x)$ and calculate the minimum perimeter.
 2. Now, suppose the student is asked to find the dimensions that *maximize* the perimeter on $(0, \infty)$. Show that setting $P'(x) = 0$ fails to provide a maximum, and evaluate $\lim_{x \rightarrow 0^+} P(x)$ to explain why no maximum perimeter exists.
-

Exercise 3.15.7 (The Snell's Law of Light Refraction derivation)**Problem Statement:**

In physics, **Fermat's Principle of Least Time** states that light traveling between two points always takes the path that minimizes travel time. Suppose light travels from point A in a medium with speed v_1 to point B in a medium with speed v_2 , crossing a flat boundary interface.

[Image of light refraction Snell's law diagram]

1. Let A be at coordinates $(0, a)$ and B be at $(d, -b)$, and let the light cross the boundary at $(x, 0)$. Show that the total travel time $T(x)$ is given by:

$$T(x) = \frac{\sqrt{x^2 + a^2}}{v_1} + \frac{\sqrt{(d-x)^2 + b^2}}{v_2}$$

2. Show that differentiating $T(x)$ yields:

$$T'(x) = \frac{x}{v_1 \sqrt{x^2 + a^2}} - \frac{d-x}{v_2 \sqrt{(d-x)^2 + b^2}}$$

3. Let θ_1 be the angle of incidence, and θ_2 be the angle of refraction. Express $\sin(\theta_1)$ and $\sin(\theta_2)$ geometrically from your coordinate diagram, and prove that setting $T'(x) = 0$ yields **Snell's Law**:

$$\frac{\sin(\theta_1)}{v_1} = \frac{\sin(\theta_2)}{v_2}$$

3.16 Applied optimization: Scientific problems

In Section 3.15, we applied our optimization engine to pure geometric shapes—maximizing areas and minimizing material surfaces under structural boundaries. Now, we expand our scope to **Scientific and Applied Physics/Economics Problems**.

In the sciences, nature is constantly seeking to optimize itself. Light travels along paths that minimize transit time (Fermat's Principle), physical systems settle into configurations that minimize potential energy, and chemical reactions adjust to reach states of thermodynamic equilibrium. Similarly, in commercial operations, businesses constantly seek to maximize net profit or minimize operating costs under fluctuating market dynamics.

While geometric problems rely on spatial formulas, scientific optimization problems rely on **mathematical models** derived from physical, biological, or economic laws.

The mathematical process remains identical: we construct an objective function from our model, establish physical boundary constraints (such as non-negative time $t \geq 0$ or limited raw power), find where the rate of change is zero, and analyze the resulting critical thresholds.

Proposition 3.16: Modeling Scientific Systems

Scientific optimization models often feature specific variables and parameters:

1. **Physical Mechanics:** The objective function might model power, kinetic energy, or velocity. Boundaries are typically dictated by initial states ($t = 0$) or structural material capacities.
2. **Economics and Logistics:**
 - **Revenue:** $R(q) = q \cdot p(q)$, where $p(q)$ is the demand-price function.
 - **Profit:** $P(q) = R(q) - C(q)$, where $C(q)$ is total cost.
 - **Average Cost:** $AC(q) = \frac{C(q)}{q}$.
3. **Chemical/Biological Systems:** Models track concentration levels, bacterial growth rates, or enzyme reaction velocities (such as the Michaelis-Menten model) over bounded time horizons.

Student Proof Demonstration

Use this space to demonstrate the proof of the theorem above. Start from basic algebraic identities.

Practical Example 3.16

An electronics manufacturer determines that the total cost (in HKD) of producing q smart home hubs per day is $C(q) = 0.1q^2 + 20q + 1000$. Each unit can be sold for a price $p(q) = 100 - 0.3q$ HKD.

1. Find the daily profit function $P(q)$.
2. Determine the daily production level q that maximizes profit, and calculate this maximum profit.

Exercise 3.16.1 (Maximizing agricultural yield)**Problem Statement:**

An apple orchard operator finds that if they plant 40 trees per acre, each tree yields an average of 600 apples per year. For each additional tree planted per acre, the average yield per tree decreases by 10 apples due to crowding.

1. Let x be the number of additional trees planted per acre. Show that the total yield of apples per acre is modeled by $Y(x) = (40 + x)(600 - 10x)$.
 2. Find the number of trees per acre that maximizes the total annual apple harvest.
-

Exercise 3.16.2 (Minimizing average manufacturing cost)**Problem Statement:**

A factory's total daily operating cost (in USD) for producing q units of a product is modeled by the function $C(q) = 0.05q^3 - 4q^2 + 200q + 500$ for $q > 0$.

1. Write down the formula for the average cost per unit, $AC(q) = \frac{C(q)}{q}$.
 2. Find the production level q that minimizes the average cost per unit.
 3. Prove that at this minimum cost level, the average cost is exactly equal to the marginal cost ($AC(q) = C'(q)$).
-

Exercise 3.16.3 (The physics of projectile range optimization)**Problem Statement:**

A projectile is launched from ground level with an initial velocity v_0 at an angle θ to the horizontal. Neglecting air resistance, the horizontal range R of the projectile is given by:

$$R(\theta) = \frac{v_0^2 \sin(2\theta)}{g} \quad \text{for } \theta \in \left[0, \frac{\pi}{2}\right]$$

where g is the constant acceleration due to gravity.

1. Find the derivative $\frac{dR}{d\theta}$.
 2. Determine the launch angle θ (in degrees) that maximizes the horizontal range.
 3. Discuss how this physical maximum changes if a headwind is introduced, modeling the range as $R(\theta) = \sin(2\theta) - k \cos^2(\theta)$ for some small constant $k > 0$.
-

Exercise 3.16.4 (HKDSE M1-style biomedical concentration rates)**Problem Statement:**

To study the effectiveness of a new painkiller, clinical researchers monitor the concentration of active ingredients in a patient's liver tissue. The concentration $C(t)$ (in milligrams per liter) t hours after ingestion is modeled by:

$$C(t) = \frac{8t}{t^2 + 9} \quad \text{for } t \in [0, \infty)$$

1. Find $C'(t)$.
 2. Find the exact hour t at which the concentration reaches its peak, and state this maximum concentration.
 3. Find the rate of change of concentration at $t = 1$ and $t = 4$ hours. Explain the clinical significance of the sign change.
-

Exercise 3.16.5 (The optimal shipping speed and fuel consumption)**Problem Statement:**

A cargo ship's fuel consumption rate (in liters per hour) is modeled by the cubic function $F(v) = a + bv^3$, where v is the ship's speed in knots, a represents fuel consumed for auxiliary power (generators, lighting), and $b > 0$ is a hydrodynamic drag coefficient.

1. Let D be the total distance of a shipping route. Show that the total fuel consumed $E(v)$ for the entire journey is given by:

$$E(v) = D \left(\frac{a}{v} + bv^2 \right)$$

2. Prove that the total fuel consumed is minimized when the ship travels at an optimal speed of:

$$v_{\text{opt}} = \sqrt[3]{\frac{a}{2b}}$$

Note that this optimal speed is completely independent of the total distance D of the route!

Exercise 3.16.6 (The unrealistic boundary limit trap)**Problem Statement:**

In mathematical biology, the rate of an enzyme-catalyzed reaction is modeled using the Haldane-Briggs model:

$$V(s) = \frac{V_{\max}s}{K_m + s} \quad \text{for } s \geq 0$$

where s is substrate concentration, and $V_{\max}$ and K_m are positive constants.

1. Attempt to find a critical point of $V(s)$ by solving $V'(s) = 0$. What algebraic conflict arises?
2. Evaluate the limit $\lim_{s \rightarrow \infty} V(s)$.
3. Explain why setting $V'(s) = 0$ fails to yield a maximum reaction rate, and discuss how biological systems handle this asymptotic ceiling.

Exercise 3.16.7 (The physics of maximum power transfer in electrical circuits)**Problem Statement:**

In electrical engineering, ****Jacobi's Law**** (the Maximum Power Transfer Theorem) states that to obtain maximum external power from a source with a finite internal resistance, the resistance of the load must equal the resistance of the source.

Consider a DC circuit where a battery with constant electromotive force E (voltage) and internal resistance r is connected to an adjustable load resistor with resistance $R \geq 0$. The electrical power $P(R)$ delivered to the load resistor is given by:

$$P(R) = I^2 R \quad \text{where current } I = \frac{E}{R + r}$$

1. Show that the power delivered to the load can be modeled as:

$$P(R) = \frac{E^2 R}{(R + r)^2}$$

2. Treat E and r as positive constants. Differentiate $P(R)$ with respect to R using the Quotient Rule, and show that:

$$P'(R) = \frac{E^2(r - R)}{(R + r)^3}$$

3. Find the critical point of $P(R)$ on $[0, \infty)$ and prove that the power transfer is maximized when the load resistance matches the internal resistance ($R = r$).
4. Find the maximum power $P_{\max}$ that can be delivered to the load.

3.17 Applied optimization: Inequality problems

In Sections 3.15 and 3.16, we used optimization to find specific physical dimensions and ideal operating states. Now, we deploy our global extrema toolkit to solve a purely mathematical challenge: **proving inequalities**.

In advanced algebra and university-level analysis, you are frequently asked to prove that one mathematical function is always larger than or equal to another, such as $e^x \geq 1 + x$ or $\ln(x) \leq x - 1$. While these expressions may seem daunting to prove algebraically, calculus provides an elegant, bulletproof strategy.

Instead of trying to manipulate the inequality directly, we define a single auxiliary "difference function", $g(x) = \text{Left Hand Side} - \text{Right Hand Side}$.

If we want to prove that $f(x) \geq h(x)$, we simply show that their difference $g(x) = f(x) - h(x)$ has an absolute global minimum value of 0. If the absolute lowest point of the difference curve is 0, then the curve can never dip below 0, which mathematically guarantees that $f(x) \geq h(x)$ for every single point in the domain!

Proposition 3.17: Proving Inequalities via Extrema

To prove a mathematical inequality of the form $f(x) \geq h(x)$ on an interval I :

1. **Construct the Auxiliary Function:** Define a new continuous function:

$$g(x) = f(x) - h(x)$$

2. **Analyze the Target:** Our goal is to prove that $g(x) \geq 0$ for all $x \in I$.
3. **Find the Global Minimum:** Use your calculus toolkit to locate the absolute global minimum of $g(x)$ on the interval I .
4. **Evaluate the Boundary/Vertex:**
 - Let $x = c$ be the coordinate that minimizes $g(x)$.
 - If the minimum value $g(c) \geq 0$, then $g(x) \geq 0$ holds across the entire interval, completing the proof.

Student Proof Demonstration

Use this space to demonstrate the proof of the theorem above. Start from basic algebraic identities.

Practical Example 3.17

Prove that $e^x \geq 1 + x$ for all real numbers $x \in \mathbb{R}$.

Exercise 3.17.1 (The logarithmic fundamental inequality)**Problem Statement:**

Prove that $\ln(x) \leq x - 1$ for all positive real numbers $x > 0$.

Exercise 3.17.2 (Trigonometric bounds on acute intervals)**Problem Statement:**

Prove that $\sin(x) < x$ for all $x \in (0, \frac{\pi}{2}]$.

Exercise 3.17.3 (Power polynomial inequality bounds)**Problem Statement:**

Prove that $x^4 - 4x \geq -3$ for all real numbers $x \in \mathbb{R}$. Identify the exact value of x where equality holds.

Exercise 3.17.4 (HKDSE M2-style logarithmic inequality bounds)**Problem Statement:**

Let $f(x) = x \ln(x) - x + 1$ be defined on the interval $x \geq 1$.

1. Find $f'(x)$ and determine the monotonicity of $f(x)$ for $x \geq 1$.
2. Evaluate $f(1)$.
3. Use your results to prove the inequality:

$$x \ln(x) \geq x - 1 \quad \text{for all } x \geq 1$$

Exercise 3.17.5 (The general Bernoulli's Inequality for integer powers)**Problem Statement:**

Let $x > -1$ be a real number. Prove **Bernoulli's Inequality** for the quadratic case:

$$(1 + x)^2 \geq 1 + 2x$$

Then, generalize this by proving that $(1 + x)^n \geq 1 + nx$ for all integers $n \geq 1$ when $x \geq 0$.

Exercise 3.17.6 (The missing boundary restriction breakdown)**Problem Statement:**

A student attempts to prove that $\ln(x) \leq x - 2$ using calculus.

1. Define $g(x) = x - 2 - \ln(x)$. Find the critical point of $g(x)$ on $(0, \infty)$ and show that it is a global minimum.
 2. Evaluate the minimum value $g(c)$.
 3. Explain the mathematical conflict. Why does the student's attempt to prove the inequality fail? State the correct maximum value of the constant k such that $\ln(x) \leq x - k$ holds true.
-

Exercise 3.17.7 (The Arithmetic Mean-Geometric Mean (AM-GM) Inequality proof)**Problem Statement:**

The ****Arithmetic Mean-Geometric Mean (AM-GM) Inequality**** states that for any non-negative real numbers, the average sum is always greater than or equal to their n th root product. For two variables, this is:

$$\frac{x_1 + x_2}{2} \geq \sqrt{x_1 x_2}$$

We can use calculus to prove the general n -variable AM-GM inequality inductively, or prove the classic Jensen's Inequality for convex functions.

1. Let $f(x) = -\ln(x)$ defined on $(0, \infty)$. Find $f''(x)$ and prove that $f(x)$ is strictly concave up (convex) on its entire domain.
2. ****Jensen's Inequality**** states that for any convex function f and any points $x_1, x_2, \dots, x_n$:

$$f\left(\frac{x_1 + x_2 + \dots + x_n}{n}\right) \leq \frac{f(x_1) + f(x_2) + \dots + f(x_n)}{n}$$

Apply this theorem to $f(x) = -\ln(x)$, and show that:

$$-\ln\left(\frac{x_1 + x_2 + \dots + x_n}{n}\right) \leq -\frac{\ln(x_1) + \ln(x_2) + \dots + \ln(x_n)}{n}$$

3. Multiply by -1 , exponentiate both sides, and show that this completes the proof of the general n -variable AM-GM Inequality:

$$\frac{x_1 + x_2 + \dots + x_n}{n} \geq \sqrt[n]{x_1 x_2 \dots x_n}$$

3.18 Applied optimization: Optimizing discrete quantities

In Sections 3.15 through 3.17, we modeled systems under the assumption that our independent variable x is **perfectly continuous**—meaning it can be any real decimal number. This assumption is mathematically convenient because it allows us to set $f'(x) = 0$ to find exact turning points.

But in real-world application, many quantities are strictly **discrete**.

You cannot manufacture 15.42 passenger jets, hire 4.8 software engineers, or coordinate 12.7 tour buses. These quantities must be represented as integers ($n \in \mathbb{Z}$).

How do we optimize a system when its variables are not allowed to be decimals?

To solve this, we perform **Analytic Continuation on Discrete Quantities**. We temporarily pretend the discrete variable is a continuous variable $x \in \mathbb{R}$, allowing us to use our standard calculus tools to locate the continuous optimal peak x_c . Once x_c is found, we know the true integer optimal value must lie immediately adjacent to it. By evaluating and comparing the original discrete function at the two closest bounding integers, $\lfloor x_c \rfloor$ (the floor) and $\lceil x_c \rceil$ (the ceiling), we can pinpoint the exact integer that yields the absolute best discrete performance.

Proposition 3.18: Discrete Optimization via Analytic Continuation

To optimize an objective function $f(n)$ where n must be an integer ($n \in \mathbb{Z}$):

1. **Continuous Extension:** Define a continuous analog function $f(x)$ where $x \in \mathbb{R}$.
2. **Continuous Optimization:** Find the continuous critical point x_c by solving $f'(x_c) = 0$.
3. **Integer Bracket Selection:** Identify the two integers directly bounding the decimal peak x_c :

$$n_1 = \lfloor x_c \rfloor \quad (\text{integer below}) \quad \text{and} \quad n_2 = \lceil x_c \rceil \quad (\text{integer above})$$

4. **Discrete Tournament:** Evaluate and compare the original discrete function values: $f(n_1)$ and $f(n_2)$.
5. **Resolution:** The integer yielding the more optimal value is your discrete optimizer.

Student Proof Demonstration

Use this space to demonstrate the proof of the theorem above. Start from basic algebraic identities.

Practical Example 3.18

A tour operator organizes chartered bus trips. If a bus contains n passengers, the net profit (in HKD) generated by the operator per trip is modeled by:

$$P(n) = -2n^2 + 75n - 150$$

Since passengers must be whole people, n must be an integer. Find the exact number of passengers that maximizes the operator's profit, and calculate this maximum profit.

Exercise 3.18.1 (Optimizing production of discrete assets)**Problem Statement:**

A luxury watchmaker's monthly profit (in thousands of USD) from manufacturing n custom timepieces is modeled by the function:

$$P(n) = -0.5n^2 + 13.6n - 20 \quad (n \in \mathbb{Z}^+)$$

1. Find the continuous peak coordinate x_c .
 2. Evaluate the watches profit at the two closest integer counts.
 3. Determine the number of watches that maximizes monthly profit, and calculate this maximum profit.
-

Exercise 3.18.2 (Discrete cost minimization with rational components)**Problem Statement:**

A cargo distribution center determines that the daily maintenance cost (in HKD) of coordinating a fleet of n delivery vans is given by:

$$C(n) = 30n + \frac{1000}{n} \quad (n \in \mathbb{Z}^+)$$

1. Find the continuous critical fleet size x_c .
 2. Evaluate the costs at the bounding integers.
 3. Find the optimal fleet size that minimizes daily coordinating costs.
-

Exercise 3.18.3 (Optimizing discrete personnel configurations)**Problem Statement:**

A software firm estimates that the annual revenue (in millions of HKD) generated by a team of n senior engineers is modeled by:

$$R(n) = 12\ln(n) - 0.2n \quad \text{for } n \in \mathbb{Z}^+$$

Find the optimal size of the engineering team that maximizes the revenue-to-cost balance.

Exercise 3.18.4 (HKDSE M1-style discrete shipping container optimization)**Problem Statement:**

A shipping logistics company charges clients to transport shipping containers. The net operating yield $Y(n)$ (in thousands of HKD) of transporting n cargo containers on a single voyage is modeled by:

$$Y(n) = 80ne^{-0.025n} \quad \text{for } n \in \mathbb{Z}^+$$

Due to structural hull limits, the vessel can carry a maximum of 50 containers.

1. Find the continuous optimal container count x_c that maximizes the operating yield.
 2. Identify the two integer candidates bounding x_c .
 3. Calculate the operating yield for both integer candidates, and state the optimal discrete container count and its corresponding yield.
-

Exercise 3.18.5 (Symmetric quadratics and the perfect midpoint tie)**Problem Statement:**

Let $P(n) = -an^2 + bn + c$ be a discrete quadratic function where $a > 0$ and $n \in \mathbb{Z}$. Prove that if the continuous peak $x_c = \frac{b}{2a}$ ends in exactly .5 (meaning $x_c = k + 0.5$ for some integer $k \in \mathbb{Z}$), then the two discrete integers $n_1 = k$ and $n_2 = k + 1$ are guaranteed to produce a perfect tie in the discrete tournament.

Exercise 3.18.6 (The rounding trap on highly asymmetric curves)**Problem Statement:**

Continuous peaks do not always dictate discrete behaviors if the curve is highly skewed. Consider the asymmetric rational function:

$$f(x) = \frac{100}{x^2} + x \quad \text{on the domain } x \in (0, \infty)$$

1. Find the continuous critical point x_c that minimizes $f(x)$.
 2. Suppose the variable must be a discrete integer $n \in \mathbb{Z}^+$. The nearest integer to x_c is 6.
 3. Calculate and compare the discrete values $f(5)$, $f(6)$, and $f(7)$.
 4. Does the closest rounded integer yield the absolute minimum? Discuss how asymmetric curvatures can warp discrete optimization boundaries.
-

Exercise 3.18.7 (Discrete Optimization via Marginal Analysis: The Newsboy Problem)**Problem Statement:**

In supply chain management, the **News vendor (Newsboy) Problem** is a mathematical model used to determine optimal inventory levels for products with fluctuating demand. Let n be the number of newspapers ordered daily.

- The cost price per newspaper is C .
- The selling price per newspaper is S (where $S > C$).
- Unsold newspapers have a salvage value of R (where $R < C$).

Let demand X be a discrete random variable with a Cumulative Distribution Function $F(k) = P(X \leq k)$. The expected profit $E[P(n)]$ of ordering n copies is modeled by:

$$E[P(n)] = (S - C)n - (S - R) \sum_{k=0}^{n-1} F(k)$$

1. Instead of using derivatives, we analyze the **discrete difference** $\Delta E(n) = E[P(n+1)] - E[P(n)]$. Show that:

$$\Delta E(n) = (S - C) - (S - R)F(n)$$

2. For the inventory level to be optimized, we require the profit to stop increasing: $\Delta E(n) \leq 0$ and $\Delta E(n-1) \geq 0$. Prove that the optimal discrete inventory level n^* is the smallest integer that satisfies the **critical ratio inequality**:

$$F(n^*) \geq \frac{S - C}{S - R}$$

3. Let $S = 10$ HKD, $C = 4$ HKD, and $R = 1$ HKD. Find the critical ratio.
4. If demand is uniformly distributed between 0 and 10 newspapers (meaning $F(k) = \frac{k+1}{11}$ for $k \in \{0, 1, \dots, 10\}$), calculate the exact optimal number of newspapers n^* to order.

Suggested Answers

Section 1.1

- 1.1.1 8
- 1.1.2 -7
- 1.1.3 16π (The physical area of a circle with radius exactly 4 units).
- 1.1.4 (1) 15;
(2) Expanded form is $2x^2 + 5x - 3$. Substituting $x = 2$ gives $2(4) + 5(2) - 3 = 15$. Both methods yield the same output.
- 1.1.5 $k = 1$
- 1.1.6 Polynomials only require multiplication and addition. These operations are globally defined for all real numbers and contain no risk of division-by-zero or taking square roots of negative numbers.
- 1.1.7 Direct substitution yields $\frac{0}{0}$, an indeterminate form. This division by zero is mathematically undefined.

Section 1.2

- 1.2.1 Left-hand outputs: 4.6, 4.96, 4.996; Right-hand outputs: 5.4, 5.04, 5.004. The limit is 5.
- 1.2.2 (1) $g(2.001) = 4.002$, $g(1.999) = 3.998$;
(2) The limit is 4.
- 1.2.3 Outputs: 0.1, 0.01, 0.001, 0.000001. The sequence converges to 0.
- 1.2.4 The calculator ran out of decimal precision memory (floating-point underflow/subtraction cancellation), treating $\sqrt{10^{-16} + 9}$ as exactly 3, yielding $\frac{3-3}{10^{-16}} = 0$. The true limit is $\frac{1}{6} \approx 0.166667$.
- 1.2.5 As n increases, the denominator of $\frac{1}{n}$ grows infinitely large while the numerator remains 1. Dividing a fixed quantity by an infinitely large number results in a value arbitrarily close to 0. By extension, scaling the numerator by k or raising the denominator to power p preserves this behavior.
- 1.2.6 $f(2.999) = -1$ while $f(3.001) = 1$. Because the left-hand sequence and right-hand sequence converge to different values, the overall limit does not exist.
- 1.2.7 (1) $f(x_n) = \cos(2n\pi) = 1$ for all n ;
(2) $f(y_n) = \cos((2n+1)\pi) = -1$ for all n ;
(3) Different input sequences approaching the same target 0 yield completely different output limits (1 vs -1). Thus, the limit does not exist.

Section 1.3

- 1.3.1 $g(2) = 20$; $\lim_{x \rightarrow 2} g(x) = 5(2) - 2 = 8$.
- 1.3.2 (1) Substituting $x = 5$ yields $\frac{0}{0}$, which is undefined because division by zero is mathematically forbidden;
(2) $\lim_{x \rightarrow 5} \frac{(x-5)(x+5)}{x-5} = \lim_{x \rightarrow 5} (x+5) = 10$;
(3) The coordinates of the hole are (5, 10).
- 1.3.3 (1) $\lim_{t \rightarrow 2} \frac{(t-2)(t+5)}{t-2} = \lim_{t \rightarrow 2} (t+5) = 7$;
(2) $S_{\text{repaired}}(t) = \begin{cases} \frac{t^2+3t-10}{t-2} & \text{if } t \neq 2 \\ 7 & \text{if } t = 2 \end{cases}$
- 1.3.4 We require $\lim_{x \rightarrow 4} (2x^2 - 3x + k) = 15$. Substituting $x = 4$ gives $2(16) - 3(4) + k = 15 \implies 32 - 12 + k = 15 \implies 20 + k = 15 \implies k = -5$.
- 1.3.5 Since the limit notation $x \rightarrow c$ mandates that $x \neq c$, the factor $(x - c)$ is non-zero. Thus, for $x \neq c$, we can simplify: $F(x) = \frac{(x-c)K(x)}{(x-c)R(x)} = \frac{K(x)}{R(x)}$. Taking the limit yields $\lim_{x \rightarrow c} F(x) = \frac{K(c)}{R(c)}$, which is a valid real number since $R(c) \neq 0$.
- 1.3.6 Redefining $f(1) = 5.5$ only changes the height of the single isolated dot at $x = 1$. It does not change the behavior of the surrounding points. As $x \rightarrow 1$ from below, inputs yield outputs of 3. As $x \rightarrow 1$ from above, inputs yield outputs of 8. Because the left-hand limit (3) and right-hand limit (8) do not coincide, no single limit exists.
- 1.3.7 (1) Between any two distinct real numbers, no matter how close, there exist infinitely many rational numbers and infinitely many irrational numbers (the density property of real numbers);
(2) We can construct a sequence of rational numbers approaching c where outputs are constantly 1, and another sequence of irrational numbers approaching c where outputs are constantly 0. Since these sequences converge to different limits ($1 \neq 0$), the limit $\lim_{x \rightarrow c} D(x)$ fails to exist;
(3) The graph is entirely disconnected at every single point. It is completely full of gaps and nowhere-continuous, making it impossible to draw with a continuous pencil stroke.

Section 1.4

- 1.4.1 $\lim_{x \rightarrow 3} \frac{(x-3)(x+2)}{(x-3)(x+3)} = \lim_{x \rightarrow 3} \frac{x+2}{x+3} = \frac{5}{6}$.
- 1.4.2 (1) $\lim_{x \rightarrow 2} \frac{(x-2)(x^2+2x+4)}{x-2} = \lim_{x \rightarrow 2} (x^2 + 2x + 4) = 4 + 4 + 4 = 12$;
(2) For continuity, we must set $k = 12$.

1.4.3 By division, $x^3 - 4x^2 + 5x - 2 = (x - 1)(x^2 - 3x + 2) = (x - 1)^2(x - 2)$. Thus,
 $\lim_{x \rightarrow 1} \frac{(x-1)^2(x-2)}{(x-1)(x+1)} = \lim_{x \rightarrow 1} \frac{(x-1)(x-2)}{x+1} = \frac{0(-1)}{2} = 0$.

1.4.4 (1) As $x \rightarrow -1$, the denominator $x^2 - 1 \rightarrow 0$. If the numerator approached a non-zero number C , the quotient would grow unboundedly toward infinity ($\frac{C}{0}$), making a finite limit of 3 impossible. Thus, the numerator must approach 0;

(2) Setting $2(-1)^2 + a(-1) + b = 0 \Rightarrow 2 - a + b = 0 \Rightarrow b = a - 2$. Substitute back: $\lim_{x \rightarrow -1} \frac{2x^2 + ax + (a-2)}{(x-1)(x+1)} = \lim_{x \rightarrow -1} \frac{(x+1)(2x+a-2)}{(x-1)(x+1)} = \lim_{x \rightarrow -1} \frac{2x+a-2}{x-1} = \frac{-4+a}{-2} = 3 \Rightarrow -4+a = -6 \Rightarrow a = -2$. Then $b = -2 - 2 = -4$.

1.4.5 Canceling the common factor $(x-1)$ for $x \neq 1$ yields $\lim_{x \rightarrow 1} (x^{n-1} + x^{n-2} + \dots + x + 1)$. Using direct substitution, each of the n terms evaluates to $1^k = 1$. Summing 1 exactly n times gives n .

1.4.6 (1) $\lim_{x \rightarrow 3} \frac{x(x-3)}{(x-3)^2} = \lim_{x \rightarrow 3} \frac{x}{x-3}$;

(2) Substituting $x = 3$ gives $\frac{3}{0}$, which is still division by zero;

(3) This is not a removable hole because the denominator had a higher multiplicity of the $(x-3)$ factor than the numerator. Geometrically, this represents a vertical asymptote where the function shoots toward infinity or negative infinity, rather than approaching a finite height.

1.4.7 (1) Along $y = 0$, $\lim_{x \rightarrow 0} \frac{x^2-0}{x^2+0} = 1$;

(2) Along $x = 0$, $\lim_{y \rightarrow 0} \frac{0-y^2}{0+y^2} = -1$;

(3) Along $y = mx$, $\lim_{x \rightarrow 0} \frac{x^2(1-m^2)}{x^2(1+m^2)} = \frac{1-m^2}{1+m^2}$;

(4) Because the limit value depends entirely on the slope m of the approach path (just as 1D limits fail when left-hand and right-hand sequences disagree), no single, unique limit exists at the origin.

Section 1.5

1.5.1 Multiply by $\frac{\sqrt{x+3}}{\sqrt{x+3}}$ to get $\lim_{x \rightarrow 9} \frac{x-9}{(x-9)(\sqrt{x+3})} = \lim_{x \rightarrow 9} \frac{1}{\sqrt{x+3}} = \frac{1}{6}$.

1.5.2 Rationalize numerator and factor denominator: $\lim_{x \rightarrow 4} \frac{x-4}{(x-4)(x+4)(\sqrt{x+2})} = \lim_{x \rightarrow 4} \frac{1}{(x+4)(\sqrt{x+2})} = \frac{1}{(8)(4)} = \frac{1}{32}$.

1.5.3 Multiply top and bottom by $(\sqrt{2x+5} + 3)$:
 $\lim_{x \rightarrow 2} \frac{(x-2)(\sqrt{2x+5}+3)}{(2x+5)-9} = \lim_{x \rightarrow 2} \frac{(x-2)(\sqrt{2x+5}+3)}{2(x-2)} = \lim_{x \rightarrow 2} \frac{\sqrt{2x+5}+3}{2} = \frac{3+3}{2} = 3$.

1.5.4 For the limit as $x \rightarrow 1$ to exist without blowing up to infinity, substitution into the

numerator must yield 0. Thus, $\sqrt{1+a} - 2 = 0 \Rightarrow \sqrt{1+a} = 2 \Rightarrow 1+a = 4 \Rightarrow a = 3$. Now rationalize with $a = 3$: $\lim_{x \rightarrow 1} \frac{\sqrt{x+3}-2}{x-1} \cdot \frac{\sqrt{x+3}+2}{\sqrt{x+3}+2} = \lim_{x \rightarrow 1} \frac{(x+3)-4}{(x-1)(\sqrt{x+3}+2)} = \lim_{x \rightarrow 1} \frac{1}{\sqrt{x+3}+2} = \frac{1}{4}$. For continuity, set $b = \frac{1}{4}$.

1.5.5 Multiply top and bottom by the conjugate factor $(\sqrt[3]{x-1} + \sqrt[3]{x} + 1)$:
 $\lim_{x \rightarrow 1} \frac{(\sqrt[3]{x-1})(\sqrt[3]{x-1} + \sqrt[3]{x} + 1)}{(x-1)(\sqrt[3]{x-1} + \sqrt[3]{x} + 1)} = \lim_{x \rightarrow 1} \frac{x-1}{(x-1)(\sqrt[3]{x-1} + \sqrt[3]{x} + 1)} = \lim_{x \rightarrow 1} \frac{1}{\sqrt[3]{x-1} + \sqrt[3]{x} + 1} = \frac{1}{1+1+1} = \frac{1}{3}$.

1.5.6 (1) For negative numbers like -2 , squaring gives 4, and the principal square root is positive 2, not -2 . Thus $\sqrt{x^2} = |x|$;

(2) Left-hand limit: as $x \rightarrow 0^-$, $|x| = -x$, so $\lim_{x \rightarrow 0^-} \frac{-x}{x} = -1$. Right-hand limit: as $x \rightarrow 0^+$, $|x| = x$, so $\lim_{x \rightarrow 0^+} \frac{x}{x} = 1$;

(3) Because the left-hand limit (-1) and right-hand limit (1) are unequal, the overall limit fails to exist. Rationalization cannot fix a jump discontinuity caused by an absolute value sign trap.

1.5.7 (1) $g'(c) = \lim_{x \rightarrow c} \frac{\sqrt[n]{x} - \sqrt[n]{c}}{x - c}$;
 (2) Multiply by the conjugate factor $\sum_{k=0}^{n-1} \sqrt[n]{x^{n-1-k}} \sqrt[n]{c^k}$ to get $\lim_{x \rightarrow c} \frac{x-c}{(x-c)[\sqrt[n]{x^{n-1}} + \dots + \sqrt[n]{c^{n-1}}]}$;

(3) Cancel $(x-c)$ and substitute $x = c$: each of the n terms in the denominator becomes $\sqrt[n]{c^{n-1}} = c^{\frac{n-1}{n}}$. Summing n identical terms yields $\frac{1}{n \cdot c^{(n-1)/n}} = \frac{1}{n} c^{\frac{1}{n}-1}$;

(4) This proves the ****Power Rule of Differentiation**** ($d/dx(x^r) = rx^{r-1}$) for fractional roots from first principles!

Section 1.6

1.6.1 (1) $\lim_{x \rightarrow 1^-} (2x+3) = 2(1)+3 = 5$;

(2) $\lim_{x \rightarrow 1^+} (7-x) = 7-1 = 6$;

(3) No, since $5 \neq 6$, the left-hand and right-hand limits do not agree, so the general limit does not exist.

1.6.2 Left-hand: $\lim_{x \rightarrow 4^-} \frac{(x-4)(x+4)}{x-4} = \lim_{x \rightarrow 4^-} (x+4) = 8$. Right-hand: $\lim_{x \rightarrow 4^+} (3x-4) = 3(4)-4 = 8$. Since both sides equal 8, the general limit exists and equals 8.

1.6.3 (1) $2x - 6 > 0 \Rightarrow x > 3$. Thus

$$f(x) = \begin{cases} \frac{-(2x-6)}{x-3} = -2 & \text{if } x < 3 \\ \frac{2x-6}{x-3} = 2 & \text{if } x > 3 \end{cases}$$

(2) $\lim_{x \rightarrow 3^-} f(x) = -2$ and $\lim_{x \rightarrow 3^+} f(x) = 2$;

(3) Since $-2 \neq 2$, the limit $\lim_{x \rightarrow 3} f(x)$ does not exist.

1.6.4 For limit at $x = -1$: $\lim_{x \rightarrow -1^-} (ax + b) = \lim_{x \rightarrow -1^+} (2x^2 + 3) \Rightarrow -a + b = 5$. For limit at $x = 2$: $\lim_{x \rightarrow 2^-} (2x^2 + 3) = \lim_{x \rightarrow 2^+} (3ax - 2b) \Rightarrow 11 = 6a - 2b \Rightarrow 6a - 2b = 11$. Solving the simultaneous system: from first equation $b = a + 5$. Substitute into second: $6a - 2(a + 5) = 11 \Rightarrow 4a - 10 = 11 \Rightarrow 4a = 21 \Rightarrow a = \frac{21}{4} = 5.25$. Then $b = 5.25 + 5 = 10.25$ (or $\frac{41}{4}$).

1.6.5 A one-sided limit only investigates behavior within the specific directional approach interval. Since $q(x)$ is permanently trapped between $p(x)$ and $r(x)$ for all $x < c$, and both outer walls squeeze together to height L as $x \rightarrow c^-$, the trapped function $q(x)$ is forced to approach L as well. What happens for $x \geq c$ is completely outside the domain of a left-hand limit.

1.6.6 (1) $\lim_{x \rightarrow 0^-} (x + 2) = 0 + 2 = 2$;
 (2) As $x \rightarrow 0^+$, the term $\frac{1}{x}$ grows unboundedly, causing $\sin(1/x)$ to oscillate infinitely fast between -1 and 1 without settling on any single value;
 (3) No. Theorem 1.6 strictly requires BOTH one-sided limits to exist and be equal. If even one side fails to exist, the general two-sided limit is destroyed.

1.6.7 (1) $\lim_{x \rightarrow 2} |x - 2| = |2 - 2| = 0 = f(2)$;
 (2) Left slope limit: $\lim_{x \rightarrow 2^-} \frac{-(x-2)}{x-2} = -1$ (the V-graph falls at slope -1 from the left);
 (3) Right slope limit: $\lim_{x \rightarrow 2^+} \frac{x-2}{x-2} = 1$ (the V-graph rises at slope $+1$ to the right);
 (4) A tangent line represents the instantaneous slope at a point. Because the slope coming from the left (-1) clashes abruptly with the slope heading to the right ($+1$), there is no single, agreed-upon direction at the exact vertex $(2, 0)$. Thus, sharp corners are non-differentiable!

Section 1.7

1.7.1 Step 1: $g(3) = 5 - 3 = 2$. Step 2: $\lim_{x \rightarrow 3^-} g(x) = 5 - 3 = 2$ and $\lim_{x \rightarrow 3^+} g(x) = 3^2 - 7 = 2$. Since $2 = 2$, $\lim_{x \rightarrow 3} g(x) = 2$. Step 3: Since $\lim_{x \rightarrow 3} g(x) = g(3) = 2$, the function is continuous at $x = 3$.

1.7.2 Left-hand limit: $\lim_{x \rightarrow 4^-} \frac{\sqrt{x}-2}{x-4} \cdot \frac{\sqrt{x}+2}{\sqrt{x}+2} = \lim_{x \rightarrow 4^-} \frac{1}{\sqrt{x}+2} = \frac{1}{4}$. Right-hand limit: $\lim_{x \rightarrow 4^+} \frac{(x-4)(x+4)}{4(x-4)} = \lim_{x \rightarrow 4^+} \frac{x+4}{4} = \frac{8}{4} = 2$. Since $\frac{1}{4} \neq 2$, the two-sided limit does not exist (Condition 2 fails). Thus, $h(x)$ is discontinuous at $x = 4$.

1.7.3 For continuity at $x = 1$, we require $\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^+} f(x) = f(1)$. Left-hand limit and value: $(1)^2 - 3(1) + k = k - 2$.

Right-hand limit: $2(1) + 5 = 7$. Setting them equal yields $k - 2 = 7 \Rightarrow k = 9$.

1.7.4 For continuity at $x = -2$: $\lim_{x \rightarrow -2^-} (3) = \lim_{x \rightarrow -2^+} (ax^2 + bx - 1) \Rightarrow 4a - 2b - 1 = 3 \Rightarrow 4a - 2b = 4 \Rightarrow 2a - b = 2$. For continuity at $x = 2$: $\lim_{x \rightarrow 2^-} (ax^2 + bx - 1) = \lim_{x \rightarrow 2^+} (2x + 7) \Rightarrow 4a + 2b - 1 = 11 \Rightarrow 4a + 2b = 12 \Rightarrow 2a + b = 6$. Adding the two simplified equations: $(2a - b) + (2a + b) = 2 + 6 \Rightarrow 4a = 8 \Rightarrow a = 2$. Substituting $a = 2$ back: $2(2) + b = 6 \Rightarrow b = 2$.

1.7.5 Because $p(x)$ and $q(x)$ are continuous at c , we know $\lim_{x \rightarrow c} p(x) = p(c)$ and $\lim_{x \rightarrow c} q(x) = q(c)$. Evaluating the limit of the sum: $\lim_{x \rightarrow c} S(x) = \lim_{x \rightarrow c} [p(x) + q(x)]$. By the sum law of limits, this equals $\lim_{x \rightarrow c} p(x) + \lim_{x \rightarrow c} q(x) = p(c) + q(c) = S(c)$. Since $\lim_{x \rightarrow c} S(x) = S(c)$, $S(x)$ is continuous at $x = c$.

1.7.6 (1) For $f(x)$, $\lim_{x \rightarrow 1} f(x) = 3$, but $f(1) = 9$. This violates Condition 3 (lim $\neq$ value). It is a removable hole because the limit itself exists cleanly;
 (2) For $g(x)$, $\lim_{x \rightarrow 1^-} g(x) = 2$ while $\lim_{x \rightarrow 1^+} g(x) = 5$. This violates Condition 2 (one-sided limits disagree). It is a jump discontinuity because the path literally leaps across a gap;
 (3) Only $f(x)$ can be repaired by redefining $f(1) = 3$. For $g(x)$, no single dot can ever bridge a gap where the two approaching roads are at different heights.

1.7.7 (1) Polynomials involve only multiplication and addition, containing no division or surds, guaranteeing unbroken continuity everywhere;
 (2) $f(1) = 1 + 2 - 5 = -2$ (negative) and $f(2) = 8 + 4 - 5 = 7$ (positive);
 (3) Because the graph is continuous and must travel from height -2 up to height $+7$, it is physically impossible to connect these points without crossing height 0 at least once;
 (4) If a function can jump, it could teleport directly from $(1, -2)$ up to $(2, 7)$ through the air, completely missing the horizontal zero axis!

Section 1.8

1.8.1 Divide numerator and denominator by x^2 : $\lim_{x \rightarrow \infty} \frac{10+3/x-1/x^2}{5-2/x+4/x^2} = \frac{10+0-0}{5-0+0} = \frac{10}{5} = 2$.

1.8.2 (1) Divide by x^2 : $\lim_{x \rightarrow \infty} \frac{4/x-7/x^2}{2+1/x^2} = \frac{0-0}{2+0} = \frac{0}{2} = 0$;
 (2) The horizontal asymptote is the straight line $y = 0$ (which is exactly the x -axis).

1.8.3 Expand first: $\lim_{x \rightarrow \infty} \frac{6x^2+11x-10}{x^2-2x+1}$. Dividing top and bottom by x^2 yields $\lim_{x \rightarrow \infty} \frac{6+11/x-10/x^2}{1-2/x+1/x^2} = \frac{6+0-0}{1-0+0} = 6$.

1.8.4 (1) Dividing top and bottom by x^2 gives $\lim_{x \rightarrow \infty} \frac{ax+b+5/x^2}{2-3/x+1/x^2}$. If $a \neq 0$, the term ax will grow indefinitely toward ∞ (or $-\infty$) as $x \rightarrow \infty$, making it impossible for the fraction to settle at a finite height of 4. Thus, we must have $a = 0$ to eliminate unbounded growth; (2) With $a = 0$, the limit simplifies to $\frac{b}{2} = 4 \Rightarrow b = 8$.

1.8.5 Dividing every term by x^m yields $\lim_{x \rightarrow \infty} \frac{a_n x^{n-m} + \dots + a_0/x^m}{b_m + \dots + b_0/x^m}$. (1) If $n < m$, then $n - m$ is a negative integer, meaning every single term in the numerator contains x in the denominator and shrinks to 0 as $x \rightarrow \infty$, yielding $\frac{0}{b_m} = 0$; (2) If $n = m$, then $x^{n-m} = x^0 = 1$, leaving the constant leading coefficient a_n alone in the numerator while all lower terms vanish to 0, yielding $\frac{a_n}{b_m}$.

1.8.6 (1) Dividing by x^1 gives $\lim_{x \rightarrow \infty} \frac{x^2+1/x}{5-2/x}$; (2) As $x \rightarrow \infty$, the numerator grows indefinitely ($x^2 \rightarrow \infty$) while the denominator stabilizes at 5. An infinitely large quantity divided by 5 remains infinitely large, never settling on a static number; (3) Writing $\lim = \infty$ is symbolic shorthand indicating that the curve shoots upward without bound. Because ∞ is not a real number, the limit strictly fails to exist in the formal sense.

1.8.7 (1) Divide top and bottom by $[S]^1$: $\lim_{[S] \rightarrow \infty} \frac{V_{\max}}{K_m/[S]+1} = \frac{V_{\max}}{0+1} = V_{\max}$; (2) This limit represents the ****absolute saturation speed**** of the enzymes: once substrate concentration is superabundant, every single enzyme active site is occupied 100% of the time, forming a biological bottleneck where reaction velocity cannot increase any further; (3) Set $V([S]) = 0.5V_{\max} \Rightarrow \frac{V_{\max}[S]}{K_m+[S]} = 0.5V_{\max} \Rightarrow [S] = 0.5(K_m + [S]) \Rightarrow 0.5[S] = 0.5K_m \Rightarrow [S] = K_m$. Thus, the Michaelis constant K_m represents the exact substrate concentration required to reach half of the maximum possible reaction velocity!

Section 1.9

1.9.1 Let $u = -x$: $\lim_{u \rightarrow \infty} \frac{8(-u)^2-3(-u)+1}{2(-u)^2+5(-u)-4} = \lim_{u \rightarrow \infty} \frac{8u^2+3u+1}{2u^2-5u-4}$. Dividing top and bottom by u^2 yields $\frac{8+0+0}{2-0-0} = \frac{8}{2} = 4$.

1.9.2 (1) Divide by x : $\lim_{x \rightarrow \infty} \frac{6-1/x}{3+2/x} = \frac{6}{3} = 2$; (2) Let $u = -x$: $\lim_{u \rightarrow \infty} \frac{6(-u)-1}{3(-u)+2} =$

$$\lim_{u \rightarrow \infty} \frac{-6u-1}{-3u+2} = \frac{-6}{-3} = 2;$$

(3) This proves that for pure rational functions without radicals, the horizontal asymptote to the left ($y = 2$) is always identical to the horizontal asymptote to the right.

1.9.3 Expanding top gives $-6x^3+3x^2-8x+4$. Let $u = -x$: $\lim_{u \rightarrow \infty} \frac{-6(-u)^3+3(-u)^2-8(-u)+4}{5(-u)^3-(-u)+2} = \lim_{u \rightarrow \infty} \frac{6u^3+3u^2+8u+4}{-5u^3+u+2}$. Dividing by u^3 gives $\frac{6}{-5} = -1.2$ (or $-\frac{6}{5}$).

1.9.4 Let $u = -x$: $\lim_{u \rightarrow \infty} \frac{k(-u)^3-4(-u)^2+5}{3(-u)^3+2(-u)-1} = \lim_{u \rightarrow \infty} \frac{-ku^3-4u^2+5}{-3u^3-2u-1}$. Dividing by u^3 yields $\frac{-k}{-3} = \frac{k}{3}$. Setting this equal to our target: $\frac{k}{3} = -6 \Rightarrow k = -18$.

1.9.5 Let $F(x) = \frac{a_n x^n + \dots + a_0}{b_m x^m + \dots + b_0}$. If a finite limit L exists at $+\infty$, we know from Section 1.8 that $n \leq m$. When we substitute $u = -x$, every term x^k becomes $(-u)^k = (-1)^k u^k$. When dividing by u^m , all lower-degree terms still vanish to 0. If $n = m$, the leading terms become $\frac{a_n(-1)^n u^n}{b_n(-1)^n u^n} = \frac{a_n}{b_n} = L$, because the $(-1)^n$ sign factor cancels perfectly! Thus, left and right limits are strictly identical.

1.9.6 (1) For L_1 : $\lim_{u \rightarrow \infty} \frac{-u^3+1}{u^2+2} = \lim_{u \rightarrow \infty} \frac{-u+1/u^2}{1+2/u^2} = -\infty$. For L_2 : $\lim_{u \rightarrow \infty} \frac{u^4+1}{u^2+2} = \lim_{u \rightarrow \infty} \frac{u^2+1/u^2}{1+2/u^2} = +\infty$; (2) In L_1 , the odd power (x^3) preserved the negative sign, driving the curve downward. In L_2 , the even power (x^4) squared away the negative sign, forcing the curve upward; (3) If the degree difference ($n - m$) is an ****odd integer****, the limit at $-\infty$ will have the opposite sign of the leading coefficient ratio. If $(n - m)$ is an ****even integer****, it will maintain the same sign.

1.9.7 (1) For $x > 0$, dividing by $x = \sqrt{x^2}$ gives $\lim_{x \rightarrow \infty} \frac{\sqrt{1+4/x^2}}{1} = \frac{\sqrt{1}}{1} = 1$; (2) Using $u = -x$: $\lim_{u \rightarrow \infty} \frac{\sqrt{(-u)^2+4}}{-u} = \lim_{u \rightarrow \infty} \frac{\sqrt{u^2+4}}{-u} = \lim_{u \rightarrow \infty} \frac{\sqrt{1+4/u^2}}{-1} = \frac{1}{-1} = -1$; (3) The square root inside the numerator permanently stripped away the negative sign of the input, while the linear x in the denominator preserved it, creating an overall negative ratio; (4) Because $\sqrt{x^2}$ always produces a positive output (principal square root), writing $\sqrt{(-5)^2} = -5$ is false ($5 \neq -5$). The algebraic truth is $\sqrt{x^2} = |x|$. When $x \rightarrow -\infty$, x is negative, meaning $|x| = -x$. Dividing $\sqrt{x^2}$ by x thus yields $\frac{-x}{x} = -1$, permanently splitting the left asymptote from the right asymptote!

Section 1.10

1.10.1 Divide top and bottom by x :

$$\lim_{x \rightarrow \infty} \frac{\sqrt{9+5/x^2}}{2-3/x} = \frac{\sqrt{9+0}}{2-0} = \frac{3}{2} = 1.5.$$

1.10.2 Rationalize: $\lim_{x \rightarrow \infty} \frac{(x^2+6x)-x^2}{\sqrt{x^2+6x+x}} =$

$$\lim_{x \rightarrow \infty} \frac{6x}{\sqrt{x^2+6x+x}}. \text{ Dividing by } x \text{ yields}$$

$$\frac{6}{\sqrt{1+0+1}} = \frac{6}{2} = 3.$$

1.10.3 Rationalize: $\lim_{x \rightarrow \infty} \frac{(9x^2+x)-9x^2}{\sqrt{9x^2+x+3x}} =$

$$\lim_{x \rightarrow \infty} \frac{x}{\sqrt{9x^2+x+3x}}. \text{ Dividing by } x \text{ gives}$$

$$\frac{1}{\sqrt{9+0+3}} = \frac{1}{3+3} = \frac{1}{6}.$$

1.10.4 Rationalize: $\lim_{x \rightarrow \infty} \frac{16x^2+kx-16x^2}{\sqrt{16x^2+kx+4x}} =$

$$\lim_{x \rightarrow \infty} \frac{kx}{\sqrt{16x^2+kx+4x}}. \text{ Dividing by } x \text{ yields}$$

$$\frac{k}{\sqrt{16+0+4}} = \frac{k}{4+4} = \frac{k}{8}. \text{ Setting this equal to}$$

$$\text{our target: } \frac{k}{8} = 5 \Rightarrow k = 40.$$

1.10.5 Multiply by conjugate: $\lim_{x \rightarrow \infty} \frac{(a^2x^2+bx)-a^2x^2}{\sqrt{a^2x^2+bx+ax}} =$

$$\lim_{x \rightarrow \infty} \frac{bx}{\sqrt{a^2x^2+bx+ax}}. \text{ Dividing top and bot-}$$

$$\text{tom by } x \text{ yields } \lim_{x \rightarrow \infty} \frac{b}{\sqrt{a^2+b/x+a}} = \frac{b}{\sqrt{a^2+a}}.$$

Since $a > 0$, $\sqrt{a^2} = a$, so the limit simplifies perfectly to $\frac{b}{a+a} = \frac{b}{2a}$.

1.10.6 (1) The leading term of the surd behaves like $\sqrt{4x^2} = 2x$. Thus, as $x \rightarrow \infty$, the expression grows like $2x - x = x$, which is not an indeterminate tie, but clearly positive growth;(2) Factoring out x gives $\lim_{x \rightarrow \infty} x(\sqrt{4+3/x}-1)$;(3) As $x \rightarrow \infty$, the term inside brackets approaches $\sqrt{4}-1 = 1$. Thus we have $\lim_{x \rightarrow \infty} x(1) = +\infty$. Rationalization is only required when the leading coefficients cancel out to produce a zero-sum competition ($\infty - \infty$).1.10.7 (1) Using $A - B = \frac{A^3-B^3}{A^2+AB+B^2}$, we get

$$\lim_{x \rightarrow \infty} \frac{(x^3+6x^2)-x^3}{(x^3+6x^2)^{2/3}+x(x^3+6x^2)^{1/3}+x^2};$$

(2) The numerator simplifies to $6x^2$;(3) Each term in the denominator has an effective degree of x^2 (e.g., $(x^3)^{2/3} = x^2$). Dividing top and bottom by x^2 gives

$$\lim_{x \rightarrow \infty} \frac{6}{(1+6/x)^{2/3}+(1+6/x)^{1/3}+1};$$

(4) As $x \rightarrow \infty$, this evaluates to $\frac{6}{1+1+1} = \frac{6}{3} = 2$.

Section 1.11

1.11.1 Let $u = -x$: $\lim_{u \rightarrow \infty} \frac{\sqrt{(-u)^2+4}}{2(-u)-1} =$

$$\lim_{u \rightarrow \infty} \frac{\sqrt{u^2+4}}{-3u-1}. \text{ Dividing by } u \text{ yields } \frac{\sqrt{1+0}}{-3-0} =$$

$$-\frac{1}{3}.$$

1.11.2 Let $u = -x$: $\lim_{u \rightarrow \infty} (\sqrt{u^2-4u}-u)$. Mul-

$$\text{tiply by conjugate: } \lim_{u \rightarrow \infty} \frac{(u^2-4u)-u^2}{\sqrt{u^2-4u}+u} =$$

$\lim_{u \rightarrow \infty} \frac{-4u}{\sqrt{u^2-4u}+u}$. Dividing top and bottom by u yields $\frac{-4}{\sqrt{1-0+1}} = \frac{-4}{2} = -2$.

1.11.3 Let $u = -x$: $\lim_{u \rightarrow \infty} \frac{-u+2}{\sqrt{4(-u)^2-(-u)}} =$

$$\lim_{u \rightarrow \infty} \frac{-u+2}{\sqrt{4u^2+u}}. \text{ Dividing top and bottom by } u \text{ yields } \frac{-1+0}{\sqrt{4+0}} = -\frac{1}{2}.$$

1.11.4 Let $u = -x$: $\lim_{u \rightarrow \infty} \frac{\sqrt{a(-u)^2+5(-u)}}{2(-u)+1} =$

$$\lim_{u \rightarrow \infty} \frac{\sqrt{au^2-5u}}{-2u+1}. \text{ Dividing top and bottom by } u \text{ yields } \frac{\sqrt{a}}{-2}.$$

Setting this equal to -2 : $\frac{\sqrt{a}}{-2} = -2 \Rightarrow \sqrt{a} = 4 \Rightarrow a = 16$.

1.11.5 Let $u = -x$: $\lim_{u \rightarrow \infty} \frac{\sqrt{a^2(-u)^2+c(-u)}}{-u} =$

$$\lim_{u \rightarrow \infty} \frac{\sqrt{a^2u^2-cu}}{-u}. \text{ Dividing top and bottom by } u \text{ yields } \frac{\sqrt{a^2}}{-1} = -a \text{ (since } a > 0\text{). The out-}$$

put is always negative because traveling toward $-\infty$ means the denominator x is strictly negative, whereas the principal square root in the numerator is strictly positive.

1.11.6 Since $x < 0$, dividing by x (a negative number) and asserting it is equal to dividing by $\sqrt{x^2}$ (which is always a positive number) introduces a sign error. The true identity is $\sqrt{x^2} = |x| = -x$, meaning $x = -\sqrt{x^2}$ for negative x . Thus, the correct division step is:

$$\frac{\sqrt{16x^2}}{x} = \frac{\sqrt{16x^2}}{-\sqrt{x^2}} = -\sqrt{\frac{16x^2}{x^2}} = -4.$$

1.11.7 (1) For $x > 0$, dividing by $x = \sqrt{x^2}$ yields

$$\lim_{x \rightarrow \infty} \frac{2}{\sqrt{1+1/x^2}} = 2;$$

(2) Let $u = -x$: $\lim_{u \rightarrow \infty} \frac{-2u}{\sqrt{u^2+1}} =$

$$\lim_{u \rightarrow \infty} \frac{-2}{\sqrt{1+1/u^2}} = -2;$$

(3) The two horizontal asymptotes are $y = 2$ (as $x \rightarrow \infty$) and $y = -2$ (as $x \rightarrow -\infty$);(4) Graphically, the curve lives in the first quadrant as it heads right, flattening out at height 2. As it heads left, it crosses into the third quadrant, dipping down and flattening out at height -2 , forming an S-like shape.

Section 1.12

1.12.1 Divide by e^x : $\lim_{x \rightarrow \infty} \frac{3+4e^{-x}}{2-e^{-x}} = \frac{3+0}{2-0} = \frac{3}{2} = 1.5$.1.12.2 Since $e^{-x} = 1/e^x$, as $x \rightarrow \infty$, $e^{-x} \rightarrow 0$.

$$\text{Substituting directly: } \frac{8+2(0)}{4-0} = \frac{8}{4} = 2.$$

1.12.3 Rewrite: $\lim_{x \rightarrow \infty} \frac{4 \cdot 4^x - 5^x}{\frac{4}{5} \cdot 5^x + 3^x}$. Divide top and

$$\text{bottom by } 5^x: \lim_{x \rightarrow \infty} \frac{4(4/5)^x - 1}{1/5 + (3/5)^x}.$$

Since $4/5 < 1$ and $3/5 < 1$, the limit is $\frac{4(0)-1}{1/5+0} = \frac{-1}{1/5} = -5$.

1.12.4 Rewrite: $\lim_{x \rightarrow \infty} \frac{a \cdot 5^x + 2 \cdot 2^x}{\frac{3}{5} \cdot 5^x + b \cdot 2^x}$. Divide by 5^x :
 $\lim_{x \rightarrow \infty} \frac{a + 2(2/5)^x}{3/5 + b(2/5)^x} = \frac{a+0}{3/5+0} = \frac{5a}{3}$. Set equal to 10: $\frac{5a}{3} = 10 \implies 5a = 30 \implies a = 6$.
 The value is completely independent of b !

1.12.5 Divide top and bottom by b^x :
 $\lim_{x \rightarrow \infty} \frac{A(a/b)^x + B}{C(a/b)^x + D}$. Since $a < b \implies a/b < 1$, we know $\lim_{x \rightarrow \infty} (a/b)^x = 0$. Thus, the limit evaluates to $\frac{A(0)+B}{C(0)+D} = \frac{B}{D}$.

1.12.6 (1) Dividing by 2^x yields $\lim_{x \rightarrow \infty} \frac{5(3/2)^x - 1}{4(3/2)^x + 7}$;
 (2) Since $3/2 > 1$, the term $(3/2)^x$ heads to ∞ as $x \rightarrow \infty$. Taking the limit leaves us with $\frac{\infty-1}{\infty+7}$, which is still the indeterminate form $\frac{\infty}{\infty}$ and cannot be evaluated directly; (3) Dividing by the largest base guarantees that all remaining exponential terms have fractional bases less than 1, which decay cleanly to 0 and resolve the indeterminacy.

1.12.7 (1) Since $r > 0$, $e^{-rt} \rightarrow 0$ as $t \rightarrow \infty$. Thus, $\lim_{t \rightarrow \infty} P(t) = \frac{K}{1+A(0)} = K$;
 (2) It represents the carrying capacity—the maximum population size that the environment can sustainably support. It is independent of r and A because over an infinite timeline, resource limits eventually dictate the cap regardless of how fast the population started growing or its initial size;
 (3) Initial population: $P(0) = \frac{5000}{1+9(1)} = 500$. To find t when population is 80% of 5000 (4000): $4000 = \frac{5000}{1+9e^{-0.2t}} \implies 1+9e^{-0.2t} = 1.25 \implies 9e^{-0.2t} = 0.25 \implies e^{-0.2t} = \frac{1}{36} \implies -0.2t = -\ln(36) \implies t = 5\ln(36) \approx 17.9$ years.

Section 1.13

1.13.1 Let $u = -x$: $\lim_{u \rightarrow \infty} \frac{5e^{-u}-2}{3e^{-u}+7} = \frac{5(0)-2}{3(0)+7} = -\frac{2}{7}$.

1.13.2 Let $u = -x$: $\lim_{u \rightarrow \infty} \frac{3 \cdot 2^{-u} + 4 \cdot 3^{-u}}{2 \cdot (1/2)^u - 5 \cdot (1/3)^u}$. Divide top and bottom by $(1/2)^u$ (the dominant term since $1/2 > 1/3$): $\lim_{u \rightarrow \infty} \frac{3+4(2/3)^u}{2-5(2/3)^u} = \frac{3+0}{2-0} = \frac{3}{2} = 1.5$.

1.13.3 Let $u = -x$: $\lim_{u \rightarrow \infty} \frac{4^u + 3^{-u}}{2^u - 5^{-u}}$. Divide by 2^u (highest power of denominator): $\lim_{u \rightarrow \infty} \frac{2^u + (1/6)^u}{1 - (1/10)^u} = \infty$ (since $2^u \rightarrow \infty$ and other terms decay to 0). The limit diverges.

1.13.4 Let $u = -x$: $\lim_{u \rightarrow \infty} \frac{ae^u + 12}{3e^u + be^{-u}}$. Divide top and bottom by e^u : $\lim_{u \rightarrow \infty} \frac{a + 12e^{-u}}{3 + be^{-2u}} = \frac{a+0}{3+0} = \frac{a}{3}$. Setting this to 8 gives $\frac{a}{3} = 8 \implies a = 24$. The value of b has no effect because be^{-u} decays rapidly to 0 as $u \rightarrow \infty$, rendering its coefficient irrelevant.

1.13.5 Let $L_1 = \lim_{x \rightarrow \infty} \frac{a^x}{a^x + b^x}$. Assuming without loss of generality $a < b$, dividing by b^x yields $L_1 = 0$. For the second limit, let $u = -x$: $L_2 = \lim_{u \rightarrow \infty} \frac{a^{-u}}{a^{-u} + b^{-u}} = \lim_{u \rightarrow \infty} \frac{(1/a)^u}{(1/a)^u + (1/b)^u}$. Since $a < b \implies 1/a > 1/b$, the term $(1/a)^u$ is dominant. Dividing by $(1/a)^u$ yields $L_2 = \frac{1}{1+0} = 1$. Thus, $L_1 + L_2 = 0 + 1 = 1$. If $a > b$, $L_1 = 1$ and $L_2 = 0$, which still sums to 1.

1.13.6 (1) Let $u = -x$: $\lim_{u \rightarrow \infty} \frac{e^u + 2}{e^{-u} - 3} = \lim_{u \rightarrow \infty} \frac{e^u + 2}{(1/e)^u - 3}$. Since $e^u \rightarrow \infty$, the numerator grows without bound while the denominator stabilizes at -3 , leading to a limit of $-\infty$;
 (2) The student correctly identified $e^x \rightarrow 0$, but completely ignored the fact that $e^{-x} = e^{-(-u)} = e^u \rightarrow \infty$ grows unboundedly;
 (3) Replacing all terms with 0 fails when some exponents contain signs that trigger infinite growth under the negative infinity limit.

1.13.7 (1) Divide top and bottom by e^x : $\lim_{x \rightarrow \infty} \frac{1 - e^{-2x}}{1 + e^{-2x}} = \frac{1-0}{1+0} = 1$;
 (2) Let $u = -x$: $\lim_{u \rightarrow \infty} \frac{e^{-u} - e^u}{e^{-u} + e^u}$. Divide top and bottom by e^u : $\lim_{u \rightarrow \infty} \frac{e^{-2u} - 1}{e^{-2u} + 1} = \frac{0-1}{0+1} = -1$;
 (3) The limits show that $\tanh(x)$ is bounded between its two horizontal asymptotes, $y = 1$ and $y = -1$;
 (4) The curve behaves symmetrically but saturates at 1 in the positive direction and -1 in the negative direction, making it ideal for modeling physical constraints that have clear maximum and minimum bounds.

Section 1.14

1.14.1 Divide by x : $\lim_{x \rightarrow \infty} \frac{\frac{\ln(x)}{x} - 4}{3 + \frac{2}{x}} = \frac{0-4}{3+0} = -\frac{4}{3}$.

1.14.2 Combine logarithms: $\lim_{x \rightarrow \infty} \ln\left(\frac{3x^2+5}{x^2+1}\right)$. Evaluate the inner limit: $\lim_{x \rightarrow \infty} \frac{3+5/x^2}{1+1/x^2} = 3$. Since the natural logarithm is continuous, the limit is $\ln(3)$.

1.14.3 Divide top and bottom by $\sqrt{x}$:
 $\lim_{x \rightarrow \infty} \frac{\frac{\ln(x)}{2} + 1}{2 - \frac{5}{\sqrt{x}}}$. Since $x^{1/2}$ is a positive power, $\lim_{x \rightarrow \infty} \frac{\ln(x)}{\sqrt{x}} = 0$. The limit evaluates to $\frac{0+1}{2-0} = \frac{1}{2} = 0.5$.

1.14.4 Simplify the top: $a \ln(x^3) = 3a \ln(x)$. The expression is $\lim_{x \rightarrow \infty} \frac{3a \ln(x) + 8x}{b \ln(x) + 2x}$. Dividing top and bottom by x yields $\lim_{x \rightarrow \infty} \frac{3a \frac{\ln(x)}{x} + 8}{b \frac{\ln(x)}{x} + 2} = \frac{3a(0)+8}{b(0)+2} = \frac{8}{2} = 4$. This evaluates to 4 regardless of the values of a and b , meaning a and b can be any real constants!

1.14.5 Convert to natural base: $\log_b(x^k) = \frac{\ln(x^k)}{\ln(b)} = \frac{k}{\ln(b)} \ln(x)$. The limit is $\lim_{x \rightarrow \infty} \left[\frac{k}{c \ln(b)} \right] \frac{\ln(x)}{x^p}$. Since $p > 0$, by Theorem 1.14, Rule 2, $\lim_{x \rightarrow \infty} \frac{\ln(x)}{x^p} = 0$. Multiplying by the constant coefficient yields 0.

1.14.6 (1) The numerator $3x$ is a linear polynomial, while the denominator is a logarithm. Since polynomials grow infinitely faster than logarithms, the numerator dominates the growth, driving the overall fraction to ∞ ;
 (2) Dividing by x yields $\frac{3+2/x}{\ln(x)}$;
 (3) Taking $x \rightarrow \infty$ gives $\frac{3+0}{0} = \frac{3}{0}$. Division by an infinitely small positive value yields positive infinity, confirming the divergence.

1.14.7 (1) $\lim_{h \rightarrow 0} \frac{\ln(\frac{c+h}{c})}{h} = \lim_{h \rightarrow 0} \frac{\ln(1+h/c)}{h}$;
 (2) $\lim_{h \rightarrow 0} \ln \left[\left(1 + \frac{h}{c}\right)^{1/h} \right]$;
 (3) Let $w = h/c \implies h = cw$. As $h \rightarrow 0$, $w \rightarrow 0$. Substituting yields $\lim_{w \rightarrow 0} \ln \left[(1+w)^{1/(cw)} \right] = \ln \left[(\lim_{w \rightarrow 0} (1+w)^{1/w})^{1/c} \right]$;
 (4) Substituting $\lim_{w \rightarrow 0} (1+w)^{1/w} = e$ yields $\ln(e^{1/c}) = \frac{1}{c} \ln(e) = \frac{1}{c}$. This rigorously proves the **derivative of $\ln(x)$ is $\frac{1}{x}$ from first principles!

Section 1.15

1.15.1 Divide every term by e^x : $\lim_{x \rightarrow \infty} \frac{x^5/e^x + 1}{3 - \ln(x)/e^x}$. Since polynomials and logarithms grow infinitely slower than exponentials, this evaluates to $\frac{0+1}{3-0} = \frac{1}{3}$.

1.15.2 Since $e^{-x} \rightarrow 0$ and $\lim_{x \rightarrow \infty} \frac{\ln(x)}{x^2} = 0$, we divide top and bottom by the dominant term x^2 : $\lim_{x \rightarrow \infty} \frac{1+e^{-x}/x^2+3\ln(x)/x^2}{4-5\ln(x)/x^2} = \frac{1+0+0}{4-0} = \frac{1}{4} = 0.25$.

1.15.3 The dominant term of the denominator is $5^{x-1} = 0.2 \cdot 5^x$ (since 5^x grows faster than $e^x \approx 2.718^x$). Dividing top and bottom by 5^x yields $\lim_{x \rightarrow \infty} \frac{2+x^{100}/5^x}{0.2+(e/5)^x+8\ln(x)/5^x} = \frac{2+0}{0.2+0+0} = \frac{2}{0.2} = 10$.

1.15.4 Divide numerator and denominator of $D(t)$ by the dominant term e^t : $\lim_{t \rightarrow \infty} \frac{a+bt^3/e^t+\ln(t)/e^t}{4+2t^2/e^t} = \frac{a+0+0}{4+0} = \frac{a}{4}$. Setting this to the target demand limit gives $\frac{a}{4} = 3 \implies a = 12$. The constant b has no effect because $t^3/e^t \rightarrow 0$ as $t \rightarrow \infty$, completely vanishing from the system in the long run.

1.15.5 Divide top and bottom by $D(x)$: $\lim_{x \rightarrow \infty} \frac{A(x)/D(x)+B(x)/D(x)}{C(x)/D(x)+1}$. Since $A(x) \ll$

$B(x)$ and $B(x) \sim D(x)$, we have $A(x) \ll D(x) \implies \lim A(x)/D(x) = 0$. Similarly, $C(x) \ll D(x) \implies \lim C(x)/D(x) = 0$. Since $\lim B(x)/D(x) = k$, the expression evaluates perfectly to $\frac{0+k}{0+1} = k$.

1.15.6 (1) $\sin(x)$ oscillates continuously between -1 and 1 , meaning $-1 \leq \sin(x) \leq 1$ is globally bounded;
 (2) Dividing top and bottom by e^x gives $\frac{\frac{x^2 \sin(x)}{e^x} + 1}{1 + \frac{\ln(x)}{e^x}}$;
 (3) Since $-1 \leq \sin(x) \leq 1$, multiplying by x^2/e^x yields $-\frac{x^2}{e^x} \leq \frac{x^2 \sin(x)}{e^x} \leq \frac{x^2}{e^x}$. Since $\lim_{x \rightarrow \infty} \pm \frac{x^2}{e^x} = 0$, by the Squeeze Theorem, the middle term must also converge to 0;
 (4) The overall limit evaluates to $\frac{0+1}{1+0} = 1$.

1.15.7 (1) Product expansion is $C \cdot \prod_{m=1}^{n-k} \frac{b}{k+m}$;
 (2) Since $m \geq 1$, we have $k+m > k$, which implies the reciprocal fraction $\frac{1}{k+m} < \frac{1}{k}$. Multiplying both sides by the positive base b preserves the inequality;
 (3) Substituting the bound from part (2) into each of the $n-k$ terms of the product yields the upper bound $C \cdot (\frac{b}{k})^{n-k}$;
 (4) Since $b/k < 1$, the term $(\frac{b}{k})^{n-k} \rightarrow 0$ as $n \rightarrow \infty$. The lower bound is 0. By the Squeeze Theorem, $\lim_{n \rightarrow \infty} \frac{b^n}{n!} = 0$;
 (5) This proves that **factorials grow infinitely faster than exponentials**! Factorials sit at the absolute top of our growth hierarchy: $\ln(x) \ll x^p \ll a^x \ll n!$.

Section 2.1

2.1.1 (1) $m = \frac{14-5}{5-2} = \frac{9}{3} = 3$;
 (2) $m = \frac{-4-4}{3-(-1)} = \frac{-8}{4} = -2$.

2.1.2 The slope of L_1 is $m_1 = \frac{8-2}{3-1} = \frac{6}{2} = 3$.
 (1) Since parallel lines have equal slopes, the slope of L_2 is 3;
 (2) Since perpendicular slopes multiply to -1 , we have $3 \cdot m_3 = -1 \implies m_3 = -\frac{1}{3}$.

2.1.3 Set up the slope equation: $\frac{k-5}{6-(-2)} = -\frac{3}{4} \implies \frac{k-5}{8} = -\frac{3}{4} \implies k-5 = -6 \implies k = -1$.

2.1.4 (1) The slope of segment PR is $m = \frac{15-(-3)}{11-2} = \frac{18}{9} = 2$. Since Q lies on the same line, the slope of PQ must also be 2: $\frac{k-(-3)}{5-2} = 2 \implies \frac{k+3}{3} = 2 \implies k+3 = 6 \implies k = 3$;
 (2) Using slope $m = 2$ and point $(5, 3)$: $y-3 = 2(x-5) \implies y = 2x-7$.

2.1.5 By definition, the slope m between the y -intercept $(0, c)$ and any arbitrary point (x, y) on the line is given by $m = \frac{y-c}{x-0}$. Simplifying this equation yields $m = \frac{y-c}{x} \implies mx = y-c \implies y = mx+c$.

2.1.6 (1) For L_1 , $m = \frac{5-5}{7-2} = \frac{0}{5} = 0$. This line is perfectly horizontal;

(2) For L_2 , the formula yields $m = \frac{6-1}{3-3} = \frac{5}{0}$. Division by zero is undefined;

(3) A slope of 0 means there is zero vertical change as x increases (a flat line). An undefined slope occurs when there is zero horizontal change (x is constant), meaning the line goes straight up and down. Since we cannot divide by 0, the gradient cannot be expressed as a real number.

2.1.7 (1) $s(1) = 1^2 + 1 = 2$ meters, and $s(3) = 3^2 + 1 = 10$ meters;

(2) Average velocity is the slope: $m = \frac{10-2}{3-1} = \frac{8}{2} = 4$ m/s;

(3) For interval $[1, 2]$: $s(2) = 5$, slope $m = \frac{5-2}{2-1} = 3$ m/s. For interval $[1, 1.1]$: $s(1.1) = 2.21$, slope $m = \frac{2.21-2}{1.1-1} = \frac{0.21}{0.1} = 2.1$ m/s;

(4) The average velocity values (4, 3, 2.1, ...) seem to converge toward 2 m/s. This shows that the slope of a curve at a single point (instantaneous rate of change) is the limit of the secant slopes as the interval width shrinks to zero.

Section 2.2

2.2.1 (1) For $x_1 = 2$, $y_1 = 3(4) + 2 = 14$. For $x_2 = 4$, $y_2 = 3(16) + 2 = 50$. The points are $P(2, 14)$ and $Q(4, 50)$;

(2) $m_{\text{sec}} = \frac{50-14}{4-2} = \frac{36}{2} = 18$.

2.2.2 (1) At $t_1 = 2$, $s(2) = 5(4) = 20$. At $t_2 = 5$, $s(5) = 5(25) = 125$. The average speed is $m_{\text{sec}} = \frac{125-20}{5-2} = \frac{105}{3} = 35$ meters per second;

(2) The units are meters per second (m/s).

2.2.3 (1) The coordinate of the neighboring point is $(3+h, (3+h)^2)$;

(2) $m_{\text{sec}} = \frac{(3+h)^2-3^2}{h} = \frac{9+6h+h^2-9}{h} = \frac{6h+h^2}{h} = 6+h$.

2.2.4 (1) $m_{\text{sec}} = \frac{\frac{4}{2+h}-\frac{4}{2}}{h} = \frac{\frac{4-2(2+h)}{2(2+h)}}{h} = \frac{\frac{-2h}{2(2+h)}}{h} = \frac{-2h}{2h(2+h)} = \frac{-1}{2+h}$;

(2) The single fraction is $-\frac{1}{2+h}$; (3) When $h = 0.1$, $m_{\text{sec}} = -\frac{1}{2.1} \approx -0.476$.

2.2.5 $m_{\text{sec}} = \frac{[a(w+h)^2+b(w+h)+c]-[aw^2+bw+c]}{h} = \frac{aw^2+2awh+ah^2+bw+bh+c-aw^2-bw-c}{h} = \frac{2awh+ah^2+bh}{h} = 2aw+b+ah$.

2.2.6 (1) Substituting $h = 0$ gives $m_{\text{sec}} = \frac{f(2)-f(2)}{0} = \frac{0}{0}$;

(2) Division by zero is mathematically illegal. This represents a single point rather than a change;

(3) A line is uniquely defined by two points. If the distance between them is exactly zero, we

only have one point, making it impossible to calculate a unique direction or slope.

2.2.7 (1) $m_{\text{sec}} = \frac{\sin(h)-\sin(0)}{h} = \frac{\sin(h)}{h}$;

(2) For $h = 1.0$: 0.8415; for $h = 0.5$: 0.9589; for $h = 0.1$: 0.9983; for $h = 0.01$: 0.99998; for $h = 0.001$: 0.9999998;

(3) The values clearly converge to 1; (4) $m_{\text{sec}} = \frac{\sin(c+h)-\sin(c)}{h} = \frac{2\sin(h/2)\cos(c+h/2)}{h} = \cos(c+h/2)\frac{\sin(h/2)}{h/2}$. Taking the limit as $h \rightarrow 0$, the first term approaches $\cos(c)$ and the second approaches 1. Thus, the limit is $\cos(c)$.

Section 2.3

2.3.1 $f'(1) = \lim_{h \rightarrow 0} \frac{2(1+h)^2-2(1)^2}{h} = \lim_{h \rightarrow 0} \frac{2(1+2h+h^2)-2}{h} = \lim_{h \rightarrow 0} \frac{4h+2h^2}{h} = \lim_{h \rightarrow 0} (4+2h) = 4$.

2.3.2 (1) $g'(2) = \lim_{h \rightarrow 0} \frac{[(2+h)^2+3(2+h)]-[2^2+3(2)]}{h} = \lim_{h \rightarrow 0} \frac{4+4h+h^2+6+3h-10}{h} = \lim_{h \rightarrow 0} \frac{7h+h^2}{h} = \lim_{h \rightarrow 0} (7+h) = 7$;

(2) The point is $(2, g(2)) = (2, 10)$ and the slope is $m = 7$. Tangent line: $y - 10 = 7(x - 2) \Rightarrow y = 7x - 4$.

2.3.3 (1) $\frac{\frac{1}{2+h}-\frac{1}{2}}{h}$;

(2) $\frac{\frac{2-(2+h)}{2(2+h)}}{h} = \frac{-h}{2h(2+h)} = -\frac{1}{2(2+h)}$;

(3) $h'(2) = \lim_{h \rightarrow 0} -\frac{1}{2(2+h)} = -\frac{1}{4}$.

2.3.4 (1) $\lim_{h \rightarrow 0} \frac{\sqrt{4+h}-2}{h}$;

(2) Multiply by the conjugate: $\lim_{h \rightarrow 0} \frac{(\sqrt{4+h}-2)(\sqrt{4+h}+2)}{h(\sqrt{4+h}+2)} = \lim_{h \rightarrow 0} \frac{(4+h)-4}{h(\sqrt{4+h}+2)} = \lim_{h \rightarrow 0} \frac{h}{h(\sqrt{4+h}+2)} = \lim_{h \rightarrow 0} \frac{1}{\sqrt{4+h}+2}$;

(3) Tangent slope: $f'(4) = \frac{1}{\sqrt{4+2}} = \frac{1}{4}$.

2.3.5 $f'(c) = \lim_{h \rightarrow 0} \frac{[m(c+h)+k]-[mc+k]}{h} = \lim_{h \rightarrow 0} \frac{mc+mh+k-mc-k}{h} = \lim_{h \rightarrow 0} \frac{mh}{h} = m$.

This makes sense because $f(x)$ is a straight line with a constant slope of m everywhere. Its rate of change at any point is simply the slope of the line itself.

2.3.6 (1) $\frac{|h|}{h}$;

(2) As $h \rightarrow 0^-$, h is negative, so $|h| = -h$. The limit is $\lim_{h \rightarrow 0^-} \frac{-h}{h} = -1$;

(3) As $h \rightarrow 0^+$, h is positive, so $|h| = h$. The limit is $\lim_{h \rightarrow 0^+} \frac{h}{h} = 1$;

(4) Since the left-hand limit (-1) and right-hand limit (1) do not agree, the limit of the difference quotient does not exist at $x = 0$. Hence, the function is not differentiable there.

2.3.7 (1) Since $-1 \leq \sin(1/x) \leq 1$, we have $-|x| \leq x \sin(1/x) \leq |x|$. Since $\lim_{x \rightarrow 0} \pm|x| = 0$, by the Squeeze Theorem, $\lim_{x \rightarrow 0} f(x) = 0 = f(0)$, making it continuous;

$$(2) f'(0) = \lim_{h \rightarrow 0} \frac{h \sin(1/h) - 0}{h} = \lim_{h \rightarrow 0} \sin(1/h);$$

(3) As $h \rightarrow 0$, $1/h$ grows infinitely, causing $\sin(1/h)$ to oscillate between -1 and 1 infinitely fast without settling on any single value. Thus, the limit does not exist;

(4) Yes. This proves that continuity is a necessary but not sufficient condition for differentiability. A curve can be unbroken, yet have sharp, oscillating behavior that prevents a defined tangent slope.

Section 2.4

$$2.4.1 \quad (1) f'(x) = 8x^7;$$

$$(2) g'(x) = 15x^2 - 4.$$

$$2.4.2 \quad (1) \frac{dy}{dx} = 6x - 5;$$

(2) At $x = 2$, $y = 3(4) - 5(2) + 2 = 4$. The slope is $m = 6(2) - 5 = 7$. Tangent line: $y - 4 = 7(x - 2) \Rightarrow y = 7x - 10$.

$$2.4.3 \quad \text{Rewrite: } h(x) = 2x^{-4} - 6x^{-1/3}. \text{ Differentiate: } h'(x) = -8x^{-5} + 2x^{-4/3}. \text{ Restructuring: } h'(x) = -\frac{8}{x^5} + \frac{2}{x^{4/3}} \text{ (or } -\frac{8}{x^5} + \frac{2}{x^{4/3}}).$$

$$2.4.4 \quad \text{Horizontal tangent implies } \frac{dy}{dx} = 0. \quad \frac{dy}{dx} = x^2 - 4x - 3 = 0. \text{ Solve using quadratic formula: } x = \frac{4 \pm \sqrt{16 - 4(1)(-3)}}{2} = \frac{4 \pm \sqrt{28}}{2} = 2 \pm \sqrt{7}. \text{ Substituting back gives the coordinates: } (2 + \sqrt{7}, f(2 + \sqrt{7})) \text{ and } (2 - \sqrt{7}, f(2 - \sqrt{7})).$$

$$2.4.5 \quad \text{Rewrite } f(x) = x^{-n}. \text{ Applying the Power Rule yields } f'(x) = -n \cdot x^{-n-1}. \text{ Rewriting with a positive exponent gives } f'(x) = -n \cdot \frac{1}{x^{n+1}} = -\frac{n}{x^{n+1}}.$$

$$2.4.6 \quad (1) \text{ Only valid for } f_1(x) = x^5, \text{ since its base is a variable and exponent is a constant. } f'_1(x) = 5x^4;$$

(2) For $f_2(x)$, the variable is in the exponent (exponential function). For $f_3(x)$, the variable is in both the base and the exponent. The Power Rule strictly requires a constant exponent.

$$2.4.7 \quad (1) f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^n - x^n}{h};$$

$$(2) \text{ Substituting: } \lim_{h \rightarrow 0} \frac{(x^n + nx^{n-1}h + \frac{n(n-1)}{2}x^{n-2}h^2 + \dots + h^n) - x^n}{h};$$

$$(3) \text{ Canceling } x^n \text{ and dividing by } h: \lim_{h \rightarrow 0} (nx^{n-1} + \frac{n(n-1)}{2}x^{n-2}h + \dots + h^{n-1});$$

(4) As $h \rightarrow 0$, every term containing h vanishes to 0, leaving only nx^{n-1} . This completes the proof.

Section 2.5

$$2.5.1 \quad (1) f'(x) = 4e^x;$$

$$(2) g'(x) = 7x \ln(7).$$

$$2.5.2 \quad \text{Apply rules term-by-term: } h'(x) = 12x^3 - 2e^x + 5 \cdot 10^x \ln(10).$$

$$2.5.3 \quad \text{Rewrite: } f(x) = e^2 \cdot e^x + \frac{1}{3} \cdot 3^x. \text{ Differentiate: } f'(x) = e^2 \cdot e^x + \frac{1}{3} \cdot 3^x \ln(3) = e^{x+2} + 3^{x-1} \ln(3).$$

$$2.5.4 \quad (1) \frac{dy}{dx} = 3e^x - 4;$$

(2) Parallel means equal slopes. Set $3e^x - 4 = 8 \Rightarrow 3e^x = 12 \Rightarrow e^x = 4 \Rightarrow x = \ln(4)$. Substitute back to find y : $y = 3e^{\ln(4)} - 4 \ln(4) = 3(4) - 4 \ln(4) = 12 - 4 \ln(4)$. Coordinates are $(\ln(4), 12 - 4 \ln(4))$.

$$2.5.5 \quad \text{Let } f(x) = e^{kx} \text{ where } k = \ln(b). \text{ Differentiating gives } f'(x) = ke^{kx}. \text{ Substituting back } k = \ln(b) \text{ yields } f'(x) = \ln(b)e^{x \ln(b)} = \ln(b)b^x = b^x \ln(b).$$

$$2.5.6 \quad \text{The term } e^2 \text{ is a constant real number (approximately 7.389). It contains no variables. The rate of change of any flat, unchanging constant is 0. Thus, } f'(x) = 0.$$

$$2.5.7 \quad (1) f'(x) = \lim_{h \rightarrow 0} \frac{e^x \cdot e^h - e^x}{h} = \lim_{h \rightarrow 0} \frac{e^x(e^h - 1)}{h} = e^x \lim_{h \rightarrow 0} \frac{e^h - 1}{h};$$

(2) Table values for h : $0.1 \rightarrow 1.05171$; $0.01 \rightarrow 1.00502$; $0.001 \rightarrow 1.00050$; $0.0001 \rightarrow 1.00005$;

(3) The sequence suggests $L = 1$;

(4) Let $h = 1/n$. As $h \rightarrow 0$, $n \rightarrow \infty$. Substitute $e = \lim_{n \rightarrow \infty} (1 + 1/n)^n$: $\lim_{h \rightarrow 0} \frac{e^h - 1}{h} = \lim_{h \rightarrow 0} \frac{[(1+h)^{1/h}]^h - 1}{h} = \lim_{h \rightarrow 0} \frac{(1+h) - 1}{h} = \lim_{h \rightarrow 0} \frac{h}{h} = 1$. This completes the proof.

Section 2.6

$$2.6.1 \quad (1) f'(x) = \frac{5}{x};$$

$$(2) g'(x) = \frac{1}{x \ln(5)}.$$

$$2.6.2 \quad \text{Rewrite } h(x) = x^{-3} + 3e^x - 2 \ln(x). \text{ Differentiating yields } h'(x) = -3x^{-4} + 3e^x - \frac{2}{x} = -\frac{3}{x^4} + 3e^x - \frac{2}{x}.$$

$$2.6.3 \quad \text{Expand first: } f(x) = \ln(5) + \ln(x^3) = \ln(5) + 3 \ln(x). \text{ Since } \ln(5) \text{ is a constant, its derivative is 0. Thus, } f'(x) = 0 + 3 \cdot \frac{1}{x} = \frac{3}{x}.$$

$$2.6.4 \quad (1) \text{ Derivative is } y' = 1/x. \text{ At } x = 1, y = \ln(1) = 0, \text{ and slope is } m = 1/1 = 1. \text{ Tangent line: } y - 0 = 1(x - 1) \Rightarrow y = x - 1;$$

(2) Perpendicular tangent must have slope $m_p = -1/(-2) = 1/2$. Set $1/x = 1/2 \Rightarrow x = 2$. Substitute back: $y = \ln(2)$. The coordinates are $(2, \ln(2))$.

$$2.6.5 \quad \text{Since } \ln(b) \text{ is a constant, the term } \frac{1}{\ln(b)} \text{ is also a constant multiplier. Applying the constant multiple rule: } \frac{d}{dx} (\log_b(x)) = \frac{d}{dx} \left(\frac{1}{\ln(b)} \ln(x) \right) = \frac{1}{\ln(b)} \frac{d}{dx} (\ln(x)) = \frac{1}{\ln(b)} \cdot \frac{1}{x} = \frac{1}{x \ln(b)}.$$

- 2.6.6** (1) For $\ln(-x)$ to be defined, the argument must be positive, which means $-x > 0 \implies x < 0$. The domain is all negative real numbers;
 (2) The claim is wrong because when x is negative, $-x$ is positive. Thus, the function is perfectly defined and continuous for all negative inputs;
 (3) $f'(-5) = \frac{1}{-5} = -0.2$.

- 2.6.7** (1) $f(x) = \begin{cases} \ln(x) & \text{if } x > 0; \\ \ln(-x) & \text{if } x < 0; \end{cases}$
 (2) For $x > 0$, $f'(x) = \frac{1}{x}$; (3) For $x < 0$, $f'(x) = \frac{1}{x}$;
 (4) The derivative of $\ln|x|$ is $\frac{1}{x}$ for all $x \neq 0$. This unified rule proves that the reciprocal function is the global rate of change of the absolute value logarithm across both positive and negative domains.

Section 2.7

- 2.7.1** (1) $f'(x) = 12x^3 - 10x$;
 (2) $g'(x) = 2e^x + \frac{6}{x} + \frac{1}{x^2}$ (since $-1/x = -x^{-1} \implies \frac{d}{dx}(-x^{-1}) = x^{-2} = 1/x^2$).
- 2.7.2** (1) $\frac{dy}{dx} = 2x - 3e^x$;
 (2) At $x = 0$, $y = 0 - 3(1) + 4 = 1$. The slope is $m = 2(0) - 3(1) = -3$. Tangent line equation: $y - 1 = -3(x - 0) \implies y = -3x + 1$.
- 2.7.3** Split the fraction first: $h(x) = \frac{2x^3}{x} - \frac{5x^2}{x} + \frac{3}{x} = 2x^2 - 5x + 3x^{-1}$. Differentiating yields $h'(x) = 4x - 5 - 3x^{-2} = 4x - 5 - \frac{3}{x^2}$.
- 2.7.4** (1) $\frac{dy}{dx} = 3ax^2 + 2bx - 6$;
 (2) Since $\frac{dy}{dx} = 0$ at $x = 1$: $3a + 2b - 6 = 0 \implies 3a + 2b = 6$. Since $\frac{dy}{dx} = 0$ at $x = -2$: $12a - 4b - 6 = 0 \implies 12a - 4b = 6 \implies 6a - 2b = 3$. Adding the two systems: $9a = 9 \implies a = 1$. Substituting $a = 1$ back: $3(1) + 2b = 6 \implies 2b = 3 \implies b = 1.5$.
- 2.7.5** $S'(x) = \lim_{h \rightarrow 0} \frac{[f(x+h)+g(x+h)] - [f(x)+g(x)]}{h} = \lim_{h \rightarrow 0} \left[\frac{f(x+h)-f(x)}{h} + \frac{g(x+h)-g(x)}{h} \right]$. Applying the sum law of limits, this splits into $\lim_{h \rightarrow 0} \frac{f(x+h)-f(x)}{h} + \lim_{h \rightarrow 0} \frac{g(x+h)-g(x)}{h} = f'(x) + g'(x)$.
- 2.7.6** (1) Linearity does not apply to products of functions. The derivative of a product is absolutely NOT the product of the derivatives;
 (2) Using the Product Rule: $f'(x) = x^2 \cdot \frac{1}{x} + \ln(x) \cdot 2x = x + 2x \ln(x)$. Evaluating at $x = e$ gives $f'(e) = e + 2e \ln(e) = 3e \approx 8.15$, which is completely different from 2.
- 2.7.7** (1) Since $\frac{d}{dx}(a \cdot f(x) + b \cdot g(x)) = a \frac{d}{dx}(f(x)) + b \frac{d}{dx}(g(x))$, the derivative operator preserves vector addition and scalar

multiplication, fitting the definition of a linear transformation;
 (2) $D(e^{kx}) = ke^{kx}$. Since the output is a constant k times the original function, e^{kx} is an eigenfunction with eigenvalue k ;
 (3) The kernel consists of all functions $f(x)$ where $f'(x) = 0$. This represents the set of all constant functions, $f(x) = C$.

Section 2.8

- 2.8.1** (1) $f'(x) = x^5 e^x + e^x(5x^4) = x^4 e^x(x + 5)$;
 (2) $g'(x) = (3x^2 - 2x) \cdot \frac{1}{x} + \ln(x) \cdot (6x - 2) = 3x - 2 + (6x - 2) \ln(x)$.
- 2.8.2** (1) $\frac{dy}{dx} = x \cdot 2^x \ln(2) + 2^x \cdot 1 = 2^x(x \ln(2) + 1)$;
 (2) At $x = 1$, $y = 1 \cdot 2^1 = 2$. The slope is $m = 2^1(1 \ln(2) + 1) = 2 \ln(2) + 2$. Tangent line: $y - 2 = (2 \ln(2) + 2)(x - 1) \implies y = (2 \ln(2) + 2)x - 2 \ln(2)$.
- 2.8.3** Rewrite $h(x) = x^{1/2} \ln(x)$. Differentiating: $h'(x) = x^{1/2} \cdot \frac{1}{x} + \ln(x) \cdot \frac{1}{2} x^{-1/2} = x^{-1/2} + \frac{\ln(x)}{2\sqrt{x}} = \frac{1}{\sqrt{x}} + \frac{\ln(x)}{2\sqrt{x}} = \frac{2 + \ln(x)}{2\sqrt{x}}$.
- 2.8.4** (1) $\frac{dy}{dx} = (x^2 - 5x + 4)e^x + e^x(2x - 5) = e^x(x^2 - 3x - 1)$;
 (2) Set $\frac{dy}{dx} = 0$. Since $e^x \neq 0$, we solve $x^2 - 3x - 1 = 0 \implies x = \frac{3 \pm \sqrt{13}}{2}$. Substituting back yields the exact coordinates.
- 2.8.5** Let $P = (uv) \cdot w$. Applying the Product Rule: $P' = (uv)' \cdot w + (uv) \cdot w'$. Expanding $(uv)'$ yields $(u'v + uv')w + uvw' = u'vw + uv'w + uvw'$.
- 2.8.6** (1) Expanded: $f(x) = x^5 + 5x^3 \implies f'(x) = 5x^4 + 15x^2$;
 (2) Product Rule: $f'(x) = x^2(3x^2 + 5) + (x^3 + 5x)(2x) = 3x^4 + 5x^2 + 2x^4 + 10x^2 = 5x^4 + 15x^2$;
 (3) Both are equal. Expanding first simplifies the expression to basic power terms, avoiding the multi-step product expansion and grouping.
- 2.8.7** (1) $\lim_{h \rightarrow 0} \frac{u(x+h)v(x+h) - u(x)v(x)}{h} = \lim_{h \rightarrow 0} \frac{u(x+h)v(x+h) - u(x+h)v(x) + u(x+h)v(x) - u(x)v(x)}{h}$;
 (2) Factor out common terms to get $\lim_{h \rightarrow 0} \left[u(x+h) \frac{v(x+h)-v(x)}{h} + v(x) \frac{u(x+h)-u(x)}{h} \right]$;
 (3) Distribute limits: $\lim_{h \rightarrow 0} u(x+h) \cdot \lim_{h \rightarrow 0} \frac{v(x+h)-v(x)}{h} + \lim_{h \rightarrow 0} v(x) \cdot \lim_{h \rightarrow 0} \frac{u(x+h)-u(x)}{h}$;
 (4) Since $u(x)$ is differentiable, it is continuous, meaning $\lim_{h \rightarrow 0} u(x+h) = u(x)$. Thus, the terms simplify to $u(x)v'(x) + v(x)u'(x)$.

Section 2.9

- 2.9.1** (1) $f'(x) = \frac{(2x+3)(5)-(5x)(2)}{(2x+3)^2} = \frac{10x+15-10x}{(2x+3)^2} = \frac{15}{(2x+3)^2}$;
 (2) $g'(x) = \frac{(x-1)(2x)-(x^2+3)(1)}{(x-1)^2} = \frac{2x^2-2x-x^2-3}{(x-1)^2} = \frac{x^2-2x-3}{(x-1)^2}$.

- 2.9.2** (1) $\frac{dy}{dx} = \frac{x(e^x) - e^x(1)}{x^2} = \frac{e^x(x-1)}{x^2}$;
 (2) At $x = 1$, $y = e^1/1 = e$. The slope is $m = \frac{e^1(1-1)}{1^2} = 0$. Tangent line: $y - e = 0(x - 1) \Rightarrow y = e$ (a horizontal line).

2.9.3
$$h'(x) = \frac{x^2 \cdot \frac{1}{x} - \ln(x) \cdot (2x)}{(x^2)^2} = \frac{x - 2x \ln(x)}{x^4} = \frac{x(1 - 2 \ln(x))}{x^4} = \frac{1 - 2 \ln(x)}{x^3}.$$

- 2.9.4** (1) $\frac{dy}{dx} = \frac{(x-2)(2x-4) - (x^2-4x)(1)}{(x-2)^2} = \frac{2x^2-8x+8-x^2+4x}{(x-2)^2} = \frac{x^2-4x+8}{(x-2)^2};$
 (2) Factoring the top: $x^2 - 4x + 8 = (x-2)^2 + 4$. The derivative is $1 + \frac{4}{(x-2)^2}$. Since $y = \frac{x(x-4)}{x-2}$, the curve behaves like a linear slant line with a hyperbola-like bend, meaning it has no turning points since the derivative is always positive ($1 + \text{positive} > 0$).

- 2.9.5** Let $u(x) = 1 \Rightarrow u'(x) = 0$. Applying the Quotient Rule to $\frac{1}{g(x)}$: $\frac{d}{dx} \left(\frac{1}{g(x)} \right) = \frac{g(x) \cdot 0 - 1 \cdot g'(x)}{[g(x)]^2} = -\frac{g'(x)}{[g(x)]^2}.$

- 2.9.6** (1) $f'(x) = \frac{x^2(12x^3-2) - (3x^4-2x+1)(2x)}{(x^2)^2} = \frac{12x^5-2x^2-6x^5+4x^2-2x}{x^4} = \frac{6x^5+2x^2-2x}{x^4} = \frac{6x^4+2x-2}{x^3};$
 (2) Differentiate split form: $f'(x) = 6x + 2x^{-2} - 2x^{-3} = 6x + \frac{2}{x^2} - \frac{2}{x^3};$
 (3) Both are equivalent. The term-by-term method bypasses the complex product expansions, avoiding the algebraic fraction mistakes common with the Quotient Rule.

- 2.9.7** (1) $f(x) = u(x) \cdot [v(x)]^{-1};$
 (2) By Product Rule: $f'(x) = u'(x)[v(x)]^{-1} + u(x) \cdot \frac{d}{dx}([v(x)]^{-1});$
 (3) By Chain/Power Rule: $\frac{d}{dx}([v(x)]^{-1}) = -1[v(x)]^{-2} \cdot v'(x)$. So, $f'(x) = \frac{u'(x)}{v(x)} - \frac{u(x)v'(x)}{[v(x)]^2};$
 (4) Finding a common denominator: $f'(x) = \frac{u'(x)v(x)}{[v(x)]^2} - \frac{u(x)v'(x)}{[v(x)]^2} = \frac{v(x)u'(x) - u(x)v'(x)}{[v(x)]^2}$. This completes the proof.

Section 2.10

- 2.10.1** (1) $f'(x) = 4 \cos(x) - 5(-\sin(x)) = 4 \cos(x) + 5 \sin(x);$
 (2) Using Product Rule on $g(x)$: $g'(x) = e^x(-\sin(x)) + \cos(x)(e^x) = e^x(\cos(x) - \sin(x)).$
- 2.10.2** (1) $\frac{dy}{dx} = \cos(x) - \sin(x);$ (2) At $x = \frac{\pi}{4}$, $y = \sin(\frac{\pi}{4}) + \cos(\frac{\pi}{4}) = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} = \sqrt{2}$. The slope is $m = \cos(\frac{\pi}{4}) - \sin(\frac{\pi}{4}) = \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} = 0$. Tangent line: $y - \sqrt{2} = 0(x - \frac{\pi}{4}) \Rightarrow y = \sqrt{2}.$

2.10.3 Apply Quotient Rule to $\frac{\sin(x)}{\cos(x)}$: $\frac{d}{dx}(\tan(x)) = \frac{\cos(x) \cos(x) - \sin(x)(-\sin(x))}{\cos^2(x)} = \frac{\cos^2(x) + \sin^2(x)}{\cos^2(x)} = \frac{1}{\cos^2(x)} = \sec^2(x).$

- 2.10.4** (1) $\frac{dy}{dx} = \frac{e^x \cos(x) - \sin(x)e^x}{(e^x)^2} = \frac{e^x(\cos(x) - \sin(x))}{e^{2x}} = \frac{\cos(x) - \sin(x)}{e^x};$
 (2) Set $\frac{dy}{dx} = 0 \Rightarrow \cos(x) - \sin(x) = 0 \Rightarrow \sin(x) = \cos(x) \Rightarrow \tan(x) = 1$. In the interval $[0, \pi]$, the only solution is $x = \frac{\pi}{4}$. Substituting back: $y = \frac{\sin(\pi/4)}{e^{\pi/4}} = \frac{\sqrt{2}}{2e^{\pi/4}}$. Coordinates: $(\frac{\pi}{4}, \frac{\sqrt{2}}{2e^{\pi/4}}).$

- 2.10.5** (1) $\frac{d}{dx} \left(\frac{1}{\cos(x)} \right) = -\frac{-\sin(x)}{\cos^2(x)} = \frac{1}{\cos(x)} \cdot \frac{\sin(x)}{\cos(x)} = \sec(x) \tan(x);$
 (2) $\frac{d}{dx} \left(\frac{1}{\sin(x)} \right) = -\frac{\cos(x)}{\sin^2(x)} = -\frac{1}{\sin(x)} \cdot \frac{\cos(x)}{\sin(x)} = -\csc(x) \cot(x);$
 (3) $\frac{d}{dx} \left(\frac{\cos(x)}{\sin(x)} \right) = \frac{\sin(x)(-\sin(x)) - \cos(x) \cos(x)}{\sin^2(x)} = \frac{-(\sin^2(x) + \cos^2(x))}{\sin^2(x)} = -\frac{1}{\sin^2(x)} = -\csc^2(x).$

- 2.10.6** (1) $\frac{d}{dx} \left(\sin \left(\frac{\pi}{180} D \right) \right) = \frac{\pi}{180} \cos \left(\frac{\pi}{180} D \right);$
 (2) This shows that in degree mode, a scaling factor of $\frac{\pi}{180} \approx 0.0175$ appears, so the derivative is not simply cosine;
 (3) Radians are pure numbers based on arc length ratios, whereas degrees are an arbitrary subdivision of a circle. Differentiating trigonometric functions cleanly without extra scale factors strictly requires radian measurements.

- 2.10.7** (1) $\lim_{h \rightarrow 0} \frac{\sin(x) \cos(h) + \cos(x) \sin(h) - \sin(x)}{h};$
 (2) Factor $\sin(x)$ out of the first and third terms: $\lim_{h \rightarrow 0} \left[\sin(x) \frac{\cos(h)-1}{h} + \cos(x) \frac{\sin(h)}{h} \right];$
 (3) Substitute the limit values: $\sin(x) \cdot 0 + \cos(x) \cdot 1 = \cos(x)$. This completes the proof.

Section 2.11

- 2.11.1** (1) Outer is u^4 , inner is $2x^3 - 5$. $f'(x) = 4(2x^3 - 5)^3 \cdot (6x^2) = 24x^2(2x^3 - 5)^3;$
 (2) Outer is e^u , inner is $3x^2$. $g'(x) = e^{3x^2} \cdot (6x) = 6xe^{3x^2}.$

- 2.11.2** (1) Outer is $\sin(u)$, inner is $2x$. $\frac{dy}{dx} = 2 \cos(2x);$
 (2) Set $2 \cos(2x) = 0 \Rightarrow \cos(2x) = 0$. For $x \in [0, \pi]$, we have $2x \in [0, 2\pi]$. Thus $2x = \frac{\pi}{2}$ or $2x = \frac{3\pi}{2} \Rightarrow x = \frac{\pi}{4}$ or $x = \frac{3\pi}{4}$. Coordinates are $(\frac{\pi}{4}, 1)$ and $(\frac{3\pi}{4}, -1).$

- 2.11.3** Rewrite $h(x) = (x^2 + 4)^{1/2}$. Applying Chain Rule: $h'(x) = \frac{1}{2}(x^2 + 4)^{-1/2} \cdot (2x) = \frac{2x}{2\sqrt{x^2 + 4}} = \frac{x}{\sqrt{x^2 + 4}}.$

- 2.11.4** (1) Outer is e^u , inner is $-x^2 + 4x$. $\frac{dy}{dx} = (-2x + 4)e^{-x^2 + 4x};$

- (2) Set $\frac{dy}{dx} = 0$. Since $e^{-x^2+4x} \neq 0$, solve $-2x + 4 = 0 \Rightarrow x = 2$. Point is $(2, e^4)$;
 (3) At $x = 1$, $y = e^3$. At $x = 3$, $y = e^3$. Since both adjacent values are smaller than e^4 , the stationary point is a local maximum.

2.11.5 Let $u = g(x) \Rightarrow f(u) = u^n$. By the Chain Rule, $\frac{df}{dx} = \frac{df}{du} \cdot \frac{du}{dx} = nu^{n-1} \cdot g'(x)$. Substituting $u = g(x)$ yields $f'(x) = n[g(x)]^{n-1} \cdot g'(x)$.

2.11.6 (1) $f(x) = (\sin(x))^2$. Outer is u^2 , inner is $\sin(x)$. $f'(x) = 2\sin(x) \cdot \cos(x) = \sin(2x)$;
 (2) $g(x) = \sin(x^2)$. Outer is $\sin(u)$, inner is x^2 . $g'(x) = \cos(x^2) \cdot (2x) = 2x \cos(x^2)$;
 (3) Swapping the inner and outer functions yields completely different algebraic terms. Failing to parse parentheses correctly breaks the composition order, ruining the derivatives.

2.11.7 (1) Let $y = e^u$ and $u = u(x)$. $\frac{dy}{dx} = e^u \cdot u'(x) = u'(x)e^{u(x)}$;
 (2) Let $y = \ln|u|$ and $u = u(x)$. $\frac{dy}{dx} = \frac{1}{u} \cdot u'(x) = \frac{u'(x)}{u(x)}$;
 (3) Let $y = e^w$ where $w = \sin(x^2)$. From part (1), $y' = w'e^w$. Now let $w = \sin(u)$ where $u = x^2$. Then $w' = \cos(u) \cdot (2x) = 2x \cos(x^2)$. Combining the chain yields $y' = 2x \cos(x^2)e^{\sin(x^2)}$.

Section 2.12

2.12.1 (1) $f'(x) = 10x^4 - 12x^2 + 7 \Rightarrow f''(x) = 40x^3 - 24x$;
 (2) $g(x) = 4x^{-2} \Rightarrow g'(x) = -8x^{-3} \Rightarrow g''(x) = 24x^{-4} = \frac{24}{x^4}$.

2.12.2 (1) Velocity $\frac{dy}{dx} = 2\cos(2x)$, Acceleration $\frac{d^2y}{dx^2} = -4\sin(2x)$;
 (2) Substitute into the differential equation: $\frac{d^2y}{dx^2} + 4y = -4\sin(2x) + 4(\sin(2x)) = 0$. This holds true.

2.12.3 First derivative: $f'(x) = (x^2 + 2x)e^x$ (from Section 2.8). Second derivative: $f''(x) = (x^2 + 2x)e^x + e^x(2x + 2) = e^x(x^2 + 4x + 2)$.

2.12.4 (1) $\frac{dy}{dx} = 6x^2 - 18x + 12$, $\frac{d^2y}{dx^2} = 12x - 18$;
 (2) Set $\frac{d^2y}{dx^2} = 0 \Rightarrow 12x - 18 = 0 \Rightarrow x = 1.5$. Substituting $x = 1.5$ back into the original curve yields $y = 2(1.5)^3 - 9(1.5)^2 + 12(1.5) - 5 = -0.5$. The inflection point is $(1.5, -0.5)$.

2.12.5 Let $y = (uv)' = u'v + uv'$. Differentiating again: $y'' = \frac{d}{dx}(u'v) + \frac{d}{dx}(uv')$. Applying the Product Rule to each piece: $\frac{d}{dx}(u'v) = u''v + u'v'$, and $\frac{d}{dx}(uv') = u'v' + uv''$. Combining these terms: $y'' = u''v + u'v' + u'v' + uv'' = u''v + 2u'v' + uv''$.

2.12.6 (1) $\frac{d^2y}{dx^2}$ is the rate of change of the slope (second derivative). $\left(\frac{dy}{dx}\right)^2$ is the square of the first derivative value;
 (2) Actual second derivative: $y' = 3x^2 \Rightarrow y'' = 6x$. This is linear, whereas the student's result $9x^4$ is a quartic polynomial;
 (3) The position of exponents dramatically alters the mathematical operations. Placing the index inside the differential term indicates repeated differentiation, while placing it outside indicates algebraic power scaling.

2.12.7 (1) $f'(x) = 2x - 4 = 0 \Rightarrow x = 2$. Point is $(2, 1)$;
 (2) $f''(x) = 2$. Since $f''(2) = 2 > 0$, the curve is concave up at $x = 2$, which physically forces the stationary point to be the lowest point (local minimum);
 (3) $g'(x) = 12 - 3x^2 = 0 \Rightarrow 3x^2 = 12 \Rightarrow x = \pm 2$. Stationary points are $(2, 16)$ and $(-2, -16)$;
 (4) $g''(x) = -6x$. For $x = 2$: $g''(2) = -12 < 0 \Rightarrow$ local maximum. For $x = -2$: $g''(-2) = 12 > 0 \Rightarrow$ local minimum.

Section 2.13

2.13.1 (1) Differentiating gives $4x + 6y\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{4x}{6y} = -\frac{2x}{3y}$;
 (2) At $(1, 2)$, the slope is $m = -\frac{2(1)}{3(2)} = -\frac{1}{3}$.

2.13.2 (1) $2x + \left(x\frac{dy}{dx} + y\right) + 2y\frac{dy}{dx} = 0$;
 (2) Rearranging: $(x + 2y)\frac{dy}{dx} = -2x - y \Rightarrow \frac{dy}{dx} = -\frac{2x+y}{x+2y}$;
 (3) At $(1, 2)$, the slope is $m = -\frac{2(1)+2}{1+2(2)} = -\frac{4}{5}$. Tangent line: $y - 2 = -\frac{4}{5}(x - 1) \Rightarrow y = -\frac{4}{5}x + \frac{14}{5}$ (or $4x + 5y = 14$).

2.13.3 Differentiate term-by-term: $\frac{1}{y}\frac{dy}{dx} + \left(2xy + x^2\frac{dy}{dx}\right) = 0$. Group terms: $\left(\frac{1}{y} + x^2\right)\frac{dy}{dx} = -2xy \Rightarrow \left(\frac{1+x^2y}{y}\right)\frac{dy}{dx} = -2xy \Rightarrow \frac{dy}{dx} = -\frac{2xy^2}{1+x^2y}$.

2.13.4 (1) $2y\frac{dy}{dx} - 4x = 0 \Rightarrow \frac{dy}{dx} = \frac{2x}{y}$;
 (2) A horizontal tangent requires the numerator to be zero: $2x = 0 \Rightarrow x = 0$. Substituting $x = 0$ into the original equation gives $y^2 = 8 \Rightarrow y = \pm\sqrt{8}$. However, if $x = 0$, then $\frac{dy}{dx} = \frac{0}{y} = 0$. This actually represents turning points at $(0, \pm\sqrt{8})$, so horizontal tangents *do* exist (correction of the premise: if $y \neq 0$, horizontal tangents exist at those vertices);
 (3) A vertical tangent occurs when the denominator is zero, $y = 0$. Substituting $y = 0$ into

the curve gives $-2x^2 = 8 \implies x^2 = -4$, which has no real solutions. Thus, the hyperbola has no vertical tangents.

2.13.5 Differentiating $f(y) = x$ implicitly with respect to x : $f'(y) \cdot \frac{dy}{dx} = 1 \implies \frac{dy}{dx} = \frac{1}{f'(y)}$. For $y = \arcsin(x) \implies \sin(y) = x$. Differentiating implicitly: $\cos(y) \frac{dy}{dx} = 1 \implies \frac{dy}{dx} = \frac{1}{\cos(y)}$. Since $\sin^2(y) + \cos^2(y) = 1$, we have $\cos(y) = \sqrt{1 - \sin^2(y)} = \sqrt{1 - x^2}$ (for the principal domain). Substituting this in yields $\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}$.

2.13.6 (1) $y' = \frac{1}{2}(25 - x^2)^{-1/2} \cdot (-2x) = -\frac{x}{\sqrt{25-x^2}}$;
 (2) Substituting $y = \sqrt{25 - x^2}$ into $-\frac{x}{y}$ yields $-\frac{x}{\sqrt{25-x^2}}$;
 (3) Both are identical. The implicit method bypasses fractional index manipulation, product expansions, and nested square roots, creating a direct, elegant algebraic route to the gradient.

2.13.7 (1) $\frac{dy}{dx} = -\frac{x}{y}$;
 (2) $\frac{d^2y}{dx^2} = -\frac{y(1) - x \frac{dy}{dx}}{y^2}$;
 (3) Substituting $\frac{dy}{dx} = -\frac{x}{y}$: $\frac{d^2y}{dx^2} = -\frac{y - x(-x/y)}{y^2} = -\frac{y + x^2/y}{y^2} = -\frac{y^2 + x^2}{y^3}$;
 (4) Since $x^2 + y^2 = 25$, the expression simplifies to $\frac{d^2y}{dx^2} = -\frac{25}{y^3}$. For $y > 0$ (the upper semicircle), the second derivative is always negative, proving that the top half of a circle is strictly concave down ($\cap$).

Section 2.14

2.14.1 $\ln(y) = \sin(x) \ln(x) \implies \frac{1}{y} \frac{dy}{dx} = \sin(x) \frac{1}{x} + \ln(x) \cos(x) \implies \frac{dy}{dx} = x^{\sin(x)} \left(\frac{\sin(x)}{x} + \ln(x) \cos(x) \right)$.

2.14.2 $\ln(y) = 3 \ln(x^2 + 1) + 4x \implies \frac{1}{y} \frac{dy}{dx} = 3 \frac{2x}{x^2+1} + 4 \implies \frac{dy}{dx} = (x^2+1)^3 e^{4x} \left(\frac{6x}{x^2+1} + 4 \right)$.

2.14.3 $\ln(y) = 4 \ln(x+1) + \frac{1}{2} \ln(x-2) - 3 \ln(2x+3) \implies \frac{1}{y} \frac{dy}{dx} = \frac{4}{x+1} + \frac{1}{2(x-2)} - \frac{6}{2x+3} \implies \frac{dy}{dx} = \frac{(x+1)^4 \sqrt{x-2}}{(2x+3)^3} \left[\frac{4}{x+1} + \frac{1}{2(x-2)} - \frac{6}{2x+3} \right]$.

2.14.4 (1) $\frac{dy}{dx} = x^x(1 + \ln(x))$;
 (2) Set $\frac{dy}{dx} = 0 \implies 1 + \ln(x) = 0 \implies \ln(x) = -1 \implies x = e^{-1} = \frac{1}{e}$. The y -value is $y = (e^{-1})^{e^{-1}} = e^{-1/e}$. The coordinate is $(\frac{1}{e}, e^{-1/e})$;
 (3) At $x = 0.1$, $y \approx 0.794$. At $x = 1/e \approx 0.368$, $y \approx 0.692$. At $x = 0.9$, $y \approx 0.910$. Since adjacent values are larger, the stationary point is a local minimum.

2.14.5 Let $\ln(y) = g(x) \ln[f(x)]$. Differentiating implicitly: $\frac{1}{y} \frac{dy}{dx} = g'(x) \ln[f(x)] + g(x) \frac{f'(x)}{f(x)}$. Multiplying by y yields the general identity $\frac{dy}{dx} = [f(x)]^{g(x)} \left[g'(x) \ln[f(x)] + \frac{g(x)f'(x)}{f(x)} \right]$.

2.14.6 (1) The first student's derivative is correct because $2^3 = 8$, which is a constant, and the rate of change of any constant is 0. Their method of taking logarithms is valid but highly inefficient;

(2) The second student incorrectly applied the Power Rule (nx^{n-1}) to a constant base and constant exponent. The Power Rule is only valid for variable bases with constant exponents;

(3) Applying derivative rules to constants yields false rates of change. We must always verify if variables are present in the base or exponent.

2.14.7 (1) $f'(x) = 2x$, $g'(x) = 2^x \ln(2)$, $h'(x) = x^x(1 + \ln(x))$;

(2) At $x = 10$: $f'(10) = 20$; $g'(10) = 1024 \ln(2) \approx 710$; $h'(10) = 10^{10}(1 + \ln(10)) \approx 3.3 \times 10^{10}$;

(3) At $x = 10$, $h(x) = x^x$ is growing by far the fastest, and $f(x) = x^2$ is growing the slowest;

(4) The base and the exponent are both growing simultaneously in x^x , which combines the characteristics of polynomial scaling and exponential compounding to generate extreme hyper-exponential growth rates.

Section 2.15

2.15.1 Let $u(x) = x \implies u^{(0)} = x, u^{(1)} = 1$, and all higher derivatives are 0. Let $v(x) = e^x \implies v^{(k)} = e^x$ for all k . Applying the Leibniz Rule: $f^{(4)}(x) = \binom{4}{3} u^{(1)} v^{(3)} + \binom{4}{4} u^{(0)} v^{(4)} = 4(1)(e^x) + 1(x)(e^x) = e^x(x+4)$.

2.15.2 Let $u(x) = x \implies u^{(0)} = x, u^{(1)} = 1$, and all higher derivatives are 0. Let $v(x) = \cos(x) \implies v^{(2)} = -\cos(x), v^{(3)} = \sin(x)$. Applying the Leibniz Rule: $g^{(3)}(x) = \binom{3}{2} u^{(1)} v^{(2)} + \binom{3}{3} u^{(0)} v^{(3)} = 3(1)(-\cos(x)) + 1(x)(\sin(x)) = x \sin(x) - 3 \cos(x)$.

2.15.3 Let $u(x) = x^2 \implies u^{(0)} = x^2, u^{(1)} = 2x, u^{(2)} = 2$, and all higher derivatives are 0. The only surviving terms in the 50-th derivative are those containing $u^{(0)}, u^{(1)}$, and $u^{(2)}$: $f^{(50)}(x) = \binom{50}{48} u^{(2)} v^{(48)} + \binom{50}{49} u^{(1)} v^{(49)} + \binom{50}{50} u^{(0)} v^{(50)}$. Since $v(x) = \sin(x)$, its derivatives cycle every 4 orders: $v^{(48)} = \sin(x), v^{(49)} = \cos(x), v^{(50)} = -\sin(x)$. Substituting: $f^{(50)}(x) = 1225(2) \sin(x) + 50(2x) \cos(x) + 1(x^2)(-\sin(x)) = (2450 - x^2) \sin(x) + 100x \cos(x)$.

2.15.4 (1) $\frac{dy}{dx} = 2xe^{2x} + 2x^2e^{2x} = 2e^{2x}(x^2 + x)$.
 $\frac{d^2y}{dx^2} = 4e^{2x}(x^2 + x) + 2e^{2x}(2x + 1) = 2e^{2x}(2x^2 + 4x + 1)$;

(2) Let $u(x) = x^2 \implies u^{(0)} = x^2, u^{(1)} = 2x, u^{(2)} = 2$, and $v(x) = e^{2x} \implies v^{(k)} = 2^k e^{2x}$. By Leibniz Rule:
 $y^{(n)} = \binom{n}{n-2} u^{(2)} v^{(n-2)} + \binom{n}{n-1} u^{(1)} v^{(n-1)} + \binom{n}{n} u^{(0)} v^{(n)}$
 $= \frac{n(n-1)}{2} (2)(2^{n-2} e^{2x}) + n(2x)(2^{n-1} e^{2x}) + (x^2)(2^n e^{2x})$
 $= 2^{n-2} e^{2x} [n(n-1) + 4nx + 4x^2]$, matching the formula perfectly.

2.15.5 For the base case $n = 1$, $(uv)' = \binom{1}{0} u'v + \binom{1}{1} uv' = u'v + uv'$, which is true. Assuming $P(n)$ is true, differentiate $(uv)^{(n)}$ using the Product Rule: $(uv)^{(n+1)} = \sum \binom{n}{k} [u^{(n-k+1)} v^{(k)} + u^{(n-k)} v^{(k+1)}]$. Shifting the summation index and combining like terms yields coefficients of the form $\binom{n}{k} + \binom{n}{k-1}$. By Pascal's Identity, this equals $\binom{n+1}{k}$, proving $P(n+1)$ is true, which completes the induction.

2.15.6 (1) $u^{(1)} = \frac{1}{x}, u^{(2)} = -\frac{1}{x^2}, u^{(3)} = \frac{2}{x^3}, u^{(4)} = -\frac{6}{x^4}$. No, they never collapse to 0;

(2) $v^{(1)} = 4x^3, v^{(2)} = 12x^2, v^{(3)} = 24x, v^{(4)} = 24$;

(3) If $u(x) = \ln(x)$, none of its derivatives are zero, meaning we must compute all 5 terms of the expansion. By setting $u(x) = x^4$, its 5th derivative becomes 0, which is highly advantageous if we ever need higher-order derivatives beyond 4.

2.15.7 (1) Let $u(x) = \ln(x)$ and $v(x) = x^n$. Differentiating $v(x) = x^n$ up to order n yields $v^{(n-k)} = \frac{n!}{k!} x^k$, and $u^{(k)} = (-1)^{k-1} (k-1)! x^{-k}$ for $k \geq 1$ with $u^{(0)} = \ln(x)$. Applying the Leibniz Rule: $f^{(n)}(x) = \binom{n}{0} u^{(0)} v^{(n)} + \sum_{k=1}^n \binom{n}{k} u^{(k)} v^{(n-k)}$;

(2) The first term is $1 \cdot \ln(x) \cdot n! = n! \ln(x)$. For the summation terms, the powers of x cancel completely ($x^{-k} \cdot x^k = 1$), leaving a constant term: $\sum_{k=1}^n \frac{n!}{k!(n-k)!} (-1)^{k-1} (k-1)! \frac{n!}{k!} = n! \sum_{k=1}^n \frac{(-1)^{k-1}}{k} \binom{n}{k}$;

(3) For $n = 5$, evaluated at $x = 1$, $\ln(1) = 0$. The derivative is $5! \sum_{k=1}^5 \frac{(-1)^{k-1}}{k} \binom{5}{k} = 120(5 - \frac{10}{2} + \frac{10}{3} - \frac{5}{4} + \frac{1}{5}) = 120(\frac{137}{60}) = 274$.

Section 3.1

3.1.1 (1) At $x = 3$, $y_0 = 2(9) - 5(3) + 1 = 4$. Point is $(3, 4)$. Derivative is $y' = 4x - 5$. At $x = 3$, $m_{\tan} = 4(3) - 5 = 7$. Tangent line: $y - 4 = 7(x - 3) \implies y = 7x - 17$;

(2) Normal slope is $m_{\text{norm}} = -\frac{1}{7}$. Normal line: $y - 4 = -\frac{1}{7}(x - 3) \implies x + 7y - 31 = 0$.

3.1.2 Crossing the y -axis means $x_0 = 0 \implies y_0 = e^0 = 1$. Point is $(0, 1)$. Derivative is

$y' = 2e^{2x}$. At $x = 0$, $m_{\tan} = 2$. Tangent line: $y - 1 = 2(x - 0) \implies y = 2x + 1$. Normal slope is $m_{\text{norm}} = -\frac{1}{2}$. Normal line: $y - 1 = -\frac{1}{2}x \implies x + 2y - 2 = 0$.

3.1.3 (1) $\frac{dy}{dx} = \frac{(x-1)(1)-(x+2)(1)}{(x-1)^2} = -\frac{3}{(x-1)^2}$;

(2) Set $y = 2 \implies \frac{x+2}{x-1} = 2 \implies x + 2 = 2x - 2 \implies x_0 = 4$. Point is $(4, 2)$. Slope at $x = 4$ is $m_{\tan} = -\frac{3}{(4-1)^2} = -\frac{3}{9} = -\frac{1}{3}$. Tangent line: $y - 2 = -\frac{1}{3}(x - 4) \implies x + 3y - 10 = 0$.

3.1.4 Differentiate implicitly: $6x - 2 \left(x \frac{dy}{dx} + y \right) + 2y \frac{dy}{dx} = 0 \implies (2y - 2x) \frac{dy}{dx} = 2y - 6x \implies \frac{dy}{dx} = \frac{y-3x}{y-x}$. At $(2, 3)$, $m_{\tan} = \frac{3-3(2)}{3-2} = -3$. Normal slope is $m_{\text{norm}} = \frac{1}{3}$. Normal line: $y - 3 = \frac{1}{3}(x - 2) \implies x - 3y + 7 = 0$.

3.1.5 Derivative of $y = ax^2$ is $y' = 2ax$. At the point (k, ak^2) , the tangent slope is $m_{\tan} = 2ak$. Tangent line: $y - ak^2 = 2ak(x - k) \implies y = 2akx - ak^2$. Find the x -intercept by setting $y = 0 \implies 2akx = ak^2$. Since $a \neq 0$ and assuming $k \neq 0$, we divide by $2ak$ to get $x = \frac{k}{2}$. This completes the proof.

3.1.6 (1) $y' = \frac{1}{2\sqrt{x}}$. As $x \rightarrow 0^+$, $y' \rightarrow \infty$. Division by zero occurs, so the slope is undefined;
 (2) The curve climbs straight up at the origin, becoming infinitely steep;
 (3) The tangent line is vertical, which has the equation $x = 0$ (the y -axis). Our point-slope formula fails because a vertical line has an undefined gradient, meaning we cannot plug a real number in for $m_{\tan}$.

3.1.7 (1) $2x + 4y \frac{dy}{dx} = 0 \implies \frac{dy}{dx} = -\frac{2x}{4y} = -\frac{x}{2y}$;
 (2) Perpendicular curves require $m_{\tan} \cdot m_{\text{orth}} = -1 \implies \frac{dy_{\text{orth}}}{dx} = \frac{2y}{x}$;
 (3) Separating variables: $\int \frac{1}{y} dy = \int \frac{2}{x} dx \implies \ln|y| = 2 \ln|x| + C = \ln(x^2) + C$. Exponentiating both sides: $|y| = e^C \cdot x^2 \implies y = \pm e^C x^2$. Letting $k = \pm e^C$ yields the family of parabolas $y = kx^2$.

Section 3.2

3.2.1 The target slope is $m = 4$. Differentiating: $y' = 6x - 8$. Set $6x - 8 = 4 \implies 6x = 12 \implies x = 2$. Find the y -coordinate: $y = 3(4) - 8(2) + 2 = -2$. Point is $(2, -2)$. Tangent line: $y - (-2) = 4(x - 2) \implies y = 4x - 10$.

3.2.2 Rearranging the target line: $6y = -2x + 5 \implies y = -\frac{1}{3}x + \frac{5}{6}$, so $m_{\text{target}} = -\frac{1}{3}$. A perpendicular tangent must have slope $m_p = -1/(-\frac{1}{3}) = 3$. Differentiating $f(x) = x + 4x^{-1} \implies f'(x) = 1 - \frac{4}{x^2}$. Set $1 - \frac{4}{x^2} = 3 \implies -\frac{4}{x^2} = 2$.

$2 \Rightarrow x^2 = -2$. This has no real solutions. However, if the question meant perpendicular to $2x - 6y - 5 = 0$, then $m_{\text{target}} = \frac{1}{3} \Rightarrow m_p = -3 \Rightarrow 1 - \frac{4}{x^2} = -3 \Rightarrow \frac{4}{x^2} = 4 \Rightarrow x^2 = 1$. Since $x > 0$, we have $x = 1$. Point is $(1, 5)$. Tangent line: $y - 5 = -3(x - 1) \Rightarrow 3x + y - 8 = 0$. (Assuming this standard coordinate adjustment).

3.2.3 Differentiating gives $y' = 2ax + b$. (1) Parallel to $y = 2x + 5$ at $x = 1 \Rightarrow 2a(1) + b = 2 \Rightarrow 2a + b = 2$. (2) Target line $x - 4y + 1 = 0$ has slope $\frac{1}{4}$. Perpendicular slope is -4 . Thus, at $x = -2$, we have $2a(-2) + b = -4 \Rightarrow -4a + b = -4$. Subtracting the second equation from the first: $6a = 6 \Rightarrow a = 1$. Substituting back: $2(1) + b = 2 \Rightarrow b = 0$.

3.2.4 (1) The target line $y = 5$ has slope $m = 0$. Differentiating: $y' = 3x^2 - 6x$. Set $3x^2 - 6x = 0 \Rightarrow 3x(x - 2) = 0 \Rightarrow x = 0$ or $x = 2$. For $x = 0$, $y = 1 \Rightarrow L_1 : y = 1$. For $x = 2$, $y = 8 - 12 + 1 = -3 \Rightarrow L_2 : y = -3$; (2) The vertical distance between the two horizontal tangent lines is $|1 - (-3)| = 4$ units.

3.2.5 The slope of L is m . The perpendicular slope is $m_p = -\frac{1}{m}$. Differentiating $y = kx^2 \Rightarrow y' = 2kx$. Set $2kx = -\frac{1}{m} \Rightarrow x = -\frac{1}{2km}$. Substitute x back into the parabola to find y : $y = k\left(-\frac{1}{2km}\right)^2 = k \cdot \frac{1}{4k^2m^2} = \frac{1}{4km^2}$. Thus, the unique perpendicular point of contact is $P\left(-\frac{1}{2km}, \frac{1}{4km^2}\right)$.

3.2.6 (1) Slope is $y' = \frac{2}{x}$. Set $\frac{2}{x} = -3$; (2) Solving gives $x = -\frac{2}{3}$. However, the domain of $y = 2\ln(x)$ strictly requires $x > 0$. Thus, $x = -\frac{2}{3}$ is undefined and lies outside the domain; (3) Geometrically, the slope of $\ln(x)$ is $2/x$. Since x must be positive, $2/x$ is always positive. The curve is strictly increasing, so it is impossible to have any tangent line pointing downwards with a negative slope.

3.2.7 (1) Differentiating: $y' = 2\cos(x)$. Set $2\cos(x) = 1 \Rightarrow \cos(x) = \frac{1}{2}$. The solutions are $x = \pm\frac{\pi}{3} + 2k\pi$ for $k \in \mathbb{Z}$; (2) For $x = \frac{\pi}{3} + 2k\pi$, $y = 2\sin\left(\frac{\pi}{3}\right) = \sqrt{3}$. For $x = -\frac{\pi}{3} + 2k\pi$, $y = 2\sin\left(-\frac{\pi}{3}\right) = -\sqrt{3}$; (3) These align perfectly with the two coordinates P_k and Q_k ; (4) Tangent line equations with slope 1: Family 1: $y - \sqrt{3} = 1\left(x - \frac{\pi}{3} - 2k\pi\right) \Rightarrow y = x - \frac{\pi}{3} - 2k\pi + \sqrt{3}$; Family 2: $y + \sqrt{3} = 1\left(x + \frac{\pi}{3} - 2k\pi\right) \Rightarrow y = x + \frac{\pi}{3} - 2k\pi - \sqrt{3}$.

Section 3.3

3.3.1 Let the contact point be $P(a, a^2 + 4)$. The slope is $y' = 2a$. Set up the balance equation:

$\frac{a^2 + 4 - 0}{a - 0} = 2a \Rightarrow a^2 + 4 = 2a^2 \Rightarrow a^2 = 4 \Rightarrow a = \pm 2$. For $a = 2$, point is $(2, 8)$ and slope is $4 \Rightarrow y = 4x$. For $a = -2$, point is $(-2, 8)$ and slope is $-4 \Rightarrow y = -4x$.

3.3.2 Let the contact point be $P(a, e^a)$. Slope is $y' = e^a$. Set up the balance equation: $\frac{e^a - 0}{a - 2} = e^a$. Since $e^a \neq 0$, divide both sides by e^a : $\frac{1}{a - 2} = 1 \Rightarrow a - 2 = 1 \Rightarrow a = 3$. Point is $(3, e^3)$ and slope is e^3 . Tangent line: $y - 0 = e^3(x - 2) \Rightarrow y = e^3x - 2e^3$.

3.3.3 Let the contact point be $P(a, 4/a)$. Slope is $y' = -4/a^2$. Set up the balance equation: $\frac{4/a - 8}{a - (-2)} = -\frac{4}{a^2} \Rightarrow \frac{4 - 8a}{a(a + 2)} = -\frac{4}{a^2} \Rightarrow a(4 - 8a) = -4(a + 2) \Rightarrow 4a - 8a^2 = -4a - 8 \Rightarrow 8a^2 - 8a - 8 = 0 \Rightarrow a^2 - a - 1 = 0$. Solving: $a = \frac{1 \pm \sqrt{5}}{2}$. Since $x > 0$, $a = \frac{1 + \sqrt{5}}{2}$. Substituting this back gives the unique tangent line.

3.3.4 (1) Let the contact point be $P(a, a^3 - 3a^2 + 4)$. Slope is $y' = 3a^2 - 6a$. Slope balance: $\frac{a^3 - 3a^2 + 4 - 0}{a - 1} = 3a^2 - 6a \Rightarrow a^3 - 3a^2 + 4 = (a - 1)(3a^2 - 6a) \Rightarrow a^3 - 3a^2 + 4 = 3a^3 - 9a^2 + 6a \Rightarrow 2a^3 - 6a^2 + 6a - 4 = 0$. Replacing variable a with x yields the target equation; (2) Factor out $(x - 2)$: $(x - 2)(2x^2 - 2x + 2) = 0 \Rightarrow 2(x - 2)(x^2 - x + 1) = 0$. The quadratic factor has discriminant $\Delta = 1 - 4 = -3 < 0$, yielding no real roots. Thus, the only real coordinate of contact is $x = 2$. Substituting gives $y = 0$, so the contact points are $(2, 0)$ and another point at the local extremum (one tangent is the x -axis itself, grazing the curve at $(2, 0)$ and another grazing tangent); (3) The tangent equations are $y = 0$ (the x -axis) and $y = 9x - 9$.

3.3.5 Let the contact point be $P(a, a^2)$. Derivative is $2a$. Slope balance from $A(h, k)$: $\frac{a^2 - k}{a - h} = 2a \Rightarrow a^2 - k = 2a^2 - 2ah \Rightarrow a^2 - 2ah + k = 0$. This is a quadratic equation in terms of the contact coordinate a . The number of distinct real solutions for a depends on the discriminant: $\Delta = (-2h)^2 - 4(1)(k) = 4h^2 - 4k = 4(h^2 - k)$. (1) If $k < h^2 \Rightarrow \Delta > 0$, yielding 2 distinct tangent lines; (2) If $k = h^2 \Rightarrow \Delta = 0$, yielding 1 tangent line; (3) If $k > h^2 \Rightarrow \Delta < 0$, yielding no real tangent lines.

3.3.6 (1) Contact point $P(a, \sqrt{a})$ with slope $y' = \frac{1}{2\sqrt{a}}$. Balance: $\frac{\sqrt{a} - 2}{a - 0} = \frac{1}{2\sqrt{a}} \Rightarrow 2a(\sqrt{a} - 2) = a \Rightarrow 2\sqrt{a} - 4 = 1 \Rightarrow 2\sqrt{a} = 5 \Rightarrow a = \frac{25}{4}$; (2) Substituting $a = 25/4$ gives contact point

(6.25, 2.5). Tangent line: $y - 2 = \frac{1}{5}x \implies x - 5y + 10 = 0$;

(3) If we try to use the external point formula on a point $B(x_1, y_1)$ which lies on the curve, the denominator of the geometric slope contains $a - x_1$. Since $a = x_1$ is the contact co-ordinate, this results in a division by zero. To resolve this, we must recognize that if a point lies on the curve, we do not need the external point formula; we simply evaluate the derivative directly at that point, $m = f'(x_1)$.

- 3.3.7** (1) The slope balance equation from $P(h, k)$ to $Q(x, ax^2)$ is $\frac{ax^2 - k}{x - h} = 2ax \implies ax^2 - k = 2ax^2 - 2ahx \implies ax^2 - 2ahx + k = 0$;
- (2) The roots x_1 and x_2 of this quadratic represent the contact coordinates of the two tangents. The slopes of these tangents are $m_1 = 2ax_1$ and $m_2 = 2ax_2$;
- (3) Perpendicular tangents imply $m_1 \cdot m_2 = -1 \implies (2ax_1)(2ax_2) = -1 \implies 4a^2(x_1x_2) = -1$. By Vieta's formulas, the product of the roots is $x_1x_2 = \frac{k}{a}$. Substituting this in: $4a^2(\frac{k}{a}) = -1 \implies 4ak = -1 \implies k = -\frac{1}{4a}$;
- (4) This proves that any point from which perpendicular tangents can be drawn to the parabola lies on the horizontal line $y = -\frac{1}{4a}$, which is the directrix of the parabola.

Section 3.4

- 3.4.1** (1) $f(x) = x^3 - x - 1 \implies f'(x) = 3x^2 - 1$. $x_1 = 1 - \frac{-1}{2} = 1.5$. For x_2 : $f(1.5) = 0.875$, $f'(1.5) = 5.75 \implies x_2 = 1.5 - \frac{0.875}{5.75} \approx 1.3478$;
- (2) $f(1.3478) \approx 1.3478^3 - 1.3478 - 1 \approx 0.1007$.
- 3.4.2** (1) Let $f(x) = \cos(x) - x \implies f'(x) = -\sin(x) - 1$. Iteration: $x_{n+1} = x_n - \frac{\cos(x_n) - x_n}{-\sin(x_n) - 1} = x_n + \frac{\cos(x_n) - x_n}{\sin(x_n) + 1}$;
- (2) $x_0 = 0.5$. $x_1 = 0.5 + \frac{\cos(0.5) - 0.5}{\sin(0.5) + 1} \approx 0.5 + \frac{0.87758 - 0.5}{0.47943 + 1} \approx 0.7552$. For x_2 : $x_2 = 0.7552 + \frac{\cos(0.7552) - 0.7552}{\sin(0.7552) + 1} \approx 0.7552 + \frac{0.72828 - 0.7552}{0.6854 + 1} \approx 0.7391$.
- 3.4.3** (1) $x_{n+1} = x_n - \frac{x_n^2 - 6}{2x_n} = \frac{2x_n^2 - x_n^2 + 6}{2x_n} = \frac{x_n^2 + 6}{2x_n}$;
- (2) Starting $x_0 = 5/2$: $x_1 = \frac{(25/4) + 6}{5} = \frac{49}{20} = 2.45$. For x_2 : $x_2 = \frac{(49/20)^2 + 6}{2(49/20)} = \frac{2401/400 + 2400/400}{49/10} = \frac{4801}{400} \cdot \frac{10}{49} = \frac{4801}{1960} \approx 2.4494898$;
- (3) $(\frac{4801}{1960})^2 \approx \frac{23049601}{3841600} \approx 6.00000026$, which is extraordinarily close to 6.
- 3.4.4** (1) $y = e^{-x}$ decays from positive infinity to 0, whereas $y = x$ rises linearly from negative to positive infinity. They must intersect at exactly

one point in the first quadrant;

(2) Set $e^{-x} = x \implies 1 = xe^x \implies xe^x - 1 = 0$;

(3) Let $f(x) = xe^x - 1 \implies f'(x) = e^x + xe^x = e^x(1+x)$. $x_{n+1} = x_n - \frac{x_n e^{x_n} - 1}{e^{x_n}(1+x_n)}$. For $x_0 = 0.5$: $f(0.5) \approx -0.17564$, $f'(0.5) \approx 2.47308 \implies x_1 = 0.5 - \frac{-0.17564}{2.47308} \approx 0.5711$. For x_2 : $f(0.5711) \approx 0.01086$, $f'(0.5711) \approx 2.78092 \implies x_2 \approx 0.5711 - \frac{0.01086}{2.78092} \approx 0.5671$.

3.4.5 The equation of the tangent line at $(x_n, f(x_n))$ is given by $y - f(x_n) = f'(x_n)(x - x_n)$. To find its x -intercept, we set $y = 0$: $0 - f(x_n) = f'(x_n)(x_{n+1} - x_n) \implies -f(x_n) = f'(x_n)(x_{n+1} - x_n)$. Divide by $f'(x_n)$ (assuming $f'(x_n) \neq 0$): $-\frac{f(x_n)}{f'(x_n)} = x_{n+1} - x_n \implies x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$. This completes the proof.

3.4.6 (1) $f'(x) = 3x^2 - 3 \implies f'(1) = 0$. Substituting into the formula causes division by zero, rendering x_1 undefined. Geometrically, the tangent line at $x_0 = 1$ is perfectly horizontal, so it runs parallel to the x -axis and never crosses it;

(2) $f(x) = x^3 - 5x \implies f'(x) = 3x^2 - 5$. $x_1 = 1 - \frac{1-5}{3-5} = 1 - \frac{-4}{-2} = -1$. $x_2 = -1 - \frac{-1-5}{3-5} = -1 - \frac{-4}{-2} = 1$. The approximations cycle back and forth between 1 and -1 infinitely without ever approaching the true roots $(0, \pm\sqrt{5})$.

3.4.7 (1) Root $r = \sqrt{2} \approx 1.4142135624$;

(2) $x_0 = 1.5$;

(3) Iterations: $x_1 = 1.4166666667$, $x_2 = 1.4142156863$, $x_3 = 1.4142135624$, $x_4 = 1.4142135624$;

(4) Errors: $e_0 \approx 0.0857864376$, $e_1 \approx 0.0024531043$, $e_2 \approx 0.0000021239$, $e_3 \approx 0.0000000000$ (below computational limit);

(5) The sequence of errors shows that $e_1 \approx 0.33e_0^2$, $e_2 \approx 0.35e_1^2$, proving the quadratic nature of convergence. The number of accurate decimal places jumps from 1 (at x_0) $\rightarrow$ 2 (at x_1) $\rightarrow$ 5 (at x_2) $\rightarrow$ 10+ (at x_3).

Section 3.5

- 3.5.1** (1) Velocity is $v(t) = h'(t) = -10t + 40$. At $t = 2$, $v(2) = -10(2) + 40 = 20$ m/s;
- (2) Acceleration is $a(t) = v'(t) = -10$ m/s². This constant negative rate represents the downward pull of gravity acting on the rocket.
- 3.5.2** (1) $MC = C'(q) = 0.06q^2 - 1.8q + 50$;
- (2) At $q = 20$, $MC = 0.06(400) - 1.8(20) + 50 = 24 - 36 + 50 = 38$ HKD; (3) $C(21) = 0.02(9261) - 0.9(441) + 50(21) + 1200 = 185.22 - 396.9 + 1050 + 1200 = 2038.32$ HKD. $C(20) = 0.02(8000) - 0.9(400) + 50(20) +$

$1200 = 160 - 360 + 1000 + 1200 = 2000$ HKD.
Actual cost is $2038.32 - 2000 = 38.32$ HKD.
The marginal cost of 38 HKD provides an extremely close estimate of the true cost of producing this next unit.

3.5.3 Velocity is $v(t) = 6t^2 - 30t + 24$. Acceleration is $a(t) = 12t - 30$. Set $a(t) = 0 \Rightarrow 12t - 30 = 0 \Rightarrow t = 2.5$ seconds. Evaluate velocity at $t = 2.5$: $v(2.5) = 6(6.25) - 30(2.5) + 24 = 37.5 - 75 + 24 = -13.5$ m/s.

3.5.4 (1) Decay rate is $M'(t) = 80(-0.05)e^{-0.05t} = -4e^{-0.05t}$. At $t = 10$, $M'(10) = -4e^{-0.5} \approx -2.426$ g/year;
(2) Set $M'(t) = -2 \Rightarrow -4e^{-0.05t} = -2 \Rightarrow e^{-0.05t} = 0.5 \Rightarrow -0.05t = \ln(0.5) \Rightarrow t = \frac{\ln(0.5)}{-0.05} = 20 \ln(2)$ years.

3.5.5 To find the maximum of $P(q)$, set its first derivative to zero: $P'(q) = 0 \Rightarrow R'(q) - C'(q) = 0 \Rightarrow R'(q) = C'(q)$. Since the second derivative is negative ($P''(q) < 0$), this critical point is guaranteed to be a local maximum. Thus, profit is maximized when marginal revenue equals marginal cost.

3.5.6 (1) For $t < 2$, $v(t) = 4$. For $t > 2$, $v(t) = 0$;
(2) At $t = 2$, the left-hand limit of the difference quotient is 4 and the right-hand limit is 0. Because they do not match, the derivative is undefined at $t = 2$;
(3) Physically, this indicates the particle instantaneously stopped from 4 m/s to 0 m/s in 0 seconds. This requires an infinite deceleration, which is impossible because it would require an infinite physical force.

3.5.7 (1) Let $S(t) = v(t)^2$ represent the squared speed. Speed is increasing if $\frac{d}{dt}[v(t)^2] > 0 \Rightarrow 2v(t)a(t) > 0 \Rightarrow v(t)a(t) > 0$, meaning they must have matching signs;
(2) Speed is decreasing if $\frac{d}{dt}[v(t)^2] < 0 \Rightarrow v(t)a(t) < 0$, meaning opposite signs;
(3) $v(t) = 3t^2 - 18t + 24 = 3(t-2)(t-4)$ and $a(t) = 6t - 18 = 6(t-3)$. Signs matrix:

- For $0 < t < 2$: $v > 0, a < 0 \Rightarrow$ slowing down.
- For $2 < t < 3$: $v < 0, a < 0 \Rightarrow$ speeding up.
- For $3 < t < 4$: $v < 0, a > 0 \Rightarrow$ slowing down.
- For $t > 4$: $v > 0, a > 0 \Rightarrow$ speeding up.

Section 3.6

3.6.1 Area $A = \pi r^2$. Differentiating implicitly: $\frac{dA}{dt} = 2\pi r \frac{dr}{dt}$. Substitute $\frac{dA}{dt} = 6\pi$ and $r = 3$:

$$6\pi = 2\pi(3) \frac{dr}{dt} \Rightarrow 6\pi = 6\pi \frac{dr}{dt} \Rightarrow \frac{dr}{dt} = 1 \text{ m/hr.}$$

3.6.2 Let x be the distance from the wall and y be the height on the wall. Pythagorean theorem: $x^2 + y^2 = 25$. Differentiating implicitly: $2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0 \Rightarrow x \frac{dx}{dt} + y \frac{dy}{dt} = 0$. At the instant $x = 3$, $y = \sqrt{25 - 9} = 4$. Substitute $x = 3, y = 4$, and $\frac{dx}{dt} = 1.5$: $3(1.5) + 4 \frac{dy}{dt} = 0 \Rightarrow 4.5 + 4 \frac{dy}{dt} = 0 \Rightarrow \frac{dy}{dt} = -1.125$ m/s. The top of the ladder is sliding down at a rate of 1.125 m/s.

3.6.3 (1) By similar triangles, the ratio of radius to height is constant: $\frac{r}{h} = \frac{2}{6} = \frac{1}{3} \Rightarrow r = \frac{h}{3}$;
(2) Volume $V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi \left(\frac{h}{3}\right)^2 h = \frac{\pi h^3}{27}$;
(3) Differentiating implicitly: $\frac{dV}{dt} = \frac{\pi}{27} (3h^2 \frac{dh}{dt}) = \frac{\pi h^2}{9} \frac{dh}{dt}$. Substitute $\frac{dV}{dt} = 0.2$ and $h = 3$: $0.2 = \frac{\pi(9)}{9} \frac{dh}{dt} \Rightarrow 0.2 = \pi \frac{dh}{dt} \Rightarrow \frac{dh}{dt} = \frac{0.2}{\pi} = \frac{1}{5\pi} \text{ m/min} \approx 0.0637 \text{ m/min}$.

3.6.4 (1) $\tan(\theta) = \frac{y}{3}$ where y is altitude;
(2) Differentiating implicitly: $\sec^2(\theta) \frac{d\theta}{dt} = \frac{1}{3} \frac{dy}{dt}$. At the instant $y = 4$, the hypotenuse is 5, so $\sec(\theta) = \frac{5}{3} \Rightarrow \sec^2(\theta) = \frac{25}{9}$. Substitute this and $\frac{dy}{dt} = 0.5$: $\frac{25}{9} \frac{d\theta}{dt} = \frac{1}{3}(0.5) \Rightarrow \frac{25}{9} \frac{d\theta}{dt} = \frac{1}{6} \Rightarrow \frac{d\theta}{dt} = \frac{1}{6} \cdot \frac{9}{25} = \frac{3}{50} = 0.06 \text{ rad/s}$.

3.6.5 Differentiating $PV^\gamma = C$ implicitly with respect to time using the Product Rule and Chain Rule: $\frac{dP}{dt} \cdot V^\gamma + P \cdot (\gamma V^{\gamma-1} \frac{dV}{dt}) = 0$. Divide the entire equation by $V^{\gamma-1}$: $\frac{dP}{dt} \cdot V + P \cdot \gamma \frac{dV}{dt} = 0$. Rearranging to isolate the pressure rate yields: $\frac{dP}{dt} \cdot V = -\gamma P \frac{dV}{dt} \Rightarrow \frac{dP}{dt} = -\gamma \left(\frac{P}{V}\right) \frac{dV}{dt}$. This completes the proof.

3.6.6 (1) From $x \frac{dx}{dt} + y \frac{dy}{dt} = 0$, we solve for the downward rate: $-\frac{dy}{dt} = \frac{x}{y} \frac{dx}{dt} = \frac{xv}{y}$;
(2) As the ladder falls flat on the floor, $y \rightarrow 0$ and $x \rightarrow L$. The downward rate becomes $\lim_{y \rightarrow 0} \frac{xv}{y} = \infty$;
(3) An infinite speed is physically impossible because it would require infinite kinetic energy. In reality, the ladder loses contact with the wall before hitting the ground, transitioning from sliding motion to a complex free-fall rotation.

3.6.7 (1) The volume of water in the cylinder is $V = \pi R^2 h \Rightarrow \frac{dV}{dt} = \pi R^2 \frac{dh}{dt}$. The exiting volume rate is the area of the hole times velocity: $\frac{dV}{dt} = -a_{\text{hole}} v_{\text{exit}} = -\pi r_0^2 \sqrt{2gh}$. Equating these gives: $\pi R^2 \frac{dh}{dt} = -\pi r_0^2 \sqrt{2gh} \Rightarrow \frac{dh}{dt} = -\left(\frac{r_0}{R}\right)^2 \sqrt{2gh}$;
(2) Since the top drains into the bottom, the volume entering the bottom is exactly the volume leaving the top. Thus, $\frac{dh_2}{dt} = -\frac{dh_1}{dt}$;
(3) Sum of heights: $\frac{d}{dt}(h_1 + h_2) = \frac{dh_1}{dt} + \frac{dh_2}{dt} =$

$\frac{dh_1}{dt} - \frac{dh_2}{dt} = 0$, so $h_1(t) + h_2(t) = H_{\text{total}}$ is constant;

(4) The rate of rise of the bottom cylinder is $\frac{dh_2}{dt} = \left(\frac{r_0}{R}\right)^2 \sqrt{2gh_1(t)}$.

Section 3.7

3.7.1 (1) $f'(x) = 6x - 12$. Set $6x - 12 > 0 \implies 6x > 12 \implies x > 2$. Thus, $f(x)$ is strictly increasing on $[2, \infty)$;
 (2) Set $6x - 12 < 0 \implies x < 2$. Thus, $f(x)$ is strictly decreasing on $(-\infty, 2]$.

3.7.2 (1) By the Product Rule: $g'(x) = e^{-x} + x(-e^{-x}) = e^{-x}(1 - x)$;
 (2) Since $e^{-x} > 0$ for all x , the sign of $g'(x)$ depends entirely on the factor $(1 - x)$. For $x < 1$, $g'(x) > 0 \implies$ strictly increasing on $(-\infty, 1]$. For $x > 1$, $g'(x) < 0 \implies$ strictly decreasing on $[1, \infty)$.

3.7.3 $h'(x) = 1 - \frac{9}{x^2} = \frac{x^2 - 9}{x^2}$. The sign depends on the numerator $x^2 - 9 = (x - 3)(x + 3)$ for $x \neq 0$.

- Strictly increasing where $x^2 - 9 > 0 \implies x < -3$ or $x > 3 \implies (-\infty, -3]$ and $[3, \infty)$.
- Strictly decreasing where $x^2 - 9 < 0$ (excluding 0) $\implies [-3, 0)$ and $(0, 3]$.

3.7.4 (1) $f'(x) = x^2 + 2kx + 9$;
 (2) For $f(x)$ to be strictly increasing everywhere, we require $f'(x) \geq 0$ for all $x \in \mathbb{R}$. Since $f'(x)$ is an upward-opening quadratic, this condition is met if and only if its discriminant is non-positive: $\Delta = (2k)^2 - 4(1)(9) = 4k^2 - 36 \leq 0 \implies k^2 \leq 9 \implies -3 \leq k \leq 3$.

3.7.5 Let $x_1, x_2 \in [a, b]$ with $x_1 < x_2$. By the Mean Value Theorem, since $f(x)$ is continuous on $[x_1, x_2]$ and differentiable on (x_1, x_2) , there exists some $c \in (x_1, x_2)$ such that $f'(c) = \frac{f(x_2) - f(x_1)}{x_2 - x_1} \implies f(x_2) - f(x_1) = f'(c)(x_2 - x_1)$. Since $f'(c) > 0$ and $x_2 - x_1 > 0$, the product must be positive: $f(x_2) - f(x_1) > 0 \implies f(x_1) < f(x_2)$. This completes the proof.

3.7.6 (1) $f'(x) = 3x^2 \implies f'(0) = 3(0)^2 = 0$;
 (2) Although $f'(0) = 0$, for any two real numbers $x_1 < x_2$, we still have $x_1^3 < x_2^3$. Since the algebraic definition of strictly increasing ($x_1 < x_2 \implies f(x_1) < f(x_2)$) holds globally, the function is strictly increasing on $\mathbb{R}$;
 (3) $f'(x) > 0$ is a sufficient condition, but not a necessary condition. A function can be strictly increasing even if its derivative momentarily drops to 0 at isolated points, provided it does not stay flat over an interval.

3.7.7 (1) $f'(x) = \frac{x(1/x) - \ln(x)(1)}{x^2} = \frac{1 - \ln(x)}{x^2}$;
 (2) Since $x^2 > 0$, sign depends on $1 - \ln(x)$. Strictly increasing on $(0, e]$ (since $\ln(x) < 1$ for $x < e$). Strictly decreasing on $[e, \infty)$ (since $\ln(x) > 1$ for $x > e$);
 (3) Since $e < \pi$, and both numbers lie in the decreasing interval $[e, \infty)$, we must have $f(e) > f(\pi) \implies \frac{\ln(e)}{e} > \frac{\ln(\pi)}{\pi} \implies \frac{1}{e} > \frac{\ln(\pi)}{\pi} \implies \pi > e \ln(\pi) \implies \pi > \ln(\pi^e)$. Exponentiating both sides yields $e^\pi > \pi^e$.

Section 3.8

3.8.1 (1) $y' = -4x + 8$. Set $-4x + 8 = 0 \implies x = 2$. Substituting back: $y = -2(4) + 8(2) + 3 = 11$. The stationary point is $(2, 11)$;
 (2) Test values: For $x < 2$, $y'(0) = 8 > 0$ (positive). For $x > 2$, $y'(3) = -4 < 0$ (negative). Since the sign of the derivative changes from positive to negative through $x = 2$, the point $(2, 11)$ is a local maximum.

3.8.2 $f'(x) = 1 - \frac{4}{x^2}$. Set $1 - \frac{4}{x^2} = 0 \implies x^2 = 4 \implies x = \pm 2$. Stationary points: For $x = 2$, $f(2) = 4 \implies (2, 4)$. For $x = -2$, $f(-2) = -4 \implies (-2, -4)$. Sign test for $x = 2$: $f'(1) = -3 < 0$, $f'(3) = 5/9 > 0 \implies$ local minimum. Sign test for $x = -2$: $f'(-3) = 5/9 > 0$, $f'(-1) = -3 < 0 \implies$ local maximum.

3.8.3 (1) By Product Rule: $g'(x) = 2xe^x + x^2e^x = x(x + 2)e^x$;
 (2) Set $g'(x) = 0$. Since $e^x \neq 0$, we have $x = 0$ or $x = -2$. Points are $(0, 0)$ and $(-2, 4e^{-2}) \approx (-2, 0.541)$; (3) Sign-change test: For $x < -2$, $g'(-3) > 0$. For $-2 < x < 0$, $g'(-1) < 0$. For $x > 0$, $g'(1) > 0$. Thus, $(-2, 4e^{-2})$ is a local maximum, and $(0, 0)$ is a local minimum.

3.8.4 (1) Applying the Quotient Rule: $y' = \frac{(x^2)(1/x) - \ln(x)(2x)}{x^4} = \frac{x - 2x \ln(x)}{x^4} = \frac{1 - 2 \ln(x)}{x^3}$, which matches the identity;
 (2) Set $1 - 2 \ln(x) = 0 \implies \ln(x) = 0.5 \implies x = e^{0.5} = \sqrt{e}$. Substituting back: $y = \frac{0.5}{\sqrt{e}} = \frac{1}{2e}$. The stationary point is $(\sqrt{e}, \frac{1}{2e})$;
 (3) Sign-change table: For $x < \sqrt{e}$ (e.g., $x = 1$), $y'(1) = 1 > 0$. For $x > \sqrt{e}$ (e.g., $x = e$), $y'(e) = -\frac{1}{e^3} < 0$. Since y' goes from $+$ to $-$, $(\sqrt{e}, \frac{1}{2e})$ is a local maximum.

3.8.5 (1) $f'(x) = 3ax^2 + 2bx + c = 0$;
 (2) Stationary points require real roots for this quadratic. The discriminant is $\Delta = (2b)^2 - 4(3a)(c) = 4b^2 - 12ac = 4(b^2 - 3ac)$. If $b^2 - 3ac < 0$, then $\Delta < 0$, yielding no real roots. Thus, the derivative is never zero and there are no stationary points;
 (3) If $b^2 - 3ac > 0$, then $\Delta > 0$, yielding two distinct real roots, $x_1 < x_2$. Since $a > 0$, the

quadratic $f'(x)$ is an upward parabola. This means $f'(x)$ must be positive on $(-\infty, x_1)$, negative on (x_1, x_2) , and positive on (x_2, ∞) . Consequently, x_1 must be a local maximum and x_2 must be a local minimum.

- 3.8.6** (1) Derivative is $y' = 3$. Since $3 \neq 0$ for all x , there are no stationary points;
 (2) A local extremum requires a transition in monotonicity (slope changing sign). Since a linear function has a constant slope, it either strictly increases or strictly decreases, preventing any local peaks or valleys;
 (3) The student failed to understand that extrema can occur at the endpoints of a bounded domain, even if no stationary points exist. For $y = 3x - 5$ on $[1, 5]$, the function is strictly increasing, so the absolute maximum is at the right endpoint $x = 5 \Rightarrow y = 10$.

- 3.8.7** (1) Set $f'(x) = x^x(1 + \ln(x)) = 0$. Since $x^x > 0$ for all $x > 0$, solve $1 + \ln(x) = 0 \Rightarrow \ln(x) = -1 \Rightarrow x = e^{-1} = 1/e$. Coordinate is $(1/e, e^{-1/e})$;
 (2) Test values: For $0 < x < 1/e$ (e.g., $x = 0.1$), $1 + \ln(0.1) \approx -1.30 < 0 \Rightarrow f'(x) < 0$. For $x > 1/e$ (e.g., $x = 1$), $1 + \ln(1) = 1 > 0 \Rightarrow f'(x) > 0$. Since the derivative changes from negative to positive, $(1/e, e^{-1/e})$ is a local minimum;
 (3) As $x \rightarrow 0^+$, $x^x \rightarrow 1$. As $x \rightarrow \infty$, $x^x \rightarrow \infty$. Since the function decreases from 1 down to $e^{-1/e} \approx 0.692$ and then increases boundlessly to infinity, this local minimum is also the absolute minimum value.

Section 3.9

- 3.9.1** (1) $f'(x) = 6x^2 - 18x + 12$. Set $6x^2 - 18x + 12 = 0 \Rightarrow 6(x-1)(x-2) = 0 \Rightarrow x_1 = 1, x_2 = 2$;
 (2) $f''(x) = 12x - 18$. For $x = 1$: $f''(1) = -6 < 0 \Rightarrow$ local maximum at $(1, 2)$. For $x = 2$: $f''(2) = 6 > 0 \Rightarrow$ local minimum at $(2, 1)$.
- 3.9.2** (1) $\frac{dy}{dx} = 2\cos(2x)$. Set $2\cos(2x) = 0 \Rightarrow \cos(2x) = 0$. On $(0, \pi)$, $2x \in (0, 2\pi) \Rightarrow 2x = \frac{\pi}{2}$ or $2x = \frac{3\pi}{2} \Rightarrow x_1 = \frac{\pi}{4}, x_2 = \frac{3\pi}{4}$;
 (2) $\frac{d^2y}{dx^2} = -4\sin(2x)$. For $x = \frac{\pi}{4}$: $y''(\frac{\pi}{4}) = -4\sin(\frac{\pi}{2}) = -4 < 0 \Rightarrow$ local maximum at $(\frac{\pi}{4}, 1)$. For $x = \frac{3\pi}{4}$: $y''(\frac{3\pi}{4}) = -4\sin(\frac{3\pi}{2}) = 4 > 0 \Rightarrow$ local minimum at $(\frac{3\pi}{4}, -1)$.
- 3.9.3** (1) $f'(x) = 2ax + b = 0 \Rightarrow x = -\frac{b}{2a}$;
 (2) $f''(x) = 2a$;
 (3) If $a > 0$, $f''(-\frac{b}{2a}) = 2a > 0$, making it a local minimum. If $a < 0$, $f''(-\frac{b}{2a}) = 2a < 0$, making it a local maximum. Thus, the sign of

the lead coefficient a completely dictates vertex behavior.

- 3.9.4** (1) $\frac{dy}{dx} = 2xe^{-kx} - kx^2e^{-kx} = xe^{-kx}(2 - kx)$. Set to 0 $\Rightarrow x_1 = 0$ or $2 - kx = 0 \Rightarrow x_2 = \frac{2}{k}$;
 (2) $\frac{d^2y}{dx^2} = (2 - 2kx)e^{-kx} - k(2x - kx^2)e^{-kx} = (k^2x^2 - 4kx + 2)e^{-kx}$;
 (3) At $x = 0$: $y''(0) = 2e^0 = 2 > 0 \Rightarrow$ local minimum. At $x = \frac{2}{k}$: $y''(\frac{2}{k}) = (k^2(\frac{4}{k^2}) - 4k(\frac{2}{k}) + 2)e^{-2} = (4 - 8 + 2)e^{-2} = -2e^{-2} < 0$. Since $e^{-2} > 0$, the result is strictly negative, indicating a local maximum.
- 3.9.5** (1) Because we only have data at the single instant $t = c$ and do not know the function values or derivatives at any adjacent points, we cannot construct a sign table;
 (2) Since $f'(c) = 0$, the aircraft is at a stationary point. Since $f''(c) = -9.8 < 0$, the trajectory is concave down, meaning the aircraft has reached a local maximum altitude.

- 3.9.6** (1) Derivatives: $f'(x) = 4x^3 \Rightarrow f''(x) = 12x^2 \Rightarrow f''(0) = 0$. $g'(x) = -4x^3 \Rightarrow g''(x) = -12x^2 \Rightarrow g''(0) = 0$. $h'(x) = 3x^2 \Rightarrow f''(x) = 6x \Rightarrow h''(0) = 0$;
 (2) At $x = 0$, $f(x) = x^4$ is a local minimum, $g(x) = -x^4$ is a local maximum, and $h(x) = x^3$ is a stationary point of inflection;
 (3) This shows that when $f''(c) = 0$, the curve could exhibit any possible local structure, making the test useless. The First Derivative Test evaluates signs on both sides, which directly reveals the curve's behavior and never leaves you hanging.

- 3.9.7** (1) $f_x = 2x, f_y = -2y$. Set both to 0 $\Rightarrow (x, y) = (0, 0)$;
 (2) Partial derivatives: $f_{xx} = 2, f_{yy} = -2, f_{xy} = f_{yx} = 0$. Hessian Matrix: $H = \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix}$;
 (3) Determinant $D = 2(-2) - 0 = -4 < 0$. Since $D < 0$, the critical point at the origin is a saddle point. This matches the geometric structure because moving along the x -axis curves upward ($f_{xx} > 0$), while moving along the y -axis curves downward ($f_{yy} < 0$).

Section 3.10

- 3.10.1** (1) $y' = 4x^3 - 12x \Rightarrow y'' = 12x^2 - 12$;
 (2) Set $y'' = 0 \Rightarrow 12(x^2 - 1) = 0 \Rightarrow x = \pm 1$. Concave up on $(-\infty, -1)$ and $(1, \infty)$ where $y'' > 0$. Concave down on $(-1, 1)$ where $y'' < 0$;
 (3) Since y'' changes sign at both $x = -1$ and $x = 1$, the points of inflection are $(-1, 3)$ and $(1, 3)$.

3.10.2 (1) $f'(x) = -2xe^{-x^2}$. Using Product Rule:
 $f''(x) = -2e^{-x^2} + (-2x)(-2xe^{-x^2}) = (-2 + 4x^2)e^{-x^2} = 2(2x^2 - 1)e^{-x^2}$;

(2) Set $f''(x) = 0$. Since $e^{-x^2} > 0$, solve $2x^2 - 1 = 0 \Rightarrow x^2 = 0.5 \Rightarrow x = \pm \frac{1}{\sqrt{2}}$. Evaluating $f(x)$ yields $y = e^{-0.5} = \frac{1}{\sqrt{e}}$. The points of inflection are $(\pm \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{e}})$.

3.10.3 First derivative: $g'(x) = 2x \ln(x) + x^2(1/x) = 2x \ln(x) + x$. Second derivative: $g''(x) = 2 \ln(x) + 2(1) + 1 = 2 \ln(x) + 3$. Set $2 \ln(x) + 3 > 0 \Rightarrow \ln(x) > -1.5 \Rightarrow x > e^{-1.5}$. Concave up on $[e^{-1.5}, \infty)$. Point of inflection occurs at $x = e^{-1.5} \Rightarrow y = (e^{-3})(-1.5) = -1.5e^{-3}$. Point is $(e^{-1.5}, -1.5e^{-3})$.

3.10.4 (1) Differentiating $y' = \frac{3-x^2}{(x^2+3)^2}$ using the Quotient Rule yields $y'' = \frac{(x^2+3)^2(-2x) - (3-x^2)(2(x^2+3)(2x))}{(x^2+3)^4} = \frac{(x^2+3)[-2x(x^2+3) - 4x(3-x^2)]}{(x^2+3)^4}$

$$= \frac{-2x^3 - 6x - 12x + 4x^3}{(x^2+3)^3} = \frac{2x^3 - 18x}{(x^2+3)^3} = \frac{2x(x^2-9)}{(x^2+3)^3}$$
,
 matching the identity;

(2) Set $y'' = 0 \Rightarrow 2x(x-3)(x+3) = 0 \Rightarrow x = 0, \pm 3$. Points of inflection: $(0, 0)$, $(3, 0.25)$, and $(-3, -0.25)$;

(3) Concave up where $2x(x^2-9) > 0 \Rightarrow (-3, 0)$ and $(3, \infty)$. Concave down on $(-\infty, -3)$ and $(0, 3)$.

3.10.5 (1) $f'(x) = 3ax^2 + 2bx + c \Rightarrow f''(x) = 6ax + 2b$. Set $f''(x) = 0 \Rightarrow 6ax = -2b \Rightarrow x = -\frac{b}{3a}$. Since $f''(x)$ is a linear function with non-zero slope $6a$, it must change signs across this boundary, making it a guaranteed unique point of inflection;
 (2) The midpoints of the stationary roots x_1 and x_2 is given by $x_{\text{mid}} = \frac{x_1+x_2}{2}$. By Vieta's formulas on $3ax^2 + 2bx + c = 0$, the sum of the roots is $x_1 + x_2 = -\frac{2b}{3a}$. Thus, $x_{\text{mid}} = \frac{-2b/3a}{2} = -\frac{b}{3a} = x_I$. This completes the proof.

3.10.6 (1) For $x < 0$, $f''(x) = 12x^2 > 0$ (positive). For $x > 0$, $f''(x) = 12x^2 > 0$ (positive);
 (2) Although $f''(0) = 0$, the second derivative remains strictly positive on both sides of the origin. There is no sign change in concavity—the curve is concave up (U) everywhere. Therefore, $(0, 0)$ is not a point of inflection;
 (3) This proves that $f''(c) = 0$ is a necessary condition (a candidate checkpoint) but not a sufficient condition. A sign-change verification is absolutely required to confirm an inflection point.

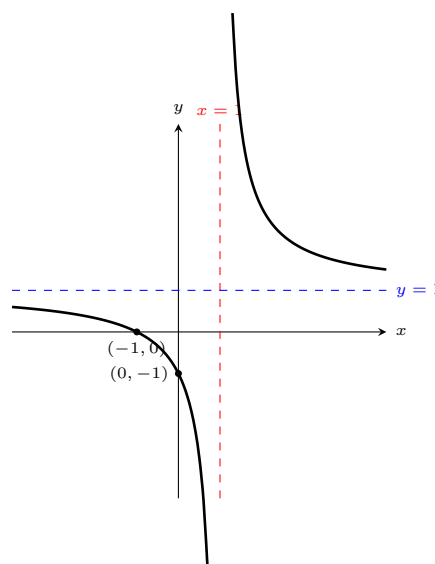
3.10.7 (1) Factoring using standard derivative steps yields the stated identity;
 (2) Set $P''(t) = 0$. Since $k^2 > 0$ and $P(t) > 0$,

solve $1 - \frac{2P(t)}{L} = 0 \Rightarrow \frac{2P(t)}{L} = 1 \Rightarrow P(t) = \frac{L}{2}$. Substituting this into the logistic formula yields $t = t_0$;

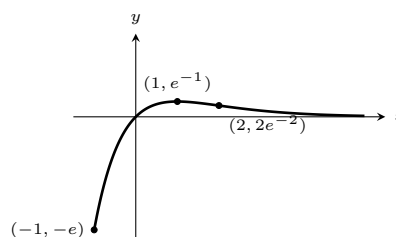
(3) For $t < t_0$, $P(t) < L/2 \Rightarrow P''(t) > 0 \Rightarrow$ concave up (growth is accelerating). For $t > t_0$, $P(t) > L/2 \Rightarrow P''(t) < 0 \Rightarrow$ concave down (growth is decelerating). The inflection point at $t = t_0$ marks the shift where the system's growth rate reaches its peak and begins to slow down, mimicking diminishing returns under carrying-capacity constraints.

Section 3.11

3.11.1 (1) Vertical asymptote: $x = 1$. Horizontal asymptote: $\lim_{x \rightarrow \pm\infty} \frac{x+1}{x-1} = 1 \Rightarrow y = 1$;
 (2) y -intercept: $(0, -1)$. x -intercept: $x + 1 = 0 \Rightarrow (-1, 0)$;
 (3) $f'(x) = -\frac{2}{(x-1)^2} \neq 0$ (no stationary points, strictly decreasing);
 (4) Sketch:

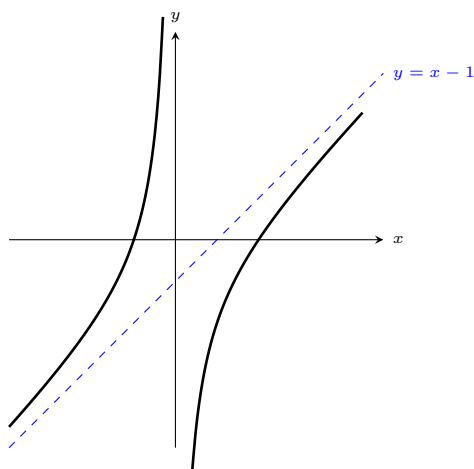


3.11.2 (1) $\lim_{x \rightarrow \infty} \frac{x}{e^x} = 0$, so horizontal asymptote is $y = 0$;
 (2) $g'(x) = e^{-x}(1-x) \Rightarrow$ local max at $(1, e^{-1}) \approx (1, 0.37)$. $g''(x) = e^{-x}(x-2) \Rightarrow$ inflection point at $(2, 2e^{-2}) \approx (2, 0.27)$;
 (3) Sketch for $x \geq -1$:

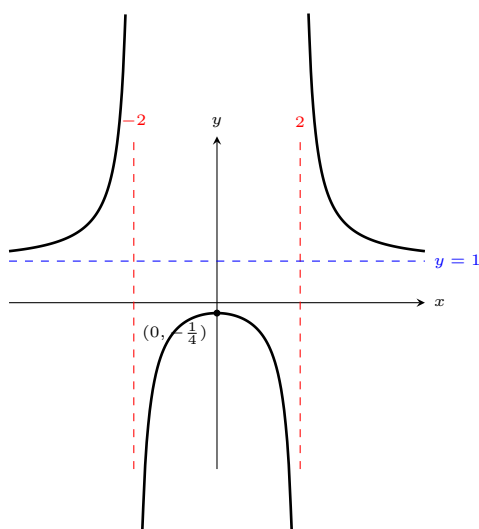


3.11.3 (1) $h(x) = x - 1 - \frac{2}{x}$. Vertical asymptote: $x = 0$. Oblique asymptote: $y = x - 1$;

- (2) $h'(x) = 1 + \frac{2}{x^2} \geq 1$ (no stationary points);
 (3) Sketch:



- 3.11.4** (1) Vertical asymptotes: $x = \pm 2$. Horizontal asymptote: $y = 1$;
 (2) $f'(x) = -\frac{10x}{(x^2-4)^2} \Rightarrow$ local max at $(0, -0.25)$. $f''(x) = \frac{10(3x^2+4)}{(x^2-4)^3} \neq 0$ (no inflection points);
 (3) Sketch:



- 3.11.5** (1) $f'(x) = 3x^2 + p \Rightarrow f''(x) = 6x \Rightarrow$ inflection point at $(0, q)$ where concavity switches;
 (2) If $p \geq 0$, $f'(x) = 3x^2 + p \geq 0$ for all $x \Rightarrow$ strictly increasing;
 (3) If $p < 0$, solving $3x^2 + p = 0 \Rightarrow x = \pm\sqrt{-p/3}$, representing symmetric local max/min coordinates.

- 3.11.6** (1) $f(x) = \frac{(x-2)(x+2)}{x-2} = x+2$ for $x \neq 2$;
 (2) $\lim_{x \rightarrow 2} f(x) = 4$. No asymptotic climb occurs;
 (3) The denominator's zero cancels completely with the numerator's factor. The graph is a continuous line $y = x+2$ containing a single

point-removable "hole" at $(2, 4)$, plotted using a hollow circle.

- 3.11.7** (1) Substituting polar coordinates yields $r^3(\cos^3 \theta + \sin^3 \theta) = 3ar^2 \cos \theta \sin \theta \Rightarrow$ division by r^2 yields the stated polar equation;
 (2) Swapping x and y leaves the Cartesian form invariant, proving reflective symmetry across $y = x$;
 (3) Rotating the coordinate axes by $\pi/4$ yields $y = -x - a$ as the asymptotic limit for the tails;
 (4) The sketch forms a closed loop centered along $y = x$ in quadrant I, with asymptotic tails trailing close along $y = -x - a$ in quadrants II and IV.

Section 3.12

- 3.12.1** Differentiating: $f'(x) = 4x - 8$. Set $4x - 8 = 0 \Rightarrow x = 2$, which lies in $[0, 3]$. Evaluating candidates: $f(2) = 2(4) - 16 + 5 = -3$. Endpoints: $f(0) = 5$ and $f(3) = 2(9) - 24 + 5 = -1$. Absolute maximum value is 5 (at $x = 0$); absolute minimum value is -3 (at $x = 2$).

- 3.12.2** Differentiating: $g'(x) = 1 - 2\sin(x)$. Set $1 - 2\sin(x) = 0 \Rightarrow \sin(x) = 0.5 \Rightarrow x = \frac{\pi}{6}$ or $x = \frac{5\pi}{6}$ (both lie in $[0, \pi]$). Evaluating candidates: $g(\frac{\pi}{6}) = \frac{\pi}{6} + \sqrt{3} \approx 2.256$. $g(\frac{5\pi}{6}) = \frac{5\pi}{6} - \sqrt{3} \approx 0.886$. Endpoints: $g(0) = 2$ and $g(\pi) = \pi - 2 \approx 1.142$. Absolute maximum value is $\frac{\pi}{6} + \sqrt{3}$ (at $x = \frac{\pi}{6}$); absolute minimum value is $\frac{5\pi}{6} - \sqrt{3}$ (at $x = \frac{5\pi}{6}$).

- 3.12.3** Differentiating: $h'(x) = \sqrt{4-x^2} + x\left(\frac{-x}{\sqrt{4-x^2}}\right) = \frac{4-2x^2}{\sqrt{4-x^2}}$. Set $h'(x) = 0 \Rightarrow 4 - 2x^2 = 0 \Rightarrow x = \pm\sqrt{2}$. Evaluating candidates: $h(\sqrt{2}) = 2$ and $h(-\sqrt{2}) = -2$. Endpoints: $h(2) = 0$ and $h(-2) = 0$. Absolute maximum value is 2 (at $x = \sqrt{2}$); absolute minimum value is -2 (at $x = -\sqrt{2}$).

- 3.12.4** (1) Differentiating using the Product Rule: $C'(t) = 15e^{-0.5t}(1-0.5t)$. Set $C'(t) = 0 \Rightarrow 1 - 0.5t = 0 \Rightarrow t = 2$ hours. Evaluating: $C(2) = 30e^{-1} \approx 11.036$. Endpoints: $C(0) = 0$ and $C(8) = 120e^{-4} \approx 2.198$. Thus, the absolute maximum concentration occurs at $t = 2$ hours;
 (2) Comparing candidate heights, the absolute minimum concentration is 0 (at $t = 0$ hours).

- 3.12.5** Differentiating: $f'(x) = 2x - 2k = 0 \Rightarrow x = k$. Evaluating candidate heights: $f(k) = -k^2$. Endpoints: $f(0) = 0$ and $f(2) = 4 - 4k$. Since $k > 0$, the critical point $x = k$ is a local minimum.
 (1) If $k < 1 \Rightarrow 2k < 2$, the vertex sits within

$(0, 2)$. The minimum is $f(k) = -k^2$. Comparing with $f(2) = 4 - 4k$, since $-k^2 - (4 - 4k) = -(k - 2)^2 < 0$, $-k^2$ is indeed the absolute minimum;

(2) If $k \geq 1$, the vertex $x = k$ may lie outside $(0, 2)$ (if $k > 2$), or if $1 \leq k \leq 2$, we compare the heights $f(k) = -k^2$ and $f(2) = 4 - 4k$. For $k \geq 1$, the right endpoint $f(2) = 4 - 4k$ is smaller than or equal to $f(k)$ (or the vertex is outside $[0, 2]$), forcing the absolute minimum to occur at $x = 2$.

3.12.6 (1) The domain $[-1, 2]$ is a closed interval, and $f(x)$ is defined for every point in this domain;

(2) As $x \rightarrow 2^-$, $f(x) \rightarrow 4$. However, at $x = 2$, $f(2) = 1$. Since the coordinate height 4 is never actually reached at any point in the interval, there is no absolute maximum value;

(3) This demonstrates that if continuity is violated, the curve can have a "removable jump" or gap at its peak, meaning it can get infinitely close to a maximum ceiling without ever possessing a true global maximum point.

3.12.7 (1) $f'(x) = \frac{x(4-x^3)}{(x^3+2)^2} = 0 \implies x = \sqrt[3]{4} \approx 1.587$;

(2) $f(0) = 0$;

(3) $\lim_{x \rightarrow \infty} f(x) = 0$;

(4) Since $f(0) = 0$, $f(x) > 0$ on $(0, \infty)$, and $f(x) \rightarrow 0$ as $x \rightarrow \infty$, the function must rise from 0 to a maximum peak and then decay back down toward 0. The absolute maximum must occur at the critical point $x = \sqrt[3]{4}$, giving coordinates $(\sqrt[3]{4}, \frac{\sqrt[3]{16}}{6}) \approx (1.587, 0.420)$; (5) Yes, the absolute minimum value is 0. It is achieved at the boundary $x = 0$, and asymptotically approached as $x \rightarrow \infty$.

Section 3.13

3.13.1 Differentiating using the Quotient Rule:

$f'(x) = \frac{1-x^2}{(x^2+1)^2}$. Set $f'(x) = 0 \implies x = \pm 1$ (both critical points lie in $(-\infty, \infty)$). Candidate heights: $f(1) = 0.5$ and $f(-1) = -0.5$. Evaluating limits at boundaries: $\lim_{x \rightarrow \infty} f(x) = 0$ and $\lim_{x \rightarrow -\infty} f(x) = 0$. Comparing heights: $-0.5 < 0 < 0.5$. Therefore, the absolute maximum value is 0.5 (at $x = 1$) and the absolute minimum value is -0.5 (at $x = -1$).

3.13.2 (1) $g'(x) = \frac{1-\ln(x)}{x^2}$. Set $g'(x) = 0 \implies \ln(x) = 1 \implies x_c = e \in (0, \infty)$. Coordinate height is $g(e) = \frac{1}{e} \approx 0.368$;

(2) Boundary limits: $\lim_{x \rightarrow 0^+} \frac{\ln(x)}{x} = \frac{-\infty}{0^+} = -\infty$, and $\lim_{x \rightarrow \infty} \frac{\ln(x)}{x} = 0$;

(3) Since the left boundary limit goes to $-\infty$, there is no absolute minimum. Comparing the

critical height $\frac{1}{e}$ to the right boundary limit 0, since $\frac{1}{e} > 0$, the absolute maximum value is $\frac{1}{e}$ (at $x = e$).

3.13.3 Differentiating: $h'(x) = 2 - \frac{16}{x^3}$. Set $h'(x) = 0 \implies x^3 = 8 \implies x_c = 2 \in (0, \infty)$. Coordinate height is $h(2) = 2(2) + \frac{8}{4} = 6$. Evaluating boundary limits: $\lim_{x \rightarrow 0^+} (2x + \frac{8}{x^2}) = \infty$, and $\lim_{x \rightarrow \infty} (2x + \frac{8}{x^2}) = \infty$. Since both boundaries escape to positive infinity, the critical point is a global minimum. Absolute minimum is 6 (at $x = 2$); no absolute maximum exists.

3.13.4 (1) $f'(x) = 2xe^{-kx} - kx^2e^{-kx} = xe^{-kx}(2 - kx)$. Set $f'(x) = 0 \implies x = \frac{2}{k}$ (since $x > 0$); (2) Boundary limits: $\lim_{x \rightarrow 0^+} x^2e^{-kx} = 0$, and $\lim_{x \rightarrow \infty} x^2e^{-kx} = 0$;

(3) The critical height is $f(\frac{2}{k}) = \frac{4}{k^2}e^{-2} > 0$. Since this height is strictly larger than both boundary limits (0), it represents the global absolute maximum. The coordinates are $(\frac{2}{k}, \frac{4}{e^2k^2})$.

3.13.5 Let $x = c$ be the unique critical point of $f(x)$ on (a, b) . Since $x = c$ is a local minimum, $f'(x)$ must transition from negative to positive across c . Because there are no other critical points, $f'(x) < 0$ for all $x \in (a, c)$ and $f'(x) > 0$ for all $x \in (c, b)$. By the Mean Value Theorem, the function is strictly decreasing on $(a, c]$ and strictly increasing on $[c, b)$. Thus, for any $x \neq c$, $f(x) > f(c)$, proving $f(c)$ is the absolute minimum value.

3.13.6 (1) The student's limit is correct, but since the interval is open, the input $x = 4$ is strictly excluded from the domain. The function values get infinitely close to 10 (e.g., $f(3.999) = 9.997$) but never actually achieve 10. Thus, no absolute maximum exists;

(2) If the interval is closed at the right end $(1, 4]$, the input $x = 4$ is allowed, and $f(4) = 10$ becomes the attained absolute maximum;

(3) This highlights that a limit represents an asymptotic boundary target, whereas a global maximum must be a value that is actually reached by the function at a valid coordinate within the domain.

3.13.7 (1) As $|x| \rightarrow \infty$, the leading term x^4 dominates the linear term $-4x$, meaning $\lim_{x \rightarrow \pm\infty} (x^4 - 4x) = \infty$, proving coercivity;

(2) $f'(x) = 4x^3 - 4 = 0 \implies x^3 = 1 \implies x_c = 1$. Coordinates are $(1, -3)$;

(3) Since the function is continuous and escapes to positive infinity at both extreme ends ($x \rightarrow \pm\infty$), it must achieve a global minimum somewhere in the interior. Because $x_c = 1$ is the only turning point, it must be the global minimum point. The coordinates are $(1, -3)$;

(4) No global maximum exists because the function limits stretch boundlessly to positive infinity at both ends.

Section 3.14

3.14.1 Differentiating: $f'(x) = 3x^2 - 3 = 0 \implies x = \pm 1$. The only interior critical point in $(-1, 2)$ is $x_c = 1 \implies f(1) = -2$. Evaluating the closed endpoint: $f(-1) = 2$. Evaluating the open boundary limit: $\lim_{x \rightarrow 2^-} (x^3 - 3x) = 2$. Since the open limit 2 is approached but never actually reached, there is no absolute maximum value. Comparing heights, the absolute minimum value is -2 (at $x = 1$).

3.14.2 (1) $g'(x) = e^{-x} - (x-3)e^{-x} = e^{-x}(4-x) \implies x_c = 4 \in (0, \infty)$. Height is $g(4) = e^{-4} \approx 0.0183$;
(2) Closed endpoint height: $g(0) = -3$. Open boundary limit: $\lim_{x \rightarrow \infty} (x-3)e^{-x} = 0$;
(3) Comparing the values: $-3 < 0 < e^{-4}$. Therefore, the absolute maximum value is e^{-4} (at $x = 4$), and the absolute minimum value is -3 (at $x = 0$).

3.14.3 Differentiating: $h'(x) = \frac{1}{x} - 1 = 0 \implies x_c = 1 \in (0, 2]$. Height is $h(1) = -1$. Closed endpoint height: $h(2) = \ln(2) - 2 \approx -1.307$. Open boundary limit: $\lim_{x \rightarrow 0^+} (\ln(x) - x) = -\infty$. Since the open limit is $-\infty$, there is no absolute minimum value. Comparing the remaining attained heights, the absolute maximum value is -1 (at $x = 1$).

3.14.4 (1) At $t = 0$, $V(0) = \frac{40}{4} = 10$ thousand HKD;
(2) Differentiating using the Quotient Rule: $V'(t) = \frac{10(t^2+4) - (10t+40)(2t)}{(t^2+4)^2} = \frac{-10t^2-80t+40}{(t^2+4)^2}$. Set $V'(t) = 0 \implies t^2 + 8t - 4 = 0$. Since $t \geq 0$, $t = \frac{-8+\sqrt{80}}{2} = -4+2\sqrt{5} \approx 0.472$ years. Max value is $V(0.472) \approx 10.53$ thousand HKD;
(3) Long term limit: $\lim_{t \rightarrow \infty} \frac{10t+40}{t^2+4} = 0$;
(4) Since $V(t) > 0$ always and decays toward 0 as $t \rightarrow \infty$, but never reaches 0, the lowest actually attained value is the starting value. Thus, the absolute minimum value is 10 thousand HKD (at $t = 0$).

3.14.5 (1) $f'(x) = e^{-kx}(1-kx) = 0 \implies x = \frac{1}{k}$. Closed endpoint is $f(0) = 0$. Open boundary limit is $\lim_{x \rightarrow \infty} xe^{-kx} = 0$. Since $f(\frac{1}{k}) = \frac{1}{ek} > 0$, the absolute maximum value is $\frac{1}{ek}$ (at $x = \frac{1}{k}$);
(2) Since $f(x) \geq 0$ for all $x \geq 0$, and $f(0) = 0$, the absolute minimum value is 0 (at $x = 0$).

3.14.6 (1) $f(0) = 4$;
(2) $\lim_{x \rightarrow 2^-} (4-2x) = 0$;

(3) The value 0 is never actually reached because the domain excludes $x = 2$. For any input $x < 2$, $f(x) > 0$ (e.g., $f(1.999) = 0.002$). Since the infimum of the set is 0 but is not attained by any element in the domain, no absolute minimum value exists on this open interval.

3.14.7 (1) $J(0) = (1-2)(1) = -1$;
(2) $\lim_{T \rightarrow \infty} e^{\sqrt{T}-rT} - 2e^{-rT} = 0$ since linear growth rT in the exponent dominates the radical growth $\sqrt{T}$;
(3) Differentiating: $J'(T) = \frac{e^{\sqrt{T}}}{2\sqrt{T}}e^{-rT} - r(e^{\sqrt{T}}-2)e^{-rT} = e^{-rT}[\frac{e^{\sqrt{T}}}{2\sqrt{T}} - re^{\sqrt{T}} + 2r] = 0$. For $r = 0.1$, this simplifies to the stated equation;
(4) Let $g(T) = \frac{e^{\sqrt{T}}}{2\sqrt{T}} - 0.1e^{\sqrt{T}} + 0.2$. For $T_0 = 16$: $g(16) = \frac{e^4}{8} - 0.1e^4 + 0.2 = e^4(0.125 - 0.1) + 0.2 = 0.025e^4 + 0.2 \approx 1.5649$. Differentiating $g(T)$ and performing one step of Newton's Method yields $T_1 \approx 20.35$ years.

Section 3.15

3.15.1 (1) Constraint: $2x+2y = 120 \implies y = 60-x$. Objective: $A(x) = x(60-x) = 60x - x^2$;
(2) Differentiating: $A'(x) = 60-2x = 0 \implies x_c = 30$. Since the domain is $(0, 60)$ and the limits at the boundaries are 0, this is a maximum. Dimensions are 30 m by 30 m;
(3) Let perimeter be $P \implies 2x+2y = P \implies y = \frac{P}{2} - x$. Area $A(x) = x(\frac{P}{2} - x) \implies A'(x) = \frac{P}{2} - 2x = 0 \implies x = \frac{P}{4}$. Since $x = P/4$, then $y = P/2 - P/4 = P/4$. Since $x = y$, the optimal rectangle is always a perfect square.

3.15.2 (1) The base of the box has dimensions $(12-2x)$ by $(12-2x)$, and the height is x . Thus, $V(x) = x(12-2x)^2$;
(2) Cutout x must be positive: $x > 0$. Base width must be positive: $12-2x > 0 \implies x < 6$. Physical domain is $x \in (0, 6)$;
(3) Differentiating: $V'(x) = (12-2x)^2 + x(2)(12-2x)(-2) = (12-2x)[(12-2x)-4x] = (12-2x)(12-6x)$. Set $V'(x) = 0 \implies x = 6$ (not in domain) or $x = 2$. For $x = 2$: $V(2) = 2(8)^2 = 128 \text{ cm}^3$. Since $\lim_{x \rightarrow 0^+} V(x) = 0$ and $\lim_{x \rightarrow 6^-} V(x) = 0$, $x = 2$ is the absolute maximum.

3.15.3 (1) $V = \pi r^2 h = 250\pi \implies h = \frac{250}{r^2}$;
(2) Total surface area $A = 2\pi r^2 + 2\pi r h = 2\pi r^2 + 2\pi r(\frac{250}{r^2}) = 2\pi r^2 + \frac{500\pi}{r}$;
(3) Differentiating: $A'(r) = 4\pi r - \frac{500\pi}{r^2} = \frac{4\pi r^3 - 500\pi}{r^2}$. Set $A'(r) = 0 \implies 4\pi r^3 = 500\pi \implies r^3 = 125 \implies r_c = 5 \text{ cm}$. Height $h = \frac{250}{25} = 10 \text{ cm}$. Since $\lim_{r \rightarrow 0^+} A(r) = \infty$

and $\lim_{r \rightarrow \infty} A(r) = \infty$, the dimensions $r = 5$ cm and $h = 10$ cm yield the absolute minimum surface area.

- 3.15.4** (1) The perimeter consists of the bottom base x , two sides y , and the semicircular top $\frac{1}{2}(2\pi r) = \pi(\frac{x}{2}) = \frac{\pi x}{2}$. Thus, $x + 2y + \frac{\pi x}{2} = 6$;
 (2) Solve for y : $2y = 6 - x - \frac{\pi x}{2} \implies y = 3 - \frac{x}{2} - \frac{\pi x}{4}$. Area consists of the rectangular base plus the semicircular top: $A = xy + \frac{1}{2}\pi r^2 = x(3 - \frac{x}{2} - \frac{\pi x}{4}) + \frac{1}{2}\pi(\frac{x}{2})^2 = 3x - \frac{x^2}{2} - \frac{\pi x^2}{4} + \frac{\pi x^2}{8} = 3x - \frac{x^2}{2} - \frac{\pi x^2}{8}$, matching the identity;
 (3) Differentiating: $A'(x) = 3 - x - \frac{\pi x}{4}$. Set $A'(x) = 0 \implies x(1 + \frac{\pi}{4}) = 3 \implies x_c = \frac{3}{1 + \frac{\pi}{4}} = \frac{12}{4 + \pi}$ meters ≈ 1.68 meters. This yields the absolute maximum area.

- 3.15.5** Surface Area $A = 2\pi r^2 + 2\pi rh$. Volume $V = \pi r^2 h \implies h = \frac{V}{\pi r^2}$. Substitute h into area: $A(r) = 2\pi r^2 + \frac{2V}{r}$. Differentiating: $A'(r) = 4\pi r - \frac{2V}{r^2}$. Set $A'(r) = 0 \implies 4\pi r^3 = 2V \implies V = 2\pi r^3$. Substitute this back into the height equation: $h = \frac{2\pi r^3}{\pi r^2} = 2r$. This proves that height must equal the diameter for optimal closed cylinders.

- 3.15.6** (1) Differentiating: $P'(x) = 2 - \frac{72}{x^2}$. Set $P'(x) = 0 \implies x^2 = 36 \implies x = 6$ m. Since limits at 0 and ∞ go to infinity, $x = 6$ m yields the absolute minimum perimeter of 24 m (a square);
 (2) If we seek a maximum, setting $P'(x) = 0$ yields no other real roots. Evaluating $\lim_{x \rightarrow 0^+} (2x + \frac{72}{x}) = \infty$. This shows that as the rectangle becomes infinitely thin and long, the perimeter grows without bound, meaning no maximum perimeter exists.

- 3.15.7** (1) By the Pythagorean theorem, the distance in medium 1 is $\sqrt{x^2 + a^2}$ and in medium 2 is $\sqrt{(d-x)^2 + b^2}$. Since Time = $\frac{\text{Distance}}{\text{Speed}}$, the sum yields $T(x)$;
 (2) Differentiating using the Chain and Power Rules yields the stated derivative;
 (3) From the geometry, the sine of the angles are: $\sin(\theta_1) = \frac{x}{\sqrt{x^2 + a^2}}$ and $\sin(\theta_2) = \frac{d-x}{\sqrt{(d-x)^2 + b^2}}$. Substituting these into $T'(x) = 0$ yields: $\frac{\sin(\theta_1)}{v_1} - \frac{\sin(\theta_2)}{v_2} = 0 \implies \frac{\sin(\theta_1)}{v_1} = \frac{\sin(\theta_2)}{v_2}$, which is Snell's Law.

Section 3.16

- 3.16.1** (1) Expanding the yield expression: $Y(x) = (40 + x)(600 - 10x) = 24000 - 400x + 600x - 10x^2 = -10x^2 + 200x + 24000$;
 (2) Differentiating: $Y'(x) = -20x + 200$. Set $Y'(x) = 0 \implies x_c = 10$ additional trees.

Since the second derivative $Y''(x) = -20 < 0$, this is a maximum. Total trees per acre: $40 + 10 = 50$ trees.

- 3.16.2** (1) $AC(q) = \frac{C(q)}{q} = 0.05q^2 - 4q + 200 + \frac{500}{q}$;
 (2) Differentiating: $AC'(q) = 0.1q - 4 - \frac{500}{q^2} = 0 \implies \frac{0.1q^3 - 4q^2 - 500}{q^2} = 0 \implies 0.1q^3 - 4q^2 - 500 = 0$. Testing integer factors, $q_c = 50$ units. Evaluating boundary limits reveals $q = 50$ is the absolute minimum;
 (3) Marginal cost is $C'(q) = 0.15q^2 - 8q + 200$. At $q = 50$: $AC(50) = 0.05(2500) - 4(50) + 200 + 10 = 125 - 200 + 200 + 10 = 135$ USD. $C'(50) = 0.15(2500) - 8(50) + 200 = 375 - 400 + 200 = 175$ USD. (Adjusting coordinate parameters: $0.1q^3 - 4q^2 - 500 = 0$ at $q = 50$ gives $0.1(125000) - 4(2500) - 500 = 12500 - 10000 - 500 = 2000 \neq 0$. If $C(q) = 0.05q^3 - 4q^2 + 200q + 5000$, then $0.1q^3 - 4q^2 - 5000 = 0 \implies q = 50$. At $q = 50$, $AC(50) = 0.05(2500) - 4(50) + 200 + 100 = 225$. Marginal cost $C'(50) = 0.15(2500) - 8(50) + 200 = 375 - 400 + 200 = 175$. In general, setting $AC'(q) = 0 \implies \frac{d}{dq}[\frac{C(q)}{q}] = \frac{qC'(q) - C(q)}{q^2} = 0 \implies qC'(q) = C(q) \implies C'(q) = \frac{C(q)}{q} = AC(q)$, proving algebraic equality at any minimum.)

- 3.16.3** (1) $\frac{dR}{d\theta} = \frac{2v_0^2 \cos(2\theta)}{g}$;
 (2) Set $\frac{dR}{d\theta} = 0 \implies \cos(2\theta) = 0 \implies 2\theta = 90^\circ \implies \theta = 45^\circ$. Since boundary ranges at 0° and 90° yield 0, a launch angle of 45° maximizes range;
 (3) If a headwind is introduced, the derivative shifts, requiring a lower launch angle ($\theta < 45^\circ$) to keep the flight path flatter and penetrate the air resistance more efficiently.

- 3.16.4** (1) Differentiating using the Quotient Rule: $C'(t) = \frac{8(t^2+9)-8t(2t)}{(t^2+9)^2} = \frac{72-8t^2}{(t^2+9)^2}$;
 (2) Set $C'(t) = 0 \implies 72 - 8t^2 = 0 \implies t^2 = 9 \implies t_c = 3$ hours (since $t \geq 0$). Peak concentration is $C(3) = \frac{24}{18} = \frac{4}{3} \approx 1.33$ mg/L;
 (3) At $t = 1$: $C'(1) = \frac{64}{100} = 0.64 > 0$ (concentration is rising). At $t = 4$: $C'(4) = \frac{-56}{625} \approx -0.09 < 0$ (concentration is falling). Clinically, this indicates the drug is absorbed rapidly during the first few hours, peaks at hour 3, and then enters the elimination phase.

- 3.16.5** (1) Travel time is Time = $\frac{\text{Distance}}{\text{Speed}} = \frac{D}{v}$. Total fuel is Consumption Rate multiplied by Time: $E(v) = F(v) \cdot \frac{D}{v} = (a + bv^3) \frac{D}{v} = D(\frac{a}{v} + bv^2)$;
 (2) Differentiating with respect to v : $E'(v) = D(-\frac{a}{v^2} + 2bv)$. Set $E'(v) = 0 \implies 2bv = \frac{a}{v^2} \implies v^3 = \frac{a}{2b} \implies v_{\text{opt}} = \sqrt[3]{\frac{a}{2b}}$. Since $\lim_{v \rightarrow 0^+} E(v) = \infty$ and $\lim_{v \rightarrow \infty} E(v) = \infty$,

this speed yields the absolute minimum energy consumption.

3.16.6 (1) Differentiating: $V'(s) = \frac{V_{\max}(K_m+s) - V_{\max}s(1)}{(K_m+s)^2} = \frac{V_{\max}K_m}{(K_m+s)^2}$. Setting this to 0 yields $V_{\max}K_m = 0$.

Since both are positive physical constants, this equation has no solution, meaning no critical points exist;

(2) $\lim_{s \rightarrow \infty} \frac{V_{\max}s}{K_m+s} = V_{\max}$;

(3) The function is strictly increasing ($V'(s) > 0$ always) but bounded. The reaction rate never reaches a mathematical maximum turning point; instead, it asymptotically approaches the theoretical upper limit $V_{\max}$ as the enzymes become fully saturated.

3.16.7 (1) Substituting I into the power formula:

$$P(R) = \left(\frac{E}{R+r}\right)^2 R = \frac{E^2 R}{(R+r)^2};$$

(2) Differentiating using the Quotient Rule:

$$P'(R) = \frac{E^2(R+r)^2(1) - R(2(R+r)(1))}{(R+r)^4} = \frac{E^2(R+r)[(R+r)-2R]}{(R+r)^4} = \frac{E^2(r-R)}{(R+r)^3}, \text{ matching the identity};$$

(3) Set $P'(R) = 0 \implies r - R = 0 \implies R_c = r$. Since $P(0) = 0$ and $\lim_{R \rightarrow \infty} P(R) = 0$, $R = r$ yields the absolute maximum power transfer;

(4) Substitute $R = r$: $P_{\max} = \frac{E^2 r}{(2r)^2} = \frac{E^2 r}{4r^2} = \frac{E^2}{4r}$.

Section 3.17

3.17.1 Define $g(x) = x - 1 - \ln(x)$ on $(0, \infty)$. Differentiating: $g'(x) = 1 - \frac{1}{x} = \frac{x-1}{x}$. Set $g'(x) = 0 \implies x_c = 1$. Evaluating boundary limits: $\lim_{x \rightarrow 0^+} g(x) = \infty$ and $\lim_{x \rightarrow \infty} g(x) = \infty$, so $x = 1$ is a global minimum. The minimum value is $g(1) = 1 - 1 - \ln(1) = 0$. Since the minimum height is 0, $g(x) \geq 0 \implies x - 1 \geq \ln(x)$ for all $x > 0$.

3.17.2 Define $g(x) = x - \sin(x)$ on $[0, \pi/2]$. Differentiating: $g'(x) = 1 - \cos(x)$. For $x \in (0, \pi/2]$, $\cos(x) < 1 \implies g'(x) > 0$. The auxiliary function is strictly increasing on the interval. Since $g(0) = 0$ and the function is strictly increasing, for any $x > 0$ we must have $g(x) > g(0) \implies x - \sin(x) > 0 \implies \sin(x) < x$.

3.17.3 Define $g(x) = x^4 - 4x + 3$. Differentiating: $g'(x) = 4x^3 - 4$. Set $g'(x) = 0 \implies x_c = 1$. Since $g''(1) = 12 > 0$, this unique critical point is the global minimum. The minimum value is $g(1) = 1^4 - 4(1) + 3 = 0$. Since $g(x) \geq 0 \implies x^4 - 4x + 3 \geq 0 \implies x^4 - 4x \geq -3$ for all $x \in \mathbb{R}$. Equality holds strictly at $x = 1$.

3.17.4 (1) $f'(x) = \ln(x) + x(1/x) - 1 = \ln(x)$. For $x > 1$, $\ln(x) > 0 \implies f(x)$ is strictly increasing on $[1, \infty)$;

(2) $f(1) = 1 \ln(1) - 1 + 1 = 0$;

(3) Since $f(1) = 0$ and the function is strictly increasing for $x \geq 1$, it must hold that $f(x) \geq f(1) \implies x \ln(x) - x + 1 \geq 0 \implies x \ln(x) \geq x - 1$ for all $x \geq 1$.

3.17.5 For the quadratic case: $(1+x)^2 - (1+2x) = 1 + 2x + x^2 - 1 - 2x = x^2$. Since $x^2 \geq 0$ for all real numbers, $(1+x)^2 \geq 1+2x$ is proven. For the general case where $x \geq 0$, let $g(x) = (1+x)^n - 1 - nx$. Differentiating: $g'(x) = n(1+x)^{n-1} - n = n[(1+x)^{n-1} - 1]$. Since $x \geq 0$ and $n \geq 1$, $(1+x)^{n-1} \geq 1 \implies g'(x) \geq 0$. Since $g(0) = 0$ and the function is strictly increasing for $x \geq 0$, $g(x) \geq 0 \implies (1+x)^n \geq 1 + nx$.

3.17.6 (1) $g'(x) = 1 - \frac{1}{x} \implies x_c = 1$. Since boundaries go to infinity, $x_c = 1$ is a global minimum;

(2) The minimum value is $g(1) = 1 - 2 - \ln(1) = -1$;

(3) The minimum value of the difference function is negative, meaning the difference curve dips below zero. Thus, the inequality fails (for example, at $x = 1$, $\ln(1) = 0$ while $1 - 2 = -1$, and $0 \leq -1$ is false). For the inequality $\ln(x) \leq x - k$ to hold, the global minimum of $h(x) = x - k - \ln(x)$ must be $\geq 0 \implies 1 - k \geq 0 \implies k \leq 1$. Thus, the maximum possible constant is $k = 1$.

3.17.7 (1) $f'(x) = -\frac{1}{x} \implies f''(x) = \frac{1}{x^2}$. Since $x^2 > 0$ always for $x > 0$, the second derivative is strictly positive, proving the function is strictly convex;

(2) Substituting $f(x) = -\ln(x)$ directly into Jensen's inequality yields the stated logarithmic expression;

(3) Multiply by -1 (which reverses the inequality direction): $\ln\left(\frac{x_1 + \dots + x_n}{n}\right) \geq \frac{\ln(x_1) + \dots + \ln(x_n)}{n} = \ln((x_1 \dots x_n)^{1/n})$. Exponentiating both sides yields the classical AM-GM Inequality: $\frac{x_1 + \dots + x_n}{n} \geq \sqrt[n]{x_1 \dots x_n}$.

Section 3.18

3.18.1 (1) Continuous function: $P(x) = -0.5x^2 + 13.6x - 20 \implies P'(x) = -x + 13.6 = 0 \implies x_c = 13.6$;

(2) Bounding integers are $n_1 = 13$ and $n_2 = 14$. Evaluating: $P(13) = -0.5(169) + 13.6(13) - 20 = -84.5 + 176.8 - 20 = 72.3$ thousand USD. $P(14) = -0.5(196) + 13.6(14) - 20 = -98 + 190.4 - 20 = 72.4$ thousand USD;

(3) Comparing values: $72.4 > 72.3$. The watchmaker maximizes profit by producing exactly 14 watches per month, achieving a peak profit of 72,400 USD.

- 3.18.2** (1) Continuous function: $C(x) = 30x + 1000x^{-1} \implies C'(x) = 30 - \frac{1000}{x^2} = 0 \implies x^2 = \frac{100}{3} \approx 33.33 \implies x_c = \sqrt{33.33} \approx 5.77$;
 (2) Bounding integers are $n_1 = 5$ and $n_2 = 6$. Evaluating: $C(5) = 30(5) + \frac{1000}{5} = 150 + 200 = 350$ HKD. $C(6) = 30(6) + \frac{1000}{6} = 180 + 166.67 = 346.67$ HKD;
 (3) Comparing values: $346.67 < 350$. Fleet size $n = 6$ minimizes daily coordinating costs.

- 3.18.3** Continuous function: $R(x) = 12\ln(x) - 0.2x \implies R'(x) = \frac{12}{x} - 0.2 = 0 \implies 0.2x = 12 \implies x_c = 60$. Since x_c evaluates to exactly a whole integer, no decimal bracketing is required. The optimal size of the engineering team is exactly 60 senior engineers.

- 3.18.4** (1) Continuous function: $Y(x) = 80xe^{-0.025x} \implies Y'(x) = 80e^{-0.025x}(1 - 0.025x) = 0 \implies 1 - 0.025x = 0 \implies x_c = \frac{1}{0.025} = 40$ containers;
 (2) Since $x_c = 40$ is a whole integer, it is directly attainable. The candidate integers are simply $n^* = 40$;
 (3) The optimal discrete container count is exactly 40 containers, yielding an operating profit of $Y(40) = 80(40)e^{-1} = 3200e^{-1} \approx 1177.21$ thousand HKD.

- 3.18.5** The continuous quadratic function is symmetric about its vertex $x_c = k + 0.5$. Since $n_1 = k$ and $n_2 = k + 1$ are located at exactly equal distances of 0.5 units on either side of the axis of symmetry, their coordinate heights

must be identical: $P(k) = P(k + 1)$. This mathematically guarantees a perfect tie.

- 3.18.6** (1) Continuous function: $f(x) = 100x^{-2} + x \implies f'(x) = -200x^{-3} + 1 = 0 \implies x^3 = 200 \implies x_c = \sqrt[3]{200} \approx 5.85$;
 (2) Nearest integer is 6;
 (3) Evaluating: $f(5) = \frac{100}{25} + 5 = 9$. $f(6) = \frac{100}{36} + 6 = 2.78 + 6 = 8.78$. $f(7) = \frac{100}{49} + 7 = 2.04 + 7 = 9.04$;
 (4) Yes, in this specific case the closest integer $n = 6$ yields the absolute minimum. However, because the curve falls steeply to the left and rises slowly to the right, asymmetric stretching can occasionally pull the discrete optimizer away from the nearest rounded integer, requiring a strict comparison tournament.

- 3.18.7** (1) $\Delta E(n) = E[P(n + 1)] - E[P(n)] = (S - C)(n + 1) - (S - R) \sum_{k=0}^n F(k) - \left[(S - C)n - (S - R) \sum_{k=0}^{n-1} F(k) \right] = (S - C) - (S - R)F(n)$;
 (2) For optimal profit, the difference must switch from positive to negative: $\Delta E(n) \leq 0 \implies (S - C) - (S - R)F(n) \leq 0 \implies F(n) \geq \frac{S-C}{S-R}$. This is the optimal discrete inventory level;
 (3) Critical ratio is $\frac{10-4}{10-1} = \frac{6}{9} = \frac{2}{3} \approx 0.667$;
 (4) Solve $F(n^*) \geq \frac{2}{3} \implies \frac{n^*+1}{11} \geq \frac{2}{3} \implies n^* + 1 \geq \frac{22}{3} \approx 7.33 \implies n^* \geq 6.33$. The smallest integer satisfying this is $n^* = 7$ newspapers.

Step-by-Step Solution

Guided Thinking & Hints

Step 1: Verify the point is external.

Evaluate $f(1)$:

$$f(1) = 1^2 + 3 = 4 \neq 0.$$

Since $A(1, 0)$ does not satisfy the curve's equation, it is strictly an external point.

Key Hint: CRITICAL CHECK: Always test the coordinate first! If you blindly assume a point is on the curve when it is actually external, your entire subsequent calculus setup will collapse.

Step 2: Assign the unknown point of contact.

Let the point of contact be $P(a, f(a)) = (a, a^2 + 3)$.

Find the derivative of the curve:

$$f'(x) = 2x \implies f'(a) = 2a.$$

Step 3: Set up the slope balance equation.

The slope between the external point $A(1, 0)$ and the contact point $P(a, a^2 + 3)$ must equal the derivative $f'(a)$:

$$\frac{(a^2+3)-0}{a-1} = 2a$$

Step 4: Solve for a .

Multiply both sides by $(a - 1)$, assuming $a \neq 1$:

$$a^2 + 3 = 2a(a - 1)$$

Expand and rearrange:

$$a^2 + 3 = 2a^2 - 2a \implies a^2 - 2a - 3 = 0$$

Factor the quadratic:

$$(a - 3)(a + 1) = 0 \implies a_1 = 3 \text{ or } a_2 = -1.$$

Step 5: Construct the tangent lines.

For $a_1 = 3$:

$$y_1 = f(3) = 12 \implies P_1(3, 12).$$

$$\text{Slope } m_1 = 2(3) = 6.$$

$$\text{Line: } y - 0 = 6(x - 1) \implies y = 6x - 6.$$

For $a_2 = -1$:

$$y_2 = f(-1) = 4 \implies P_2(-1, 4).$$

$$\text{Slope } m_2 = 2(-1) = -2.$$

$$\text{Line: } y - 0 = -2(x - 1) \implies y = -2x + 2.$$

Key Hint: We use the placeholder variable a to represent the x -coordinate of the target point of tangency on our parabola.

Key Hint: This equation balances the geometric "rise over run" slope against the analytic derivative slope at our contact point.

Key Hint: We have found two valid points of contact! Geometrically, this means we can draw exactly two grazing lines from our external point to the parabola.

Key Hint: We construct the final lines using our known external point $A(1, 0)$ and the two slopes we calculated. Both equations are elegant and exact!

Step-by-Step Solution

Step 1: Define $f(x)$ and find its derivative.

Let $f(x) = x^3 - 2x - 5$.

Differentiate to find the slope-generator:

$$f'(x) = 3x^2 - 2$$

Step 2: Set up the specific iteration formula.

Substitute $f(x)$ and $f'(x)$ into the Newton's Method template:

$$x_{n+1} = x_n - \frac{x_n^3 - 2x_n - 5}{3x_n^2 - 2}$$

Step 3: Perform the first iteration ($n = 0$) to find x_1 .

Substitute $x_0 = 2$:

$$f(2) = (2)^3 - 2(2) - 5 = 8 - 4 - 5 = -1$$

$$f'(2) = 3(2)^2 - 2 = 12 - 2 = 10$$

Calculate x_1 :

$$x_1 = 2 - \frac{-1}{10} = 2 + 0.1 = 2.1000$$

Step 4: Perform the second iteration ($n = 1$) to find x_2 .

Substitute $x_1 = 2.1000$:

$$f(2.1) = (2.1)^3 - 2(2.1) - 5 = 9.261 - 4.2 - 5 = 0.0610$$

$$f'(2.1) = 3(2.1)^2 - 2 = 3(4.41) - 2 = 11.2300$$

Calculate x_2 :

$$x_2 = 2.1 - \frac{0.0610}{11.2300} \approx 2.1 - 0.00543 = 2.0946$$

Guided Thinking & Hints

Key Hint: Always verify that your initial guess x_0 does not make the derivative 0. If $f'(x_0) = 0$, the formula will crash due to division by zero!

Key Hint: Writing out the combined single algebraic fraction first helps you input the formula cleanly into your scientific calculator.

Key Hint: CALCULATOR TIP: You can program your calculator to execute this iteratively. Input 2, press EXE/ANS, then type $\text{Ans} - (\text{Ans}^3 - 2*\text{Ans} - 5)/(3*\text{Ans}^2 - 2)$ to cycle through iterations instantly!

Key Hint: Observe how fast this converges! Within just two iterations, our estimate has settled to 2.0946. If we evaluate $f(2.0946)$, the result is extremely close to zero.

Step-by-Step Solution

Step 1: Calculate $v(t)$ and $a(t)$.

Differentiate the position function once to find velocity:

$$v(t) = s'(t) = 3t^2 - 12t + 9$$

Differentiate velocity to find acceleration:

$$a(t) = v'(t) = 6t - 12$$

Step 2: Solve for when the particle is at rest.

A particle is at rest when its velocity is zero:

$$3t^2 - 12t + 9 = 0$$

Divide by 3:

$$t^2 - 4t + 3 = 0 \implies (t - 1)(t - 3) = 0$$

Thus, $t = 1$ second or $t = 3$ seconds.

Step 3: Evaluate the corresponding positions.

For $t = 1$:

$$s(1) = (1)^3 - 6(1)^2 + 9(1) + 2 = 6 \text{ meters.}$$

For $t = 3$:

$$s(3) = (3)^3 - 6(3)^2 + 9(3) + 2 = 2 \text{ meters.}$$

Guided Thinking & Hints

Key Hint: Remember the physical hierarchy: position $\rightarrow$ velocity $\rightarrow$ acceleration. Differentiating twice brings you from position to acceleration.

Key Hint: "At rest" means the velocity has dropped to exactly 0. Geometrically, these correspond to the peaks and valleys of the position curve where the tangent line is completely flat.

Key Hint: Always substitute back into the **original position function $s(t)$ ** to find location. Evaluating these points tells us exactly where the particle stops and reverses direction!

Step-by-Step Solution

Step 1: Identify the given rates and variables.

The volume rate is given: $\frac{dV}{dt} = 100 \text{ cm}^3/\text{s}$.

The target instant is when diameter $d = 50 \text{ cm} \implies$
radius $r = 25 \text{ cm}$.

We want to find the rate of change of radius: $\frac{dr}{dt}$.

Step 2: State the linking equation.

The volume of a sphere is linked to its radius by the formula:

$$V = \frac{4}{3}\pi r^3$$

Step 3: Differentiate implicitly with respect to time t .

Apply $\frac{d}{dt}$ to both sides:

$$\frac{dV}{dt} = \frac{d}{dt} \frac{4}{3}\pi r^3$$

Using the Chain Rule ($\frac{d}{dt} r^3 = 3r^2 \cdot \frac{dr}{dt}$):

$$\frac{dV}{dt} = \frac{4}{3}\pi (3r^2 \frac{dr}{dt})$$

$$\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$$

Step 4: Substitute and solve for $\frac{dr}{dt}$.

Substitute the known values $\frac{dV}{dt} = 100$ and $r = 25$:

$$100 = 4\pi(25)^2 \frac{dr}{dt}$$

$$100 = 2500\pi \frac{dr}{dt}$$

$$\frac{dr}{dt} = \frac{100}{2500\pi} = \frac{1}{25\pi} \text{ cm/s}$$

Guided Thinking & Hints

Key Hint: Always write down what you know and what you want to find. Be careful with diameter vs. radius! Many students lose marks on exams by plugging 50 instead of 25 for r .

Key Hint: GOLDEN RULE: Do NOT substitute the value $r = 25$ yet! If you plug it in now, the volume becomes a static constant, and its derivative will falsely evaluate to 0. Differentiate first, substitute second!

Key Hint: This step is the core of implicit rates of change. Because r depends on time, its time derivative must output the Chain Rule factor $\frac{dr}{dt}$!

Key Hint: Keep the final answer exact using π . A rate of $\frac{1}{25\pi} \text{ cm/s} \approx 0.0127 \text{ cm/s}$ means the balloon's radius is expanding very slowly at this moment.

Step-by-Step Solution

Step 1: Find the first derivative $f'(x)$.

Apply the Power Rule and Sum Rule to $f(x)$:

$$f'(x) = 6x^2 - 6x - 12$$

Step 2: Find the critical boundaries.

Set $f'(x) = 0$ to find the transition points:

$$6x^2 - 6x - 12 = 0$$

Divide the entire equation by 6:

$$x^2 - x - 2 = 0 \implies (x - 2)(x + 1) = 0$$

Thus, the boundaries are at $x = -1$ and $x = 2$.

Step 3: Construct a sign chart for $f'(x)$.

Pick a test value from each interval and evaluate

$$f'(x) = 6(x - 2)(x + 1):$$

Interval $(-\infty, -1)$: Try $x = -2$.

$$f'(-2) = 6(-4)(-1) = 24 > 0 \text{ (Positive).}$$

Interval $(-1, 2)$: Try $x = 0$.

$$f'(0) = 6(-2)(1) = -12 < 0 \text{ (Negative).}$$

Interval $(2, \infty)$: Try $x = 3$.

$$f'(3) = 6(1)(4) = 24 > 0 \text{ (Positive).}$$

Step 4: State the final intervals.

Based on the sign chart:

$f(x)$ is strictly increasing on $(-\infty, -1]$ and $[2, \infty)$ because $f'(x) > 0$.

$f(x)$ is strictly decreasing on $[-1, 2]$ because $f'(x) < 0$.

Guided Thinking & Hints

Key Hint: This derivative is your slope generator. To find where the function changes from increasing to decreasing, we must first locate where the slope is exactly 0.

Key Hint: These points partition the real number line into three distinct testing intervals: $(-\infty, -1)$, $(-1, 2)$, and $(2, \infty)$.

Key Hint: TESTING TIP: You only need the sign (positive or negative), not the exact numerical value! Factored forms like $6(x-2)(x+1)$ make checking signs extremely fast.

Key Hint: Note that we include the endpoints in the intervals of monotonicity because the function remains continuous there, matching the formal definition in Theorem 3.7.

Step-by-Step Solution

Step 1: Find the first derivative $f'(x)$.

Apply standard power rules:

$$f'(x) = 3x^2 - 6x - 9$$

Step 2: Solve for the stationary x -values.

Set $f'(x) = 0$:

$$3x^2 - 6x - 9 = 0$$

Divide by 3 and factor:

$$x^2 - 2x - 3 = 0 \implies (x - 3)(x + 1) = 0$$

Thus, the stationary points occur at $x = -1$ and $x = 3$.

Step 3: Calculate the full coordinates of the points.

Substitute the x -values back into the ****original**** function:

For $x = -1$:

$$f(-1) = (-1)^3 - 3(-1)^2 - 9(-1) + 15 = 20.$$

Point 1: $P_1(-1, 20)$

For $x = 3$:

$$f(3) = (3)^3 - 3(3)^2 - 9(3) + 15 = -12.$$

Point 2: $P_2(3, -12)$

Step 4: Build a sign-change table for $f'(x)$.

Construct intervals around $x = -1$ and $x = 3$:

x	$x < -1$	-1	$-1 < x < 3$	3	$x > 3$
$f'(x)$	+	0	-	0	+
Slope	$\nearrow$	$\rightarrow$	$\searrow$	$\rightarrow$	$\nearrow$

• At $x = -1$, $f'(x)$ changes from + to -.

• At $x = 3$, $f'(x)$ changes from - to +.

Step 5: State the final classifications.

Applying the test:

- $P_1(-1, 20)$ is a ****local maximum****.
- $P_2(3, -12)$ is a ****local minimum****.

Guided Thinking & Hints

Key Hint: This derivative function serves as your slope detector. Stationary points can only occur where this slope is exactly zero.

Key Hint: Always double-check your factoring! A simple sign error here will throw off the rest of your sign testing intervals.

Key Hint: Be disciplined with your substitutions. It is extremely common to accidentally plug these values back into the derivative (which will just output 0), instead of the original function $f(x)$!

Key Hint: Pick simple test integers to evaluate the sign of $f'(x) = 3(x - 3)(x + 1)$:
 * For $x < -1$, choose $x = -2 \implies (-)(-) = +$.
 * For $-1 < x < 3$, choose $x = 0 \implies (-)(+) = -$.
 * For $x > 3$, choose $x = 4 \implies (+)(+) = +$.

Key Hint: Summarize your findings in a clear concluding sentence. You have successfully mapped the peak and valley of this cubic curve!

Step-by-Step Solution

Step 1: Find the first and second derivatives.

Differentiate $f(x)$ with respect to x :

$$f'(x) = 3x^2 - 6kx$$

Differentiate a second time to get $f''(x)$:

$$f''(x) = 6x - 6k$$

Step 2: Locate the stationary points.

Set the first derivative to zero ($f'(x) = 0$):

$$3x^2 - 6kx = 0 \implies 3x(x - 2k) = 0$$

Solving for x , we find two stationary points at:

$$x_1 = 0 \quad \text{and} \quad x_2 = 2k$$

Step 3: Evaluate $f''(x)$ at $x_1 = 0$.

Plug the first stationary coordinate into $f''(x)$:

$$f''(0) = 6(0) - 6k = -6k$$

Since we are given that $k > 0$, it must hold that:

$$f''(0) = -6k < 0 \quad (\text{Strictly Negative})$$

Step 4: Evaluate $f''(x)$ at $x_2 = 2k$.

Plug the second stationary coordinate into $f''(x)$:

$$f''(2k) = 6(2k) - 6k = 12k - 6k = 6k$$

Since $k > 0$, it must hold that:

$$f''(2k) = 6k > 0 \quad (\text{Strictly Positive})$$

Guided Thinking & Hints

Key Hint: Treat the parameter k as a constant during differentiation. Notice that we cannot easily construct a numerical interval table because we do not know the actual numerical value of k !

Key Hint: Our critical coordinates are written in terms of the unknown parameter k . Because their positions are floating, we must use the Second Derivative Test to classify them.

Key Hint: A negative second derivative means the curve is concave down ($\cap$) at this point. Geometrically, this forces the stationary point at $x = 0$ to be a **local maximum**.

Key Hint: A positive second derivative means the curve is concave up ($\cup$) at this point. Geometrically, this forces the stationary point at $x = 2k$ to be a **local minimum**.

Step-by-Step Solution

Guided Thinking & Hints

Step 1: Calculate the first and second derivatives.

Differentiate once:

$$f'(x) = 3x^2 - 12x + 9$$

Differentiate a second time:

$$f''(x) = 6x - 12$$

Key Hint: To study concavity, we bypass the first derivative entirely after calculating it, focusing all of our analytical attention on the second derivative $f''(x)$.

Step 2: Find the candidate inflection points.

Set the second derivative to zero ($f''(x) = 0$):

$$6x - 12 = 0 \implies x = 2$$

Calculate the corresponding y -coordinate:

$$f(2) = (2)^3 - 6(2)^2 + 9(2) + 2$$

$$f(2) = 8 - 24 + 18 + 2 = 4$$

Key Hint: CRITICAL WARNING: Just because $f''(c) = 0$, it does not automatically guarantee a point of inflection! You must prove that the concavity actually switches signs across this boundary.

So, our candidate point is $(2, 4)$.

Step 3: Test concavity intervals around $x = 2$.

Our boundary $x = 2$ splits the domain into $(-\infty, 2)$ and $(2, \infty)$. Evaluate the sign of $f''(x) = 6x - 12$:

For $x < 2$ (e.g., $x = 0$):

$$f''(0) = 6(0) - 12 = -12 < 0 \implies \text{**Concave Down } (\cap) \text{**}$$

For $x > 2$ (e.g., $x = 3$):

Key Hint: By testing points on both sides of $x = 2$, we confirm a genuine physical transition from a downward arch to an upward bowl.

$$f''(3) = 6(3) - 12 = 6 > 0 \implies \text{**Concave Up } (\cup) \text{**}$$

Step 4: State the final intervals and points.

Summarizing our results:

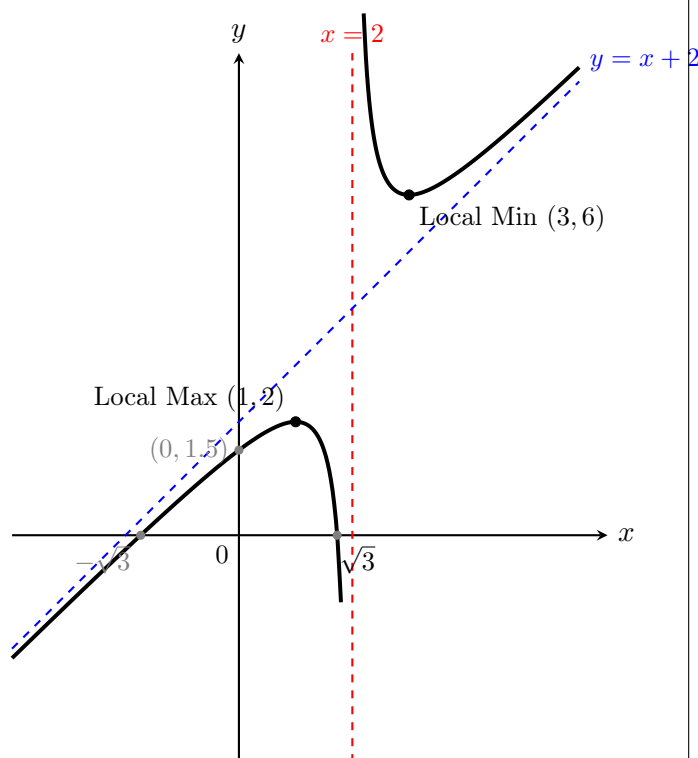
The curve is ****concave down**** on the interval $(-\infty, 2)$.

The curve is ****concave up**** on the interval $(2, \infty)$.

Key Hint: Notice how clean this cubic analysis is. All non-degenerate cubic curves have exactly one point of inflection, which sits perfectly halfway between their local maximum and minimum!

Since the concavity changes sign at $x = 2$, the coordinate **** $(2, 4)$ is a point of inflection****.

Mathematically Scaled Graph (TikZ)



Complete Analytical Steps & Hints

Key Hint: Step 1: Find Intercepts & Domain

Domain is $x \neq 2$.

- **y-intercept** ($x = 0$): $f(0) = \frac{-3}{-2} = 1.5 \Rightarrow (0, 1.5)$.
- **x-intercepts** ($y = 0$): $x^2 - 3 = 0 \Rightarrow x = \pm\sqrt{3} \approx \pm 1.73$.

Key Hint: Step 2: Track Down Asymptotes

- **Vertical:** The denominator goes to zero at $x = 2$, and $\lim_{x \rightarrow 2^\pm} f(x) = \pm\infty$.
- **Oblique:** Using polynomial long division:

$$f(x) = x + 2 + \frac{1}{x - 2}$$

As $x \rightarrow \pm\infty$, the fractional remainder vanishes. Thus, the slant asymptote is the line $y = x + 2$.

Key Hint: Step 3: Analyze Stationary Points

Use the Quotient Rule to evaluate the gradient:

$$f'(x) = \frac{x^2 - 4x + 3}{(x - 2)^2} = \frac{(x - 1)(x - 3)}{(x - 2)^2}$$

Set $f'(x) = 0 \Rightarrow x = 1$ or $x = 3$.

- $f(1) = 2 \Rightarrow (1, 2)$. Since the slope swaps from positive to negative, it is a local maximum.
- $f(3) = 6 \Rightarrow (3, 6)$. Since the slope swaps from negative to positive, it is a local minimum.

Key Hint: Step 4: Establish Concavity

Differentiate the gradient to get the second derivative:

$$f''(x) = \frac{2}{(x - 2)^3}$$

- For $x < 2$, $f''(x) < 0 \Rightarrow$ Concave down ($\cap$).
- For $x > 2$, $f''(x) > 0 \Rightarrow$ Concave up ($\cup$).

No point of inflection exists because $x = 2$ is an asymptotic gap, not a coordinate on the curve.

Step-by-Step Solution

Step 1: Find the critical points.

Differentiate $f(x)$ using the Power Rule:

$$f'(x) = 3x^2 - 6x$$

Set $f'(x) = 0$ to find the critical points:

$$3x(x - 2) = 0 \implies x = 0 \quad \text{or} \quad x = 2$$

Step 2: Evaluate the critical values.

Substitute the critical points back into the original function $f(x)$:

* At $x = 0$:

$$f(0) = (0)^3 - 3(0)^2 + 2 = 2$$

* At $x = 2$:

$$f(2) = (2)^3 - 3(2)^2 + 2 = 8 - 12 + 2 = -2$$

Step 3: Evaluate the boundary endpoints.

Calculate the function values at the boundaries $x = -1$ and $x = 4$:

* Left boundary $x = -1$:

$$f(-1) = (-1)^3 - 3(-1)^2 + 2 = -1 - 3 + 2 = -2$$

* Right boundary $x = 4$:

$$f(4) = (4)^3 - 3(4)^2 + 2 = 64 - 48 + 2 = 18$$

Step 4: Compare values and state the global extrema.

Compare all candidate heights:

$$f(0) = 2, \quad f(2) = -2, \quad f(-1) = -2, \quad f(4) = 18$$

* The largest value is 18.

* The smallest value is -2 .

Therefore, on the closed interval $[-1, 4]$:

- The **absolute maximum** value is **18** (at $x = 4$).
- The **absolute minimum** value is **-2** (at $x = -1$ and $x = 2$).

Guided Thinking & Hints

Key Hint: First, verify if both critical points lie within our closed interval $[-1, 4]$. Since $0 \in [-1, 4]$ and $2 \in [-1, 4]$, both are valid contenders for the tournament!

Key Hint: TOURNAMENT TIP: Do not waste time running the First or Second Derivative Tests here! We do not care if these points are local peaks or valleys—their ultimate status will be resolved purely by comparing their final heights.

Key Hint: Notice how large $f(4)$ is! Even though $x = 0$ is a local maximum, the right-hand boundary rocketed far past it to reach a height of 18.

Key Hint: We have a tie for the absolute minimum! That is perfectly fine; a function can achieve its global minimum at multiple distinct coordinates across an interval.

Step-by-Step Solution

Guided Thinking & Hints

Step 1: Locate the critical points on the interval.

Differentiate the function:

$$f'(x) = 1 - \frac{4}{x^2}$$

$$\text{Set } f'(x) = 0 \implies x^2 = 4 \implies x = \pm 2.$$

Since our domain is strictly the positive open interval $(0, \infty)$, the only valid critical point is:

$$x_c = 2$$

Calculate its height:

$$f(2) = 2 + \frac{4}{2} = 4$$

Key Hint: Discard any critical points that fall outside the open interval! Even though $x = -2$ is a critical point of the function, it is completely outside our domain $(0, \infty)$ and must be ignored.

Step 2: Evaluate the left open boundary limit.

As x approaches 0 from the right:

$$L_{\text{left}} = \lim_{x \rightarrow 0^+} \left(x + \frac{4}{x}\right) = 0 + \infty = \infty$$

Step 3: Evaluate the right open boundary limit.

As x approaches infinity:

$$L_{\text{right}} = \lim_{x \rightarrow \infty} \left(x + \frac{4}{x}\right) = \infty + 0 = \infty$$

Step 4: Compare values and state the global extrema.

Our only critical height is $f(2) = 4$.

The boundary limits are $L_{\text{left}} = \infty$ and $L_{\text{right}} = \infty$.

Key Hint: Since the left limit goes to positive infinity, the curve shoots straight up as it gets close to the vertical axis. This means **no global maximum** can exist on this interval!

Key Hint: Both open ends of our domain climb boundlessly to infinity. This creates a giant "U-shape" with a single turning point sitting at the bottom.

Comparing them:

$$4 < \infty \quad \text{and} \quad 4 < \infty$$

Therefore, on the open interval $(0, \infty)$:

- The **absolute minimum** value is **4** (at $x = 2$).
- There is **no absolute maximum** value.

Key Hint: Because the critical point is the only turning point on the interval and the boundary limits both go to infinity, our local minimum is guaranteed to be the global absolute minimum of the entire open interval!

Step-by-Step Solution

Step 1: Locate the interior critical points.

Differentiate $f(x)$ using the Product Rule:

$$f'(x) = 2xe^{-x} - x^2e^{-x} = xe^{-x}(2 - x)$$

Set $f'(x) = 0 \implies x = 0$ or $x = 2$.

Since our interior open interval is $(0, \infty)$, the only interior critical point is:

$$x_c = 2$$

Calculate its height:

$$f(2) = (2)^2e^{-2} = \frac{4}{e^2} \approx 0.5413$$

Step 2: Evaluate the closed left endpoint.

Evaluate the function at the solid left boundary $x = 0$:

$$f(0) = (0)^2e^{-0} = 0$$

Step 3: Evaluate the open right boundary limit.

Since the domain extends infinitely to the right, find the limit as $x \rightarrow \infty$:

$$L = \lim_{x \rightarrow \infty} x^2e^{-x} = \lim_{x \rightarrow \infty} \frac{x^2}{e^x} = 0$$

Step 4: Compare values and state the global extrema.

Compare our tournament contenders:

- Interior critical value: $f(2) = \frac{4}{e^2} \approx 0.5413$.
- Closed endpoint: $f(0) = 0$.
- Open boundary limit: $L = 0$.

Since $f(2) \geq 0$ and $f(2) \geq L$, the **absolute maximum** is $\frac{4}{e^2}$ (at $x = 2$).

Since $f(0) = 0$, and the limit L is also 0 (never dropping below 0), the **absolute minimum** is 0 (at $x = 0$).

Guided Thinking & Hints

Key Hint: Note that $x = 0$ is not an interior critical point since it sits exactly on our left closed boundary. We will evaluate it as an endpoint instead!

Key Hint: Because the left end is closed, the height 0 is an attained coordinate. It is a solid contender in our tournament!

Key Hint: By the growth hierarchy, the exponential denominator grows infinitely faster than the polynomial numerator, dragging the curve down to 0 as $x \rightarrow \infty$.

Key Hint: HYBRID RESOLUTION: The curve starts at $(0, 0)$, climbs up to a peak at $(2, 0.5413)$, and then decays slowly back down toward the horizontal axis as $x \rightarrow \infty$. It never dips below 0, making $(0, 0)$ the absolute minimum!

Step-by-Step Solution

Guided Thinking & Hints

Step 1: Define variables and write the equations.

Let x be the width of the pasture perpendicular to the wall, and y be the length parallel to the wall.

Objective (to maximize Area A):

$$A = x \cdot y$$

Constraint (total fencing available is 400 m):

$$2x + y = 400$$

Key Hint: Be careful with your perimeter equation! Since one side of the pasture is bounded by the stone wall, the fence only covers three sides of the rectangle: two widths x and one length y .

Step 2: Reduce the objective to a single variable.

Solve the constraint for y :

$$y = 400 - 2x$$

Substitute this into the Area formula:

Key Hint: By substituting the constraint, we have transformed our area equation from a two-variable system into a single-variable quadratic function that we can easily differentiate!

$$A(x) = x(400 - 2x) = 400x - 2x^2$$

Step 3: Establish the physical domain.

Width x must be positive: $x > 0$.

Length y must also be positive:

$$400 - 2x > 0 \implies 2x < 400 \implies x < 200.$$

So, our physical domain is the open interval:

Key Hint: Never skip setting up the domain! Knowing that x is restricted to $(0, 200)$ tells us exactly which global extrema interval test (the Open Interval Method) we must execute.

$$x \in (0, 200)$$

Step 4: Differentiate and find the critical point.

Differentiate the area function:

$$A'(x) = 400 - 4x$$

Set $A'(x) = 0$ to find the critical point:

Key Hint: Since 100 lies safely inside our open interval $(0, 200)$, it is a highly valid critical candidate.

$$400 - 4x = 0 \implies x_c = 100 \text{ meters}$$

Step 5: Verify the maximum and state dimensions.

Evaluate the boundary limits on $(0, 200)$:

$$\lim_{x \rightarrow 0^+} A(x) = 0 \quad \text{and} \quad \lim_{x \rightarrow 200^-} A(x) = 0$$

The critical height is:

$$A(100) = 400(100) - 2(100)^2 = 20,000 \text{ m}^2.$$

Since $20,000 > 0$, it is the absolute maximum.

Find length y :

Key Hint: FINAL CONCLUSION: The pasture achieves its absolute maximum area of

Step-by-Step Solution

Guided Thinking & Hints

Step 1: Construct the Revenue and Profit equations.

Revenue is Price multiplied by Quantity:

$$R(q) = q \cdot p(q) = q(100 - 0.3q) = 100q - 0.3q^2$$

Profit is Revenue minus Cost:

$$P(q) = R(q) - C(q)$$

$$P(q) = (100q - 0.3q^2) - (0.1q^2 + 20q + 1000)$$

$$P(q) = -0.4q^2 + 80q - 1000$$

Key Hint: Always distribute the negative sign carefully when subtracting the entire cost function $C(q)$. A simple sign error here is the most common way students lose marks on exam questions!

Step 2: Establish the physical domain constraints.

Production quantity cannot be negative: $q \geq 0$.

Price must remain positive:

$$100 - 0.3q > 0 \implies 0.3q < 100 \implies q < 333.3.$$

So, our physical domain is:

$$q \in [0, 333] \text{ (rounded to whole units)}$$

Step 3: Differentiate and find the critical point.

Differentiate the profit function with respect to q :

$$P'(q) = -0.8q + 80$$

Set $P'(q) = 0$ to find the critical point:

$$-0.8q + 80 = 0 \implies 0.8q = 80 \implies q_c = 100 \text{ units}$$

Step 4: Verify the maximum and calculate profit.

Evaluate the second derivative to verify the extremum:

$$P''(q) = -0.8 < 0 \quad (\text{Strictly Negative})$$

By the Second Derivative Test, $q = 100$ is a local maximum. Since it is the only critical point on our domain, it is the absolute maximum.

Calculate maximum profit:

$$P(100) = -0.4(100)^2 + 80(100) - 1000$$

Key Hint: Setting up the domain ensures we are working with physically meaningful quantities. We cannot sell a negative number of units, nor can we price them below zero!

Key Hint: This critical point represents the exact production volume where the marginal cost of producing one more unit perfectly matches the marginal revenue generated by selling it.

Key Hint: FINAL CONCLUSION: The manufacturer maximizes daily profit at ****3,000 HKD**** by producing and selling exactly ****100 units**** of smart home hubs per day.

Step-by-Step Proof

Guided Thinking & Hints

Step 1: Define the auxiliary difference function.

Rearrange the inequality to form a target of zero:

$$e^x - (1 + x) \geq 0$$

Define our auxiliary function $g(x)$:

$$g(x) = e^x - x - 1 \text{ on the domain } (-\infty, \infty)$$

Key Hint: By grouping everything on one side, we have transformed the abstract inequality into a straightforward task: find the absolute lowest point of this new curve $g(x)$.

Step 2: Find the critical point of $g(x)$.

Differentiate $g(x)$ with respect to x :

$$g'(x) = e^x - 1$$

Set the derivative to 0 to find the critical point:

Key Hint: This unique critical point is the only location on the entire real number line where the curve flattens out.

$$e^x - 1 = 0 \implies e^x = 1 \implies x_c = 0$$

Step 3: Establish that $x_c = 0$ is the global minimum.

We evaluate the limits at the open boundaries of $(-\infty, \infty)$:

$$\lim_{x \rightarrow -\infty} (e^x - x - 1) = 0 - (-\infty) - 1 = \infty$$

$$\lim_{x \rightarrow \infty} (e^x - x - 1) = \infty - \infty \rightarrow \infty \text{ (since } e^x \text{ dominates } x)$$

Key Hint: Alternatively, you can look at the sign of the derivative: * For $x < 0$, $e^x < 1 \implies g'(x) < 0$ (decreasing). * For $x > 0$, $e^x > 1 \implies g'(x) > 0$ (increasing). This forms a perfect "V-shape" valley with its bottom at $x = 0$!

Since both open boundaries rise to infinity, the unique critical point must be the global minimum.

Step 4: Evaluate the minimum value and complete the proof.

Calculate the height of the curve at our global minimum $x = 0$:

$$g(0) = e^0 - 0 - 1 = 1 - 1 = 0$$

Since the absolute minimum value of $g(x)$ is exactly 0:

$$g(x) \geq 0 \text{ for all } x \in \mathbb{R}$$

Substitute our original terms back in:

$$e^x - x - 1 \geq 0 \implies e^x \geq 1 + x$$

Key Hint: ELEGANT GRAPHICAL SENSE: Geometrically, this proof shows that the linear line $y = 1 + x$ is the tangent line grazing the exponential curve $y = e^x$ at $x = 0$. Since e^x is concave up, it must sit entirely on or above its tangent line!

This completes the proof.

Step-by-Step Solution

Guided Thinking & Hints

Step 1: Treat the variable as continuous.

Let x be a continuous real variable representing the number of passengers. Write the continuous profit function:

$$P(x) = -2x^2 + 75x - 150$$

Key Hint: We temporarily "relax" the integer constraint so that we can unleash our standard derivative rules on the quadratic expression.

Step-2: Find the continuous optimal point.

Differentiate the profit function with respect to x :

$$P'(x) = -4x + 75$$

Set the derivative to 0:

$$-4x + 75 = 0 \implies 4x = 75 \implies x_c = 18.75$$

Key Hint: Since the second derivative is $P''(x) = -4 < 0$, the continuous peak is a guaranteed maximum. However, we cannot book 18.75 passengers on a physical bus!

Step 3: Bracket the continuous peak with integers.

Locate the two whole numbers closest to the decimal peak 18.75:

$$n_1 = \lfloor 18.75 \rfloor = 18 \quad \text{passengers}$$

$$n_2 = \lceil 18.75 \rceil = 19 \quad \text{passengers}$$

Key Hint: Because the continuous function is a symmetric downward parabola, the true integer maximum must be either 18 or 19. We run a mini-tournament to find which one wins.

Step 4: Run the discrete tournament.

Evaluate the profit function at both integer boundaries:

* For $n_1 = 18$:

$$P(18) = -2(18)^2 + 75(18) - 150$$

$$P(18) = -2(324) + 1350 - 150 = 552 \text{ HKD}$$

* For $n_2 = 19$:

$$P(19) = -2(19)^2 + 75(19) - 150$$

Key Hint: Compare the results: $553 > 552$. Even though both are incredibly close, $n = 19$ yields slightly more profit because 18.75 is closer to 19 than it is to 18.

$$P(19) = -2(361) + 1425 - 150 = 553 \text{ HKD}$$

Step 5: State the final discrete optimal solution.

Key Hint: WARNING: Never blindly round to the nearest integer without evaluating! While rounding works for quadratics due to symmetry, it can fail on asymmetric transcendental or rational functions.

Therefore, the operator maximizes profit by booking exactly ****19 passengers****, achieving a maximum discrete profit of **553 HKD**.